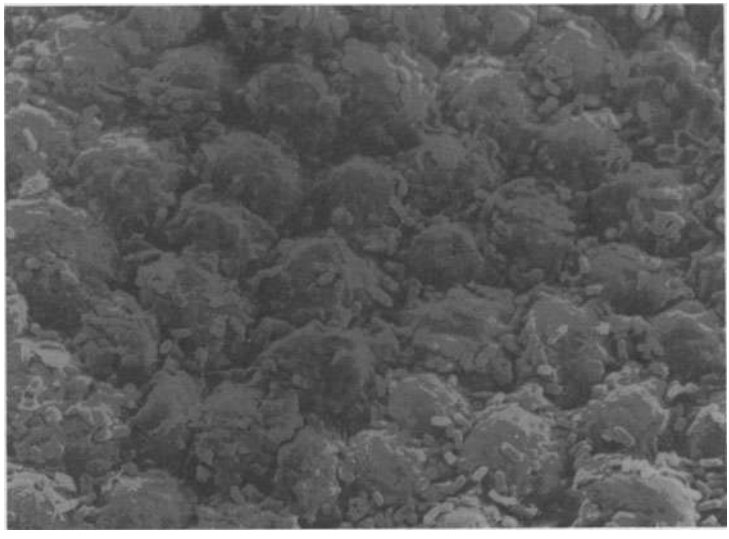
BASIC FOOD MICROBIOLOGY



Micrograph of bacteria growing on alfalfa sprouts.

BASIC FOOD MICROBIOLOGY

Second Edition

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Preface

The second edition of *Basic Food Microbiology* follows the same general outline as the highly successful first edition. The text has been revised and updated to include as much as possible of the large body of information published since the first edition appeared. Hence, foodborne illness now includes listeriosis as well as expanded information about *Campylobacter jejuni.*

Among the suggestions for altering the text was to include flow sheets for food processes. The production of dairy products and beer is now depicted with flow diagrams.

In 1954, Herrington made the following statement regarding a review article about lipase that he published in the *Journal of Dairy Science:* "Some may feel that too much has been omitted; an equal number may feel that too much has been included. So be it."

The author is grateful to his family for allowing him to spend the time required for composing this text. He is especially indebted to his partner, Sally, who gave assistance in typing, editing, and proofreading the manuscript.

The author also thanks all of those people who allowed the use of their information in the text, tables, and figures. Without this aid, the book would not have been possible.

1

General Aspects of Food

BASIC NEEDS

Our basic needs include air that contains an adequate amount of oxygen, water that is potable, edible food, and shelter. Food provides us with a source of energy needed for work and for various chemical reactions. Food also supplies chemicals for growth, for repair of injured or worn-out cells, and for reproduction. Food consumption can be a pleasurable experience, and a time for meeting with family and friends. Food is so necessary for our existence that the search for food has been the main occupation of human beings throughout history.

FOOD NEEDS

In the United States, supermarket shelves are so well-stocked that an uninformed observer might assume that our search for an adequate food supply has been successful. However, it is believed that 12.5 percent of the earth's people receive considerably less food than they need; as many as 50 percent might be receiving a marginal level of food (Kahn 1981). The reasons for this widespread hunger problem include unequal distribution of food and money, as well as cultural, religious, and superstitious beliefs.

The main problem facing the world today is its increasing population (Fig. 1.1). In recent years the birth rate has declined, but so has the death rate. The present world population is over 5 billion, and the annual increase is estimated to be 60 to 80 million people. Predictions for the future food supply range from very pessimistic, with famines expected to start in the near future, to the very optimistic viewpoint that there will be plenty of food for a population of almost limitless numbers. The most widely held opinion seems to be that, unless the rate of population increase can be substantially reduced, the demand for food will eventually outrun the supply, regardless of the efficiency of food production.

Although world agricultural production has been increasing, the vagaries of nature (floods, droughts, freezing, and other adverse climatic conditions) could cause a severe setback. With the expected increase in

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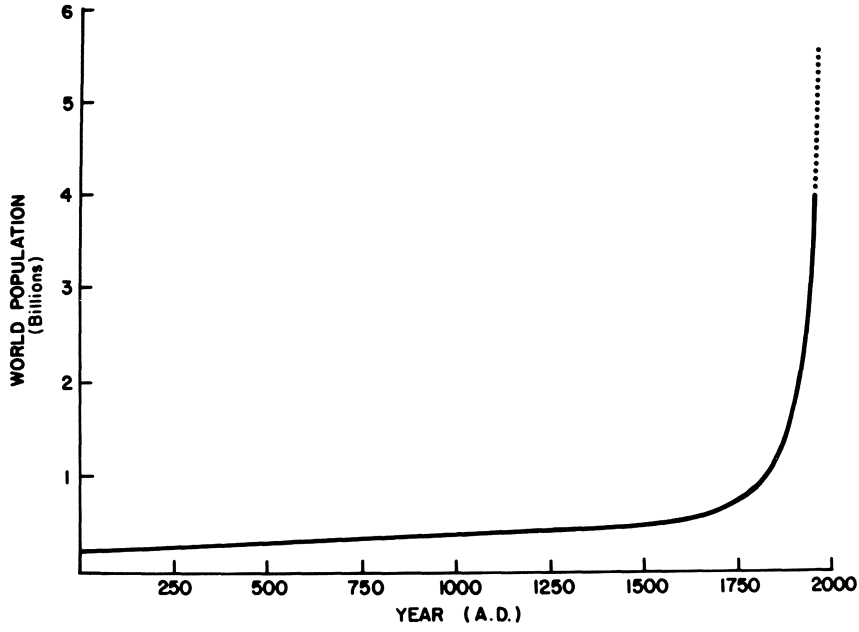

Figure 1.1. Estimated world population. Will we be able to feed everyone in the future?

population, the search for food will continue to be the primary endeavor of many food scientists, including food microbiologists.

SOURCES OF FOODS

Our supply of food depends upon the photosynthetic reaction between solar energy and plants that contain chlorophyll. Through photosynthesis, carbon dioxide and water are converted to glucose. Further cellular reactions produce the various organic compounds—carbohydrates, fats, proteins, and vitamins.

Both the amount of food and its dietary quality can be increased. To do this, we can improve the utilization of our land and fishery resources, upgrade the quality of plant proteins, convert waste materials into edible foods, and prevent losses or deterioration of our food supplies.

Land Resources

Increasing the productivity of land resources will require utilization of more land and higher yields of foods. However, there is a finite

amount of land, and not all of the land that is available can be used for agriculture. Even less can be used for crop production. Experts have estimated that the amount of land used for agriculture can be doubled. This will not necessarily double production. These added lands are of marginal quality and will require increased water, fertilizer, technology, and energy to make them productive.

One problem in American food production is the loss of good agricultural land for the construction of highways, airports, shopping centers, housing, and factories. With an increasing population, there will be an even greater demand for land for these purposes. The United States is now losing about 3 million acres of arable land per year to development.

Fishery Resources

Oceans and seas cover more than 70 percent of the earth's surface, yet in the United States less than 10 percent of our protein comes from fishery resources. It is likely that this resource could be utilized more effectively as a source of food for the future. As on land, the principal food-producing organisms in the sea are plants (phytoplankton). These plants use the photosynthetic process to produce food for herbivorous animals in the sea. These, in turn, furnish food for small carnivorous animals, larger fish, and ultimately human beings.

Microorganisms as Food

The use of microorganisms in food products is not a new idea. The action of yeast in the fermentations producing wine and beer and the leavening of doughs has been known for at least 4,000 or 5,000 years. The nature of the action in these fermentations did not become established until the latter part of the nineteenth century when the relation of living yeast cells to fermentation was discovered. Microorganisms are used in the fermentation of various foods and are consumed as part of these foods. This is especially evident in cheese. *Penicillium roqueforti,* the blue mold of Roquefort cheese, and *Penicillium camemberti,* the white mold of Camembert cheese, are consumed with the cheese. Therefore, the concept of using microorganisms as part of the food supply should not be completely objectionable.

Although we can synthesize amino acids and polypeptides commercially, microorganisms can produce not only these substances, but proteins, antibiotics, vitamins, steroids, and many other products. The development of microorganisms as a food source will involve many disciplines, but food microbiologists will play an especially important role.

Bacteria, yeasts, or molds cannot create foods, but they can grow on

cellulosic compounds that would otherwise be wasted. Algae can utilize solar energy to produce food. The production of microbial protein is discussed in Chapter 9.

Wastes as Food

For every kilogram of food produced, between 5 and 10 kg of waste materials are left in the field or at the processing plant. These substances are considered wastes because their economic value is such that it is not profitable to utilize them. Surprisingly, not long ago, fat was removed from milk, fish, and oilseeds for human use, while the more valuable protein was wasted or fed to animals. Hence, the "waste" products of today may have food value in the future.

Some wastes are not readily usable because they are seasonal, diluted with water, or require transportation to amass a large quantity for processing. With waste disposal becoming an ever-increasing problem, and food shortages becoming more critical, it is difficult to believe that the problems involved in utilizing acceptable wastes cannot be resolved.

Processes such as reverse osmosis and ultrafiltration can be used to concentrate dilute wastes (Moon 1980). Alternative systems for concentration of wastes in the meat industry have been discussed (Hansen 1983). Progress is being made in the recovery of wastes and in the utilization of these materials in human foods and in animal feeds (Cherry, Young, and Shewfelt 1975; Cooper 1976; Kamm et al. 1977; Knorr 1983; Toyama 1976).

Legal Aspects

U.S. food laws and regulations must be considered before any new or novel food can be sold. The U.S. Department of Agriculture (USDA) regulates red meat and poultry processing operations. In all other cases, the U.S. Food and Drug Administration (FDA) decides what is an acceptable food or food ingredient. Besides the dictates of these federal agencies, the food laws of states and local jurisdictions must be followed.

If an ingredient is to be used in a food, it cannot create a health hazard. The Food, Drug, and Cosmetic Act provides that a food shall be deemed to be adulterated if it consists in whole or in part of any filthy substances. What will be the FDA's reaction to single-cell protein obtained from microorganisms grown on unconventional substrates? If a food shortage does develop, we may need to reevaluate the aesthetic aspects of our foods. The health criteria for processed wastes have been discussed (Taylor et al. 1974).

Preventing Losses

The data on losses of plant and animal products are neither adequate nor reliable enough to allow us to conduct investigations to fully determine the causes of the losses. We do know that deterioration, waste, and loss occur in almost every step from production to consumption. Since we already have a shortage of food and will need more food in the future, we must make an increased effort to protect our food supply from further losses. If losses could be reduced or prevented, our food supply would increase with no additional utilization of land or sea resources.

The main losses of foods result from the action of microorganisms, insects, rodents, birds, nematodes, and the enzymes inherent in the food. One study (Ennis, Dowler, and Klassen 1975) estimated that 30 percent of crops worldwide are lost to pests. Albrecht (1975) suggested that pests destroy 25 percent of all crops in the United States. He believed that government regulations prevent adequate pest control. Schweigert (1975) estimated that food losses from production to consumption range from 20 to 50 percent. The monetary loss from producer to consumer is reported to be more than $30 billion in the United States.

Reasons for Food Preservation

Preservation can be defined as a process by which foods are treated to retard decay or spoilage. There are many reasons for preserving foods. Several plant foods are harvested only once each year. To have a supply of these foods throughout the year, rather than only at harvest time, preservation is necessary. In case of a crop failure caused by a natural disaster such as drought, wind, hail, flood, fire, freezing, or insect and disease infestations, or by human disasters, such as war, the preservation of previously produced excess food becomes paramount.

With preservation, one can obtain a more varied diet because a crop can then be used throughout the year and because crops native to only a small area can be transported and used anywhere in the world. One of the reasons developing countries have food shortages is that they do not have facilities for preservation and transportation of foods. Thus, certain areas have a temporary surplus of food while other areas have a shortage.

Flesh foods deteriorate rapidly if held at ambient temperatures. In some countries, although fish is plentiful in the coastal areas, there is a protein shortage inland because refrigeration and rapid transit are lacking to transport the fish protein without spoilage.

Preservation allows the holding of foods so that they can be used as ingredients for mixed foods. Many of our convenience foods are combinations of various foods. Some systems used to preserve food also destroy

many of the organisms and toxic factors that are hazards in food products.

METHODS OF FOOD PRESERVATION. The chief methods of food preservation can be listed in four basic categories: asepsis (preventing entry of microorganisms into foods); removing microorganisms; inhibiting growth by controlling the environment; and destroying microorganisms. Various systems have evolved from these basic procedures (Table 1.1).

Most methods of preserving food are merely modifications of systems used in ancient times. The addition of salt as a chemical preservative, fermentations, smoking, and cold storage have been practiced for over 2,000 years. Comparatively, canning might be considered a modern method, although the process of preserving food by putting it into a closed container and heat-treating it was patented by Appert in 1810. A much newer system for preserving food involves the use of radiation; at this time, however, irradiation has been approved by the FDA only for specific purposes (see Chapter 12).

FOOD HAZARDS

From the beginning of life until death, a person is subjected to potentially harmful environments. During one's lifetime, some hazards disappear and others take their place, so that the problems of safety are not static. The ingestion of food is no exception. Food may serve as a carrier of chemical and biological substances, either added or acquired as contaminants from soil, water, air, food handlers, equipment, and other sources. The possible subtle relationships between these substances in foods and physical vigor, mental alertness, longevity, resistance to infection, and the onset of degenerative diseases are not fully understood. Since we do not have all the answers regarding the safety of all possible substances, there are many controversies concerning the overall safety of

TABLE 1.1. FOOD-PRESERVATION METHODS

Asepsis	Gas or vacuum packing
Centrifugation	Acidification
Filtration	Fermentation
Refrigeration	Fumigation
Freezing	Pasteurization
Drying	Cooking
Freeze-drying	Canning
Chemicals	Radiation
Smoking	

food. Although absolute safety of food is an ideal goal, for all practical purposes, it is unattainable. Since a human being is a biological system, and since biological systems vary, a food that causes no ill effects in one person may cause problems in another person.

Wodicka (1977) listed six principal categories of food hazards: microbiological hazards, malnutrition, environmental contaminants, naturally occurring toxins, pesticides, and conscious food additives. Drug residues or filth in foods might be hazardous when ingested. In addition, mutagens and carcinogens may be formed when certain foods are heated (Grose et al. 1986; Jägerstad et al. 1983). Many chemicals that can be ingested relatively safely at low levels may be hazardous when ingested at high levels. Even a high-fiber diet may reduce the absorption of essential vitamins and minerals from the digestive tract.

Naturally Occurring Toxins

Several books and articles have been written concerning toxic agents naturally present in foods (Coon 1975; Elton 1981; Gori 1979; Hatfield and Brady 1975; Hironi 1981, Jadhav, Sharma, and Salunkhe 1981; Lewis and Endean 1983; McMichael 1984; Munro 1976; Onoue et al. 1983; Oppenheimer 1985; Panasiuk and Bills 1984; Shupe and James 1983; Swain, Truswell, and Loblay 1984; Taylor 1982, 1985; van der Hoeven et al. 1983; Wilson, McGann, and Bushway 1983). These naturally occurring toxins include estrogens, tumorigens, carcinogens, cyanogens, as well as seafood toxins, fungal toxins (mycotoxins), nutritional inhibitors, and antigens that produce allergies. The quantities of these substances in foods are usually low and, during processing, some of these substances are altered to reduce their potency.

Problems resulting from the natural toxicants are often due to people's eating raw foods or too much of only one type of food, or mistaking toxic plants for similar, edible plants. Potatoes contain the alkaloid solanine, a potent cholinesterase inhibitor that interferes with the transmission of nerve impulses. The amount of potatoes an average person eats each year contains enough solanine to be fatal if consumed in one dose.

At high levels, even polyunsaturated fats reportedly increase the incidence of tumors and gallstones, increase the body's requirement of vitamin E, and cause premature aging in laboratory animals. When excessively heated, these fats are reported to contain toxic substances. One product, malonaldehyde, is carcinogenic. Potentially carcinogenic lipid peroxides are easily formed from polyunsaturated fats by autoxidation.

Many substances not generally considered toxic may present a potential hazard to some people. For example, some consumers of milk develop severe distress of the digestive system because of an enzyme deficiency that results in an intolerance to lactose.

Microorganisms

Data compiled by the Centers for Disease Control (CDC 1981a, 1981b, 1983) show that microbiological hazards are by far the most common type of food hazard (Table 1.2). Since not all foodborne illnesses are reported, the CDC data are not exact, but they are the most complete data presently available. Each year, more than 60 percent of foodborne outbreaks are the result of bacterial etiologies, while less than 30 percent are due to chemicals from various sources. The CDC has discontinued listing foodborne outbreaks caused by unknown etiologies. In the past, such outbreaks accounted for 25 to 30 percent of the total number.

Although not listed as causing any foodborne outbreaks, mycotoxins, the toxins produced by molds, have received much attention in the past ten to fifteen years. They are considered to be naturally occurring toxins in foods.

Besides microorganisms, there are other biological entities that present a potential health hazard. *Trichinella spiralis* (the agent of trichinosis), *Taenia solium* (pork tapeworm), *Taenia saginata* (beef tapeworm), *Ascarias* (a roundworm) and *Entamoeba histolytica* (which causes amoebic dysentery) are a few of the agents that have been found in foods.

ROLE OF THE MICROBIOLOGIST

The food microbiologist is concerned with the biochemical reactions of microorganisms in and on foods. These reactions result in spoilage, public health hazards, and fermentation products. Determining the numbers and types of microorganisms associated with food, and knowing the sources of microorganisms, factors affecting their multiplication, and systems that can be used for their control are important to food microbiologists. However, we cannot look only at these aspects, but must also examine other facets of foods, such as their chemical and physical charac-

TABLE 1.2. CONFIRMED FOODBORNE OUTBREAKS, 1978–1980

	1978		1979		1980	
Cause	No.	%	No.	%	No.	%
Bacterial	105	68.2	119	69.2	136	61.5
Viral	5	3.2	6	3.5	12	5.4
SUBTOTAL	110	71.4	125	72.7	148	66.9
Parasitic	7	4.5	11	6.4	7	3.2
Chemical	37	24.0	6	20.9	66	29.9
TOTAL	154		142		221	

SOURCE: Data from CDC (1981a, 1981b, 1983)

teristics and the various attributes that are referred to as quality. We can sterilize a food to destroy all microorganisms, but if the process makes the food inedible or depletes its nutritional value, then the sterilization process is not satisfactory.

With an understanding of food science, a food microbiologist can better relate his or her role to the very important endeavor of providing all people with an adequate supply of safe, wholesome foods. The role of the food microbiologist in the food industry was discussed by Bauman (1982) and Winslow (1982).

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2

Estimating the Number of Microorganisms

An important aspect of food microbiology is the examination of food or other materials for microorganisms.

NUMBERS OF MICROORGANISMS IN FOOD

The number of microorganisms in a food as determined by the aerobic plate count (APC) is variable because of the original contamination, increase or decrease of microorganisms during processing, recontamination of processed product, and growth or death during storage, retailing, and handling. The microbial flora are changing constantly. In foods such as refrigerated fresh meat, the microbial numbers increase during storage, whereas in dried or frozen foods, the viable organisms tend to decrease in number. The APCs for a food may vary from less than 10 to over 100,000,000 microorganisms per gram, depending upon the product, how long it was stored, and the temperature of storage. The logarithms of the range of APCs reported for various foods are listed in Table 2.1. The microbial contents of various foods were reported by Pizzo, Purvis, and Waters (1982).

The usual range of organisms in most animal products is 1,000 to 10,000 per gram. Ground meat is more contaminated than whole cuts of meat because of the type of meat that is used in the product, the extra handling during grinding, and the release of meat juices that allow bacteria to multiply. Foods that receive a heat treatment generally have lower microbial numbers than foods not heated. Even then, poor-quality ingredients, poor sanitation, unsatisfactory heating, recontamination, or poor handling and storage, cause some heated products to have high numbers of microorganisms.

An estimate of the number of microorganisms in or on foods is needed in order to determine if a product meets the microbial levels expressed in specifications, guidelines, or standards. Spoilage of some foods is imminent when the APC reaches very high numbers (10^7–10^8/g). Hence, the microbial count can be used to help predict the shelf life of

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TABLE 2.1. AEROBIC PLATE COUNTS OF VARIOUS FOODS

Food	Overall Range[a]	Usual Range[a]
Animal Products		
Beef (steaks, roasts)	2–6	4
Beef (ground)	3–8	5–7
Pork sausage	4–6	5
Ham	1–8	4
Bacon	3–7	4
Dry sausage	3–7	4–5
Chicken carcasses (cm^2)	2–7	3–4
Fish (fresh)	2–8	4–5
Fish (smoked)	1–7	2–4
Fish sticks or crab cakes	2–6	3–4
Shrimp (raw)	2–7	4–5
Shrimp (raw, breaded)	2–8	4–6
Milk (raw, grade A)	2–5	3
Milk (pasteurized)	2–4	2
Milk (dry)	1–6	2–3
Butter	3–5	4
Plant Products		
Raw		
Almonds	0–4	3
Beans or peas	3–7	4–5
Broccoli or kale	6–7	
Carrots, potatoes, or spinach	4–7	
Corn or cucumbers	5–7	
Tomatoes	3–7	
Frozen		
Asparagus, beans, or peas	2–5	
Corn	2–7	
Squash	2–4	
Dried		
Carrots	2–4	
Garlic	4–6	
Parsley	2–5	
Spices		
Cinnamon	1–5	2–3
Cloves	2–3	3
Ginger	2–7	
Nutmeg	2–4	
Oregano	2–6	3–4
Pepper	6–7	7
Sage	3	
Mixed dried		
Soup (meat-type)	3–5	4
Soup (vegetable-type)	2–5	3–4
Salads		
Chicken or ham	1–7	3–5
Green	3–8	5–6
Macaroni	3–6	4–5
Shrimp	3–7	6
Tuna	2–6	3–4

[a] Reported in logarithms of bacteria per gram.

certain foods. To a limited extent, the microbial numbers might be used to evaluate the potential safety of foods. The count also might indicate if the product was produced under sanitary conditions, or if the product was mishandled during harvesting, processing, or storage. In general, as the microbial count increases, the quality of the food is reduced. This generalization does not apply to fermented foods, since microorganisms are used in their production. There are cases in which the number of microorganisms in a food has little or no relationship to potential shelf life, spoilage, or a health hazard. Other factors to be considered include the type of food, the type of microorganisms present, and the storage conditions.

To ensure production of food with a low number of microorganisms, the producer must assay not only the final food product but also such things as ingredients, processing equipment, packaging, and environmental samples. These determinations aid in the evaluation of general sanitary practices prevailing during processing and handling of food, and the potential sources of contamination. The determination of microbial numbers is needed to evaluate the effectiveness of methods of preservation.

The presence of particular types of microorganisms, especially potential pathogens or toxin producers, is more important than the estimate of the total number of microorganisms. In general, the main difference in these analyses is that specific types of microorganisms are determined with selective or differential media rather than with noninhibitory media. Thus, for purposes of simplicity, this discussion will be limited to total number estimations. Some of the special procedures are discussed with specific organisms in later chapters of this text.

Although the term *total count* has been used, no single method or medium is capable of detecting all of the microorganisms in a food. Thus, the counts that are obtained are merely estimates of the actual microbial population. Errors of ± 90 percent in counts are not unusual when the level is 10,000 to 100,000 per gram (Collins and Lyne 1976). Besides the errors, many assumptions are involved in microbial estimations. Also, there are factors that affect the growth of microorganisms and influence the results when the viability of the cells is involved in the enumeration technique. With all of these considerations, it is essential that the technician doing the testing does not further influence the results by using poor technique.

For microbial analysis, a sample and a system for estimating the number of microorganisms in the sample are needed. After the data from the evaluation are obtained, the information must be reported and, when necessary, follow-up checks should be made. If the report is for manage-

ment, an interpretation of the results might be included. What do they mean? Are the levels of microorganisms acceptable, or are they too high?

THE SAMPLE

If the samples are not delivered to the laboratory, it might be necessary to establish a sampling procedure. The samples of food might be obtained from the processing line, from warehouse storage, or from retail shelves. Food is processed as liquid, solid, mixed solid and liquid, or semisolid, and in many shapes and sizes. Since there are many variables in the food and many places of sampling, several sampling plans will be needed. Sampling suggestions have been made for various factors in food products (AOAC 1985; APHA 1984; Barrow 1983; FDA 1978; Jones 1979; Kilsby and Pugh 1981; Montagna 1982; Rao and Koehler 1979; Roberts, MacFie, and Hudson 1980; Schutz 1984).

The sampling plan should reflect the ultimate use of the analysis, the potential health hazard of the food, or potential for spoilage. If the results are needed to satisfy the requirements of a microbiological standard, the sampling plan as outlined in the standard should be followed. If the results are for the producer's information, a less restrictive sampling plan can be used. Sampling plans have been suggested for microbiological standards (Biltcliffe et al. 1983; ICMSF 1974; Martin 1979) and for salmonellae (health hazard) testing (Olson 1975). Further discussion of these sampling plans is presented in appropriate chapters of this text.

A sample will yield significant and meaningful information only if it represents the mass of material being examined, is collected in a manner that protects it against microbial contamination, and is protected from changes in the population that might occur between collection and analysis.

Representative Samples

The need for a representative sample cannot be overemphasized. The results of the analysis can be no more reliable than the sample on which they were based. Usually microorganisms are not distributed homogeneously, so thorough mixing of the product prior to sampling is important. Thorough mixing is not as easy for nonliquid foods as for liquid foods.

The size of the particles being sampled may influence the sampling procedure, since many particles of a product such as powdered milk can be obtained; but if the product were sides of beef, a different procedure would be necessary.

Sampling material in motion, such as on a production line, tends to minimize variables and gives a more representative sample than sampling material at rest, such as in stacks in a warehouse or on retail shelves. With on-line sampling, automatic sampling devices might be considered. These devices usually give a more random and reliable sample and at less cost than manual sampling of the product.

If cases or containers are stacked as a lot, the person collecting the samples must randomly select containers throughout the entire pile. If only containers around the edges or in front of the stack are selected, he or she is introducing a bias into the results of the analysis.

The laboratory analysis is usually more expensive than obtaining the sample, so cutting corners in sampling is not the way to save money.

Number of Samples

The number of samples needed, or the frequency of sampling, depends upon many factors. The uniformity or homogeneity of the product, the size of the many particles, previous knowledge of the material, and experience will help dictate the amount of sampling needed. Either too few samples or too many samples waste product, laboratory material, and labor.

For microbiological standards, the number of samples to be obtained and analyzed is included in the standard. One of the prime considerations that influence the number of samples to be analyzed is the potential health hazard of the foods. Statistical sampling schemes will help ensure that the samples give an acceptable assessment of the microbial conditions of the food, ingredient, or other substance being analyzed.

Aseptic Collection of Samples

Aseptic technique is needed when samples are collected. To prevent possible contamination, if the samples are in individual containers, such as cans, bottles, or boxes of food, they should be taken directly to the laboratory for analysis. On the other hand, if the product is in bulk or in containers of impractical size to submit directly to the laboratory, representative portions must be transferred to sterile containers using aseptic technique.

Since there is little interest in bacteria associated with sampling devices or sample containers, the instruments must be sterile. If possible, the instruments should be sterilized in the laboratory rather than at the place of sampling. After the sampling equipment is cleaned, the preferred methods of sterilization are (1) steam at 121.5°C in an autoclave for 15 to 30 min (the time for exposure depends on how bulky the mate-

rial is and how closely the material is packed in the chamber), or (2) a hot air oven. The suggested conditions for hot air sterilization vary from 1 to 3 hr at 160° to 180°C. If protected from recontamination, the sterilized instruments may be stored. Alternative systems for sterilizing are needed when neither an autoclave nor a hot air oven is available. These include (1) expose to steam (100°C) for 1 hr and use the same day, (2) immerse in water at 100°C for 5 min and use immediately, (3) immerse in 70 percent alcohol and flame to burn off alcohol immediately before use, or (4) flame with hydrocarbon (propane or butane) torch so that all working surfaces contact the flame before use. Using the alternative systems has been questioned. According to the FDA (1978), alcohol flaming is unsatisfactory because the instrument does not get hot enough to be effectively sterilized, and the flaming alcohol creates a fire hazard. The FDA recommends using a propane torch. Tansey (1973) suggested using a heavy-duty butane lighter rather than an unwieldy torch.

When obtained, the sample should be placed in a sterile container. A wide-mouth screw-capped jar is recommended (APHA 1984; FDA 1978); however, plastic bags or other acceptable containers can be used.

The methods of sampling and the types of instruments needed are determined by the substance to be sampled.

LIQUIDS AND SMALL PARTICLES. These foods can be mixed and sampled with sampling tubes, dippers, teaspoons, tablespoons, spatulas, or similar instruments.

LARGE MATERIALS. If these substances can be cut, they may be sampled with a knife or cheese trier. For many materials, such as animal carcasses or processing equipment, the surface is sampled.

SURFACES. Since microorganisms are on the surfaces of equipment as well as on foods such as animal carcasses, the sampling and analysis of surfaces are important. The system to use for sampling depends upon the type of surface, the amount of contamination, and the use of the data that are obtained. Some of the surface sampling systems that have been used are listed in Table 2.2. Each system has certain advantages and disadvantages. No single method is the best for all of the diverse surfaces of foods and equipment. Hence, several have been proposed and compared (Dewhurst, Rawson, and Steele 1986; Dickens, Cox, and Bailey 1986; Goulet et al. 1983; Lee and Fung 1986; Scott, Bloomfield, and Barlow 1984; Speers, Lewis, and Gilmour 1984). Microorganisms become attached to surfaces, which makes them difficult to recover for analysis. Excision and maceration of tissue yield higher numbers of microorganisms than do systems such as swabbing, rinsing, or contact agar or tape

TABLE 2.2. SURFACE SAMPLING METHODS

Swab	Contact systems
Cotton	Agar-syringe
Alginate	Agar-sausage
Glass sampler	Agar plate (RODAC)
Cylinder template	Tape
Scrape	Membrane filter pad
Excise tissue	Agar spray
Wash-rinse	Drip or exuded juice
Vacuum probe	Abrasive discs

(Anderson et al. 1987; Lillard and Thompson 1983; Morgan, Krautil, and Craven 1985).

AIR. The two general methods for air sampling are solid and liquid impingement. The systems for solid impingement include the settling plate, slit samples, the sieve or Anderson sampler, the centrifugal sampler, and the membrane sampler. Except for the settling plate, specific volumes of air are sampled. Systems of air sampling have been evaluated and compared (Lundholm 1982; Macher and First 1983; Placencia et al. 1982).

Holding of Sample

For best results, the sample should be analyzed immediately. When this is not possible, the sample should be refrigerated to prevent growth of any microorganisms. Alternatively, the sample can be packed in ice. If shipment to another city is necessary, or if the sample is a frozen product, dry ice should be placed in the package. Refrigeration is preferred to freezing, because freezing may cause death or damage to some cells, which may then give erroneous results when the sample is analyzed.

Preparation of Sample

Many methods of analysis require some preparation of the sample. The main consideration is to get the bacteria into a homogeneous suspension so they can be pipetted. If a food is a liquid such as milk, an aliquot can be mixed and pipetted, but if the food is a solid, such as hamburger, it is necessary to blend the food with a diluent to obtain a suspension. The rinse or wash samples from surfaces are treated as liquid samples, while swabs are placed in sterile diluent and shaken to suspend the bacteria.

SOLID FOOD. Solid food is generally mixed with a sterile diluent in a sterile mechanical blender or other system to obtain a 1:10 dilution of the food (Fig. 2.1). This 1:10 dilution also is referred to as 1/10 or 10^{-1} dilution. A 1:10 dilution means that in 10 g of the mixture, there is 1 g of food, or in 1 g of the mixture, there is 0.1 g of food, with associated organisms. Thus, if 1 g of the 1:10 dilution is analyzed, the microbial count is that of 0.1 g of food. To report the count as the number per gram, it is multiplied by 10. The first dilution of the food may be 1:2 or 1:4, such as for shellfish (APHA 1970; Cook and Pabst 1984).

As an alternative to blending, a sterile plastic bag containing the sample and diluent is placed in a device called a stomacher (Fig. 2.2). In the stomacher, the compression and shearing forces of the pounding result in a homogeneous suspension of sample and microorganisms (Deibel and Banwart 1982; Purvis et al. 1987; Sharpe and Jackson 1972; Thomas and McMeekin 1980; Thrasher and Richardson 1980).

Various systems might be used to make further dilutions of the blended food sample. One is the loop-tile method (Hudson, Roberts, and Whelehan 1983).

DILUENTS. Several diluents have been suggested and used. Although AOAC (1985) recommends the use of Butterfield's buffered phosphate, 0.1 percent peptone water is also accepted. Peptone water is easy to prepare, and it protects the organisms during dilution and plating. One disadvantage of this preparation is that, if the prepared dilution is allowed to remain at room temperature for extended periods, the organisms will multiply. Not more than 20 min should elapse between the first dilution in phosphate buffer until the last plate is poured in the series (APHA

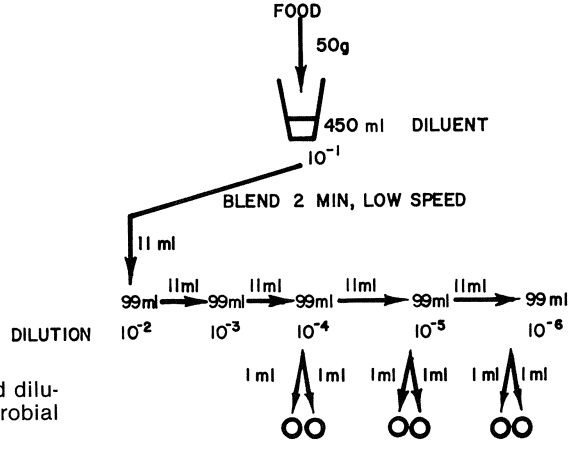

Figure 2.1. Suspension and dilution of food sample for microbial analysis.

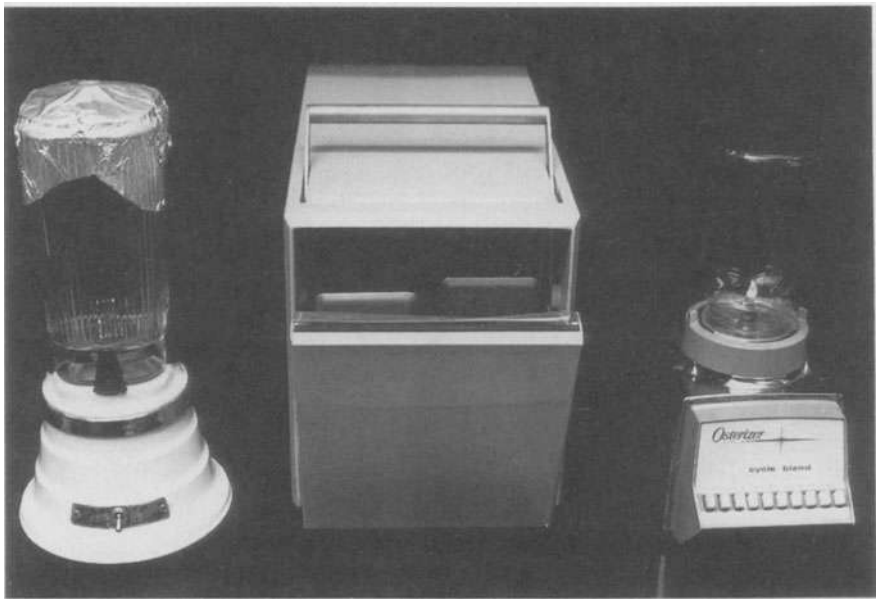

Figure 2.2. Systems for blending food samples with diluents. *From left to right:* Waring blender, stomacher, Osterizer.

1984). An increase in count up to 10 percent can be expected in this 20-min period.

According to Harrewijn (1975), some pertinent aspects to be considered are the composition, temperature, and pH of the diluent; anaerobic or aerobic condition; carryover of inhibitors with the food; and any treatments needed to allow the recovery of cells injured during food processing or preparation of the sample. The recovery of injured waterborne coliforms is aided by diluents containing peptone or milk (McFeters, Cameron, and Le Chevallier 1982).

The soaking of mustard seeds in a sterile diluent for 10 min prior to analysis resulted in an increased aerobic plate count (Cowlen and Marshall 1982). Perhaps other such samples should be soaked before analysis.

DILUTIONS NEEDED. For the plate count, only plates with less than 250 or 300 colonies are considered to be countable (FDA 1978; Tomasiewicz et al. 1980). Hence, for moderately to highly contaminated foods, dilutions are needed beyond the 1:10 ratio of the original suspension.

A 1:10 dilution of a 1:10 dilution is a 1:100 dilution. This 1:100 dilution is prepared by aseptically transferring 10 ml of the 1:10 dilution

to a screw-capped bottle containing 90 ml of sterile diluent (or 11 ml transferred to 99 ml). The bottle is shaken (twenty-five times through a 30-cm arc in 7 sec) to distribute the organisms homogeneously. Further dilutions can be made in this manner as far as needed.

The dilutions needed to estimate the number of microorganisms in a food can be determined by experience or by the requirements of standards, guidelines, or specifications. If 50,000 organisms per gram are allowed in a specification, a 1:1000 dilution can be used for the plate count. If fewer than fifty organisms are observed on the incubated plate, the food is within the limit, but if more than fifty colonies are observed, it does not meet the requirement. Usually two or three dilutions are analyzed to increase the chances of obtaining an acceptable plate to count; for the most probable number (MPN), at least three dilutions are needed.

ANALYSIS

Several procedures can be used to estimate a microbial population (Table 2.3). Not all of these procedures are readily adaptable to all foods, however. The ideal test should be accurate, rapid, inexpensive, and useful for most types of samples.

TABLE 2.3. SYSTEMS TO ESTIMATE THE MICROBIAL LOAD OF FOOD

Direct microscopic count (DMC)	Electrical
Breed clump count	Conductance
Electronic particle count	Impedance
Pour plate (APC, SPC)	Capacitance
Spread plate	Voltage drop
Spiral plate	Spectrophotometric (optical density)
Drop plate	Adenosine triphosphate (ATP)
Plate loop	Reductase tests
Roll tube	Easicult-TTC
Oval tube	Respiration rates
Burri strip/slant	*Limulus* amoebocyte lysate
Little plate	Chemical indicators (decomposition products)
Tube dilution	pH
Most probable numbers (MPN)	Agar droplets
Membrane filter	Millipore sampler
Hydrophobic grid (HGMF)	Bactoscan
Direct epifluorescent filter technique (DEFT)	Microcalorimetry
Microtiter—Spot plate	Flow cytometry
Dry rehydratable film	
Petrifilm	

NOTE: For the systems not discussed in text, see the following references: Ackland, Manvell, and Bean 1984; Bailey and May 1979; Ginn, Packard, and Fox 1984; King and Mabbitt 1984; Kramer 1977; O'Toole 1974; Schoon et al. 1970; Sharpe and Kilsby 1971; Swientek 1981, 1983.

Total Cell Counts

Some systems make no differentiation of living or dead cells. All microbial cells are counted. Two of these are the direct microscopic count and the electronic particle count.

DIRECT MICROSCOPIC COUNT (DMC). With this method, the results are obtained sooner than with most other procedures because no incubation period is needed for the cells to metabolize and multiply.

Liquid foods may be determined directly, but solid foods must be put into a suspension (1:10 dilution) before analysis. A counting chamber can be used, but for food, usually a portion (0.01 ml) of the material (measured with a standard loop or micropipet) is spread uniformly over a prescribed area on a glass slide (usually 1 cm^2).

For products such as eggs or cream, xylene or another suitable solvent is added prior to staining to remove the fat from the material. After drying, the slide is then fixed by dipping in ethyl alcohol for 1 to 2 min before staining.

Several stains have been suggested for and used in the DMC. The stained films are examined with a microscope, using the oil-immersion objective. The number of fields to be examined and counted is inverse to the number of cells and clumps observed in each field.

To calculate the organisms per gram of food, the diameter of the field that is examined must be known. The diameter (d) is measured with a stage micrometer to the nearest 0.001 mm. Since the field is a circle, the area can be calculated ($A = \pi r^2$).

The average number of cells or clumps per field is calculated and divided by the area of the field to obtain the number per mm^2. To determine the number of cells or clumps per cm^2, the number per mm^2 must be multiplied by 100 (since there are 100 mm^2 per cm^2). The resultant number is then multiplied by the dilution factor, which, in the case of liquid food (milk) is 100 (0.01 ml was used), or 1,000 for solid food (0.01 ml of a 1:10 dilution).

Values of the DMC. The DMC is a rapid method because an estimate of the bacterial load can be obtained in a short time. This is of value when on-the-spot alterations or adjustments are needed in the processing operation to remedy any problems.

Other values of the DMC that have been suggested include the following: (1) little work is required; (2) the test is not too difficult; (3) very little apparatus or equipment is needed except for a microscope; (4) the prepared slide can be stored and maintained as a permanent record; (5) some idea as to the type of organism (cocci or rods) is obtained; (6) counts represent organisms in the original product (if it has been treated, such

as by heat); (7) preservatives can be added to the sample for holding prior to analysis, for shipment, or to hold for futher study, so that organisms do not multiply; (8) only a small amount of sample is needed, which is of value if the product is expensive.

Since both living and dead cells are counted, there is some question as to the value of the microscopic count. However, high numbers of cells, whether living or dead, in pasteurized products indicate poor quality of product before processing, survival or multiplication of bacteria during processing, or recontamination or growth, or both, after processing.

The value of the DMC is limited to samples with high cell loads. However, with increased technology and efforts in sanitation, the bacterial load in many foods has been reduced. The DMC has little or no value for foods with low microbial loads.

Besides being used to evaluate the microbial content of a food, the DMC can be used to evaluate the number of body cells (leucocytes or lymphocytes) in milk. This is especially valuable in indicating mastitis in cows.

The value of the DMC depends upon the type of food and the type of organisms associated with the food. For products that have received a treatment such as heat to control the microorganisms, it is doubtful that the DMC could predict shelf life of the product. It is also doubtful that the DMC would have any value in determining the public health hazard of the product.

Assumptions and Errors. In any microbiological method of analysis, many assumptions and errors are made. There are errors inherent in the analytical procedure, as well as those introduced by the technician doing the test.

Assumptions are also made regarding the sample. It is assumed that (1) the sample is representative of the entire lot of product; (2) the subsample used for analysis is representative of the sample; (3) cells are distributed homogeneously in the sample as well as the subsample and, if not originally homogeneous, that operations such as mixing, blending, or shaking have produced a homogeneous mixture; (4) the weighing or measuring of the subsample, diluents, and aliquots is accurate; and (5) the sample or subsample has been handled so that there is no contamination or multiplication of cells during sampling or analysis.

Errors in the DMC can occur during the preparation and staining of the cells, counting the cells or clumps, or in the calculations involved in converting the raw data into the count per gram of product.

It is easier to spread the sample uniformly in circles than in squares. If the material is not spread uniformly, the cells will not be distributed homogeneously. It is assumed that the smear on the slide will dry into

flat layers of uniform density. However, the smears have been found to vary in thickness from one area to another. For some foods, such as liquid egg, the film on the slide may be of such thickness that it obscures many bacterial cells, which cannot then be counted. Organisms that are lightly stained are difficult to discern, and with an unevenly stained background, it is difficult to distinguish dirt or other particles from bacterial cells. Improper illumination with resulting eye strain and fatigue is another cause of errors in cell counting. The staining process itself can also wash some cells from the slide or can result in the counting of precipitated stain as cells.

For other errors, see APHA (1978).

ELECTRONIC PARTICLE COUNT. The electronic counter is based on the principle that cells are poor electrical conductors as compared to an electrolyte solution. A dilute suspension of cells in saline or other suitable electrolyte is drawn through a minute aperture conducting an electric current between two electrodes—usually platinum. Each cell passing through the aperture displaces an equal volume of the electrolyte solution and causes a momentary increased impedance to the flow of electric current. The resulting voltage pulse is proportional to the size or volume of the particle passing through the aperture. These pulses are amplified and counted. They simultaneously appear on the screen of an oscilloscope.

Since background particles, such as those that occur in foods, also would produce pulses as they pass the aperture, or could clog the aperture, the particles must be removed. Although electronic particle counters are used successfully to count somatic cells in milk (Dijkman et al. 1979), blood cells, and mammalian cells, much work needs to be done before they are useful in determining microorganisms in foods. However, Kogure and Koike (1987) reported satisfactory results when a particle counter was used to determine the bacterial biomass of seawater.

Viable Counts

Several methods exist to estimate the number of viable microorganisms in foods. Most of the systems are based on the plate count or tube dilution methods.

PLATE COUNT (POUR). The standard plate count (SPC) has been the usual technique for estimating the living microorganisms in foods. The procedure is relatively simple. Appropriate dilutions are plated immediately by transferring a measured aliquot to a sterile petri plate and add-

ing sterile melted and cooled (42° to 45°C) agar. The type of agar used for the SPC is noninhibitory and nutritious, unless specific microbial types are to be determined. For the aerobic plate count the media usually used are plate count agar or tryptone glucose extract agar, but various other agars have been employed. The medium and inoculum should be mixed thoroughly to distribute the cells uniformly. After solidification of the agar, the prepared plates are inverted (turned upside down) to prevent condensation of moisture on the agar surfaces, and then incubated. The temperature and time of incubation will vary, depending upon the type of cells that are being determined (psychrotrophs, mesophiles, or thermophiles). A temperature of 32°C for three days is used for eggs and egg products, while 35°C for 48 ± 2 hr is listed for frozen, chilled, or prepared foods (AOAC 1985). Huhtanen (1968) found the highest bacterial counts in raw milk when the plates were incubated at 27°C. However, these counts were not significantly different from those obtained at a range from 10° to 30°C.

During the incubation period, growth and multiplication of cells will occur until a visible colony is formed. These colonies are then counted on the plates that contain from 30 to 300 colonies (AOAC 1985; FDA 1978). Cowell and Morisetti (1969) furnished evidence that greater precision is obtained from plates containing from 80 to 320 colonies. A range of 25 to 250 colonies was suggested in a later study (Tomasiewicz et al. 1980). The number of colonies is multiplied by the dilution factor and reported as the number of colony-forming units (CFU) per gram of food.

Desirable Characteristics. The pour plate procedure is simple, can cover a large concentration range, and, at present, is probably the most precise method for determining those bacteria that will grow in an agar medium (Gilchrist et al. 1973). Besides these virtues, the organisms can be recovered for further study. The results should reflect the level of viable microorganisms in the food at the time of sampling.

The data obtained from the pour plate should reveal information such as the source of microorganisms, potential shelf life, or possible public health hazards of the product. With the present aerobic plate system, the source of microorganisms generally is not determined (Blankenagel 1976).

In most foods, microbial growth causes undesirable changes. Hence, the plate count might be used as an indicator of potential shelf life or of incipient spoilage. No relationship was found to exist between the bacterial count and potential shelf life of iced shrimp (Cobb et al. 1973), or pasteurized milk (Watrous, Barnard, and Coleman 1971). The usual plate count system does not differentiate types of organisms that cause spoilage.

It is generally agreed that any potential health hazard is not determined by the aerobic plate count. Some people believe that a high microbial count indicates improper handling with possible pathogens being present. Quite often the reverse is true, and low-count products contain potential pathogens. Microbial toxins can be present after the bacteria are destroyed by processing.

Undesirable Characteristics. There are many facets of the pour plate system that are undesirable. Of most concern are time, expense, technical requirements, information obtained, and accuracy.

The prepared plates must be incubated so that the organisms can produce a visible colony prior to counting. This incubation period may range from two to ten days. For highly perishable products, or for determining production or processing conditions, it is desirable to obtain the results as soon as possible. If a ten-day incubation period is needed, the potential shelf-life of a food can be determined more easily by incubating the food directly.

Since the pour plate system is so common in the United States, we might not realize that it is rather expensive compared to other methods. In some countries, other, less expensive methods are used in preference to the plate count.

The pour plate method seems simple to do, but a trained technician is needed to perform the test. The accuracy of the pour plate depends upon the ability of the technician as well as on assumptions and errors inherent in the technique.

Assumptions and Errors. The same assumptions and errors in sampling as discussed for the DMC apply equally to the pour plate. The technical ability and concern of the technician during cleaning of glassware, preparation of dilutions and media, sampling, plating, counting, and calculating can influence the reported CFU.

Two major assumptions of the pour plate system are that (1) microorganisms are in suspension as dissociated single-cell units so that each colony on the plate arises from an individual cell; and (2) all cells planted in the medium will multiply and produce a visible colony. Neither assumption is accurate.

Quite often bacteria grow in chains or clusters. Mixing, shaking, and other procedures do not always separate these chains or clumps into individual cells. Hence, when plated, a colony may arise from not only one, but several bacterial cells.

The environment in which the organisms are placed (medium, temperature, oxygen) as well as previous treatments of the cells (sublethal heating, freezing, radiation) and even the presence of other types of microorganisms influence the ability of the cell to multiply and produce a

visible colony. No one environmental condition will support the growth of all of the types of microbial cells that might be present in a food product. Hence the bacterial count should be referred to as CFU per gram of food.

The main value of a plate count is to be able to compare the results of various samples taken at different times from the different laboratories. This is possible only when the results are reproducible. It is important that standardized procedures be followed so that results can be compared.

PLATE COUNT (SURFACE). In this system, the sterile melted and cooled agar is poured in sterile petri plates. After solidification, the plates are preincubated overnight. The incubation dries the surface of the agar so that, when planted, the organisms do not coalesce. Before using, the dried agar surface should be observed for any possible contamination.

Aliquots of dilutions are added to the dry surface and uniformly spread over the agar by means of a sterile glass rod, bent into the shape of a hockey stick. Various amounts of aliquots have been suggested. Inasmuch as we usually work with dilutions in the order of 10, it is much easier to calculate the results per gram of product if 0.1-ml aliquots are used.

For simplification, calibrated loops can be used in place of pipets for preparing dilutions as well as for inoculating the pour plate or the surface of the spread plate. In the plate loop system, a calibrated loop is fitted into the barrel of a repeating syringe. The loopful of sample is flushed onto the agar surface in a petri plate with sterile diluent in the syringe. According to O'Connor (1984), the precision and accuracy of the plate loop system are within acceptable limits.

With the drop plate method, 0.02 ml of inoculum is allowed to drop onto the surface so that it spreads over an area 1.5 to 2.0 cm in diameter. Six to eight drops are placed on an agar surface in a petri plate, with no further manual spreading. After inoculation, the plates are inverted and incubated, and the resultant colonies counted as with the pour plate method.

Automated devices for distributing the samples over the agar surface have been described and evaluated (Gilchrist et al. 1973; Gilchrist et al. 1977; Jarvis, Lach, and Wood 1977; Kramer, Kendall, and Gilbert 1979; Tilton and Ryan 1978; Trotman and Byrne 1975). One type of automatic plating system is the spiral plater (Fig. 2.3) (Gilchrist et al. 1973; Gilchrist et al. 1977). This system was adopted as an official method by the AOAC (1981). With this mechanical device, a stylus dispenses the sample, or dilution, in a spiral, in varying amounts, on an agar surface from the center

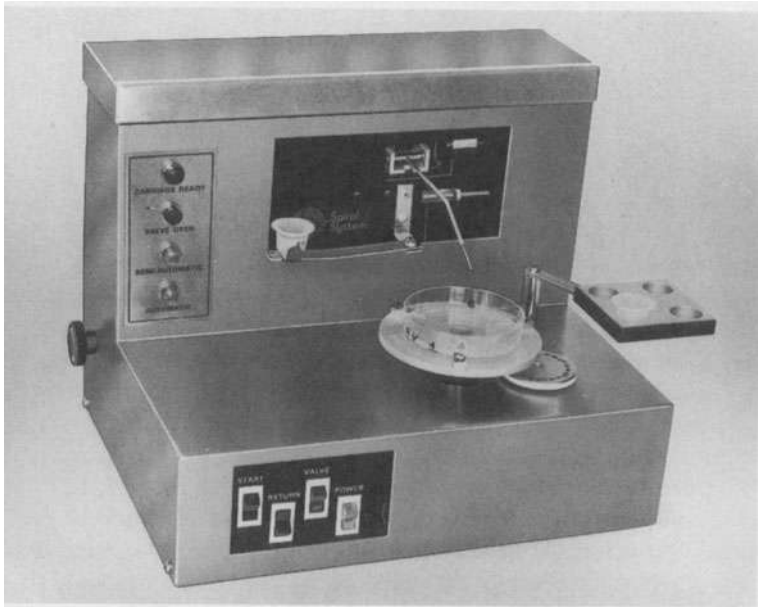

Figure 2.3. The spiral plater.

of the dish to the outer edge. By varying the amount of inoculum, the equivalent of three dilutions can be plated on one agar surface. After incubation of the inoculated plates, a laser colony counter, developed for the spiral plater, follows the spiral from the outer edge toward the center, counts the colonies, and determines the CFU for the inoculum. Also, the colonies can be counted manually with the use of a spiral grid system.

Surface vs. Pour Plates. The desirable aspects listed for the pour plate are equally applicable to the surface plate.

It is well recognized that higher counts are obtained by surface spread plates than by pour plates. The possibility of heat-sensitive organisms being damaged by hot agar during the preparation of pour plates is overcome by using the spread plate technique. Obligate aerobic organisms will grow faster on the surface than in the depth of agar in pour plates. Surface colonies are always detectable sooner and are much larger and easier to count than are colonies in a pour plate.

The main advantage of surface plates as compared to pour plates is that surface plating can be automated. With automated analyses such as the spiral plate system, both work and materials are saved. One problem with the spiral plate system is that the stylus is easily clogged if food

particles are present in the sample. Hence, filtration of the sample may be needed to remove these particles prior to plating. Hoben and Somasegaran (1982) found the drop plate method to be more economical than either the spread or pour plate systems.

The undesirable characteristics of the spread plate are similar to those discussed for the pour plate. With the spread plate system, some of the organisms might cling to the glass rod used for spreading. Treating of the glass rod with silicone helps to overcome the problem. Reportedly, precision with the pour plate is better than with the spread plate.

Dry, Rehydratable Film. As an alternative to the petri plates used in the aerobic plate count systems, plastic films with a dry, rehydratable medium coated upon them (Petrifilm SM) have been developed. The dry medium contains nutrients, a cold water–soluble gel, and 2,3,5-triphenyltetrazolium chloride that is reduced by microbial growth from white to red. The prepared samples or dilutions are added at the rate of 1.0 ml per plate. The sample is spread over an area of about 20 cm^2 by applying pressure with a plastic spreader on the overlay film. The liquid in the sample rehydrates the medium, then the gel is allowed to solidify before the prepared films are incubated for bacterial growth. Reportedly, the Petrifilm SM system was a satisfactory alternative to the aerobic plate count for poultry (Bailey and Cox 1987), pasteurized fluid milk (Senyk et al. 1987), and ground beef (Smith, Fox, and Busta 1985). Petrifilm methods were adopted as official first action by the AOAC (Brickey et al. 1986).

ROLL TUBE. The basic idea of the roll tube is the same as for the pour plate method, except that screw-capped test tubes or bottles are used in place of petri plates. Test tubes are sterilized with 2 to 4 ml of plate count agar (with 2 percent agar). When the melted agar is cooled to 42° to 45°C, 0.1 ml of the appropriate dilution of the sample is added and the tube rolled in cool water in a horizontal position until the agar is solidified in a thin layer on the inner wall of the tube.

The roll tubes are incubated upside down so that any water that condenses collects below the inoculated agar and does not smear the colonies. After incubation, the colonies that develop are counted with the aid of a low-power magnifier. Multiplying the colony count by the dilution factor yields the number of organisms per gram of food.

Although the basic idea of the roll tube is similar to the plate count, there are obvious differences. Since test tubes are used rather than petri plates, the cost of the procedure may be lower or higher, depending upon the relative cost of these items. Less plate count agar is used in the roll tube method.

Hartman (1968) stated that the roll tube requires less space, materials, and time, with less risk of contamination and less desiccation of the media in the tubes than in plates during long incubation periods. He also reported that there is no waiting for agar to solidify to invert and incubate such as in the pour plate system. There are machines for rolling the tubes.

It would seem that the colonies would be more difficult to discern and count in the roll tube than in the pour or spread plate techniques. In his review, Hartman (1968) did not find counting of the colonies to be a problem in the roll tube. Devices are available to assist in the counting of colonies in roll tubes. The roll tube technique can be used to determine anaerobic types of microorganisms in foods (Gray and Johnson 1976).

BURRI STRIP OR SLANT. This method involves the spreading of a sample over an agar slant with a calibrated loop. Test tubes can be used, but the oval tube gives a larger surface for the growth of colonies. The agar surface must be dry to prevent colonies from coalescing. After incubation (32° or 37°C for 24 hr) in a horizontal position, the surface is examined for microbial growth. Colonies may be counted or comparisons can be made as to the extent of growth that occurs so that high- and low-count products can be distinguished.

The Burri slant method is a simple test for the evaluation of plant sanitation.

LITTLE PLATES. Since Frost introduced the little plate system in 1916, many modifications to the system have been proposed. The original procedure was to mix 0.1 ml of milk with about 2 ml of nutrient agar, and this was spread uniformly over a 4 cm^2 area on a glass slide. After incubation for 3 to 8 hr in a moist chamber, the slides were air-dried, flame-fixed, and stained for counting. The colonies were observed and counted with a microscope.

Modifications have been suggested in the procedure, such as the types of slide used, the method of inoculation and incubation, as well as type of stains. A similar procedure was described by Postgate (1969) to distinguish viable cells from dead cells, since to observe colonies on the slide, the cells must be viable.

This system is a more rapid method than the plate count, since only 3 to 8 hr of incubation are used. Besides being rapid, an estimate of the viable number of cells is obtained, which is not the case with DMC. The little plate, slide plate, and microplate methods give results comparable those for the plate count.

MEMBRANE FILTERS. When fluids are filtered through a membrane filter (MF), all particles, bacteria, or cells larger than the pores are retained on the filter surface.

The procedure has been useful for analyzing processed water, various beverages, or air when the microbial count is relatively low. Such low contamination is difficult to evaluate with the APC. More recently, MF systems have been used to analyze foods with relatively high numbers of bacteria. Prefilters are used to remove food particles that might clog the MF. In some cases, surfactants and enzymes, such as proteases, are used to degrade the food so that it can be filtered (Bourgeois et al. 1984; Entis, Brodsky, and Sharpe 1982; QALL 1981).

The retained microorganisms can be cultured by aseptically transferring the filter onto a nutrient agar or one that is selective, differential, or both. After incubation for 6 to 8 hr, the microcolonies can be counted with a microscope similarly to that used in the little plate or microplate method. After incubation for 24 to 48 hr, the colonies can be counted similarly to the APC.

The bacterial cells can be stained with 0.1 percent toluidine blue (O'Toole 1983a, 1984). After destaining the filter and making it transparent, the dye retained by the cells is determined with a spectrophotometer. Reportedly, this reading is related to the number of cells on the filter.

A membrane filter with hydrophobic material in a grid pattern is called a hydrophobic grid membrane filter (HGMF). The grids are compartments of equal and known size, and the hydrophobic material deters the spreading of colonies. After the organisms are grown on the filter, the number of squares containing colonies is enumerated and converted to a most probable number. The results can be determined manually or with an automated counting system (sample analyzer). A disposable filter unit has been developed for the HGMF (Tsuji and Bussey 1986). The HGMF system was given official status by the AOAC (AOAC 1983; Entis 1986).

In one system, the microorganisms on the filter are subjected to a fluorescent dye, acridine orange, which stains viable cells, and then observed with an epifluorescent microscope. This direct epifluorescent filter technique (DEFT) was reviewed by Pettipher (1986). The DEFT is a rapid method and is especially useful for samples of food containing high numbers of organisms (Pettipher 1987; Qvist and Jakobsen 1985; Shaw et al. 1987). The method was not suitable for heated samples (Hunter and McCorquodale 1983; Rodrigues and Kroll 1986). However, a double staining system using acridine orange and janus green B allowed the differentiation of viable and heat-killed cells (Rodrigues and Kroll 1986). A modified Gram-staining procedure using acridine orange as the counterstain allows the differentiation of Gram-positive and Gram-

negative cells (Rodrigues and Kroll 1985). Microorganisms on DEFT slides can be counted automatically (Pettipher 1986).

TUBE DILUTION. The tube dilution method is essentially the aseptic inoculation of a series of tubes of sterile nutrient broth with a series of dilutions of the food. After incubating the inoculated tubes, the broth is observed for turbidity, which indicates growth of organisms. If no turbidity is evident, it is assumed that no microorganisms were present or were able to multiply. With broth that appears turbid due to inoculated food, growth can be detected by streaking on an agar surface and observing growth after a few hours of incubation, or by spreading some turbid broth on a slide and looking for microorganisms with the aid of a microscope.

If the tube with the 1:100 dilution showed growth and the tube with 1:1000 had no growth, there were between 100 and 1,000 organisms in the food. Sometimes this rough estimate is all that is needed. It gives only an estimate of the range of bacteria that are present.

MOST PROBABLE NUMBERS (MPN). By using several tubes at each dilution and recording the positive (showing growth) tubes and negative (no growth) tubes, you get a more accurate estimate of the number of organisms present. In the tube dilution example, if you inoculated 10 tubes with 1 ml of the 1:1,000 dilution, there would be as much total inoculum as in the 1:100 tube which showed growth. Theoretically, one or more of the 10 tubes with the 1:1,000 dilution also should be turbid. The relationship of positive and negative tubes has been determined mathematically and MPN tables have been derived (Tables 2.4 and 2.5). To use the MPN system, at least three dilutions are needed. Ideally, the least dilute tubes should all be positive and the most dilute tubes (of the three dilutions) should all be negative. This is not always the case, so the rule that has been established is to select the highest dilution in which all portions tested are positive (no lower dilution giving negative results), and the two succeeding dilutions are then chosen. The more tubes that are used in each dilution, the more accurate is the estimate, but for reasons of convenience, three-tube or five-tube series are adopted. After selecting the three series of dilutions, consult the appropriate MPN table, obtain a most probable number that satisfies the number of positive tubes, and multiply this by the dilution factor to obtain the MPN per gram of product.

Assumptions and Errors (MPN). The assumptions and errors due to sampling and diluting apply to the MPN technique. It is assumed that a single viable cell inoculated into a tube of broth will multiply so that a change

TABLE 2.4. MOST PROBABLE NUMBER (MPN) PER GRAM OF SAMPLE

Number of Positives			Program Values		Two-sided 95% Conf. Limits		One-sided Upper 95% Limit
1.0	0.1	0.01	MPN	St. Error	Lower	Upper	
0	0	0	< 0.03	—	—	—	—
1	0	0	0.36	0.36	0.05	2.54	1.85
1	1	0	0.74	0.52	0.18	2.94	2.36
1	1	1	1.12	0.64	0.36	3.47	2.89
2	0	0	0.92	0.65	0.23	3.67	2.94
2	1	0	1.47	0.85	0.47	4.55	3.80
2	1	1	2.05	1.02	0.77	5.46	4.66
2	2	0	2.11	1.05	0.79	5.61	4.79
2	2	1	2.76	1.24	1.15	6.64	5.76
2	2	2	3.48	1.42	1.56	7.74	6.80
3	0	0	2.31	1.33	0.74	7.17	5.98
3	1	0	4.27	2.14	1.60	11.38	9.72
3	1	1	7.49	3.35	3.12	17.99	15.63
3	2	0	9.33	4.17	3.88	22.41	19.47
3	2	1	14.94	6.10	6.71	33.25	29.23
3	2	2	21.46	8.11	10.23	45.02	39.97
3	3	0	23.98	17.41	5.78	99.49	79.15
3	3	1	46.22	17.47	22.03	96.96	86.07
3	3	2	109.89	38.87	54.94	219.82	196.65
3	3	3	>110.00	—	—	—	—

SOURCE: Data Courtesy of Robert J. Parnow (personal communication).
NOTE: Standard error, upper and lower 95% confidence limits, and one-sided upper 95% confidence limits when three dilutions are used with three tubes in each dilution at levels of 1.0, 0.1. and 0.01 g per tube.

such as turbidity or production of acid or gas can be observed. Because dilution to extinction is necessary, good aseptic technique is needed, since any contamination during inoculation of the tubes of broth could result in growth. The MPN is less precise than the agar plating methods (Pike et al. 1972).

Some people become confused when 1 g of sample is added to a tube with 9 ml of broth for the MPN series. They feel that since this is a 1:10 dilution, somehow it has to be considered when the dilution factor for the MPN is determined. It does not make any difference if there are 8, 9, 10, or 11 ml of nutrient media per tube. The only consideration is the amount of original sample that is added to the tube (0.01, 0.001, 0.0001 g, or whatever).

Advantages of the MPN. The MPN is, in some ways, easier or simpler to do than the plate count. Broth can be dispensed into tubes with an automatic pipetter. Selective or differential media can be used so that certain types of organisms can be determined. The MPN is particularly useful

TABLE 2.5. FREQUENTLY OCCURRING MOST PROBABLE NUMBER (MPN) PER GRAM OF SAMPLE

Number of Positives			Program Values		Two-sided 95% Conf. Limits		One-sided Upper 95% Limit
1.0	0.1	0.01	MPN	St. Error	Lower	Upper	
0	0	0	<0.18	—	—	—	—
1	0	0	0.19	0.19	0.03	1.34	0.98
1	1	0	0.40	0.28	0.10	1.60	1.28
1	1	1	0.60	0.35	0.19	1.86	1.55
2	0	0	0.44	0.31	0.11	1.76	1.41
2	1	0	0.68	0.39	0.22	2.11	1.76
2	1	1	0.92	0.46	0.34	2.45	2.09
2	2	0	0.93	0.46	0.35	2.48	2.12
2	2	1	1.17	0.52	0.49	2.81	2.44
2	2	2	1.42	0.58	0.64	3.16	2.78
3	0	0	0.77	0.44	0.25	2.39	1.99
3	1	0	1.07	0.54	0.40	2.85	2.44
3	1	1	1.36	0.61	0.57	3.27	2.84
3	2	0	1.38	0.62	0.57	3.32	2.88
3	2	1	1.69	0.69	0.76	3.76	3.31
3	2	2	2.02	0.76	0.96	4.24	3.76
3	3	0	1.72	0.70	0.77	3.83	3.37
3	3	1	2.05	0.77	0.98	4.30	3.82
4	0	0	1.27	0.64	0.48	3.38	2.89
4	1	0	1.68	0.75	0.70	4.04	3.50
4	1	1	2.11	0.86	0.95	4.70	4.13
4	2	0	2.16	0.88	0.97	4.81	4.23
4	2	1	2.64	1.00	1.26	5.54	4.92
4	2	2	3.17	1.12	1.58	6.34	5.67
4	3	0	2.70	1.02	1.29	5.66	5.03
4	3	1	3.25	1.15	1.62	6.50	5.81
4	3	2	3.86	1.29	2.01	7.42	6.68
4	4	0	3.35	1.18	1.68	6.70	5.99
4	4	1	3.98	1.33	2.07	7.65	6.89
5	0	0	2.31	1.03	0.96	5.55	4.82
5	1	0	3.29	1.34	1.48	7.32	6.44
5	1	1	4.56	1.72	2.17	9.56	8.49
5	2	0	4.93	1.86	2.35	10.34	9.18
5	2	1	6.99	2.47	3.50	13.98	12.50
5	2	2	9.43	3.14	4.91	18.12	16.32
5	3	0	7.92	2.80	3.96	15.84	14.17
5	3	1	10.86	3.62	5.65	20.87	18.79
5	3	2	14.05	4.44	7.56	26.11	23.64
5	3	3	17.49	5.27	9.68	31.58	28.72
5	4	0	12.99	4.33	6.76	24.97	22.48
5	4	1	17.23	5.45	9.27	32.02	28.99
5	4	2	22.11	6.67	12.24	39.92	36.31
5	4	3	27.80	8.02	15.79	48.95	44.70
5	4	4	34.54	9.58	20.06	59.49	54.51
5	5	0	23.97	7.58	12.90	44.55	40.33
5	5	1	34.76	10.48	19.25	62.77	57.08
5	5	2	54.22	15.65	30.79	95.48	87.18
5	5	3	91.78	25.46	53.28	158.09	144.86
5	5	4	160.94	43.06	95.26	271.90	249.92
5	5	5	>1,600	—	—	—	—

SOURCE: Data Courtesy of Robert J. Parnow (personal communication).
NOTE: Standard error, upper and lower 95% confidence limits, and one-sided upper 95% confidence limits when three dilutions are used with five tubes in each dilution at levels of 1.0, 0.1 and 0.01 g.

for samples with only a few organisms and can be used to detect organisms in samples larger than 1 g.

Estimations Based on Metabolism

Microbial metabolism is used in general microbiology to determine fermentation of sugars, starch hydrolysis, production of hydrogen sulfide, indole, or reduction of nitrate. The metabolic products that are produced can be determined and used to estimate microbial populations or the quality of the food.

REDUCTASE TESTS. Organisms obtain energy from chemical reactions involving either organic or inorganic compounds. This involves an oxidation-reduction reaction; the energy source becomes oxidized, while another compound is reduced. Oxygen may or may not be involved because oxidation-reduction reactions concern electron transfers. When a compound loses an electron it becomes oxidized, and another compound which accepts this electron is reduced.

Compounds vary in their oxidation-reduction potential, which is the tendency for a compound to give up electrons. Since these reactions consist of electron transfers, they can be measured electrically with a potentiometer and are expressed by the electrical unit, the volt. The oxidation-reduction potential also is called the redox potential.

Besides being determined potentiometrically, the redox potential can be determined with indicators or dyes. Many compounds undergo color changes when oxidized or reduced. If such a compound is added to a substrate containing metabolizing bacteria, electrons may be transferred to the indicator, and its color will be altered.

Since the color change of the indicator depends on the metabolic rate of a microbial culture, the larger the number of cells, the sooner the indicator will show a color change. The reduction time is inversely proportional to the number of cells present (Fig. 2.4). Although several oxidation-reduction indicators could be used, methylene blue, resazurin, and the tetrazoliums are the ones used most often in food analysis. The reductase tests usually are called dye reduction tests, apparently because the dye methylene blue is used. However, resazurin and the tetrazoliums are not dyes, but indicators (Conn 1961).

During reduction, methylene blue becomes colorless. This dye has been used to determine the bacterial quality of milk and dairy products such as ice cream (Anderson and Whitehead, 1974). Also, it has been suggested as a means to predict the sterility of heated food (Hall 1971) and to estimate the number of bacteria in ground beef (Emswiler et al. 1976).

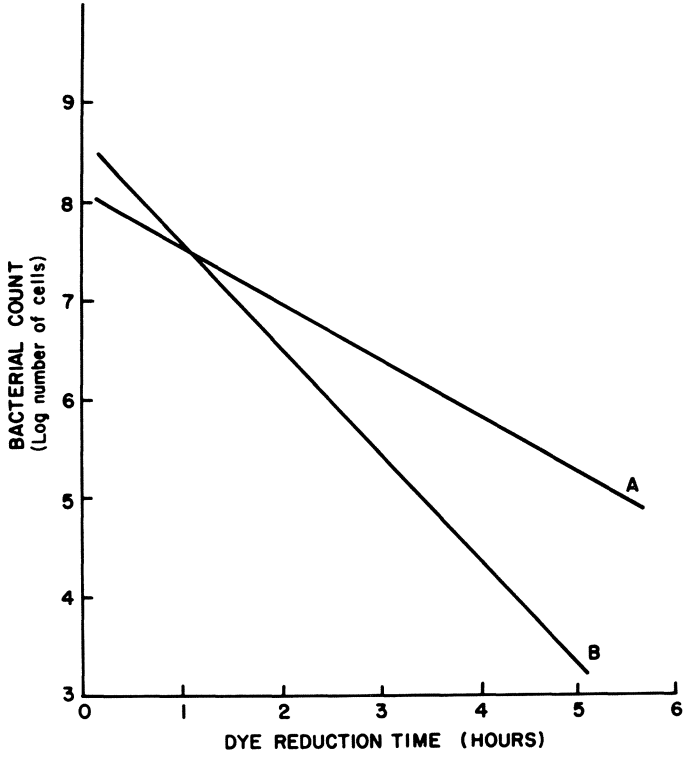

Figure 2.4. Relationship between reduction time and microbial load. The slope of the line depends upon the types of microorganisms that are present.

Two color changes occur during the reduction of resazurin. The blue color goes through various shades of purple and mauve to pink. This color change is not due to an electron transfer, but to the loss of a loosely bound oxygen atom. This color change to pink is not reversible by atmospheric oxygen. The second color change results in the indicator becoming colorless and is reversible by atmospheric oxygen. The resazurin test is a simple, relatively rapid, inexpensive system to determine the quality of fresh scallop meat (Webb et al. 1972) and frozen shrimps (Kümmerlin 1982).

The tetrazolium salt most often used for testing food is 2,3,5-triphenyltetrazolium chloride (TTC), since it is less toxic to bacteria than are other tetrazolium salts. The TTC is colorless in the oxidized state but forms intensely colored pink to red pigments (formazans) when reduced. The red color that develops when TTC is sprayed onto a surface indicates

sites of bacterial activity. This indicator can be used to distinguish bacterial colonies from food particles in an SPC.

Comparison of Reductase Tests to Viable Count Tests. The reductase test generally gives an estimate of the bacterial contamination in a shorter time than the SPC. The information obtained from reductase tests can, at best, be used to obtain a rough estimate of the number of microorganisms present in or on a food.

Not all organisms cause a lowering of the redox potential at the same rate. If a clump or chain of bacteria is plated in agar, a single colony will develop, but the metabolic activity in the reductase test will be the sum of the total number of cells in the clump or chain. This will result in a more rapid color change in the indicator than the plate count would suggest.

For a cell to be counted in the SPC, it must multiply and form a visible colony. Cells may be metabolizing, but not reproducing. These cells could cause a color change in the redox indicator and not be included in the SPC.

Methylene blue and tetrazolium are inhibitory to certain microorganisms. Sometimes tetrazolium is added after the organisms have grown, such as by flooding an agar surface, due to its potential for inhibiting the cells. It has been suggested that reducing enzymes naturally present in foods can cause color changes of these indicators. In this case, the reductase test would indicate more contamination than is present.

CHEMICAL INDICATORS OF DECOMPOSITION. Everyone makes organoleptic evaluations of food by sight, smell, taste, or touch. The food industry relies on these organoleptic tests to determine certain quality attributes of foods. This type of analysis is very subjective, and many arguments can develop between seller and buyer. Chemical indicators can be used to evaluate the quality of food in a more objective manner. Food is composed of various chemical compounds that are subject to biochemical changes. These changes may be desirable or undesirable, depending upon the food, the microorganisms that are present, and the end products of the reaction. Decomposition of a food with resulting quality deterioration is an undesirable change.

The main reactions occurring in foods are catalyzed by enzymes. These enzymes may be tissue enzymes naturally present in the food, they may be produced by microorganisms associated with the food, or they may be added to catalyze a desirable reaction (see Chapter 9). Some chemical reactions, such as oxidation, occur in foods without specific enzymes to catalyze them. The extent of change occurring in the food may or may not be related to the number of microorganisms present.

The type and amount of metabolic products formed depends upon

the kind of food (protein, carbohydrate or fat), the type of microorganism (proteolytic, saccharolytic, or lipolytic), the availability of oxygen (aerobic-oxidation, decay, or oxidative rancidity; anaerobic-fermentation, putrefaction, or hydrolytic rancidity), the temperature (psychrotrophic, mesophilic, or thermophilic organisms) and the types of inhibitors that might be present.

Criteria for Chemical Indicators. For a chemical to be a useful indicator, it must meet the following criteria: (1) it must be absent or at very low levels in sound food; (2) it should be produced by the predominant spoilage flora and not used as a nutrient; (3) it should not be detected quantitatively with simple and rapid tests and the tests should never yield false positive results; (4) it should not have a useful function in the food; and (5) it should preferably be able to distinguish poor quality from poor processing operations.

Possible Chemical Indicators. Fields, Richmond, and Baldwin (1968) presented a comprehensive review of chemical indicators. Some potential chemicals for estimating the microbial or other quality of food are listed in Table 2.6.

Because of variations in a food and its microbial flora, none of these chemical indicators is entirely satisfactory. However, the presence of certain indicators in some foods does correlate with the microbial count or organoleptic evaluation. In general, a group of compounds such as volatile reducing substances, total volatile acids, or bases gives a better indication of quality than a single indicator such as ammonia, indole, or alcohol. One problem with many of these indicators is that by the time there is a significant change in the amount that is present, deterioration of the food is very evident. Perhaps incubation of the food for a few hours before analysis could be used to develop the presence of indicators more rapidly. However, this higher temperature could alter the dominant spoilage flora so that the metabolic products would differ from those expected to develop under normal storage conditions.

TABLE 2.6. POTENTIAL CHEMICAL INDICATORS OF FOOD QUALITY

Ammonia	Indole
Trimethylamine	Ethanol
Dimethylamine	Furfural
Total volatile bases	Hydrogen sulfide
Total volatile acids	Total reducing substances
Free fatty acids	Volatile reducing substances
Water insoluble acids (oleic, palmitic)	Histamine
Organic acids (acetic, lactic, pyruvic, succinic)	Hypoxanthine
	Diacetyl
	Acetylmethylcarbinol

PHYSICAL TESTS. Microbial growth causes alterations in foods, such as acid content or pH. As the pH is varied, the water-binding capacity of protein changes. This difference can be determined by the extract release volume test.

The fluorescence of liquid egg is related to mustiness and growth of certain bacteria, while pyoverdine, a fluorescent pigment produced by pseudomonads, has been determined in frozen whole egg and on poultry carcasses.

pH. Since basic compounds such as ammonia and amines are formed during deterioration of protein foods, the pH will tend to rise. If carbohydrates are present and fermented, the pH will tend to decline. Shelef and Jay (1970) recommended blending 10 g of meat with water and titrating to pH 5.0 with 0.02*N* HCl. If more than 2.0 ml of acid is required, the meat is in some form of incipient spoilage.

Extract Release Volume (ERV). As meat deteriorates, there is an increase in the amount of water retained and a decrease in ERV. Shelef (1974) reported the relationship between pH and ERV. Regardless of the microbial quality of the meat, the maximum ERV occurred at pH 5.5.

ADENOSINE TRIPHOSPHATE (ATP). During metabolism, cells form high-energy phosphate bonds stored in ATP. Not only microbial cells, but also other living cells contain ATP.

When an animal dies and the muscle glycogen is utilized by anaerobic glycolysis, the amount of ATP decreases. It has been established that when bacterial cells are killed, the ATP disappears. In starved cells, the ATP falls to low levels before the loss of viability is evident. Since all microbial cells contain ATP, it should be possible to estimate the number of cells by quantitating the ATP in a system.

The method for determining ATP is based on the firefly reaction as shown in Figure 2.5. A purified extract from the firefly, containing luciferin and luciferase, when reacted with ATP in the presence of magnesium ions, causes a light emission. Crude extracts from the firefly, when reacted with adenosine diphosphate (ADP) have caused light emission. When the purified firefly extract is added to bacterial ATP, the light emission can be measured with a photometer. For the assay, it is necessary to eliminate the nonbacterial ATP and then to release the bacterial ATP to react with the luciferin-luciferase system.

One method used to eliminate nonbacterial ATP is to rupture the somatic cells and hydrolyze the released ATP (Anon. 1974; Thore et al. 1975). Another method is the centrifuging and/or filtration of a food to separate the bacterial cells from food cells (Jarvis 1982; Picciolo et al. 1981; Stannard and Wood 1983).

After elimination of food ATP, the microbial ATP can be released

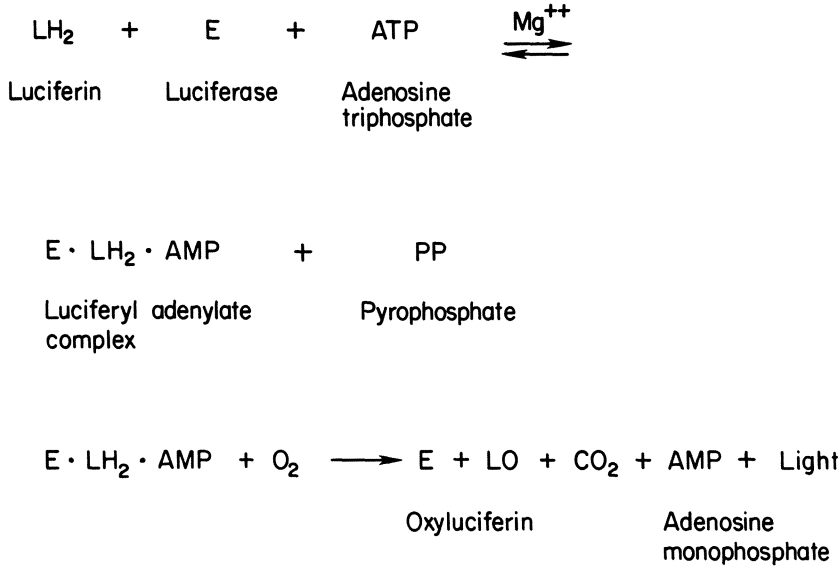

Figure 2.5. Reactions involved in adenosine triphosphate determinations.

from the cells and assayed with the luciferin-luciferase system, and an estimate of the number of cells is calculated.

The ATP system is relatively simple and rapid. The system is reportedly reliable and acceptable for food and water analysis (Jarvis 1982; Picciolo et al. 1981). Firefly bioluminescence was reviewed by McElroy and DeLuca (1983).

MEASUREMENT OF GAS PRODUCTION. When organisms metabolize compounds, carbon dioxide is produced as a metabolic product and oxygen is consumed. One method for determining CO_2 production uses ^{14}C-labeled glucose and measures the reactivity of the released $^{14}CO_2$. The detection time of $^{14}CO_2$ is proportional to the logarithm of the original inoculum (Waters 1972). This procedure has been used for determining various bacteria in water and food (Hatcher et al. 1977; Lampi et al. 1974; Mafart et al. 1978; Rowley, Previte, and Srinivasa 1978).

Headspace gas of bottled fruit juice was analyzed for CO_2 by an infrared system (Threlkeld 1982). Infrared analysis detected 89 percent of the samples that were positive by conventional methods. Most of the false negative samples lacked the presence of fermentative organisms.

IMPEDANCE. The metabolic activity of cells changes the chemical composition of a medium, a process that results in altering the impedance (Firstenberg-Eden and Zindulis 1984). The impedance is the opposi-

tion to the flow of alternating electrical current and can be measured by a sensitive meter. This system has been used to estimate microorganisms in cooked food (Rowley et al. 1979), cereal grain (Sorrells 1981), milk (Firstenberg-Eden 1984; Firstenberg-Eden and Tricarico 1983; Gnan and Luedecke 1982; Martins et al. 1982), meat (Firstenberg-Eden 1983; Martins and Selby 1980), frozen vegetables (Hardy et al. 1977), and orange juice (Weihe, Seibt, and Hatcher 1984).

OTHER INSTRUMENTATION OR METHODS. Besides those systems described, other systems have been used by microbiologists to develop rapid methods for microbial detection. A pyrolysis-gas chromatography-mass spectrometry experiment was used to analyze Martian soil for organic compounds that would indicate life on Mars (Klein 1976; Simmonds 1970). Sanders and Parkes (1970) attempted to correlate infrared analysis of swabbings from broiler skin with bacterial numbers.

Gas chromatography or gas-liquid chromatography has been used to detect metabolic products of microorganisms which can be used to differentiate bacterial species or estimate the count or quality (Carlsson 1973; Staruszkiewicz and Bond 1978).

Limulus amoebocyte lysate forms a gel in the presence of small amounts of endotoxin from Gram-negative bacteria. This assay has been suggested as a system to estimate Gram-negative bacteria in various samples such as meat (Jay 1981; Seiter and Jay 1980), milk (Hansen, Mikkelsen, and Møller-Madsen 1982), fish (Sullivan et al. 1983), cooked turkey (Dodds, Holley, and Kempton 1983), and drinking water (Sykora et al. 1980).

It is not possible to discuss all of the procedures that might be used to determine microorganisms in foods. Some other procedures were described by Bordner (1981), Feldman (1982), Goldschmidt and Fung (1978, 1979), Jarvis (1982), Newsom (1978), and Southern (1979).

Although agar is the usual solidifying agent in solid media, various substitutes have been suggested (Cranston 1983; Kang et al. 1982; Lin and Casida 1984; Shungu et al. 1983).

Comparison of Methods

There are many procedures that can be used to estimate the total microbial population of a food product. The test that is selected depends upon the use of the information that is obtained. If the results are to be used to satisfy a microbiological standard, then the specified method, usually the standard plate count, must be used. If the results are for internal quality control, some other method that is simpler, faster, or less expensive might be used. The procedure selected will depend on the accu-

racy, reliability, or precision that is needed. The laboratory equipment and personnel, both now and in the future, will probably dictate the type of analysis that can be conducted. The cost of supplies, materials, instruments, and labor must be compared with other factors to determine which method or methods give the desired information with ease, simplicity, speed, and low cost.

In selecting the test, not only precision but also accuracy must be considered. Precision is an index of the random error in a group of determinations and is not related to accuracy. The term *accuracy* considers the relationship of the determined value and the actual value. Since there is no way to know the actual or true number of microorganisms in a sample, it is difficult to assess the accuracy of a method. Quite often, the results of a method are compared to those obtained with the APC. The results obtained with the APC may not be accurate, but the method yields acceptable reproducibility or precision.

Rapid tests such as those based on metabolic products and instrumentation are valuable for control purposes. Time is important in analyzing highly perishable foods. Also, it is desirable to keep the inventory of processed food at a reasonable level. If unacceptable product is shipped from storage, it might need to be recalled. A recall of a product and the bad publicity that often results is not desirable for a processor.

If microbial inhibitors are present in the food, these will be carried over into the growth medium. In these cases, as the sample is diluted, higher counts might be obtained due to dilution of the inhibitors. If a food containing glucose is added to a medium designed to determine lactose fermentation, erroneous results may be obtained due to fermentation of the added glucose.

In processed foods, there may be cells that are sublethally injured. These are of special importance when the types of organisms are determined with selective media.

Comparisons of various methods of estimating microbial counts have been published (Greenwood et al. 1984; O'Toole 1983b). Perhaps food microbiologists have emphasized the so-called total count too much, since spoilage organisms and potential pathogens are more important in a food than are other types of organisms. However, at the present time, the total count is usually the first microbial characteristic determined for almost any food.

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3

Microorganisms Associated with Food

The significance of microorganisms in foods depends upon several conditions: (1) the numbers found; (2) the types of microorganisms; (3) the type of food; (4) the treatments to which the food has been exposed; (5) the processing or storage treatments the food will receive; (6) whether the food is to be eaten as is or heated; and (7) the individuals who might consume the food. In this text, it is necessary to limit the discussion of microorganisms in foods to bacteria, fungi, and viruses.

Microorganisms may have at least one of four functions in a food. They may have a useful function, cause spoilage, be a health hazard, or be inert. The inert microorganisms do not find an environment favorable for growth. In most cases of foodborne illness, spoilage, or useful activity, the microorganisms grow and multiply. Organisms that cause foodborne illness are of more concern than are other types of microorganisms.

Spoilage is the result of undesirable changes in the odor, color, flavor, texture, or appearance of food. Some organisms that do not directly cause changes in a food may alter the flora so that spoilage organisms can grow. An example is the bacteriophages that attack useful organisms, allowing undesirable organisms to grow and cause spoilage.

Useful organisms are those which produce desirable changes in food, such as converting milk to cheese, sugar to alcohol, and cabbage to sauerkraut. These changes are referred to as *fermentations.* The microorganism does not always have to be present to have a useful function, since enzymes can be separated from the organism, and the enzymes used to produce the desired reaction. Another function of useful organisms is the production of single-cell protein which can be used as food. Spoilage organisms and useful organisms are similar in that they both produce changes in the food. The difference between them is that one is desirable and the other is undesirable.

Although it is easy to establish categories for microorganisms, it is difficult to place an organism in only one category, since it may have different functions in different foods. An organism may spoil one food but be inert in another, because of the different characteristics of foods.

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However, we do tend to associate certain organisms with particular functions in food, and these will be considered as much as possible.

DETERMINING TYPES OF MICROORGANISMS

The basic systems of microbial analysis discussed in Chapter 2 can be used to determine the specific types of microorganisms that are present in or on a food. However, differential or selective media are substituted for the noninhibitory, nonselective media. Quite often, prior to determining whether certain types of organisms are present, an enrichment procedure is used to increase the probability of detection. After these enrichment processes, the organisms are detected on differential or selective agar.

Various tests are used to differentiate organisms isolated from selective or differential agars. The usual staining reactions and morphological characteristics are determined. The required biochemical tests depend upon the microorganisms being considered. Various commercial kits and systems for determining biochemical reactions have been evaluated (Costigan and Hollick 1984; Cox and Mercuri 1979; Cox, Bailey, and Thomson 1983; Cox et al. 1984; Ferraro, Edelblut, and Kunz 1981; Fung, Goldschmidt, and Cox 1984; Griffiths and Phillips 1982; Izard et al. 1984; Kelly and Latimer 1980; Odlaug et al. 1982). Serological reactions can utilize cellular, flagellar, or other antigens. The development of monoclonal antibodies (Løvborg 1984) and immunoassays such as radioimmunoassay (RIA), enzyme immunoassay (EIA) (Milby and Zare 1984), and enzyme-linked immunosorbent assay (ELISA) are used to detect specific microorganisms as well as toxins and other antigenic agents. The production of pigment, antimicrobial resistance pattern, phage typing, and bacteriocin typing can be useful for the differentiation of strains of some organisms (Anderson and Engley 1978; Kaneko and Hashimoto 1982).

In the future, it is likely that genetic relatedness will become the system to identify microorganisms (Krieg and Holt 1984). This includes deoxyribonucleic acid (DNA) and ribonucleic acid (RNA) homology (Palleroni 1983), as well as DNA colony hybridization and gene-specific DNA probes (Claus, Moseley, and Falkow 1983).

Besides the genetic information in the chromosome, many bacteria contain extrachromosomal elements called plasmids. The genetic material in the plasmids is not absolutely essential for the cells to survive, but it may give the cell a selective advantage in certain environments.

When a bacterium that contains plasmids divides, the plasmids are replicated so there is at least one copy of the plasmids in each of the

daughter cells. Under certain conditions, the plasmids from one cell can be transferred to another cell by conjugation.

The plasmids have been separated into several classes according to the characteristics they confer upon the cells. Class F plasmids, the fertility plasmids, have the ability to transfer genetic material by conjugation. Other plasmids are involved with antibiotic, chemical, or radiation resistance; enzyme coding, so that metabolic reactions different from the usual ones occur; virulence and production of toxins, such as enterotoxins; and various other traits (Helinski 1976; Mach and Grimes 1982; Murray et al. 1983; Taylor, Levine, and Kouvelos 1982; Veltkamp 1979).

MICROBIAL TYPES IN FOOD

The microbial analysis of food products yields many diverse types of microorganisms. However, we are concerned with the predominant types and those which may cause spoilage or be a health hazard.

Various genera of bacteria have been isolated from red meat shortly after slaughter of the animal. Lee, Fung, and Kastner (1982) reported the prevalent genera on hot-boned beef as *Staphylococcus, Micrococcus, Brevibacterium,* and *Bacillus.* Others that have been isolated from red meat include *Lactobacillus, Corynebacterium, Flavobacterium, Pseudomonas, Alcaligenes, Streptococcus, Pedicoccus, Acinetobacter, Microbacterium,* and *Aerococcus.* Quite often coliforms, salmonellae, and *Clostridium perfringens* are found on red meats. The temperature of holding and the packaging materials influence the types of organisms that become dominant on fresh meat. Species of *Micrococcus* and *Lactobacillus* are important on cured meat. Also, *Bacillus, Vibrio,* coliforms, fecal streptococci, and various yeasts can be isolated from cured meats.

The same genera found on red meat tend to be present on poultry and fishery products. Although the potential pathogens, such as *Salmonella* and *Campylobacter,* have been isolated from poultry, generally less than 25 percent of the carcasses are contaminated (Cunningham 1982; Kraft et al. 1982; McKinley and Avens 1981; Norberg 1981). Fishery products, especially those from the oceans, contain *Vibrio* species (Sobsey et al. 1980; Sochard et al. 1979).

Animal products such as milk and eggs tend to have bacteria similar to those of meat and poultry (Coghill 1982; Cousin 1982; Johnston and Bruce 1982; Sashihara et al. 1979; Schwab et al. 1982).

Many types of microorganisms are associated with plant products. The most common bacteria in vegetables are *Lactobacillus* and *Leuconostoc.* The genus *Erwinia* is important in vegetable spoilage. Other genera include *Aer-*

omonas, Alcaligenes, Bacillus, Corynebacterium, Enterobacter, Flavobacterium, Klebsiella, Pseudomonas, Serratia, and *Xanthomonas.* Molds on vegetables include *Alternaria, Aspergillus, Aureobasidium, Botrytis, Chaetomium, Cladosporium, Epicoccum, Fusarium, Mucor, Penicillium, Phoma,* and *Rhizopus* (Andrews et al. 1982; Geeson 1979; Webb and Mundt 1978). During processing, the initial microflora are distributed throughout the product and are supplemented with other microorganisms.

The important organisms on fruits are molds (*Penicillium, Alternaria, Aspergillus*). *Lactobacillus* species cause spoilage in fruit juices.

Molds such as *Aspergillus, Fusarium,* and *Penicillium* are found on dry products such as grain and peanuts.

Several genera of yeasts are important in the spoilage of foods, especially fruits and foods containing sugar. Yeasts are also useful in fermentation and the production of single-cell protein.

BACTERIA

Compared to the total number of species of bacteria, relatively few have any importance in food, and most of these are the common types that are discussed in general microbiology. Individual species of bacteria are discussed with their functions in later chapters, so we will be concerned mainly with the genera that are important.

In *Bergey's Manual of Determinative Bacteriology* (Buchanan and Gibbons 1974), the bacteria are grouped according to the Gram reaction, morphology, and relation to growth with or without oxygen. In essence, that system is used by Krieg and Holt (1984) as well as in this text for bacteria found in food. Various morphological types of bacteria are shown in Figures 3.1 and 3.2.

Aerobic/Microaerophilic, Motile, Helical/Vibroid Gram-Negative Bacteria

In this group only one genus, *Campylobacter,* is important in food.

CAMPYLOBACTER. These Gram-negative, oxidase-positive organisms have a singular polar flagellum at one or both ends and are motile with a typical corkscrew motion. Unlike most bacteria, campylobacters use amino acids or tricarboxylic acid intermediates rather than carbohydrates as a source of energy. The species of concern is *C. jejuni.* This organism is found on and in animal products and is a cause of gastroenteritis in human beings. (See Chapter 6.)

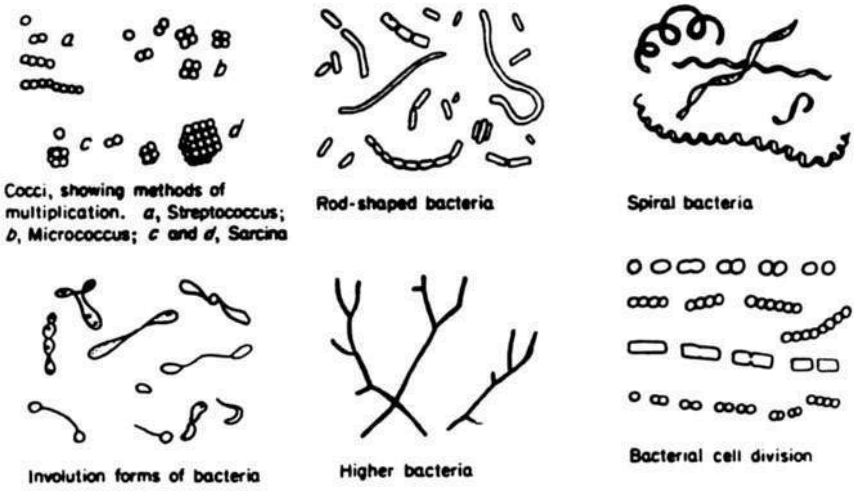

Figure 3.1. Morphological forms of bacteria.
Courtesy of Weiser, Mountney, and Gould (1971).

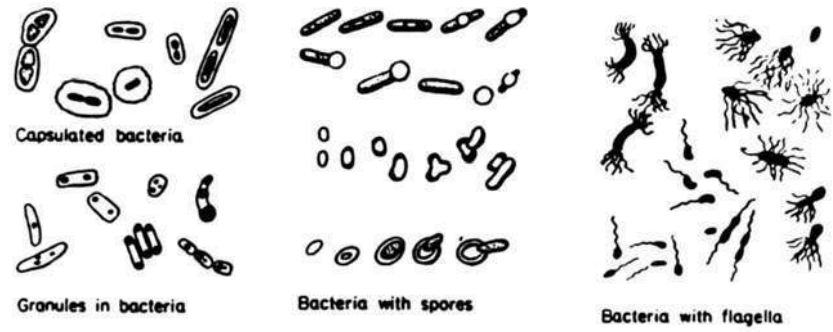

Figure 3.2. Structures of Bacteria.
Courtesy of Weiser, Mountney, and Gould (1971).

Gram-Negative, Aerobic Rods, and Cocci

Included in this group are the family Pseudomonadaceae with the genera *Pseudomonas* and *Xanthomonas;* the family Halobacteriaceae with the genera *Halobacterium* and *Halococcus;* the family Acetobacteriaceae with the genera *Acetobacter* and *Gluconobacter;* the family Neisseriaceae with the genus *Acinetobacter* and four other genera (*Alcaligenes, Alteromonas, Brucella,* and *Flavobacterium*) of uncertain status (Krieg and Holt 1984).

PSEUDOMONAS. This genus is divided into four sections on the basis of RNA homology and a fifth section which contains species whose relationships to the others are generally unknown (Krieg and Holt 1984). Oyaizu and Komagata (1983) divided the pseudomonads into nine groups, primarily on the basis of cellular fatty acid composition.

The pseudomonads are straight to curved, motile (polar flagella) rods. They are widely distributed in nature and are found on both animal and plant products. Some pseudomonads are pathogenic for plants (Wong and Preece 1980). Others, such as *P. aeruginosa,* produce toxic substances and are opportunistic pathogens for human beings (DeBell 1979; Homma 1978). Some *Pseudomonas* species have been suggested as causative agents of foodborne illness, but there is incomplete proof that this is so (Bryan 1973).

The pseudomonads are noted for their biochemical activity, being able to attack a wide variety of organic compounds, including aromatic types, and some synthetically produced chemicals. Some species (*P. aeruginosa* and *P. cepacia*) have been found to grow in distilled water. The pseudomonads produce catalase and most produce oxidase. They can produce acids oxidatively from glucose, maltose, or both.

Some species of pseudomonads produce pyoverdine or fluorescein, which are water-soluble fluorescent pigments (Leisinger and Margraff 1979). These pigments can be observed in spoiled foods by using an ultraviolet light. They are usually yellow-green but may appear blue or orange, depending on the species and environmental factors.

The pseudomonads produce enzymes that catalyze proteolytic (McKellar 1982; Patel, Bartlett, and Hamid 1983) and lipolytic (Roy 1980) reactions that contribute to spoilage of refrigerated fresh animal products (Shaw and Latty 1982). Some pseudomonads produce pectolytic enzymes that can cause soft rot of fleshy vegetables (Cuppels and Kelman 1980; Palleroni and Holmes 1981).

The psychrotrophic pseudomonads are found in almost all types of refrigerated and frozen foods. Since they are not very heat resistant, they are not found in heat-processed foods unless the food is recontaminated after heating. They are not very resistant to drying or gamma irradiation. *P. aeruginosa* is notoriously resistant to quaternary ammonium compounds.

Several recent reports of selective media for *Pseudomonas* have been published (Gould et al. 1985; Katoh and Itoh 1983; Wu and Thompson 1984).

XANTHOMONAS. The organisms in this genus are plant pathogens. They are straight, motile (polar flagellum) rods and are catalase positive.

A medium to isolate a strain of *X. campestris* from walnuts was described by Mulrean and Schroth (1981).

Being plant pathogens, these organisms have been involved with various types of rots of these products. *X. campestris* produces xanthan gum, which is used in the food industry.

HALOBACTERIACEAE. The two genera *Halobacterium* and *Halococcus* are extreme halophiles requiring about 15 percent salt for growth. They are found in solar salt (produced by evaporation of seawater).

Their halophilic character prevents them from growing in most foods. However, in foods preserved by salting, these organisms can produce a red pigment, bacteriorubein.

The high salt concentration is necessary for activity of the enzymes, stability of membranes and ribosomes, and synthesis of protein. A low salt concentration results in *Halobacterium* changing from rods to spherical forms.

ACETOBACTER. These cells are ellipsoidal to straight or slightly curved rods. Young cultures are Gram negative, while older cultures are Gram variable. They are motile (peritrichous flagella) or nonmotile.

The *Acetobacter* are noted for the oxidation of ethanol to acetic acid. They oxidize acetate and lactate to CO_2 and H_2O. They are found on fruits and vegetables and are involved in the souring of fruit juices and alcoholic beverages (beer and wine).

A selective medium for differentiating *Acetobacter* and *Gluconobacter* was described by Cirigliano (1982).

GLUCONOBACTER. *G. oxydans* represents this genus. This organism is ellipsoidal to rod-shaped, Gram negative to weakly Gram positive in older cultures. The cells occur singly, in pairs, or in chains. They are strictly aerobic and oxidize ethanol to acetic acid.

G. oxydans is found in various food products such as vegetables, fruits, baker's yeast, beer, wine, cider, and vinegar. It is involved in spoilage, causing the souring of fruits.

ACINETOBACTER. These organisms are very short, plump rods or coccobacilli, predominantly in pairs or short chains. They are strictly aerobic, mesophilic saprophytes.

They are found in soil and water and also in animals and human beings. Organisms identified as *Acinetobacter* have been isolated from various types of raw and prepared foods, including beef and poultry carcasses (Lahellec et al. 1975). They are potential spoilage organisms. Meth-

ods for enumeration of *Acinetobacter* were described by LaCroix and Cabelli (1982) and Spino and Geldreich (1981).

ALCALIGENES. The four species in this genus are motile (four to eight peritrichous flagella) rods, coccal rods, or cocci. They are strict aerobes and oxidase positive.

They are widespread in nature, being found in soil, water, decaying matter, and the intestinal tract of animals. These organism are involved with the spoilage of protein foods (eggs and dairy products).

ALTEROMONAS. This genus comprises Gram-negative, motile (polar flagellate), nonfermentative bacteria with a guanine + cytosine (GC) ratio between 43.2 and 48 mol percent (Baumann et al. 1972). This ratio is listed as 38:50 mol percent by Krieg and Holt (1984).

These organisms are common in marine environments and on fish, causing fish spoilage (Chan et al. 1978; Gillespie 1981; Gray and Stewart 1980). *Alteromonas* was isolated from ground beef (Parker and Levin 1983).

BRUCELLA. These coccobacilli or short rods are nonmotile. *Brucella melitensis* (pathogenic for goats and sheep), *B. abortus* (pathogenic for cattle), and *B. suis* (pathogenic for pigs) can affect people, causing brucellosis (undulant fever). The sources of infection are raw milk or dairy products, uncooked meat or sausage products, discharges from infected animals, and human carriers.

Since these organisms are susceptible to the heat treatment used for pasteurization of milk and cooking of meat, they should be of limited importance in food.

There are between 100 and 200 cases of brucellosis in the United States annually. The majority of cases involve meat-packing-plant workers, and most of these are involved with hog-slaughter operations. Most of the other cases involve livestock producers, veterinarians, and government inspectors.

FLAVOBACTERIUM. This genus contains diverse bacteria. The species includes nonmotile rods that produce yellow, orange, or greenish-yellow pigment (Hayes 1977). Environmental factors (substrate and temperature) affect the synthesis of pigment and resultant hue. These organisms prefer temperatures below 30°C, although some strains can grow at 37°C.

Flavobacterium species have been isolated from water, soil, animals, humans, and various food products. They can produce discoloration on some foods. The organisms have been found on thawing frozen vegeta-

bles, fresh vegetables, refrigerated fish and shellfish, meat, meat products, poultry, and in cannery environments.

Gram-Negative, Facultative Anaerobic Rods

In this group, the family Enterobacteriaceae contains several genera, including *Escherichia, Shigella, Salmonella, Citrobacter, Klebsiella, Enterobacter, Erwinia, Serratia, Edwardsiella, Proteus,* and *Yersinia.* The family Vibrionaceae includes *Vibrio* and *Aeromonas.* There are other genera in these families and in this section, but they are not very important in food microbiology.

ESCHERICHIA. The principal species is *E. coli.* The organisms are members of the coliform group, which are indicators of fecal contamination (see Chapter 7). *E. coli* may cause spoilage of food, and certain strains are enteropathogenic or enterotoxigenic.

E. coli is found in soil and water, on plants, in the intestinal tract of animals, and in various foods, especially animal products and foods handled by people. Since the cells are heat sensitive, their presence in heat-pasteurized or cooked products indicates recontamination after the treatment.

SHIGELLA. The four species in this genus are nonmotile, nonspore-forming rods. The normal habitat is the intestinal tract of human beings and other primates. Rarely are these organisms isolated from other animals.

The shigellae are the causative agent of shigellosis, a foodborne gastroenteritis. The number of foodborne illnesses due to these organisms is not great, but the illness can be severe. Additional information about *Shigella* and shigellosis is in Chapter 6 and in Dodd and Jones (1982).

SALMONELLA. The genus *Salmonella* is divided into five subgenera (I–V). These subgenera are differentiated on the basis of minor biochemical differences. Subgenus I includes the majority of serotypes of this genus, especially the more important serotypes, and subgenus III includes the organisms previously called the Arizona group. Most of the different types were differentiated on the basis of serological reactions, so they are called serotypes or serovars. The number of known serovars is approaching 1900.

The salmonella organisms are catalase positive and oxidase negative, with both respiratory and fermentative metabolism. Most of the serotypes are motile (peritrichous flagella), do not ferment lactose, and grow well in simple media containing glucose, inorganic nitrogen, and mineral salts. Although salmonellae do not ferment sucrose, strains of *S. arizonae*

contain a plasmid that gives them the ability to ferment this sugar (Bartlett and Trust 1980).

The salmonellae are said to be ubiquitous, being worldwide and found in or on soil, water, sewage, animals, humans, processing equipment, feed, and various food products. The natural habitat is the intestinal tract of humans and animals. It is logical that humans, animals, and their environments are the primary sources of salmonellae. Some serotypes seem to be localized in a region or a country, but with national and international travel and trade, the organisms are easily disseminated.

The serotypes of *Salmonella* are an important cause of foodborne illness. Further information about these organisms is found in Chapter 6.

CITROBACTER. The species in this genus are motile (peritrichous flagella) rods that utilize citrate as the sole carbon source. Most ferment lactose. They are members of the coliform group of indicator organisms. They are normal intestinal inhabitants and are found on various foods, especially animal products. These organisms are listed as causing enteritis in humans, but proof of transmission by foods is inconclusive (Bryan 1973). They have been associated with various human infections. Citrobacter can also cause spoilage of foods.

Cross-reactions with antisera of other Enterobacteriaceae indicate that there are close relationships between *Citrobacter, Salmonella,* and *Escherichia.*

KLEBSIELLA. The klebsiellae are nonmotile, encapsulated rods. They are found on grain and fresh produce and in frozen foods. They are members of the coliform group, can cause food spoilage, and are potential health hazards, causing gastroenteritis. According to Bryan (1973), proof that transmission of these organisms is by means of food is inconclusive.

ENTEROBACTER. Organisms in the *Enterobacter* genus are similar to the klebsiellae, except they are motile (peritrichous flagella).

They can be found in soil, water, sewage, the intestinal tract of people and animals, and in various food products. The organisms are important in food as a potential health hazard and as indicators of spoilage.

ERWINIA. Organisms in this genus are small, mostly motile (peritrichous flagella), predominantly single, straight rods. They are oxidase negative and catalase positive.

The organisms cause various plant diseases and soft rots. *E. carotovara* is the main spoilage organism of stored vegetables (Chapter 8). At least one species is associated with humans and animals.

SERRATIA. The cells of *Serratia* are motile (peritrichous flagella) rods. They are important as potential food-spoilage organisms. *S. marcescens* is noted for the production of red pigment and is used as a biological marker (Yu 1979).

EDWARDSIELLA. These organisms have peritrichous flagella. They produce indole from tryptophan and produce H_2S on triple sugar iron agar.

Although these organisms cause enteritis in humans, proof that transmission is by means of food is not conclusive (Bryan 1973). *E. tarda* has a wide geographical distribution and infects numerous animal species.

PROTEUS. Organisms in this genus are highly motile (peritrichous flagella) rods that occur singly, in pairs, or in short chains. On moist agar surfaces, some *Proteus* species tend to swarm, which makes it difficult to pick isolated colonies for further study. Swarming is more evident at 20°C than at 37°C.

The organisms are widely distributed. They are common inhabitants of the gastrointestinal tract, and are found in sewage, soil, in decomposing animal protein, and in various foods.

The importance of *Proteus* in food includes potential health hazards and spoilage. Although strains of *Proteus* can cause enteric infection in humans, according to Bryan (1973), proof of transmission by food is inconclusive.

YERSINIA. The cells in this genus are ovoid to rod shaped. Included in this genus is *Y. enterocolitica,* which is associated with gastroenteritis, meseteric lymphadenitis, terminal ileitis, and pseudoappendicitis (Bercovier et al. 1980). This organism has been isolated from red meat, chicken, shellfish, milk, ice cream, and vegetables. These organisms may be more prevalent in food than reported.

VIBRIO. This genus is in the family Vibrionaceae. The cells are short, motile, oxidase positive, curved, or straight rods. Some strains fail to grow without the presence of NaCl, the optimum concentration being 3.0 percent.

Strains of *V. costicola* can tolerate NaCl concentrations of 23 percent. This species has been found in cured meats and curing brines (Gardner 1980–1981). A medium for isolating these salt-tolerant organisms was described by Gardner (1973). Methods for enumeration, phage typing, and characterizing *V. cholerae* were discussed by various researchers (Bockemuhl and Meinicke 1976; Morris et al. 1976; Salles and Momen 1981).

V. cholerae and *V. parahaemolyticus* are important pathogens, causing

gastroenteritis in humans. *V. cholerae* is found in the intestinal tract of people and animals, in water, and occasionally in food. *V. parahaemolyticus* is found in the ocean, in seafoods, and in the intestinal contents of infected humans. Various other vibrios have been incriminated in outbreaks of foodborne illness (Roberts and Gilbert 1979).

AEROMONAS. The cells are rods with rounded ends to coccoid. They are motile (polar flagella), and are oxidase and catalase positive.

These organisms are frequently mistaken for members of the family Enterobacteriaceae because of their similarity in growth and biochemical characteristics. A positive oxidase test and nitrate reduction help differentiate the *Aeromonas* from Enterobacteriaceae.

The main habitat of *Aeromonas* is water. Some strains cause disease (hemorrhagic septicemia) in fish, eels, and frogs. Some strains cause enteritis in humans. One source of this infection may be fish or other seafood. *Aeromonas* may play a role in spoilage of fish and other animal products.

Gram-Negative, Anaerobic Straight, Curved, and Helical Rods

In this category, the family Bacteroidaceae contains twelve genera, including *Bacteroides.* These organisms are nonmotile, straight rods. Being nonsporeforming obligate anaerobes, they probably cause no changes in foods. However, they are found in fecal material in very high numbers. Hence, they might be useful as indicators of fecal contamination of food as well as water (Fiksdal et al 1985).

Gram-Positive Cocci

The Gram-positive cocci include aerobic or facultative anaerobic bacteria in the family Micrococcaceae with the genera *Micrococcus* and *Staphylococcus.* Other Gram-positive cocci include the genera *Streptococcus, Leuconostoc, Pediococcus,* and *Aerococcus.*

MICROCOCCUS. These spherical cells are strict aerobes, catalase positive, occur singly or in pairs, and characteristically divide in more than one plane to form irregular clusters, tetrads, or cubical packets. They can grow in the presence of 5 percent salt.

The micrococci are found in soil, water, and dust, and on the skin of people and other animals. They are found in several types of foods, especially milk and dairy products, on animal carcasses, and in meat products. They are important as potential spoilage organisms.

STAPHYLOCOCCUS. These nonmotile cells occur singly, in pairs, or in irregular clusters. They are facultative anaerobes with respiratory and fermentative metabolism. Most strains can grow in 7.5 percent to 15 percent salt. The organisms usually are sensitive to chlorine, chloramine, iodine, and iodophors. Although usually sensitive to heat, they are moderately resistant to radiation.

Both *S. aureus* and *S. epidermidis* are commonly found on the skin and mucous membranes of humans and warm-blooded animals. They are potential pathogens, being either the primary pathogen or secondary invader.

The staphylococci are found in many types of food products. In general, they are not able to compete very well with other organisms. *S. aureus* produces enterotoxins, which are one of the main causes of foodborne illness. *S. aureus* is found in pimples, boils, acne, wound infections, and in the nose. It is easily transferred to foods by careless food handlers. This species is differentiated by its ability to produce coagulase (which clots blood plasma) and a heat-stable nuclease. *S. aureus* is described further in Chapter 6.

Other species have been found in food (Devriese et al. 1983; Schleifer and Fischer 1982). *S. hyicus* reportedly produces an enterotoxin (Hoover, Tatini, and Maltais 1983).

STREPTOCOCCUS. The bacterial species in this genus have various characteristics and functions. These organisms have been divided into several groups (Bridge and Sneath 1983). The groups of primary importance in food are the lactic streptococci and enterococci. Researchers have suggested that organisms in the latter group be put into the genus *Enterococcus* (Collins et al. 1984; Schleifer and Kilpper-Bälz 1984).

These spherical cells occur in pairs or chains. With the exception of some strains, they are not motile. They are facultative anaerobes with a fermentative metabolism, fermenting glucose primarily by the hexose diphosphate pathway and producing mainly lactic acid. Thus, they are called homofermentative.

The fermentation of carbohydrates to lactic acid is desirable in products such as cultured milk, cheese, and sauerkraut, but it is undesirable in products such as fresh milk. They can utilize sugar in an alcohol fermentation, resulting in lower alcohol and higher acid than is wanted. Some strains utilize citric acid and form acetoin and diacetyl. The ability to produce diacetyl in milk is desirable in the manufacture of cultured sour cream, buttermilk, and butter.

Various functions such as lactose metabolism, proteolytic activity, and hemolysin and bacteriocin production are mediated by plasmids (Anderson and McKay 1977; Gasson 1983; McKay 1983; Oliver, Brown, and Clewell 1977).

The streptococci are widely distributed, being found in air, water, sewage, soil, on plants, in the intestinal tract of people and animals, and in various food products.

Some species and strains are somewhat heat resistant, surviving 60°C for 30 min. The enterococci can survive freezing and frozen storage in food products. This makes these organisms acceptable as potential indicator organisms in frozen foods.

Although some pathogenic types may be transmitted to foods, there is more interest in the streptococci as possible indicators of fecal contamination, as useful fermentative organisms, and as potential spoilage bacteria.

LEUCONOSTOC. These spherical to lenticular cells occur in pairs or chains. They often have complex nutrient requirements such as vitamins, amino acids, and a fermentable carbohydrate. They ferment glucose to lactic acid, ethanol, and CO_2.

The organisms are important in fermentation and spoilage of foods. They are not pathogenic. Although they have physiological differences, Garvie (1983) has suggested that *L. cremoris, L. dextranicum,* and *L mesenteroides* belong to a single deoxyribonucleic acid (DNA) homology group and should be considered subspecies of *L. mesenteroides.*

PEDIOCOCCUS. These homofermentative cocci occur as single cells, pairs, or tetrads. They have rather complex nutritional requirements, such as vitamins and amino acids, which makes them useful to assay for these nutrients. These organisms are found in pickles, sauerkraut, beer, wine, and other fermenting materials.

AEROCOCCUS. This genus has only one recognized species, *A. viridans,* which used to be a *Pediococcus (P. homari)* (Buchanan and Gibbons 1974). The genus is very similar to *Pediococcus.* On blood agar, the colonies are surrounded by a green zone, hence the name viridans (green).

The aerococci have been found in air and dust, human infections, meat-curing brines, and on raw and processed vegetables.

Endospore-forming Rods and Cocci

This group includes the genera *Bacillus, Clostridium, Sporolactobacillus,* and *Desulfotomaculum.* Although *Desulfotomaculum* is listed with dissimilatory sulfate or sulfur-reducing bacteria (Krieg and Holt 1984), the discussion of this genus is included with endospore-forming bacteria. The spores that these organisms produce are different from the vegetative cells (Fig. 3.3) so there is another entity to consider. The spores are an inactive or dormant state of the organisms.

Figure 3.3. Spores formed in the center of the rod: *Bacillus subtilis* at left, *Clostridium sporogenes* at right.
Courtesy of Weiser, Mountney, and Gould (1971).

The organisms go through certain stages to change from a vegetative cell to a spore and back to a vegetative cell. The stages we can consider are cell sporulation, germination, and outgrowth.

SPORULATION. The mechanisms that trigger spore formation are not fully known. Most strains of clostridia produce spores when incubated in a good medium under anaerobic conditions, 3° to 8°C below their optimum growth temperature.

Muhammed, Morrison, and Boyd (1975) stated that demonstrating the complete sporulation requirements of an organism is impossible because some nutrients essential for spore formation also may be required for growth.

For all species, not all cells sporulate, regardless of the conditions. There is no information as to why some cells in a culture sporulate and others do not. However, there is agreement that spore formation begins after the exponential growth phase during the stationary phase. Spore formation can be arbitrarily divided into seven stages: (1) development of axial chromatin filament; (2) spore septation; (3) engulfment of the spore protoplast; (4) cortex formation; (5) coat formation; (6) maturation; and (7) the free spore stage. The spore consists of a core surrounded by several layers of mucopeptide and proteinaceous outer coats (Aronson and Fitz-James 1976).

An organism can initiate sporulation without completing the process and reverting to a vegetative cell. However, there is a stage at which the cell becomes irreversibly engaged in sporulation and is committed to completing the process. The composition of the substrate and the temperature of sporulation can affect the resistance characteristics of the resultant spores (Bayliss, Waites, and King 1981).

The main characteristic that is important to food microbiologists is the resistance of spores to heat, radiation, chemicals, desiccation, and freezing (Gould 1977). Quite commonly, the heat resistance of spores is about 10^5 times, and radiation resistance is about 10 times more than that of the corresponding vegetative cells. Spores can be stored for long periods and retain their ability to germinate and produce vegetative cells.

GERMINATION. Since spores are dormant, they must be converted to vegetative cells to be important in food spoilage or toxin production.

The conversion of the heat-resistant spore to vegetative cells would allow less severe thermal processes for food preservation.

Spores may need a conditioning treatment prior to germination. This process, called activation, may be induced by aging, heating, radiation, altering the pH, or with chemicals. Low numbers of viable spores do not always germinate immediately. Thus, canned foods that apparently pass short storage tests to determine potential spoilage, can show spoilage after extensive storage.

Activation is a reversible process and, if conditions do not allow germination, the spore reverts to its dormant state. Although some spores may germinate without heat shock, low levels of heat treatment, such as 65° to 85°C for 10 to 30 min, will activate most spores and induce germination. The heat treatment used depends upon the species and strain. Spores of species such as *C. botulinum* type E strains are heat sensitive and should not be heated above 70°C. Higher temperatures (10° to 115°C) for 3 to 10 min can be used to activate spores of the thermophilic *B. stearothermophilus.*

During germination, the bright, refractile spores become dark; dipicolinic acid is released and the cortex disintegrates. Although respiration in the spore is undetectable, there is an abrupt onset of respiration during germination. There is activity of a variety of enzymes, typical of the vegetative form. The spores lose their resistance to heat, desiccation, chemical agents, electric shock, and hydrostatic pressure. The germinating spores show a temporary rise in resistance to ultraviolet light and ionizing radiation followed by a rapid fall in resistance.

Although pasteurized milk supported germination of *B. cereus,* raw milk supported little or none (Wilkinson and Davies 1973). This helps to explain the spoilage defect due to *B. cereus* in pasteurized milk but not in raw milk. This inability to germinate in raw milk might be due to natural inhibitory agents found in raw milk that are inactivated by pasteurization.

A review of bacterial spore germination by Smoot and Pierson (1982) is suggested for further information on this subject.

OUTGROWTH. The development of a vegetative cell through the first cell division is called outgrowth. Outgrowth proceeds by the swelling of the spore, emergence from the spore coat, elongation, and cell division. Germination and outgrowth can be followed by determining the synthesis of RNA, proteins, and DNA, in that order.

Outgrowth occurs when germination takes place in a substrate capable of supporting vegetative growth. If the germinating medium is not sufficient to support growth, either development is stopped or the outgrowing cell may form a second spore with no intervening cell division.

This cycle of spore—cell—spore, is called a microcycle. Microcycle sporulation can be induced by dilution of an acceptable medium or by suspending germinated spores in a glucose-free medium.

The effects of various factors on the germination and outgrowth is important in food microbiology. It is the ability to determine viable from nonviable spores that allows us to establish thermal processes that will destroy spores in food. Also, if we could cause all of the spores to germinate prior to heat processing, the thermal treatment could be reduced.

BACILLUS. These organisms are usually Gram-positive rods, but older cultures may appear as Gram negative. The majority are motile, produce catalase, and produce acid but not gas from glucose.

The cells in this genus vary from strict aerobes to facultative anaerobes. The nutrient requirements vary from simple to complex. There are psychrotrophs, mesophiles, and thermophiles. Thus, for growth, the minimum temperature varies from $-5°C$ to about 45°C. The maximum for some species is 25°C and for others up to 75°C. The minimum pH for growth varies from pH 2.0 for *B. acidocaldarius* to pH 7.5 to 8.0 for *B. alcalophilus*. The salt tolerance is 2 percent or less for some species. Others can grow in 25 percent salt. Due to the diversity of the species, there have been suggestions that the genus should be divided. Five genera have been proposed, but this differentiation has not been adopted.

Bacillus can be found in soil, water, fecal material, decaying materials, and in various foods. Ingredients can serve as a source of *Bacillus*. Spices, flour, starch, and sugar have been incriminated as sources of spores for contamination of fermented sausages, bread, and canned food. The resistance of the spores of *Bacillus* to various agents makes these organisms important in food preservation. Some species, such as *B. subtilis*, decompose pectin and polysaccharides of plant tissue, causing spoilage of fresh plant products. Low-acid canned foods are spoiled by *B. stearothermophilus*, and *B. coagulans* causes spoilage of tomato products. *B. cereus* is involved in foodborne gastroenteritis and *B. anthracis* causes anthrax of both animals and man.

Besides potential spoilage and health hazards, some species of *Bacillus* can be useful in foods. The bacilli are a source of proteolytic enzymes that might be used to clot milk for cheese production. Some bacilli have been suggested for use in the production of single-cell protein. Some species are insect pathogens, which makes them useful in food production.

CLOSTRIDIUM. The species in this genus are divided into four groups on the basis of spore position and gelatin liquefaction (Buchanan and Gibbons 1974). The cells are usually Gram positive, at least in the early

stages of growth. They are catalase negative and, except for a few aerotolerant species, are strictly anaerobic.

The critical tolerance of NaCl is 2.5 to 6.5 percent. Sodium nitrate at 0.5 to 1.0 percent inhibits these organisms. The lethal chlorine concentration is 2.5 μg/ml.

The principal metabolic products of a species can be determined by chromatography (column, thin-layer, or gas). In conjunction with other tests, gas chromatography is a valuable aid in differentiating the clostridia. The analysis of soluble cellular proteins with polyacrylamide gel electrophoresis was used to differentiate species of clostridia (Cato et al. 1982). Anaerobic methods of analysis were reviewed by Anderson and Fung (1983).

The primary source of clostridia is soil. They are found in the intestinal tract of people and animals, as well as in various foods.

The species in this genus include two involved with foodborne illness, several that cause food spoilage, and many of no concern to food microbiologists. Some are free-living nitrogen-fixing organisms, some cause serious illness (tetanus, gas gangrene), and others are used to produce commercial chemicals such as butyric acid, butanol, acetone, and enzymes.

The spores of some species are very heat resistant and may survive the heat treatment of canned foods. If the surviving spores can germinate and the vegetative cells grow, spoilage will occur. If *C. botulinum* spores survive the heat treatment, germination, and outgrowth of the spores may result in the production of potent toxins.

DESULFOTOMACULUM. The species in this genus are similar to the clostridia. However, they are Gram negative and have a higher DNA base composition (G+C is 41–46 mol percent as compared to 23–43 mol percent). The clostridia do not reduce sulfate, but these organisms reduce sulfur compounds (sulfates, sulfites, and other reducible sulfur compounds) to H_2S.

D. nigrificans is a thermophilic spore former that causes sulfide spoilage of canned foods.

Regular, Nonsporing, Gram-Positive Rods

In this group, the genera *Lactobacillus* is the most important to food microbiologists. Other genera in food are *Brochothrix, Kurthia,* and *Listeria.*

LACTOBACILLUS. These organisms are straight to curved rods occurring singly or in chains. The rods vary from long and slender to short coccobacilli. (Figs. 3.4 and 3.5.) Generally they are nonmotile. Although considered to be Gram positive, as the culture ages, the cells may become

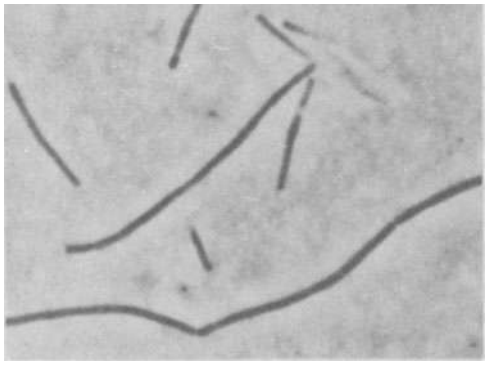

Figure 3.4. *Lactobacillus bulgaricus,* the high-acid-producing long rod of sour milk preparations such as yogurt (magnification = × 2,250).
Courtesy of Pederson (1979).

Gram-negative. Growth is enhanced by 5 to 10 percent CO_2. These organisms generally have complex nutrient requirements. Both homofermentative and heterofermentative types are in this genus.

Lactobacilli are found in plant and animal material, in various places in the body of human beings and other warm-blooded animals (including the intestinal tract) and in various foods.

In food microbiology, the lactobacilli are useful but also cause spoilage. They are useful in fermentations in which lactic acid production is desirable, whether in plant or animal products (sauerkraut, pickles, olives, fermented sausages). Some lactobacilli form slime and spoil sugar cane. Others cause green discolorations of sausage products. Vacuum-packaged meat becomes sour due to lactobacilli. Even in fermented vegetables, they may be a problem, causing pink sauerkraut or bloater formation in fermented cucumbers. The lactobacilli can cause spoilage of vinegar-preserved products, such as catsup and mayonnaise.

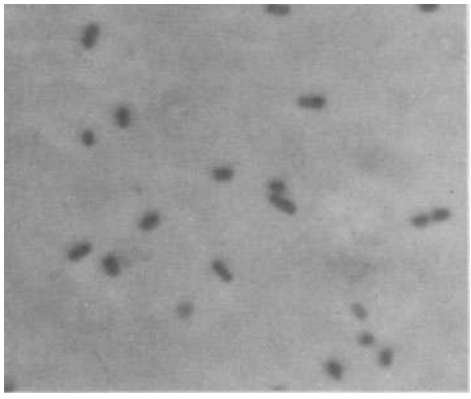

Figure 3.5. *Lactobacillus brevis,* the common heterofermentative species of the genus (magnification = × 2,250).
Courtesy of Pederson (1979).

Due to their complex nutrient requirements, lactobacilli are used in assays of food for vitamins, amino acids, and other nutrients. The lysine-excreting mutants of lactobacilli have been suggested for use in food and feed enrichment (Sands and Hankin 1974). Vandercook and Smolensky (1976) suggested using *L. plantarum* to detect adulteration of orange juice with imitation orange beverages. The imitation juice does not support the growth of this organism as well as does real orange juice.

BROCHOTHRIX. The species *B. thermosphacta* was previously named *Microbacterium thermosphacta.* It is a Gram-positive, facultatively anaerobic psychrotroph. It is associated with red meat and meat products, and may, in certain cases, cause spoilage (a sweet off-odor) of these foods.

KURTHIA. These Gram-positive cells are regular, unbranched rods in young cultures but become coccoid in older cultures by fragmentation of the rods. The cells are strict aerobes.

These organisms are found in intestinal contents, stagnant water, fresh and spoiling meat and meat products, meat-processing plants, and milk. According to Gardner (1969), *K. zopfii* is not known to cause spoilage of refrigerated meat, but its presence indicates that the meat was exposed to temperatures higher than refrigerator temperatures during processing, distribution, or retailing. In meat held at 2°C, *Kurthia* is overgrown by *Pseudomonas* species and other organisms. Although present in food, their importance might be only as an indicator of mishandling.

Irregular, Nonsporing, Gram-Positive Rods

This group of bacteria includes the genera *Corynebacterium, Arthrobacter, Brevibacterium,* and *Propionibacterium.*

CORYNEBACTERIUM. This genus is divided into three sections: (1) human and animal parasites and pathogens; (2) plant pathogens; and (3) nonpathogenic.

The coryneform bacteria are characterized by their pleomorphism. The *Corynebacterium* cells are straight to slightly curved rods but have a tendency to form club and pointed shapes. Generally they are not motile and are Gram positive. The best growth is aerobic, but they can grow in anaerobic conditions.

These organisms are widely distributed in nature. They are found in water, soil, plants, and animals. These organisms can be found in food products derived from both animals and plants. They have been associated with spoiling food, but it is doubtful they are primary spoilage organisms.

ARTHROBACTER. Organisms in this genus show considerable pleomorphism. The coccal form may appear as spheres, ovoid, or slightly elongated. When large cocci are transferred to a fresh medium, from one to three (seldom four) germination tubes arise from the cells. These develop into rods which vary in size and shape. Upon aging, the rods change almost completely into cocci. The cocci are Gram positive and the rods have Gram-positive granules surrounded by Gram-negative cellular materials.

The organisms are found in soil, in and on meat and poultry products, in milk products, dairy waste, activated sludge, and brewery and fish slime.

According to Buchanan and Gibbons (1974), the organism referred to as *Brevibacterium linens,* which is important in cheese, is related to and should perhaps be placed in the genus *Arthrobacter.*

BREVIBACTERIUM. Although this genus is listed in *Bergey's Manual* (Buchanan and Gibbons 1974), no species are recognized. However, Suzuki and Komagata (1983) listed several species of *Brevibacterium,* including *B. linens. Brevibacterium linens* is important in flavor production in cheese, especially limburger cheese.

PROPIONIBACTERIUM. Generally these organisms are Gram-positive rods, but they may be diphtheroid, club-shaped, coccoid, elongate, bifid, or branched. They are anaerobic to aerotolerant.

Propionibacters are found on people and in the intestinal tract of people and animals. During fermentation, propionbacters produce propionic acid and acetic acid, with lesser amounts of other organic acids. The species of most interest in food microbiology is *P. freudenreichii* subsp. *shermani.* This is used in the manufacture of Swiss cheese. Due to the production of propionic acid and CO_2, the organism is responsible for the characteristic flavor and the eyes in this cheese. These bacteria synthesize large quantities of vitamin B_{12} and produce propionic acid and can be used in the commercial production of these compounds.

Mycobacteria

Mycobacterium tuberculosis causes tuberculosis. So that this disease could be controlled, pasteurization of milk was inaugurated. Today, tuberculosis is primarily an airborne disease, rather than foodborne. *Mycobacterium* strains are found in raw milk, oysters, pork, and vegetables sprayed with sewage effluent (Hosty and McDurmont 1975; Thoen, Jarnagin, and Richards 1975; Van Donsel and Larkin 1977).

Streptomycetes and Their Allies

Some *Streptomyces* may be useful. *Streptomyces* produce extracellular α-galactosidase (Lyons, Pridham, and Hesseltine 1969). According to these workers, this enzyme may be useful during beet-sugar processing (aids in crystallization of sucrose) and be used for determining raffinose and for eliminating the agent in beans that induces flatulence. *Streptomyces* was suggested as a possible source of single-cell protein, protease enzymes, and glucose isomerase.

The Rickettsias and Chlamydias

In this grouping, the family Rickettsiaceae includes the genus *Coxiella.*

COXIELLA. These cells are short rods, occasionally appearing as diplobacilli or spheres. There are no flagella or capsules. The cells are Gram negative, but under certain conditions they may appear to be Gram positive. They are resistant to chemicals and elevated temperatures that normally inactivate *Rickettsia* species. These organisms will not grow on agar media but require host cells.

Since these organisms do not grow outside a host cell, they are not important in food spoilage. However, food can serve as a carrier of the organisms so that they may infect humans. *Coxiella burnetii* is the causative agent of Q fever. Infected cows, sheep, and goats shed the organism in their milk, which is the food source for human infection if raw milk is consumed. This organism is more heat resistant than the vegetative cells of most pathogens. Therefore, the time and temperature for pasteurization of milk are determined primarily to destroy these organisms. Heating at 62.8°C for 30 min or 71.7°C for 15 sec is adequate to eliminate *C. burnetii* from milk.

MOLDS

Molds are part of a larger group of microorganisms called fungi. The exact number of fungi is not known. Some fungi have not been isolated and identified, while others have been given more than one name. The fungi are ubiquitous, but soil, air, water, and decaying organic matter are prime sources.

Fungi are heterotrophic organisms that lack the definite root, stem, or leaves of higher plants. They possess a thallus and are called thallophytes. They are differentiated from the algae and higher plants by their lack of chlorophyll. Hence, they are saprophytic or parasitic. They differ

from bacteria by their more complex structure and greater size. The fungi may be multicellular or unicellular.

The fundamental structural units of molds are filaments or tubes called hyphae. By formation of crosswalls or septa, some hyphae form chains of cells that are septate. Others may not form septa and the hyphae are nonseptate or coenocytic. The septa have pores that allow the movement of cytoplasm from one cell to another. As the hyphae elongate, they intertwine. A mass of these intertwined branched hyphae is called a mycelium. Part of the mycelium grows into the substrate and absorbs food. This is known as the vegetative mycelium. The mycelium that remains in the air above the substrate and bears spores is called the aerial or reproductive mycelium. When the spores find a proper substrate, the cycle is repeated.

Molds can reproduce sexually, asexually, or by both systems. A fungus that has a sexual phase is known as a perfect fungus, while one that has no sexual phase is an imperfect fungus.

There are several types of spores formed by the fungi. The asexual fungi produce spores directly from or by the mycelium. These are called thallospores, conidiospores, or sporangiospores. There are three types of thallospores: blastospores, chlamydospores, and arthrospores. Sexual spores include oospores, zygospores, ascospores, and basidiospores.

Enumeration

With bacteria, one cell is usually responsible for the growth of a colony. However, a mold colony may result from a single cell, a spore, a piece of mycelium, or a number of cells. Thus there is a poor one-to-one relationship for molds. The problems involved with estimating growth by evaluation of the mycelium were discussed by Calam (1969) and Sutton and Starzyk (1972).

In the past, media for enumerating molds and yeasts were acidified to pH 4 or 5. The low pH inhibited bacteria and allowed the fungi to grow. However, fungi that have experienced a sublethal treatment might not grow on an acidified medium. Hence, media incorporating antibiotics rather than acidification to inhibit bacteria have been developed and tested for fungal enumeration (Baggerman 1981; Beuchat 1979; Henson et al. 1982; Mossel, Vega, and Put 1975; Nelson 1972). Dichloran has been added to media to inhibit the spreading of mold colonies (Henson 1981). The addition of sodium thiosulfate and sodium tetrathionate to media reduced the effect of heavy metal toxicity on the growth of fungi (Wainwright and Grayston 1983). Higher fungal counts were obtained using the spiral plater than the pour plate method (Zipkes, Gilchrist, and Peeler 1981). Microscopic methods are used to enumerate mold filaments

in tomato products as well as in other canned fruits and vegetables (AOAC 1985; Bandler and Cichowicz 1981). Determining the fungal chitin by conversion to glucosamine is useful in estimating fungal mycelia in tomato products (Bishop et al. 1982). These and other methods used for fungal analysis were reviewed by Jarvis et al. (1983).

Fungi in Food

Several types of fungi have been isolated from food. The extent of contamination is influenced by the prevalence in the environment. *Penicillium* and *Aspergillus* enjoy favorable conditions throughout the year, while some other fungi are limited to warm temperatures. *Cladosporium* and *Alternaria* are prevalent during the summer and early autumn.

Although important in all types of foods, molds are more apt to cause spoilage or be a health hazard in foods such as grain, flour, nuts, or fruit, which do not support the growth of bacteria due to low water activity or low pH.

Throughout the world, fungi rank second only to insects in causing the loss of stored products (Christensen and Kaufman 1974). In countries that have implemented rodent and insect controls, fungi destroy more stored products than does any other agent.

Importance of Molds

We usually consider that the appearance of mold on a food is an indication of spoilage. However, even before growth is evident to the naked eye, these organisms can be working to cause degradation of foods.

Although molds associated with food are not considered to be pathogenic, some produce mycotoxins. These toxic substances pose a potential health hazard to humans.

Molds can be useful in the processing of many foods such as cheese. Their enzyme systems can be isolated and used in many food processes. Some fungi are used to convert wastes into usable food (protein) for animal food or human use. Also, they are a source of vitamins.

The production of antibiotics by molds is well known. Antibiotics have been of great value in medicine, and they have been investigated for use as food preservatives.

Obviously many inert molds are associated with foods. They simply cannot grow due to an unsatisfactory environment or the overgrowth by bacteria that cause spoilage before the molds get a chance to multiply.

Zygomycetes

These fungi produce asexual spores (sporangiospores, arthrospores, and conidiospores) or sexual spores (zygospores). The hyphae contain no septa, but older hyphae may have septa.

The organisms of interest in food microbiology are in the order Mucorales. The family Mucoraceae include the genera *Mucor* and *Rhizopus*. The family Thamnidiaceae includes the genus *Thamnidium*.

MUCOR. Like other dimorphic fungi, species of *Mucor* can develop as either individual spherical cells that multiply by budding (as yeasts), or they form typical mycelia of coenocytic hyphae (Fig. 3.6). In normal aerobic conditions, growth is usually filamentous. Stoloniferous growth does not occur in mucor, which differentiates this genus from *Rhizopus*.

Organisms in this genus occur in soil and manure, and in fruits, vegetables, stored grain, and other foods. *Mucor* species are used in the Orient in food fermentations. *M. pusillus* produces an extracellular protease that has milk clotting activity. *Mucor* is a spoilage organism of berries.

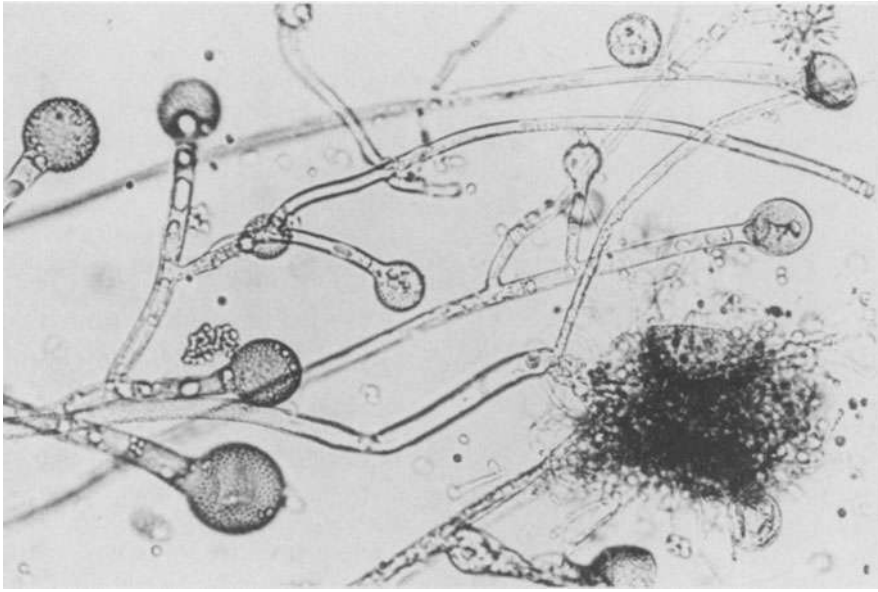

Figure 3.6. Hyphae and sporangia of *Mucor*.
Courtesy of Continental Can Co.

RHIZOPUS. These organisms have coenocytic mycelia and spread by stolons. Usually they reproduce asexually with sporangiospores or arthrospores.

These are common spoilage organisms of various stored foods. *Rhizopus stolonifer* is the common bread mold. Pectinolytic enzymes are secreted by *Rhizopus* species. The degradation of pectin results in the soft rot of various plant products. Due to the heat stability of the pectinolytic enzyme, the infection of fruit before heat processing can result in postprocess softening.

Rhizopus produces high yields of fumaric acid from fermentable sugars. Species are used in the production of fermented foods such as tempeh.

THAMNIDIUM. Organisms in this genus have coenocytic mycelia. They are found on meat as well as on soil and animal excrement. They grow on refrigerated meat, causing a defect referred to as whiskers. A process using *T. elegans* to tenderize meat was patented by Williams (1957). However, Kotula, Campano, and Kinsman (1982) found that the lipolytic and proteolytic enzymes of this organism had little effect on meat at 4°C, the typical temperature at which beef is held in coolers.

Ascomycetes

Both molds and yeasts are present in this class of fungi. Ascomycetes develop sexual ascospores in a saclike structure called an ascus. Besides this sexual phase, they may grow an extension of the hyphal tip or have an imperfect state in which asexual spores are produced. Those fungi for which no perfect state has been observed are listed in Deuteromycetes (Fungi Imperfecti). When a sexual phase is found for a fungus in Fungi Imperfecti, it is given a name according to the structure and form of the perfect state. This name takes precedence over the imperfect, asexual state.

There are between 1,900 and 2,000 genera in Ascomycetes. Only a few are of importance in foods.

BYSSOCHLAMYS. Besides asexual reproduction, the two species of importance (*B. fulva* and *B. nivea*) sexually produce heat-resistant ascospores. These organisms can grow at relatively low pH levels and in reduced oxygen tension. They produce strong pectolytic enzymes. With this combination of properties, they have been implicated in the spoilage of canned fruit and fruit juice. Spoilage may or may not be accompanied by gas. If gas is formed, the can may bulge slightly.

Besides spoilage, *B. fulva* produces a mycotoxin that is toxic to brine

shrimp, chicken embryos, and rats (Kramer, Davis, and Diener 1976). Chu, Nei, and Leung (1973) studied a renninlike enzyme (byssochlamyopeptidase A) that might be useful in milk clotting for cheese manufacture.

CLAVICEPS. The species *Claviceps purpurea* is of interest to food microbiologists because it produces toxic alkaloids on cereals. These contain a tetracyclic ring called lysergic acid. When ingested, these alkaloids cause numerous symptoms, especially hallucinations. With improved grain handling, the illness is rare in humans. The last major outbreak was in 1951.

NEUROSPORA. The name *Neurospora* is derived from the characteristically ribbed ascospores. The ascospores of *N. crassa* remain viable for many years. Heat shock for 20 min at 60°C aids in the germination of the spores.

In asexual reproduction, branched chains of pink conidia develop from upright stalks. Budding of the terminal conidia of the chain produces further conidia. When the terminal conidia form two buds, the chain branches.

Neurospora crassa and *N. sitophila* are two well-known species of this genus. Wild type strains of *Neurospora* have simple nutrient requirements. They have been used as tools in genetics and biochemical research. *Neurospora* can be found in warm, humid environments. *N. sitophila* causes problems in bakeries and is the red or pink bread mold. *N. sitophila* is used in the fermentation of red or orange ontjom, a fermented peanut press cake used in the Orient.

Deuteromycetes

This heterogeneous group of fungi has branching, septate hyphae and reproduces asexually by conidia or sclerotia. There are various systems used to classify the Fungi Imperfecti. Rather than attempt to segregate them, those of importance in food microbiology are discussed in alphabetical order.

ALTERNARIA. These organisms are characterized by muriform, dark-colored spores. The aerial mycelia are described as wooly, gray, brown, and olive-green (Fig. 3.7).

Alternaria is one of the most prevalent of the molds that cause spoilage of tomatoes in the field, attacking injured or weakened tissue. Due to darkening of the tissues, the defect is called black rot. Late-harvested tomatoes are particularly susceptible. These organisms cause defects of var-

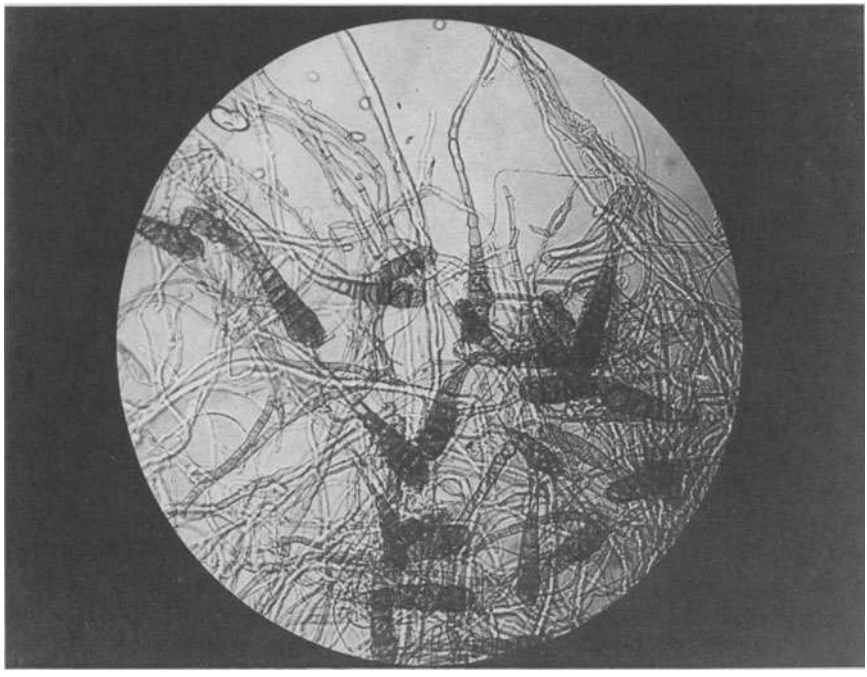

Figure 3.7. Hyphae and spores of *Alternaria*, a cause of rot of tomatoes.
Courtesy of Continental Can Co.

ious plant products, including internal mold of peppers (Halfon-Meiri and Rylski 1983). Also they are involved in the development of rancid flavors in dairy products.

A. alternata produces the enzyme β-D-galactosidase which, according to Macris (1982), is useful in reducing the lactose level in dairy products.

The *Alternaria* may be a health hazard because it produces mutagenic substances (Scott and Stoltz 1980).

ASPERGILLUS. There are more than 100 species in this genus. These organisms, along with penicillia, are known as storage fungi of grain. They discolor infected grain and reduce or destroy germination of the seed. These organisms can cause spoilage of a wide range of food products.

Certain aspergilli are very useful in food microbiology. Species such as *A. oryzae* are used to break down rice starch to glucose in the manufacture of sake and similar alcoholic beverages. Some strains are used in the production of shoyu (soy sauce) and miso (bean paste).

Strains of *Aspergillus* are used in the commercial production of citric, gluconic, and gallic acids. Almost all of the commercial citric acid is produced by *A. niger* growing in sucrose solutions. The organisms are a source of amylase and pectinolytic enzymes. A proteolytic enzyme of *Aspergillus* is able to clot milk and might be a substitute for rennet in cheesemaking.

These fungi may be useful as a source of protein. The mycelium of *A. niger* grown on brewery wastes contained 29 percent crude protein and might be used as a feed supplement (Hang, Splittstoesser, and Woodams 1975). Reade and Gregory (1975) described the use of *A. fumigatus* to convert the starchy root cassava to microbial protein for food or feed.

Several species of *Aspergillus* produce substances that are toxic to other biological systems Strains of *A. flavus* and *A. parasiticus* produce aflatoxins. There is considerable information that implicates aflatoxins with human illness.

The aspergilli are common contaminants of organic materials and soils. They are found on fruits, vegetables, stored grain, peanuts, and other food products. *Aspergillus* is shown in Figure 3.8.

BOTRYTIS. The asexual form of *Botrytis* reproduces by conidia. The conidiophores develop from a sclerotium and are irregularly branched. The conidia occur on short sterigmata. The positions and numbers of sterigmata cause the conidia to appear in grapelike clusters (Fig. 3.9).

B. cinerea is the common species of *Botrytis*. It is the gray mold of various plants and plant products, especially lettuce, tomato, strawberry, raspberry, and grape. It can be classed as a field mold, since it is a common soil contaminant and attacks fruits and vegetables in the field. It enters fruits through cracks and injured areas.

CLADOSPORIUM. These organisms are common in the soil. They can grow on connective tissue or the fat covering of meat when it is refrigerated for several days. This results in black spots on the meat. *C. carpophilum* causes peach scab, numerous dark circular lesions on the fruit. The organisms are associated with stored grains and dairy products.

COLLETOTRICHUM. Molds in this genus are involved with spoilage of foods. *C. circinans* causes onion smudge. Colored onions are resistant. *C. phomoides* causes anthracnose rot of tomatoes. Invasion by this organism is not dependent upon an injury to the fruit.

FUSARIUM. The conidia produced by these organisms have various shapes. The colonies may be fluffy and spreading, with colors varying

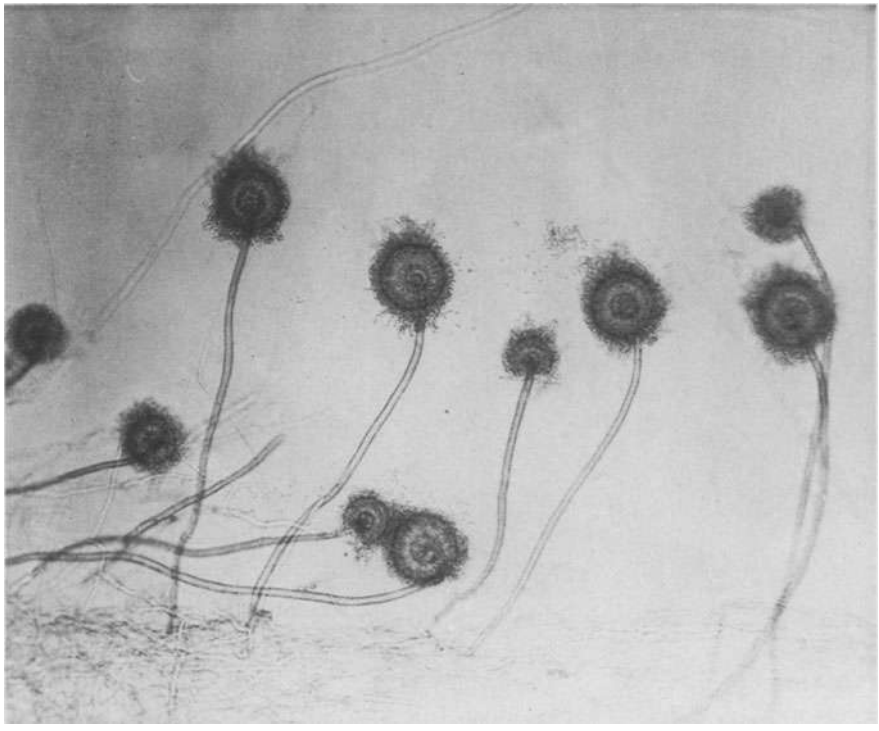

Figure 3.8. *Aspergillus,* showing mycelia and conidial heads.
Courtesy of Continental Can Co.

from white, through shades of pink, red, brown, yellow, orange, and blue, to purple.

The fusaria are widespread in nature, being found in soil, decaying material, and foods. Some fusaria are associated with plant diseases. A disease in rice due to *F. moniliforme* led to the discovery of gibberellic acid, a plant growth stimulant.

Fusaria cause tomato rot, entering the fruit through insect or other damage of the skin. *F. solani* causes a decay of potatoes called powdery rot, dry rot, or white rot.

The fusaria produce mycotoxins that affect various animals and possibly human beings. Animals refuse to eat feed that is highly infected with fusaria.

GEOTRICHUM. This yeastlike fungus grows rapidly at room temperature, forming white to cream-colored colonies. The septate, branching

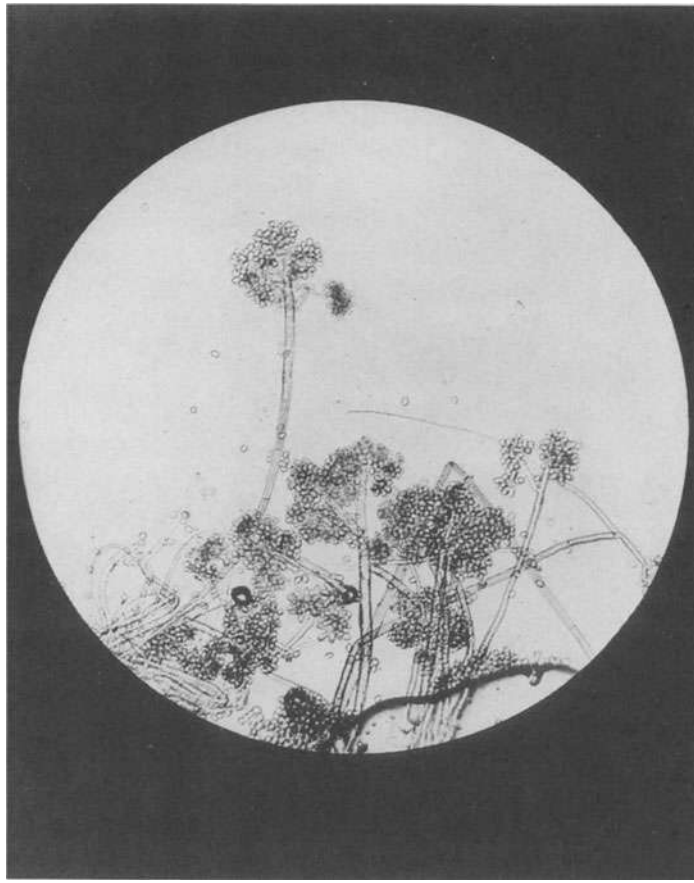

Figure 3.9. Mycelia and spores of *Botrytis.*
Courtesy of Continental Can Co.

hyphae fragment into chains of rectangular, barrel-shaped, or spherical arthrospores that readily break apart. These arthrospores are a means of reproduction. *Geotrichum* is depicted in Figure 3.10.

Geotrichum is a spoilage organism. It has been called dairy mold, because it is found growing on dairy products. It also causes watery rot, a common spoilage of tomatoes.

G. candidum is called machinery mold because it will grow on equipment on which are attached food particles or juices. As the food product is processed, it becomes contaminated with the mold. Hence, the mold is found in many types of processed foods. The mold is killed by heat used in thermal processing of food, but the hyphae can be deter-

Figure 3.10. *Geotrichum*—machinery mold.
Courtesy of Continental Can Co.

mined by microscopic examination. The presence of these mold filaments in processed foods has been considered to be an adulterant and an indication of inadequate sanitation in the processing plant. However, Splittstoesser et al. (1980) found little correlation between the aerobic plate count and the incidence of the mold in frozen blanched vegetables.

PENICILLIUM. There are many species of *Penicillium.* They are closely related to the aspergilli. The penicillia are characterized by branching of the conidiophore to form a brushlike conidial head (Fig 3.11). The organisms in this genus can be divided into three groups on the basis of the type of conidiophore branching. The conidia may be white, green, gray-green, blue-green, or yellow-green.

These organisms are widely distributed in nature and are found on many foods. They are important spoilage organisms of various fruits.

This is one of the storage fungi of grain. The species most often encountered are *P. cyclopium* and *P. viridicatum.* This is probably because they are able to grow at relatively low levels of moisture and temperature.

Figure 3.11. *Mycelia* and conidial heads of *Penicillium.*
Courtesy of Continental Can Co.

P. martensii and *P. viridicatum* are associated with blue-eyes, a blue-green discoloration of corn germs. Species of *Penicillium* are found growing on the fatty layer or connective tissue of meat that is stored in a refrigerator for several days, and on moldy bread.

Species of *Penicillium* are useful in various ways. Antibiotics produced by penicillia include penicillin. Some species are used in cheese manufacture. *P. camemberti* and *P. caseicolum* are important in Camembert, Brie, and similar cheeses, while *P. roqueforti* is used in Roquefort, Gorgonzola, and blue-veined cheeses. Schwimmer and Kurtzman (1972) suggested using *P. crustosum* to remove caffein from coffee. The penicillia produce enzymes such as glucose oxidase as well as proteins that can be used by the food industry.

Certain species of penicillia can be a health hazard. Some species have been associated with pulmonary and urinary tract infections. Penicillia produce mycotoxins. *P. islandicum, P. citrinum,* and *P. citreoviride* were involved in yellow rice disease which caused the deaths of several people.

Although *P. roqueforti* is used in cheese processing, toxin-producing strains of these species have been isolated (Scott et al. 1977), and roque-

fortine, a neurotoxin, has been isolated from blue cheese (Scott and Kennedy 1976).

SCOPULARIOPSIS. This genus produces conidia in brushlike clusters similar to *Penicillium.* The colonies appear cottony and may be cream-colored, yellow, brownish, chocolate brown, or almost black. They are never green like the *Penicillium.* One characteristic of *S. brevicaulis* is that it produces a poisonous gas (diethylarsine) with a garliclike odor when grown in the presence of arsenical compounds.

Scopulariopsis is found on all types of decaying matter and grows well on high-protein substrates.

Being proteolytic, *Scopulariopsis* is involved with the development of off-flavors in dairy products (especially Camembert cheese), and spoilage of meat. It has been detected on ham (both on the surface and in deeper tissues), stored eggs, and peanuts.

SPOROTRICHUM. The colonies are usually white but may be yellow, gray, pink, red, or green. It is a common soil inhabitant. *S. carnis* grows at low temperatures ($-5°$ to $-8°$C) and can cause a defect, called white spot, of refrigerated meat.

S. thermophile has an optimum growth temperature near 40°C. It produces cellulase and has been suggested for use in converting cellulose to simple carbohydrates (Coutts and Smith 1976). The use of *S. pulverulentum* as a potential food source was studied by Eriksson and Larsson (1975).

TRICHODERMA. This genus has irregularly branched conidiophores. The conidia are produced in slime balls.

Species are common in soil and on organic matter. The nitrogen content of the cells is about 5 percent, of which 60 to 70 percent is protein. Since the species *T. viride* is cellulolytic and has a high protein content, it has been considered for converting cellulosic wastes into foods.

YEASTS

The yeasts have been defined as fungi in which the usual dominant, or conspicuous, form is unicellular. The unicellular form gives yeasts an advantage over the mycelial form of molds. There is a greater surface-to-volume ratio that allows a higher metabolic activity. Also, the unicellular form is more readily distributed than is the mycelial form. Some yeasts produce a true mycelium by fission of cells that remain attached. A

pseudomycelium, formed by budding, has constrictions between the cells.

There are some 350 species recognized and classified into thirty-nine genera (Kreger-van Rij 1969; Lodder 1970). The classification of yeasts is based on morphological, cultural, sexual, and physiological characteristics.

On the basis of method of reproduction, the yeasts can be separated into four groups. Only two of these groups contain yeasts involved with foods. One group produces sexual ascospores in asci and is in the class Ascomycetes. These are the so-called true yeasts. The yeasts in the other group form no sexual spores and have no sexual life cycle. These are the false yeasts, Fungi Imperfecti, or Deuteromyces. All yeasts can reproduce asexually, and this is the only method for about 50 percent of them. The usual asexual (vegetative) reproduction is by budding (Fig. 3.12). Some yeasts reproduce by fission or by an intermediate system called bud fission. If the new cell or bud appears at the short end of the mother cell, it is called polar budding. If a bud forms at both ends, it is bipolar budding. The appearance of buds any place on the mother cell is multilateral budding.

Nearly all of the yeasts that produce sexual ascospores and are associated with food are in the class Ascomycetes, order Endomycetales, and family Saccharomycetaceae. For most yeasts, the maximum number of spores per ascus is four, but a few yeasts can produce eight. Strains of *Kluyveromyces* may produce a large number of spores (up to 100) per ascus. These ascospores have many shapes, including spheroidal, ovoid,

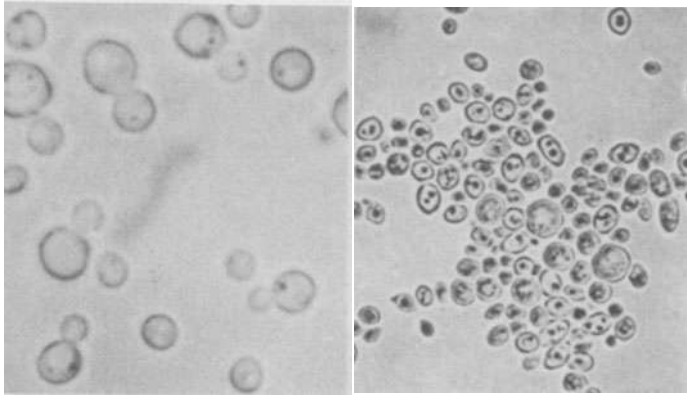

Figure 3.12. Cells of *Saccharomyces cerevisiae,* showing budding.
Courtesy of Amerine, Berg, and Cruess (1979).

reniform, elliptical, bean-shaped, cylindrical, and needle-shaped. Some spherical or oval spores have a ledge in the middle or to one side, which makes them appear saturn-shaped or helmet-shaped.

The production of ascospores may play a role in survival or in adaptation to altered environmental conditions. However, the main purpose is the rearrangement of heritable qualities.

The vegetative cells may have shapes similar to those of ascospores. Young yeast cells may be small and spherical and, as they grow older and larger, they may assume other shapes. The older cells become misshapen and scarred due to budding. Aged cells tend to become shriveled.

Some molds are dimorphic and have a yeast-like phase. Some yeasts produce a pseudomycelium or true mycelium. Hence, these organisms may be confused when a culture is examined, especially for borderline genera. The mold *Geotrichum* resembles the yeast *Trichosporon.* They both form a true mycelium and arthrospores. However, *Trichosporon* can reproduce by budding.

Most yeast cultures are cream, tan, or gray. However, some are yellow, pink, red, green, or brown.

The physiology of yeasts was discussed by Kreger-van Rij (1969) and MacMillan and Phaff (1973). Information about fermentation, assimilation, and nutritive requirements aids in identification of yeast species. This information was compiled by Lodder (1970) and is too extensive to repeat. The heat resistance of various yeasts has also been documented (Put et al. 1976; Put and De Jong 1982).

Yeasts are associated with nearly all types of food products. Foods such as fresh vegetables, meat, poultry, and cheese often contain yeasts, but in these foods, bacteria usually outgrow the yeasts. When bacterial inhibitors are added, yeasts can dominate. Osmophilic yeasts such as *Saccharomyces rouxii* can tolerate high sugar concentrations. These yeasts are found in foods such as honey, molasses, sugar, and fruit. Salt-tolerant yeasts grow as films on brined food and on salted food and ham.

Certain strains of yeasts are very important in the baking industry, in chemicals, in single-cell protein, in sewage disposal, and in the fermentation of alcoholic beverages.

Methodology

Sublethally stressed yeast is recovered at maximum levels with a medium at pH 8.0 or over (Nelson 1972). Therefore, the use of potato dextrose agar (PDA) acidified to pH 3.5 is not satisfactory for enumeration of yeasts in many food products. To inhibit bacteria and to allow yeasts to grow, a combination of antibiotics is used in place of acidification.

Fluorescent microscopy can be used to detect yeasts in certain products (Andrews 1982; Aries 1976; Cranston and Carver 1974).

Classification

In this text, the genera and species of yeasts are those listed in Lodder (1970). As more information accumulates regarding the relationships of yeast such as DNA base composition and serological reactions, there may be some revisions in the classification (Price, Fuson, and Phaff 1978).

The yeasts are divided into Ascomycetes (the ascosporogenous yeasts) and Deuteromycetes (the asporogenous yeasts).

ASCOMYCETES. These yeasts produce sexual ascospores as well as reproduce asexually. Unless otherwise noted, asexual reproduction is by budding. Certain genera such as *Citeromyces, Dekkera,* and *Endomycopsis* have been isolated from food and may be involved with spoilage or fermentation. Due to space limitations, these are not discussed in this text.

Debaryomyces. The vegetative cells are often spherical to globose and reproduce by multilateral budding. The sexual ascospores, produced by conjugation of the mother cell and bud, are spherical or oval. There are usually one or two spores per ascus. Fermentation is weak, slow or absent. Nitrate is not assimilated.

D. hansenii has a high salt tolerance, being able to grow in media with 18 to 20 percent salt. It has been isolated from brined food and salted meat products and can use nitrate as the sole source of nitrogen. *Debaryomyces* species are perhaps the most widely distributed film-forming yeasts associated with food brines.

Species of *Debaryomyces* have been associated with spoiled mushrooms, cheeses, cider, white wine, tomato puree, sausage, and salted beans.

Hanseniaspora. These yeast cells are diploid, lemon-shaped, ovoid, or long ovoid. They are the ascospore stage of the genus *Kloeckera.* They have a vigorous fermentation, but have a low tolerance to alcohol (about 4 to 6 percent). All species require inositol and pantothenate for growth. Due to this need, they can be used to assay for these compounds.

These yeasts are found in the soil of orchards and vineyards and in all insects. They have been isolated from grape berries and grapes and have been associated with fermenting spoiled fruit.

Hansenula. This genus reproduces by multilateral budding and ascospores. A pseudomycelium or a true mycelium may be formed. These yeasts assimilate nitrate.

Species of *Hansenula* have been isolated from foods such as grain, fruit, shrimp, and brines of fermenting cucumbers and olives.

Some species and strains are useful. According to Batra and Millner (1974), certain species are used in India to produce kanji (a beerlike beverage) and murcha (rice wine). A strain of *Hansenula* grows on methanol to produce single-cell protein (Levine and Cooney 1973).

Kluyveromyces. These organisms reproduce by multilateral budding. The cells may be spherical, ellipsoidal, cylindrical, or elongate. Some species produce a red pigment, but the colonies are grayish-white, brownish-cream, yellowish-gray, sometimes tinged with pink. The organisms show a vigorous fermentation. They grow at temperatures of from 5° to 46°C.

This genus has a widespread distribution. They are found in various foods such as fruits, preserves, corn meal, grape must, milk, and other dairy products. The species that ferment lactose, such as *K. lactis,* are found in dairy products. Some species are osmophilic. Organisms of this genus cause spoilage of foods, especially figs and dairy products. Certain strains have been suggested as a source of the enzyme polygalacturonase.

Pichia. The cells have many shapes. Asexual reproduction is by multilateral budding, with most species forming a pseudomycelium. There is limited formation of a true mycelium.

The colonies are slimy or pasty and colored yellowish-white, yellowish-brown, tan, white, cream, grayish-white, or red.

These are common yeasts found in various sources. Their association with foods includes both fermentations and spoilage. Foods in which these yeasts are found and can cause spoilage include fruit, beer, dairy products, sorghum, and wine. They can occur as films on brined foods.

Saccharomyces. This genus is considered to be a rather heterogeneous group of organisms. The cells are spheroidal, ellipsoidal, cylindrical, or elongate. Vegetative reproduction is by multilateral budding. Pseudomycelia may be formed, but true mycelia are absent. On agar, the colonies are generally white or cream colored, with a typical yeasty odor. Cowan and Bryant (1981a, 1981b) developed a serological classification of brewing yeasts using a microimmunoelectrophoresis technique.

The name *Saccharomyces* means sugar fungus. All species have a vigorous fermentation. Some species are among the most useful yeasts, due to the production of alcohol (brewing) and carbon dioxide (baking). Strains of *S. uvarum* and *S. cerevisiae* are most often used in these fermentations. Besides these important uses, species and strains have been used to remove glucose from egg white prior to drying, to remove the mucilage layer from coffee beans during processing, to assay for vitamins (biotin, pyridoxine, pantothenic acid, thiamin), as a source of enzyme (invertase),

as autolyzed yeast extract for flavoring in place of meat extracts, as a source of ergosterol, and as a source of single-cell protein.

Sinai et al. (1974) fed brewer's yeast (*S. cerevisiae*) to Rhesus monkeys and found that the yeast significantly enhanced the monkeys' resistance to respiratory and enteric infections. They believed the resistance was due to the stimulation of phagocytes.

The organisms in this genus have a widespread distribution. They are associated with a variety of foods and can cause spoilage in fruit and fruit products, sugar, syrup, honey, vinegar, mayonnaise, salad dressing, dairy products (buttermilk, cheese), and fermenting foods such as cucumbers for pickles.

S. aceti, found in wine, can convert alcohol to acetic acid. *S. bailii* var *osmophilus* is found in substances with a high concentration of sugar, alcohol, sulfur dioxide or acetic acid. *S. bailii* var *bailii* has been associated with spoiled wine, vinegar, salad dressing, and mayonnaise. Two sugar-tolerant (osmophilic) species, *S. rouxii* and *S. bisporus,* are the main spoilage organisms of dates and can cause spoilage of honey, syrup, and other foods with up to 60 percent sugar (Restaino et al. 1983).

Occasionally, species of *Saccharomyces* exhibit pathogenic behavior (Eschete and West 1980).

Schizosaccharomyces. The cells are spheroidal to cylindrical. Vegetative reproduction is by fission. The absence of budding distinguishes this genus from other yeasts. The cells may form a rudimentary true mycelium that can break up into arthrospores. There are four to eight spores per ascus.

The name *Schizosaccharomyces* indicates the close relationship to *Saccharomyces* (fermentation of glucose to alcohol), but also a difference (fission as compared to budding).

As with other yeasts, these species are widespread. They have been associated with spoilage of prunes, figs, raisins, and wine.

Yang (1973) suggested using *S. pombe* in the fermentation to produce wine from grapes with a low pH. The yeast converts malic acid to lactic acid, resulting in a less acid wine than when *Saccharomyces cerevisiae* is used. *S. malidevorans* is noted for its ability to convert malic acid to lactic acid.

DEUTEROMYCETES. These yeasts do not produce sexual spores. Unless otherwise stated, these asporogenous yeasts reproduce by budding. Certain genera, such as *Brettanomyces* and *Kloeckera,* are found in foods and may be involved in spoilage or fermentation.

Candida. This is a rather large genus of yeasts. All species form pseudomycelia; by fission, some form true mycelia.

Organisms in this genus are widespread. Besides being found in soil,

water, and air, they are found in plants, insects, higher animals, humans, sewage, on processing equipment, and in food products.

These organisms have been involved with spoilage of various foods such as frankfurters, fresh fruits, vegetables, dairy products, brines, and alcoholic beverages. *C. vini* (formerly *Mycoderma vini*) forms "flowers" on wine and causes spoilage. *C. mycoderma* is a film-forming yeast on fermenting olive and cucumber brines. The psychrophilic strains tend to become dominant in refrigerated fruit juices.

This genus contains strains of yeast that are very useful in food industries and in associated industries. They are used as food yeasts, a source of lipid, vitamins, invertase, lactose, and lysine. *C. utilis* is used as a fodder yeast for animals and is a potential food yeast for humans.

There are potentially pathogenic species of *Candida* for both people and animals. However, there is no information that food is the source of these infective agents. *Candida* species are found in the intestinal tract, and it has been suggested that they may cause diarrhea in malnutrition (Gracey et al. 1974).

Rhodotorula. The cells (spheroidal, ovoidal, elongate) reproduce by multilateral budding. There may be a rudimentary pseudomycelium. The organisms form red or yellow carotenoid pigments. These organisms are widespread and are involved in the spoilage of a wide range of foods.

Species of this genus may be useful as a source of lipids, cystine, and methionine and for degrading wastes in the production of single-cell proteins.

Torulopsis. Torulopsis has been defined as a heterogeneous group of imperfect yeasts that cannot be placed in the homogeneous genera. The cells are spherical or oval to elongate and reproduce by multipolar budding. *Torulopsis* can be differentiated from *Candida* by not being able to form a pseudomycelium, from *Cryptococcus* by not producing starch, and from *Rhodotorula* by not producing carotenoid pigments.

The colonies are usually white to cream and are usually glossy or shiny but may be dull. The tolerance to sodium chloride varies from 2 percent to 21 percent ($^{w}/_{v}$) for different species. At 30°C, *T. halonitratophilia* is an obligate halophile. There are osmophilic species in this genus.

Species of *Torulopsis* are found in various types of foods. They are credited with causing surface slime on cottage cheese, spoilage of refrigerated beef, cream, butter, sweetened condensed milk, and various food brines.

Torulopsis magnoliae has been suggested for use in the production of glycerol. *T. ernobii* is a source of lipase. *T. utilis* is a source of single-cell protein. *Torulopsis* strains have been found in sourdough, apparently aiding in the leavening of this product.

Trichosporon. The cells of *Trichosporon* species are of various shapes. Pseudomycelia and true mycelia may be formed. The presence of arthrospores differentiates this genus from *Candida.*

These organisms are found on various foods such as fresh shrimp, crab, butter, cheese, fruit, fruit juice, and rice. They are part of the flora used in the production of idli (Batra and Millner 1974), and may be a source of protein.

VIRUSES

Viruses are obligate intracellular parasites. They generally have a limited host range. However, nearly all forms of life are susceptible to some type of virus. To survive and replicate, a virus must contact and invade an acceptable host cell. If a host cell is not available, the virus may become inactivated. This inactivation is influenced by temperature, humidity, pH, and substrate composition as well as other factors.

There is some doubt that viruses are alive. Cliver (1971) stated that they are not alive, but borrow life from the host cell and direct it to produce more viruses.

Viruses have a genetic system with either RNA or DNA. This is surrounded by a protein coat, called the capsid. The capsid protects the genetic material from nucleases, serves as the vehicle of transmission from one host cell to another, and forms a bond with the host cell during the attachment stage of infection.

For a virus to infect a host cell, there must be an attachment site on the host cell. The attached virus injects its nuclear material into the host. This can result in the viral genetic material causing the cell to manufacture new viruses, or the genetic material may attach to the cellular chromosome and the cell becomes lysogenic. If new viral particles are produced, usually the cell is lysed upon release of the new viruses. Then these viruses must contact another acceptable host.

If the virus material becomes attached to the chromosome of the cell, the genetic material of the virus is reproduced along with that of the cell (Campbell 1976). This lysogenic state may continue for an indefinite period. Eventually some disturbance triggers the viral genetic material to become activated, and the cell produces new viruses, which are subsequently released.

Although the viruses are inactive as far as the reaction on food is concerned, they are important. Since they are obligate parasites, and some cause disease in human beings, the presence of viruses can be considered potentially harmful. They can be put into the category of an environmental condition that allows spoilage, since they, as bacteriophages,

can attack useful fermentative organisms, resulting in an unsatisfactory product being developed. Since they are involved with genetic transfer (transduction), the viruses might be considered helpful in the development of better strains of useful microorganisms. However, transduction may be undesirable when it results in unacceptable mutant strains that are less sensitive to inhibitors, are more virulent, or lose the characteristics that made them useful. Transduction is known to occur in only a few groups of bacteria.

Intestinal Viruses

The primary viruses that can infect people and are foodborne are the intestinal or enteric viruses. Cliver (1971) included the enteroviruses, adenoviruses, reoviruses, and the agents that cause infectious hepatitis and epidemic viral diarrheas as intestinal viruses.

The enteroviruses are spherical and about 20 to 30 nm. They have single-stranded RNA. These viruses are stable at pH 3.0. This tolerance to low pH allows them to survive passage through the stomach. They replicate in the intestinal tract, are shed in the feces and, if they contaminate food that is subsequently ingested, the process is repeated. Wilner (1973) included poliovirus, coxsackievirus A, coxsackievirus B, and echovirus as enteroviruses.

The adenoviruses are hexagonal, about 70 to 90 nm in diameter. They contain double-stranded DNA. They are acid stable. There are thirty-three serotypes of adenoviruses that infect human beings (Wilner 1973).

The reoviruses are hexagonal, and about 70 to 80 nm in diameter. They contain double-stranded RNA. The reoviruses, like the other intestinal viruses, are stable at pH 3 to 5.

The agents that cause infectious hepatitis have been described (Provost et al. 1975). They are spherical, about 27 nm in diameter. Viral diarrheal agents have been isolated or characterized. However, these agents pass through bacterial filters and can be transmitted through several people and still cause diarrhea. If the agent were merely a chemical toxin, this transmission would dilute it beyond the ability to cause illness.

Since these viruses are intestinal, they are found in feces of infected persons and resultant sewage. Present sewage treatment does not inactivate all of the viruses from the effluent or the sludge (Goddard, Bates, and Butler 1982; Irving and Smith 1981; Larkin, Tierney, and Sullivan 1976). When sewage wastes are used to irrigate or fertilize crops, the vegetation is contaminated. Larkin, Tierney, and Sullivan (1976) found that poliovirus persisted on radishes and lettuce for up to thirty-six days. Such foods are eaten raw, so any contaminating viruses that are not removed by washing are ingested. When sewage contaminated with viruses is

dumped into coastal water, the shellfish growing in these waters tend to bioaccumulate these viruses (Metcalf et al. 1980). Shellfish have been the source of enteric viruses in several outbreaks of gastroenteritis (Eyles, Davey, and Huntley 1981).

The procedure for the detection of viruses in or on foods consists of several parts. First, the viruses are separated from a food suspension by differential filtration or centrifugation. Then, the viruses are concentrated by centrifugation or by filtration. Bacterial cells are destroyed in the viral concentrate by adding antibiotics. Aliquots of the treated viral suspension are planted on a monolayer of susceptible host cells. After incubation, the monolayer of cells is observed for plaques (holes) produced by lysis of cells due to the viruses. The number of plaque-forming units (PFU) is determined by counting the plaques and multiplying by the calculated dilution factor. Several modified and improved methods for concentrating viruses from water have been suggested (Guttman-Bass and Armon 1983; Keswick et al. 1984; Lipson and Stotzky 1983; Shields, Berenfeld, and Farrah 1985; Wait and Sobsey 1983). Absorption, concentration, and detection of viruses in fecal samples were described by Pontefract and Bergeron (1985). The methods used to concentrate and detect viruses were reviewed by Cliver, Ellender, and Sobsey (1983a, 1983b) and by Richman et al. (1984). Enzyme-linked immunosorbent assays have been described for the detection of viruses (DeBokx and Maat 1979; Locarnini et al. 1979; Marco and Cohen 1979). Fortunately, the viral contamination of food is very low to nonexistent (Kostenbader and Cliver 1977; Larkin 1981).

Bacterial Viruses (Bacteriophages)

The taxonomy of bacteriophages was discussed by Ackermann and Eisenstark (1974). The phages can have at least four relationships with their host cells. These are as follows: (1) Virulent phages cannot combine with the host's chromosomes to establish a lysogenic relationship; (2) pseudolysogenic phages can establish a carrier state in the host and may act as transducing agents; (3) defective temperature phages form particles with morphological attributes of a normal phage, but cannot replicate; and (4) true temperate phages are able to replicate and lysogenize their host cells.

The genetic material of temperate phages can become attached to the chromosome of the host cell. The properties of the phage are modified and it becomes a prophage. The prophage acts as a bacterial gene, dividing the bacterial chromosome and being transmitted to daughter cells. A bacterium that carries prophage is called lysogenic. Agents such as UV light, X-rays, organic peroxides, nitrogen mustard, and mitomycin C can

cause the release of phage material from the chromosome, with resultant production of phage and lysis of the cell when the phage is released.

Bacteriophages can be stored by freezing with liquid nitrogen, using or not using glycerol (19 percent) as a protective additive, and holding at −196°C. They can be held for about two years in a broth lysate stored at 4°C, with only slight loss of titer. Freeze-drying or aerosolization results in a loss of titer of phage (Cox, Harris, and Lee 1974). The loss occurs during rehydration of the phage.

Bacteriophages can be enumerated by determining the clearance of susceptible bacterial cultures in broth or by the formation of plaques on lawns of the bacteria. In some cases, the phages must be concentrated prior to enumeration (Seeley and Primrose 1982).

Perhaps the most important function of phages in foods is their role in destroying cultures of lactic acid bacteria during the fermentation of dairy products.

There is a possibility that phages could be used to control unwanted bacteria in various food products. The main problems appear to be the host specificity of the phages and the development of lysogenic bacteria. The lysogenic bacteria can continue metabolizing and cause spoilage of foods.

Phages can be useful in epidemiological work. Since phages are rather host specific, a set of phages can be devised and used for typing bacterial species or strains. Phage typing can aid in tracing the origin of contamination or infection of foodborne outbreaks of gastroenteritis.

One reason that phages can lyse an entire bacterial culture is that they have a high rate of replication. If a bacterium has a generation time of 20 min, in 1 hour there will be eight cells. A phage may require from 40 to 80 min for replication, but the burst size (number of phages released on lysis of cell) may be from 2 to over 100. It is obvious that with a large burst size, in only a short time the phages will outnumber susceptible bacterial cells.

Fungal Viruses

Lemke and Nash (1974) reviewed the fungal viruses. According to them, viruses have been reported for more than sixty species of fungi. They believe the metabolism and genetics of fungi may be influenced by these viruses. All of the fungal viruses have double-stranded RNA, and most have a polyhedral shape. In a review of viruslike particles of yeast, Bruenn (1980) questioned referring to these agents as viruses. According to him, no extracellular particles are present and they lack the criterion of infectivity normally associated with viruses.

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4

Factors That Affect Microbial Growth in Food

Food microbiologists need a knowledge of the factors that influence microbial growth. Desirable growth conditions are needed for enumeration, fermentations, or the production of single-cell protein. Undesirable conditions are used for food preservation.

Discussions concerning environmental conditions for growth attempt to relate the macroenvironment of the system to the microenvironment in which a microorganism exists. This has been necessary because little or no information exists about the microenvironments in foods and other products. There are many microenvironments with different conditions in a food product. The microenvironments in a liquid food are less heterogeneous than are those in a solid food. Some microorganisms that could not grow in the measured macroenvironment might be able to grow in a microenvironment of the food.

The microenvironments in food products are changing constantly. The reactions catalyzed by enzyme systems naturally inherent in some foods result in heat being generated, atmospheric oxygen used, and carbon dioxide and other gases given off. These changes in the food affect the processes of microbial systems. The environment determines the conditions under which the microorganisms exist, and microorganisms influence the conditions prevailing in the environment. In any food environment, certain microbial species survive and become dominant. Organisms that lack the ability to withstand stresses induced by an unfavorable environment will succumb. Microorganisms can respond rapidly to changes in the growth conditions.

With optimum conditions for growth, some bacteria will reproduce every 15 to 20 min. Although they may grow in conditions above or below the optimum, the lag phase or the generation time may be longer (Fig. 4.1). If the stresses become too severe or increase in number, the organisms might not grow or might fail to survive.

The optimum conditions for microbial growth are influenced by the enzyme systems. The enzymes are organic catalysts that increase the rate of reactions. The rates of metabolic reactions are influenced by the environmental conditions that affect the activity of the enzymes.

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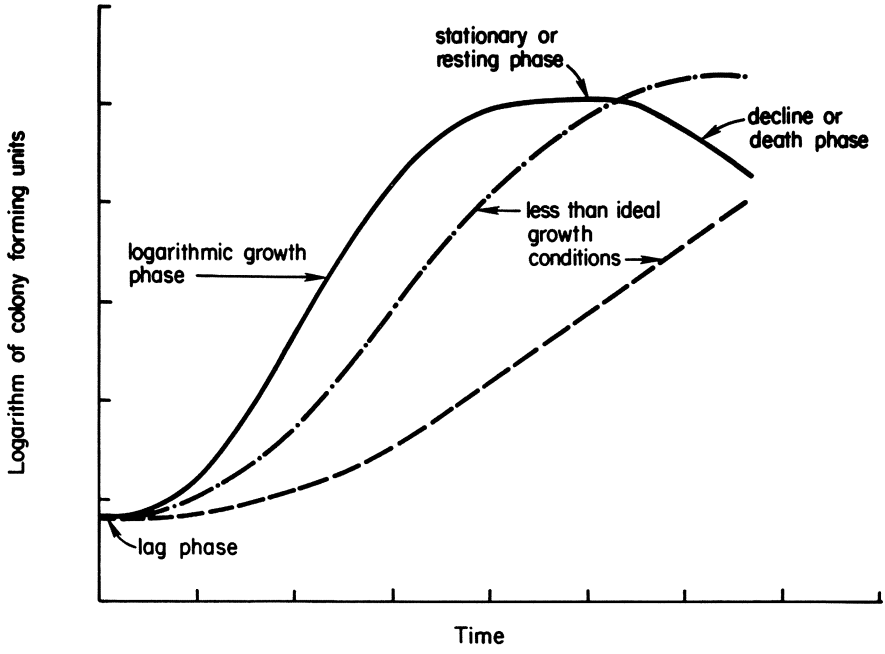

Figure 4.1. Bacterial growth curves. Under less-than-optimum conditions, the lag phase, generation time, or both may be extended.

The conditions that affect the metabolism and multiplication of microorganisms include nutrients, water, pH, inhibitors, oxygen, light, temperature, and time. Since organisms are part of the environment and can alter it, microbial interactions (the effect of one organism on another) are important. The previous stresses (sublethal heating, freezing, or radiation) that a microorganism has endured will affect its ability to cope with the environment.

NUTRIENTS

All biosystems require certain chemicals and chemical reactions in order to survive and reproduce. The microbial cell must be able to obtain a source of energy and synthesize cellular protoplasm from its environment. Organisms require a source of carbon and nitrogen, growth factors, such as vitamins, minerals, and water. The ability of organisms to utilize compounds and synthesize cellular components depends upon the enzyme systems that the organism can make, according to its genetic

code. Since the genetic code can change (mutate) there may be slight or important differences between strains of a species of microorganism. The substrate and other environmental factors affect the amount of enzymes produced by microorganisms. Some enzymes of microorganisms are listed in Table 4.1 and are discussed more fully in Chapter 9.

Energy and Carbon Sources

In general microbiology a distinction is made between the types of energy sources and carbon sources of microorganisms. The organisms of primary concern in food microbiology are the chemoorganotrophs that use organic compounds as their energy and carbon sources. By definition, these microorganisms are heterotrophic.

Nitrogen Source

Amino acids are needed to produce cellular proteins, including enzymes. A number of organisms, such as *Escherichia coli,* can utilize nitrogen in the form of nitrates or ammonia to produce amino acids. Other organisms need one or several amino acids supplied to them in the substrate.

Other Growth Factors

Some organisms need growth factors, such as vitamins, purines, or pyrimidines. For some fastidious organisms, metabolites must be supplied as preformed molecules. In general, the Gram-positive bacteria require more preformed growth factors than do other microorganisms. Vitamins function in coenzyme systems as listed in Table 4.2.

The obligate parasites, such as viruses, require a living organism to supply the machinery to synthesize their components.

TABLE 4.1. SOME ENZYMES ASSOCIATED WITH MICROORGANISMS

Enzyme	Substrate	Major Product
Invertase	Sucrose	Glucose, fructose
Maltase	Maltose	Glucose
Ribonuclease	Ribonucleic acid	Nucleotides
Lactic dehydrogenase	Lactic acid	Pyruvic acid
Proteinases	Proteins	Peptides, amino acids
Amylase	Starch	Dextrins, maltose, glucose
Lipases	Fats	Glycerol, fatty acids
Catalase	Hydrogen peroxide	Water, oxygen
Urease	Urea	Carbon dioxide, ammonia

TABLE 4.2. FUNCTIONS OF VITAMINS

Vitamin	Coenzyme	Function-Reaction
Biotin	Biotin coenzyme	Carboxyl transfer, deamination, CO_2 fixation
Pantothenic acid	Coenzyme A	Acyl transfer or exchange
Folic acid	Tetrahydrofolate	Transfer of one-carbon units, methylation, purine, and pyrimidine synthesis
Thiamin (B_1)	Thiamin pyrophosphate	Accepts carboxyl groups from decarboxylation of keto acids; TCA cycle, lipid metabolism
Riboflavin (B_2)	Flavin mononucleotide (FMN)	Dehydrogenation, electron transfer-oxidations
	Flavin adenine dinucleotide	Dehydrogenation
Nicotinamide (B_5) (Niacin)	Nicotinamide adenine dinucleotide	Dehydrogenation, energy generation
	Nicotinamide adenine dinucleotide phosphate	Dehydrogenation, hydrogen acceptor, synthesis of fatty acids
Pyridoxine (B_6)	Pyridoxyl phosphate	Deamination, racemization, transamination, decarboxylation of amino acids
Cobamide (B_{12})	Cobamide coenzyme, deoxyadenosyl (B_{12})	Transfer of methyl groups

It would be difficult to list the essential metabolites of specific microorganisms. There are shifts in the nutritional requirements of the strains comprising a culture. Not only do species of organisms have different essential metabolites, but strains within the species differ in their requirements.

Elements and Minerals

Certain elements or minerals found in cellular components are needed in trace amounts by microorganisms. Somewhat larger amounts of sodium, potassium, calcium, and magnesium are needed, as compared to iron, copper, manganese, zinc, cobalt, and molybdenum.

Phosphorus and sulfur are also needed by microorganisms. Trace elements enhance the activity of some enzyme systems and are needed for the production of toxins and other secondary metabolites.

Foods as Nutrients

Foods contain carbohydrates, fats, proteins, vitamins, minerals, water, and other factors which microorganisms need for growth. Since they are derived from biological systems, foods vary in their composition. The components of the soil can affect the chemical composition of plants and

the food derived therefrom. Variations in composition are also due to geography, climate, season, maturity, processing methods, variety, and market conditions. The type of diet eaten by an animal also can affect the composition of the food derived from this source.

From their general composition, foods are classed as protein, carbohydrate, or fat. The protein foods are primarily of animal origin, the carbohydrate foods of plant origin, and fats or oils are derived from animals and plants.

Fresh meats contain essentially no carbohydrate (less than 1 percent), since most is converted to lactic acid by glycolytic reactions during rigor. The vitamin and mineral content of most foods is sufficient for the growth of most organisms. The composition of foods influences the types of microorganisms that will grow, as well as the products that are formed during microbial growth and spoilage.

MOISTURE

Some microorganisms can remain alive in a dried condition but cannot carry out their normal metabolic activities or multiply without water. Microorganisms can grow only in aqueous solutions. They cannot grow in pure water or in the absence of water. Water dissolves more substances than any other solvent. In this capacity, water is used to bring nutrients into the cells and to dispel waste products. Water is involved in the chemical reactions that break down substrates to usable molecules. These reactions include the hydrolysis of the peptide bonds in protein, the ester bonds in fats, and the conversion of polysaccharides to monosaccharides. Water is a donor of hydrogen ions, contributes to the regulation of cellular pH and temperature, and is the lubricant of tissue (Beall 1983).

The water in a food is both bound and free. Bound water is held by physical forces to macromolecules and is not available to act as a solvent or to participate in chemical reactions. Hence, it is not available to microorganisms for metabolic activity.

When solutes are dissolved in water, the freezing point is lowered, the boiling point is increased, the vapor pressure of the water is reduced, and there is potential development of an osmotic pressure.

Water Activity

This is an index of the availability of water for chemical reactions and microbial growth. Water activity (a_w) is the ratio of the vapor pressure (VP) above a material or solution (P) and that of pure water (P_o) at the same temperature. The values of water activity range from 0 to 1.

The *VP* of a pure liquid depends upon the rate of escape of molecules from the surface. The escape of water to the air is measured by the equilibrium relative humidity (*ERH*). Thus, *VP* and *ERH* are related. When pure water is altered by the addition of a solute, the concentration of water is decreased and the rate of escape from the surface is diminished.

The water activity of a solution is defined in terms of *VP* and *ERH* by the formula

$$a_w = \frac{P}{P_o} = \frac{ERH}{100}$$

Many systems have been discussed for calculating, predicting, or determining the water activity of various substances, including foods (Favetto et al. 1983; Ferro Fontan and Chirife 1981; Labuza et al. 1976; Northolt and Heuvelman 1982; Nunes, Urbicain, and Rotstein 1985; Stamp et al. 1984; Teng and Seow 1981; Troller 1983a, 1983b; Wiebe et al. 1981). Volatile compounds in a food may result in erroneous measurements of a_w when an electric hygrometer is used (Favetto et al. 1984). The a_w of food generally decreased as the temperature was raised from 23° to 27°C (Scott and Bernard 1983).

Water Activity and Microbial Growth

Microorganisms have a maximum, optimum, and minimum a_w for growth. Since the a_w of pure water is 1.00, and microorganisms cannot grow in pure water, the maximum or upper limit for microbial growth is an a_w somewhat less than 1.00. Some foods have a_w values greater than 0.995, which are rounded to 1.00. However, microorganisms can grow in these foods.

The prevention of microbial growth is of importance to food microbiologists. Hence, the minimum a_w at which growth can occur has been of most interest. In general, for growth, bacteria require a higher a_w than yeasts, and yeasts require a higher a_w than molds. The minimum a_w values for the growth of various microorganisms are listed in Table 4.3. Even different strains of the same species appear to have different minimum a_w for growth. The limiting a_w for growth is influenced by other environmental factors which should be at optimum values for the organism being tested if the absolute minimum water activity is to be determined. If these factors are not optimum during testing, the resultant minimum a_w for growth of an organism will appear to be higher than the true value.

The solute used to lower the a_w affects the minimum a_w for growth of microorganisms. Osmotolerant organisms grow at a lower a_w when the solute is sugar rather than salt. The reverse is true for halophilic or halo-

TABLE 4.3. APPROXIMATE MINIMUM WATER ACTIVITIES FOR GROWTH OF MICROORGANISMS

Organism	Minimum a_w	Organism	Minimum a_w
Most spoilage bacteria	0.90–0.91	*Staphylococcus albus*	0.88–0.92
Acinetobacter	0.95–0.98	*S. aureus*	0.83–0.92
Aeromonas	0.95–0.98	*Streptococcus*	0.92–0.98
Alcaligenes	0.95–0.98	*Vibrio parahaemolyticus*	0.94–0.98
Arthrobacter	0.95–0.98	Halophilic bacteria	0.75
Bacillus	0.90–0.99	Most yeasts	0.87–0.94
B. cereus	0.92–0.95	Osmophilic yeasts	0.60–0.78
Citrobacter	0.95–0.98	Most molds	0.70–0.80
Clostridium botulinum	0.90–0.98	Xerophilic molds	0.60–0.70
Type A	0.93–0.95	*Aspergillus*	0.68–0.88
Type B	0.93–0.96	*A. glaucus*	0.70–0.75
Type E	0.94–0.97	*A. flavus*	0.78–0.90
C. perfringens	0.93–0.97	*A. halophilicus*	0.68
Corynebacterium	0.95–0.98	*A. niger*	0.80–0.84
Enterobacter	0.95–0.98	*Botrytis cinerea*	0.93
Escherichia coli	0.94–0.97	*Debaryomyces*	0.87–0.91
Flavobacterium	0.95–0.98	*Fusarium*	0.80–0.92
Klebsiella	0.95–0.98	*Hansenula*	0.89–0.90
Lactobacillus	0.90–0.96	*Mucor*	0.80–0.93
Leuconostoc	0.96–0.98	*Penicillium*	0.78–0.90
Micrococcus	0.90–0.95	*Rhodotorula*	0.89–0.92
M. roseus	0.90–0.93	*Saccharomyces cerevisiae*	0.90–0.94
Pseudomonas aeruginosa	0.96–0.98	*S. rouxii*	0.62–0.81
P. fluorescens	0.94–0.97	*Xeromyces bisporus*	0.60–0.61
Salmonella	0.93–0.96		

tolerant microorganisms. Organisms tend to grow better in a substrate in which the a_w is altered by desorption rather than adsorption (Labuza, Cassil, and Sinskey 1972).

The minimum a_w for most bacteria is 0.90 to 0.91, except for *S. aureus* and the halotolerant bacteria. The minimum a_w for these latter bacteria is listed as 0.75, which also is the approximate a_w of a saturated NaCl solution (Table 4.4). The growth of *S. aureus* at various a_w levels is shown in Figure 4.2. As the a_w is reduced below 0.99, a reduced growth rate, as well as less total growth is evident.

A very high a_w may limit the growth rate. A typical curve relating the growth rate to a_w is shown in Figure 4.3. The optimum growth for most organisms is slightly below 1.00, and for halophiles, or xerotolerant organisms, the optimum is lower. The halophile *Vibrio costicola* did not grow at a_w values above about 0.98 (Forsyth and Kushner 1970). The optimum a_w for the growth of *Penicillium* species varied from 0.93 to 0.98 (Hocking and Pitt 1979).

TABLE 4.4. RANGES OF EQUILIBRIUM RELATIVE HUMIDITIES OF SATURATED SOLUTIONS AT 20°C

Chemical	Range of ERH	Chemical	Range of ERH
LiBr	6.6	$NaNO_2$	65.6–66.0
$NaOH \cdot H_2O$	7.0–8.9	KI	68.2–70.0
$LiCl \cdot H_2O$	10.2–15.0	$NaNO_3$	74.0
$K(C_2H_3O_2)$	20.0–23.0	NaCl	75.0–77.5
$CaCl_2 \cdot 6H_2O$	32.3	$Na_2S_2O_3$	78.0
$MgCl_2 \cdot 6H_2O$	32.5–33.6	$(NH_4)_2SO_4$	80.0–81.5
CrO_3	35.0–38.7	KCl	80.3–88.2
$Zn(NO_3)_2 \cdot 6H_2O$	42.0	$ZnSO_4 \cdot 7H_2O$	89.0–90.5
$K_2CO_3 \cdot 2H_2O$	43.0–44.0	$BaCl_2$	90.0–91.0
KNO_2	45.0–49.1	K_2HPO_4	92.0
$LiNO_3 \cdot 3H_2O$	47.0	KNO_3	92.0–95.0
$Mg(NO_3)_2 \cdot 6H_2O$	54.9	$NH_4H_2PO_4$	93.0–93.9
$Na_2Cr_2O_7 \cdot 2H_2O$	55.2	$Na_2HPO_4 \cdot 12H_2O$	95.0–96.2
$NaBr \cdot 2H_2O$	55.3–59.3	K_2SO_4	96.4–98.2
NH_4NO_3	64.9	$CuSO_4 \cdot 5H_2O$	98.0
$Mg(C_2H_3O_2)_2 \cdot 4H_2O$	65.0	$K_2Cr_2O_7$	98.3

OTHER ASPECTS. Other aspects of water activity besides the minimum activity for growth are important. These aspects include the germination of spores, toxin production, resistance to heat, and survival of microorganisms.

Pitt and Christian (1968) listed the a_w requirements of thirty-six molds for germination and for the formation of asexual and sexual sporulation. Often a higher water activity is needed for sporulation than for germination and growth. Sexual sporulation usually requires a higher a_w than does asexual sporulation. According to a review by Sperber (1983), the a_w requirement for bacterial sporulation is the same as for growth, but germination usually can occur at lower a_w values.

The production of enterotoxin by *S. aureus* requires a higher a_w than for growth (Smith et al. 1983). Higher a_w levels are needed for the production of mycotoxins than for growth by several species of *Aspergillus* and *Penicilliium* (Beuchat 1981, 1983).

In general, lowering the a_w of the substrate increases the heat resistance of microorganisms. However, at extremely low levels of a_w, the resistance may be decreased. Bacterial spores are most heat resistant at a_w values of 0.2 to 0.4.

Generally, the lower the a_w, the longer the organisms survive during storage. The tolerance to low a_w values results from the ability of the organism to accumulate compatible solutes within the cell. These solutes include proline and γ-aminobutyric acid for bacteria, and polyalcohols for fungi.

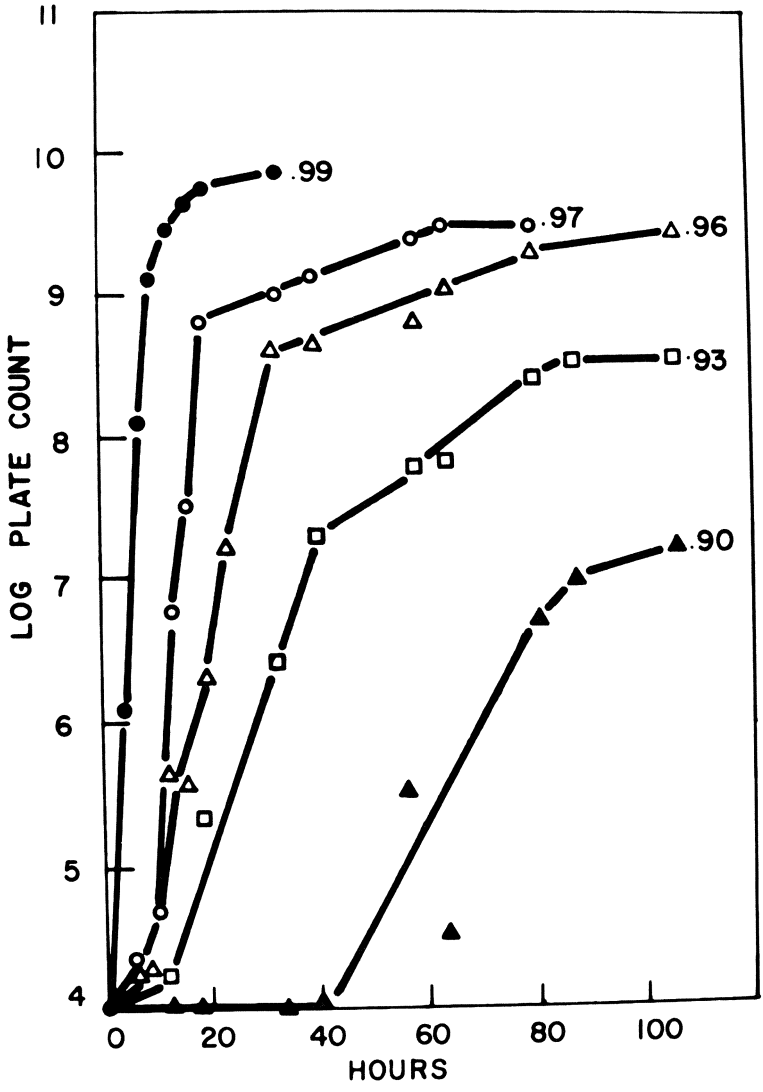

Figure 4.2. Growth of *Staphylococcus aureus* at various levels of water activity.
Courtesy of Troller (1971).

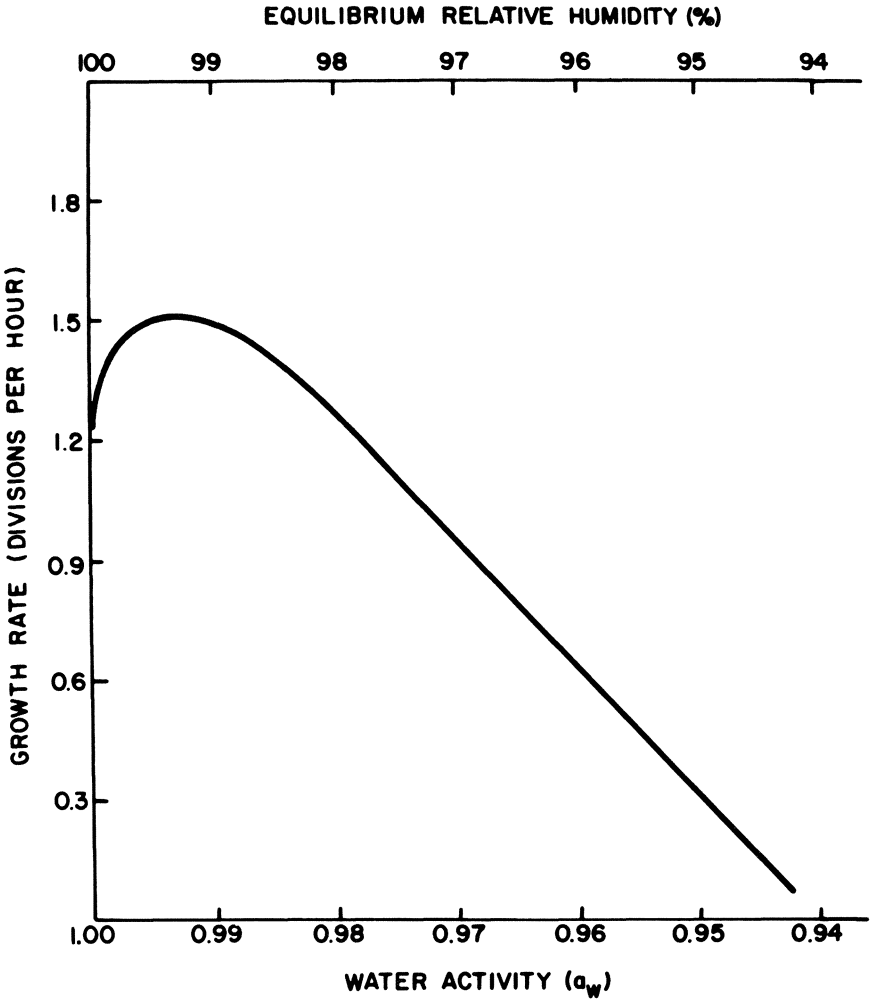

Figure 4.3. Effect of water activity on bacterial growth rate.

Water Activity of Food

The a_w of food can be lowered by removing water (dehydration), by adding solutes (sugar, salt), or by freezing. The usual a_w levels of various foods are listed in Table 4.5. Fett (1973) reported variations of a_w of ± 0.003 to ± 0.011, depending upon the system of measurement; other researchers reported that different methods of measurement may vary ± 0.02 a_w units (Labuza et al. 1976). Dewpoint hydrometer measurements are reproducible to 0.003 a_w and accuracy of 0.006 a_w (Northolt 1979).

TABLE 4.5. APPROXIMATE WATER ACTIVITIES OF SELECTED FOODS

Food	a_w	Food	a_w
Fresh fruit or vegetables	0.97–1.00	Jelly	0.82–0.94
Fresh poultry or fish	0.98–1.00	Rice	0.80–0.87
Fresh meats	0.95–1.00	Fruitcake	0.80–0.87
Pudding	0.97–0.99	Flour	0.67–0.87
Eggs	0.97	Cake icing	0.76–0.84
Juices—fruit and vegetable	0.97	Molasses	0.76
Bread	0.95–0.96	Honey	0.54–0.75
White	0.94–0.97	Dried fruits	0.51–0.89
Crust	0.30	Chocolate candy	0.69
Cheese, most types	0.91–1.00	Rolled oats	0.65–0.75
Parmesan	0.68–0.76	Toffee	0.60–0.65
Cured meat	0.87–0.95	Caramels	0.60–0.65
Baked cake	0.90–0.94	Noodles	0.50
Refrigerated biscuit dough	0.94	Dried whole egg	0.40
Maple syrup	0.90	Biscuits	0.30
Salted egg yolk	0.90	Dried vegetables	0.20
Soft, moist pet food	0.83–0.91	Cereals	0.10–0.20
Nuts	0.66–0.84	Crackers	0.10
Jam	0.75–0.80	Sugar	0.10

Fresh foods, such as fruit, vegetables, meat, poultry, and fish have a_w values of 0.98 to over 0.99, which will allow the growth of most microorganisms.

The rate of growth of bacteria is greater than that of yeasts or molds. Hence, with foods of high a_w, bacteria will outgrow the fungi (yeasts and molds) and cause spoilage, while at a_w values that restrict or prevent bacterial growth, the fungi grow and become dominant. Exceptions to these generalizations include fruits spoiled by fungi because of the acidity of the product restricting bacterial growth. Products that have low a_w due to sugar content (jams, jellies, or honey) will be subject to attack by osmophilic yeasts, while products that contain high salt concentrations (low a_w) will be spoiled by halotolerant or halophilic bacteria. Dried foods generally have a_w values below 0.75. A safe a_w level for storage is usually considered to be 0.70 or less. In foods protected by low a_w, enzymatic changes can occur, but at a slow rate.

SORPTION ISOTHERMS. The moisture content of foods is often determined and reported. The relationship between moisture and a_w values is desirable. This relationship is described by a sorption isotherm that has the shape of a sigmoid curve (Fig. 4.4). Numerous mathematical equations have been used to describe sorption isotherms (Aguerre, Suarez, and Viollaz 1983; Ferro Fontan et al. 1982; King et al. 1983).

The curve may be an isotherm of adsorption (a moistening process), or an isotherm of desorption (a drying process). The isotherms vary with

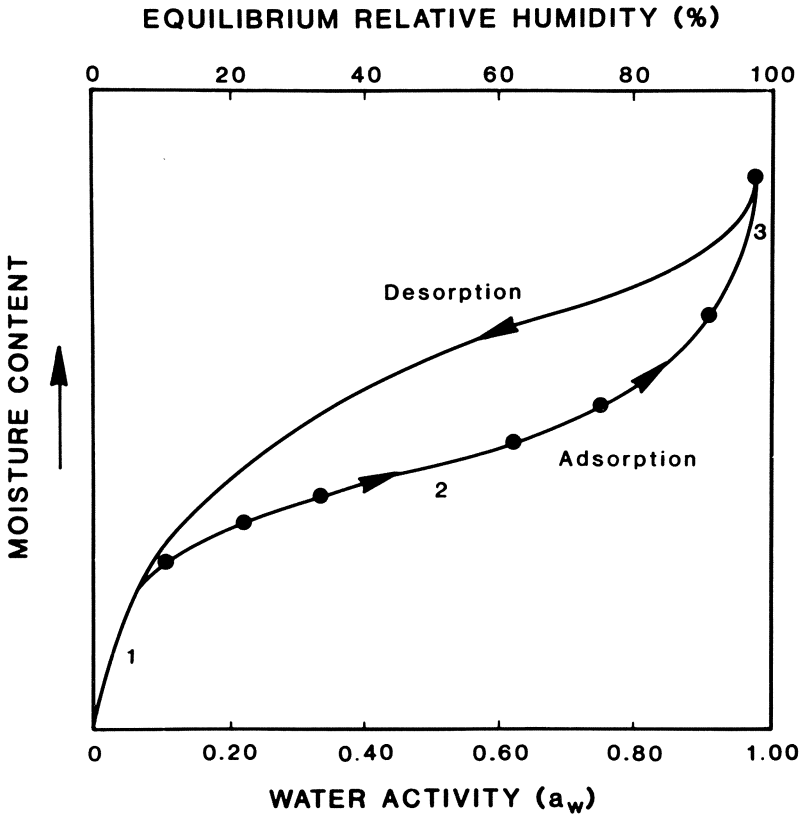

Figure 4.4. Typical sorption isotherm. 1-Monomolecular film of water, no growth of organisms; 2-some added water to monomolecular film, slight growth in upper range; 3-water activity is sufficient for microbial growth.

temperature for different products. The adsorption and desorption isotherms shown in Figure 4.4 have a hysteresis "loop." According to Troller (1983a), equilibrium will occur, given sufficient time, so that only one line is evident.

Three areas are apparent in most isotherms. In the lowest part of the curve, the water is in a monomolecular layer on the food product, or so-called bound water, and is not available for biological activity. In the middle of the curve, additional layers of water are added to the monomolecular film, and there is a rapid increase in a_w with a moderate increase in moisture content. In the upper regions of this section of the curve, some water becomes available to support the growth of some molds and yeasts. In the upper part of the isotherm, the moisture content increases rapidly

with comparatively small increases in water activity, and the food contains free water that is available for bacterial as well as fungal growth.

If food is not stored in moisture-proof containers, it will equilibrate with the relative humidity of the surrounding air according to its sorption or desorption isotherm. The direction will be determined by the moisture in the food and the relative humidity of the air. Dry foods may pick up moisture to the extent that molds will be able to grow, while foods with high a_w stored in low-humidity air may lose moisture, causing shrinkage and other organoleptic changes (Labuza and Contreras-Medellin 1981). The rate of equilibration depends upon the relative humidity of the air, the initial a_w of the food, and temperature.

In mixtures of foods such as cereals and raisins, or marshmallow topping on a cookie base, the moisture from one food will transfer to the drier product. For raisins in cereals, this results in the cereal becoming less crisp and the raisins becoming hard. Coating the raisins with beeswax, oils, or sugar has been used to retard the transfer of moisture.

Since many of our foods are combinations of various components, the a_w values and isotherms of the ingredients should be considered in order to avoid potential problems either with organoleptic quality or with possible microbial growth. Although a product may have an a_w not conducive to growth of microorganisms, syneresis may occur, increasing the a_w in microenvironments, thus allowing microbial growth.

Case hardening, the phenomenon in which the moisture content of the exposed surface is different from that in the unexposed portion, may act as a barrier for moisture sorption or desorption. This may influence the measurement of the a_w of a food so that, although the determined a_w of the food would appear in a safe range, the interior of the food will support microbial growth.

Foods that can be eaten without rehydration but are microbiologically stable without refrigeration are called intermediate moisture foods. These foods are discussed in Chapter 11.

pH

In many texts, pH is defined as the negative logarithm of the hydrogen ion concentration. The pH is actually a function of apparent concentrations or activity of the electrolytes in solution. Thus, pH is the negative logarithm of the activity of the hydrogen ions. Since pH is usually determined by means of a pH meter, it might be defined as a value calculated from the electromotive force (emf) of an electromotive cell in which the substance being tested is one of the electrolytes. In the food industry, the determination and maintenance of pH are important in quality control and processing requirements.

Effect on Microorganisms

Microorganisms have a minimum, optimum, and maximum pH for growth. Most bacteria show optimum growth at a pH near 7.0, while others are favored by an acid environment, probably due to the inhibition of other organisms, thus eliminating microbial competition. The acid formers (*Lactobacillus* and *Streptococcus*) can tolerate moderate acidity, while proteolytic types (*Pseudomonas*) can grow in moderately alkaline substrates. In general, molds can grow at lower pH values than yeasts, and yeasts are more tolerant of low pH values than are bacteria. Bacteria usually grow faster than yeasts in a neutral or slightly acid substrate, but at pH 5.0 or less, yeasts can compete with or outgrow bacteria.

Microorganisms can grow in a wide pH range (Table 4.6). The variations in pH values for growth may be due to different strains of a species or different species in a genus, the type of substrate, the acid or base used to adjust the pH, or other factors. There is an interrelationship between pH and other environmental factors.

Alteration of the pH of a substrate may relate indirectly to growth. For example, the availability of metallic ions is altered. Although coexisting in the free form at low pH, magnesium and phosphorus form an insoluble complex at higher pH values. In an alkaline medium, ferric, zinc, and calcium ions become insoluble.

The pH of the substrate may influence cell permeability. At low pH, the membrane becomes saturated with hydrogen ions, thus limiting passage of essential cations. Conversely, at a high pH, saturation of the membrane with hydroxyl ions will limit the passage of essential anions.

The toxicity of adverse pH values is due partly to the penetration of undissociated molecules of acid or basic substances into the cells. At low pH values, undissociated weak acids can enter the cell, then ionize and alter the internal pH. In alkaline solutions, undissociated weak bases can enter the cell. If the internal pH is changed to basic pH values, amino acid transferase RNA is inhibited and protein synthesis is stopped. The pH of the substrate influences the activity of enzyme systems and the products of metabolism of microorganisms.

ALTERATION OF pH BY MICROORGANISMS. The environment influences microorganisms and the microorganisms influence the environment. During growth, metabolic products are formed. These can be either acidic or alkaline, depending upon the substrate, the organisms involved, and the time allowed for growth. The initial reaction of most organisms is acidic because of the breakdown of carbohydrates and the formation of organic acids. The alteration of pH by production of acids is used in the food-fermentation industries. The lactic acid bacteria tend

TABLE 4.6 APPROXIMATE pH RANGES FOR MICROBIAL GROWTH

	pH		
Organism	Minimum	Optimum	Maximum
Bacteria (most)	4.5	6.5–7.5	9.0
Acetobacter	4.0	5.4–6.3	—
Bacillus subtilis	4.2–4.5	6.8–7.2	9.4–10.
Clostridium botulinum	4.8–5.0	6.0–8.0	8.5–8.8
C. perfringens	5.0–5.5	6.0–7.6	8.5
C. sporogenes	5.0–5.8	6.0–7.6	8.5–9.0
Erwinia carotovora	4.6	7.1	9.3
Escherichia coli	4.3–4.4	6.0–8.0	9.0–10.
Gluconobacter oxydans	4.0–4.5	5.5–6.0	—
Lactobacillus (most)	3.0–4.4	5.5–6.0	7.2–8.0
L. acidophilus	4.0–4.6	5.5–6.0	7.0
L. plantarum	3.5	5.5–6.5	8.0
Leuconostoc cremoris	5.0	5.5–6.0	6.5
L. oenos	—	4.2–4.8	—
Pediococcus cerevisiae	2.9	4.5–6.5	7.8
Propionibacterium	4.7	6.2–7.0	7.5
Proteus vulgaris	4.4	6.0–7.0	8.4–9.2
Pseudomonas (most)	5.6	6.6–7.0	8.0
P. aeruginosa	5.6	6.6–7.0	8.0–9.0
Salmonella (most)	4.5–5.0	6.0–7.5	8.0–9.6
S. typhi	4.0–4.5	6.5–7.2	8.0–9.0
S. choleraesuis	5.0	7.0–7.6	8.2
Serratia marcescens	4.6	6.0–7.0	8.0
Staphylococcus aureus	4.0–4.7	6.0–7.0	9.5–9.8
Streptococcus lactis	4.1–4.8	6.4	9.2
Vibrio	5.5–6.0	—	9.0
V. cholerae	—	8.6	—
V. parahaemolyticus	4.8–5.0	7.5–8.5	11.0
Yeasts	1.5–3.5	4.0–6.5	8.0–8.5
Hansenula	—	4.5–5.5	—
Kluyveromyces	1.5–2.0	—	—
Pichia	1.5	—	—
Saccharomyces cerevisiae	2.0–2.4	4.0–5.0	—
S. rouxii	1.5	3.5–5.5	8.5–10.5
Molds	1.5–3.5	4.5–6.8	8.0–11.
Aspergillus niger	1.2	3.0–6.0	—
A. oryzae	1.6–1.8	—	9.0–9.3
Botrytis cinerea	2.5	—	7.4
Mucor	—	3.0–6.1	9.2
Penicillium	1.9	4.5–6.7	9.3
Rhizopus nigricans	—	4.5–6.0	—

to lower the pH by production of lactic acid, while proteolytic types such as pseudomonads tend to raise the pH by production of ammonia or other basic chemicals.

Bacillus coagulans grew in tomato juice, raising the pH from 4.5 to 5.0 (Anderson 1984). Molds growing on tomatoes raised the pH to levels that allowed bacterial growth in the food (Mundt and Norman 1982).

SURVIVAL. The pH for survival of cells is somewhat different from that for growth. Some organisms show greater survival at slightly acid pH levels 5.6 to 6.5 and optimum growth at 6.8 to 7.2. Microorganisms can survive at pH levels too acid or too basic for metabolism and growth. The effect of pH on survival during heating is very evident.

pH of Food

The pH of a food, along with other environmental factors, will determine the types of microorganisms that are able to grow and dominate, and eventually cause spoilage, a desired fermentation, or a potential health hazard. Besides the microbial aspects of the pH of foods, the acids in foods play other important roles.

The food may be naturally acid, acids may be added, or acids may be produced in the food by enzymatic action with or without microbial growth.

The pH of a food is determined by the balance between the buffering capacity and the acids or alkaline substances it contains. Since protein has a high buffering capacity, protein foods have greater buffering than do fruits or vegetables. This is important in the fermentations producing lactic acid because the production of small amounts of acid in sauerkraut or pickle processes will lower the pH significantly.

The approximate pH values for selected foods are listed in Table 4.7. The pH values are only approximations because there is considerable variation in the pH of some foods and the pH of food products can change during ripening, processing, or storage. The pH values of various foods were reported by Sapers, Phillips, and Divito (1984).

Foods have been categorized according to their pH as follows:

High-acid foods	pH below 3.7
Acid foods	pH 3.7–4.6
Medium-acid foods	pH 4.6–5.3
Low or nonacid foods	pH over 5.3

EGG WHITE. This is one of the most alkaline biological solutions. The albumen of a freshly laid chicken egg is approximately pH 7.6. When the egg is stored in air, the carbon dioxide from carbonic acid in the albumen is released through the egg shell. When this occurs, the pH increases and has been reported to reach levels of 9.4 to 9.7. Although the high pH is desirable from a microbiologist's viewpoint, these high pH levels are associated with thinning of albumen and a decrease in the strength of the vitelline (yolk) membrane, or a general decrease in egg quality. Egg quality is maintained at an albumen pH near 8.2, which is above the

TABLE 4.7. APPROXIMATE pH RANGES OF SELECTED FOODS

Food	pH Range	Food	pH Range
Egg white	7.6–9.5	Cabbage	5.2–6.3
Shrimp	6.8–8.2	Turnip	5.2–5.6
Crab	6.8–8.0	Spinach	5.1–6.8
Scallops	6.8–7.1	Asparagus	5.0–6.1
Cod, small	6.7–7.1	Cheeses, most	5.0–6.1
Cod, large	6.5–6.9	Camembert	6.1–7.0
Catfish	6.6–7.0	Cottage	4.1–5.4
Soda crackers	6.5–8.5	Gouda	4.7
Maple syrup	6.5–7.0	Bread	5.0–6.0
Milk	6.3–6.8	Carrots	4.9–6.3
Brussels sprouts	6.3–6.6	Beets	4.9–5.8
Whiting	6.2–7.1	Bananas	4.5–5.2
Haddock	6.2–6.7	Dry sausages	4.4–5.6
Cantaloupe	6.2–6.5	Pimientos	4.3–5.2
Dates	6.2–6.4	Tomato juice	3.9–4.7
Herring	6.1–6.6	Mayonnaise	3.8–4.0
Butter	6.1–6.4	Tomatoes	3.7–4.9
Honey	6.0–6.8	Jams	3.5–4.0
Mushrooms	6.0–6.5	Apricots	3.5–4.0
Cauliflower	6.0–6.7	Apple sauce	3.4–3.5
Lettuce	6.0–6.4	Pears	3.4–4.7
Egg yolk	6.0–6.3	Grapes	3.3–4.5
Corn, sweet	5.9–6.5	Cherries	3.2–4.7
Oysters	5.9–6.6	Pineapple	3.2–4.1
Celery	5.7–6.0	Peaches	3.1–4.2
Peas	5.6–6.8	Rhubarb	3.1–3.2
Turkey	5.6–6.0	Strawberries	3.0–4.2
Chicken	5.5–6.4	Grapefruit	2.9–4.0
Halibut	5.5–5.8	Raspberries	2.9–3.7
Beans, lima	5.4–6.5	Apples	2.9–3.5
Potatoes, Irish	5.4–6.3	Plums	2.8–4.6
Walnuts	5.4–5.5	Oranges	2.8–4.0
Pork	5.3–6.4	Cranberries	2.5–2.8
Beef	5.3–6.2	Lemons	2.2–2.4
Onions	5.3–5.8	Limes	1.8–2.0
Sweet potatoes	5.3–5.6		

maximum for the growth of many microorganisms. Storage in an atmosphere of CO_2 or oiling the egg shell maintains the pH at an intermediate level.

RED MEAT. The pH of living animal tissue is near neutral (pH 7.0–7.2). The circulating blood brings nutrients and oxygen to the cells and removes the waste products of metabolism. When an animal is slaughtered, blood no longer circulates, anaerobic conditions develop, and any metabolic products accumulate. The inherent tissue enzymes ferment the muscle glycogen to lactic acid, which lowers the pH. When glycogen sup-

plies are abnormally low in the muscle prior to slaughter, less acid is produced, with a higher-than-normal ultimate pH. The temperature at which a beef carcass is held determines the rate of glycolysis and pH changes.

Immediately after slaughter, the pH of most beef muscles is 6.9 to 7.2 (Hamm 1982). After 24 hr postmortem, most muscles from well-rested animals have a pH of 5.4 to 5.6 (Currie and Wolfe 1980; Tarrant and Sherington 1980). It is generally believed that pH 5.3 is the limit below which glycolysis is inhibited, even though some glycogen may be present. The pH of most pork muscles with normal glycolysis ranges from 5.3 to 5.8 (Warriss 1982).

It might be expected that a higher glycogen content is present in the muscles of well-fed, rested, unexcited animals than in starved, tired, or stressed animals.

Preslaughter stress conditions, such as fighting among hogs, have been associated with a rapid pH drop and low ultimate pH. This rapid lowering of the pH while the muscles are still warm results in pale, soft, exudative (PSE) pork. Preslaughter exercise or stress causes muscles to be deficient in glycogen and results in ultimate higher pH values (above pH 6.0). This meat is called dark, firm, dry (DFD) pork, or dark cutting beef. With a higher-than-normal ultimate pH, these muscles have a tendency to spoil more readily than do muscles with a lower pH.

Subjecting beef carcasses to an electrical current of various intensities promotes tenderness of the muscles as well as a rapid drop in pH (Martin et al. 1983). Also, a rapid pH reduction was noted when beef muscle was pressurized (Elkhalifa et al. 1984).

The pH of commercial dry sausages was reported as ranging from 4.4 to 5.6 (Acton and Dick 1976). Fermentation is responsible for the lower pH values of sausages as compared to fresh meat. The addition of acid (acetic, lactic, citric) results in a low pH of pickled meats.

Microbiologically, a low pH in fresh meat is desired. The minimum pH for the growth of most pseudomonads that spoil meat is 5.6. If meat has an ultimate pH less than 5.6, it would be expected to have a longer shelf life. On the other hand, higher pH values have been associated with enhanced water-holding capacity, nutrition, juiciness, tenderness, and increased emulsifying properties and binding strength of meat, which are desirable attributes.

CHICKEN. The pH of chicken muscle varies similarly to that of red meats. Postslaughter chicken muscle is approximately pH 7.0, which is reduced during glycolysis to 5.5 to 5.9. A rapid glycolysis at high temperature can result in tough meat. The pH of meat from heat-stressed birds was 5.46, from cold-stressed, 5.51, and from control birds, 5.53 (Lee et al.

1976). This indicates that this type of stress has little influence on the final pH of chicken meat.

SEAFOODS. The pH of fish (7.0–7.3) is lowered during rigor to pH 5.5 to 6.5, depending on the species of fish and the initial amount of glycogen in the muscle. Fish muscle generally has less glycogen than poultry or red meat.

The pH of cod was found to be lower at the end of the spawning season, and large cod had a lower pH than small cod (Love et al. 1972). They found that a low pH was related to gaping in cod. This is a condition in which the tissue that holds the muscle segments together becomes weakened and causes slits to appear across the fillets.

According to Love and Haq (1970), haddock has a lower pH than whiting. These researchers postulated that the difference may be in their eating habits. The whiting chases food, while the haddock tends to graze. The haddock, being a less active fish, probably does not struggle as much as a whiting when caught, so there would be greater carbohydrate reserve, which results in more lactic acid production postmortem and a lower pH. Of all the fish for which data are available, the halibut has the lowest ultimate pH and the best keeping quality.

The pH of canned crab is usually pH 6.8 to 7.4 but may be higher. Motohiro and Inoue (1970) observed that the pH of fresh king crab was 6.8 for the hard shell and 7.4 for the paper shell intermolt stages.

A pH range of 7.1 to 8.2 was determined on brown shrimp obtained from commercial fishing boats at landing (Cobb et al. 1973). The upper limit, pH 8.2, may seem rather high, since pH 7.95 has been indicative of shrimp spoilage. At the time of spoilage, the pH of brown shrimp was found to be 7.3 to 8.5 (Cobb et al. 1973). This indicates that pH 7.95 has little, if any, significance. At a pH of about 8.5, the bacterial numbers were estimated at 10^5 per gram (Flores and Crawford 1973). This level of microorganisms usually does not indicate spoilage.

FRUITS AND VEGETABLES. Fruits have a lower pH than do most foods. Unripe fruit generally has a lower pH than ripe fruit. As the fruit matures, the moisture and sugars increase, and the acids decrease or are diluted. The ripening temperature influences the ultimate pH, generally in a direct relationship. The pH of fruit influences not only the growth of microorganisms, but also quality factors, such as softening and discoloration of the canned product. Since the pH is low, fruits are usually spoiled by mold growth. The pH of the tomato is influenced by variety, maturity, geographic area of growth, and storage conditions. Vegetables generally have a higher pH than fruits and are therefore subject to bacterial spoilage.

OXIDATION-REDUCTION POTENTIAL

When a substance is oxidized, it loses electrons, and these electrons must be accepted by another substance which then becomes reduced. The oxidation-reduction (OR) potential is a measure of the tendency of a reversible system to give or receive electrons.

The OR potential is referred to in Chapter 2 in regard to the use of reductase tests to estimate microbial numbers in foods. By use of indicators, the OR potential of a system can be estimated. The indicator should give a distinct color to the solution at the pH of the system being studied. Unfortunately, some indicators behave as both OR and pH indicators. They may give an intense color in an alkaline solution, but in neutral or acid solutions yield only a dull tint. The usual method of measuring OR is with a platinum redox electrode attached to a pH/mV meter.

Eh is a measure of intensity, not of capacity of a system. The capacity is the poising ability, which acts in a manner similar to buffers in maintaining pH at a constant value. A system is said to be poised if it resists change in potential.

Effect on Microorganisms

In microbial cultures, the simultaneous oxidations and reductions are the sources of energy for cell processes. Since energy is needed by the cell to function normally, OR reactions and OR potentials are important.

The broad classifications of organisms into aerobes and anaerobes, as discussed in general microbiology, is based on their apparent tolerance to oxygen during growth. Strictly aerobic organisms grow only in the presence of free atmospheric oxygen. Strictly anaerobic organisms survive and increase only in the absence of free oxygen. Facultative anaerobes can grow with or without free oxygen. Some facultative organisms prefer growth in atmospheric oxygen; others do not. Microaerophilic organisms cannot multiply in either entirely aerobic or anaerobic conditions. They grow best in a limited amount of free oxygen.

In discussing growth-limiting effects on anaerobic bacteria, Hentges and Maier (1972) cited references to show that some obligate anaerobic types of organisms failed to grow if exposed to oxygen in the air during the manipulation of diluting and plating. These obligate anaerobes rarely are reported in foods, probably because not enough care is taken to find them. If they are such strict anaerobes, they probably cause few, if any, serious problems in foods.

In one study, anaerobes were classed as intolerant, moderately tolerant, or tolerant, on the basis of their tolerance to atmospheric oxygen (Rolfe et al. 1978). When exposed to oxygen, intolerant types survived less than 2 hr, moderately tolerant for 4 to 8 hr, and tolerant anaerobes

at least 48 hr. Of the eleven anaerobic species the researchers tested, *Clostridium perfringens* was the most oxygen tolerant.

From preliminary studies, Hentges and Maier (1972) stated that the OR potential is unimportant regarding the multiplication of *Bacteroides* species. When oxygen is excluded from the environment, *Bacteroides* are unaffected by a high OR potential of the medium. Walden and Hentges (1975) studied the growth and survival of *Bacteroides fragilis, Clostridium perfringens,* and *Peptococcus magnus* at low and high Eh with and without oxygen. Even at an Eh of −50 mv these organisms did not grow in the presence of oxygen. No inhibition was observed in the absence of oxygen even at an Eh of +325 mv. This indicates that oxygen, rather than a positive potential, was the limiting factor for growth of these anaerobes. Of these three organisms, *P. magnus* was the most sensitive to oxygen and *C. perfringens* the least sensitive.

Other researchers reported that *B. fragilis* was not affected by an Eh of +300 mv in the absence of oxygen, but cell death occurred with oxygen at +250 mv (Onderdonk et al. 1976). They suggested that dissolved oxygen has an inhibitory effect on *B. fragilis,* which may be independent of changes in Eh alone. Similar effects of oxygen at low Eh were noted for sporulation and growth of *Clostridium butyricium* (Douglas, Hambleton, and Rigby 1973), and *C. botulinum* (Lund and Wyatt 1984).

Many anaerobic clostridia (*C. botulinum, C. histolyticum, C. sporogenes*) do not need a negative Eh for growth. These organisms will grow at Eh levels from +85 to +160 mv, especially in the absence of oxygen.

Due to complexities in maintaining and measuring OR potentials, there is not as much information on this parameter as on pH. Also, the results vary considerably from one experiment to another, between strains of species, and between different researchers.

Effect of Organisms on Eh

It has been well established that, as organisms metabolize, a lower Eh value is produced in the medium or substrate. The lowering of the potential has been ascribed to the consumption of oxygen or the production of reducing substances.

With aerobic bacteria, there is a slight lowering of the OR potential during the lag phase of growth due to the consumption of oxygen. As the bacteria enter the log phase of growth, this oxygen usage increases and causes a rapid drop in Eh. As the Eh becomes negative, the growth rate of the bacteria decreases. An overall lowering of the potential is between 400 and 500 mv.

At the beginning of growth, facultative anaerobes alter the Eh of the substrate in a manner similar to aerobes, but the rate of reduction may be somewhat slower. After the bacteria enter the log phase of growth, the

Eh decreases very rapidly. This overall potential drop may be from 700 to 800 mv or more. As the culture reaches the stationary phase, and as the age of the stationary phase increases, the observed OR potential shifts to more positive values.

Oxygen is removed from media prior to inoculation and growth of anaerobes. As a result, the initial Eh is lower than that for aerobes or facultative anaerobes. During the germination and emergence of spores or the lag phase of growth, the Eh is lowered by 500 to 700 mv (Douglas and Rigby 1974). There is little further reduction during the lag phase of growth. Although the total reduction of Eh is generally less for anaerobes than for facultative anaerobes, a lower Eh is attained due to the lower initial Eh of the substrate.

Redox Potential of Food

The OR potential of a food depends upon the composition (oxydizing and reducing agents present), the pH, the oxygen tension in the atmosphere, and the access that the atmosphere has to the food. Due to the complications of determining the Eh of foods and the variations obtained due to atmospheric influences, the OR potentials of only a few foods are available.

Harrison (1972) stated that in a conglomerate mixture with different systems, not in equilibrium, an overall redox potential cannot be conceived. It is evident that most foods are conglomerate mixtures. Regardless, there have been attempts to measure and compare the redox potential of foods. Living cells tend to have a low OR potential due to -SH groups in animal products and ascorbic acid and reducing sugars in plants. In dairy products, the redox potential is related to fat oxidation. Thus, there is an inverse relation of OR potential and keeping quality. The contamination of milk with cupric or ferric ions tends to increase the OR potential, resulting in less shelf life of the product.

The OR potential of cheese depends on the type, with Emmenthal (Swiss) cheese varying from -200 to -50 mv and Cheddar cheese from $+220$ to $+340$ mv (Mossel and Ingram 1955).

The Eh of meat has been reported to range from -150 to $+250$ mv. There is an oxygen requirement for postmortem tissue. The myoglobin of muscle can bind oxygen to form oxymyoglobin, or it can be oxidized by oxygen to form metmyoglobin. To oxygenate the pigment requires about 5 μl of oxygen per gram of meat, while oxidation to metmyoglobin requires about 13 μl of oxygen per gram of meat. Other oxygen requirements are tissue respiration, lipid oxidation, tissue fluids with low oxygen tension dissolving O_2 and bacterial demands. DeVore and Solberg (1974) found that tissue respiration accounted for 80 percent of the oxy-

gen uptake of postmortem meat, while bacterial demands were insignificant.

The Eh of plant products varies from +383 to +436 mv for fruit juices, +74 mv for spinach, +225 mv for barley, and −470 mv for wheat germ (Mossel and Ingram 1955). The Eh of cherries and peaches was determined to be +179 mv and +175 mv, respectively (Ross, Bartlett, and Hard 1953). The addition of ascorbic acid lowered the Eh of the fruit. The Eh of canned foods ranged from −18 mv for sliced carrots packed in glass to −446 mv for beef stew packed in cans (Montville and Conway 1982).

The oxygen content of the atmosphere and the access of the atmosphere to the food influences the OR potential. Large pieces of foods, such as carcasses, have less total surface area than do smaller-sized foods. Food in a deep vat has less surface exposed proportionately than does food in shallow vats. Food at rest in a vat has a lower Eh than does liquid being stirred or mixed. Food that is packaged in a material impermeable to oxygen should have a lower Eh than unpackaged food. Vacuum packaging will reduce or prevent the access of free oxygen to the food.

Relationships of Food OR and Microorganisms

The relatively high Eh of fruit juices favors the growth of aerobic microorganisms. Since these products have a low pH, the aerobic yeasts and molds are the predominant organisms that cause spoilage.

In meat, the surfaces exposed to the atmosphere allow the growth of aerobic bacteria, but in deeper tissue, the Eh is lower so that anaerobes can grow. In prerigor meat, the Eh is sufficiently high to prevent the growth of anaerobic types, but during rigor, the Eh is reduced to the extent that allows clostridia to grow. The Eh of fresh raw milk inhibits the growth of *C. botulinum.* The heating of milk lowers the Eh so that *C. botulinum* can grow (Kaufmann and Marshall 1965).

Although molds are considered to be strictly aerobic, some types can grow in low levels of oxygen. One of the reasons that *Penicillium roqueforti* can grow in the inside of Roquefort-type cheese is its tolerance for low pO_2. *Byssochlamys fulva* grows in grape products at extremely low levels of oxygen.

INHIBITORY SUBSTANCES

When you hear inhibitory substances in foods mentioned, you might envision some chemical preservative added by the processor. However, microbial inhibitors were present long before human beings appeared on earth.

Action of Inhibitors

Generally, inhibitors affect microorganisms by acting on the whole cell, cell wall, or cell membranes, by interfering with the genetic mechanism of the cell, by interfering with the enzyme systems of the cell, or by binding essential nutrients.

Many microbial inhibitors have been reported to be naturally present in foods. The factors involved with the growth of microorganisms, such as pH, that have been discussed in this chapter, are partially due to chemicals naturally present in foods. Besides these, there are other inhibitors in both animal and plant products.

Inhibitors in Animals Products

Microbial inhibitors have been found in egg white, milk, and various tissues from animals.

EGG WHITE. The growth of bacteria in egg white is limited by the presence of lysozyme, enzyme inhibitors (such as ovomucoid), avidin, conalbumin, and a high pH (Table 4.8).

Lysozyme. In 1922, Fleming reported a lytic agent in egg white and other biological systems, which was named lysozyme. This enzyme has been referred to as muramidase, N-acetylmuramide glycanohydrolase, glucohydrolase, mucopeptide glycohydrolase and β-glucosaminidase. To help alleviate confusion about the name of this and other enzymes, the Commission on Enzymes has assigned numbers to the various enzymes. Lysozyme is 3.2.1.17.

Lysozyme is found in egg white, milk, many tissues and secretions of animals, some molds, and in the latex of various plants. This widespread distribution has made it a very popular substance for study. Lysozymes from different animal species and from different organs of the same ani-

TABLE 4.8. MICROBIAL INHIBITORS IN EGG WHITE

Inhibitor	Effect on Microorganisms	Egg White Solids (%)
Lysozyme	Lysis of cell walls (Gram-positive bacteria)	3.5
Ovomucoid	Enzyme inhibitor	11.0
Conalbumin (ovotransferrin)	Chelates metal, ions, especially ferric ions	13.0
Ovoflavoprotein	Sequesters riboflavin	
Avidin	Binds biotin	0.05
pH	Alkaline conditions restrict growth, accentuate iron chelation	

mal differ chemically and immunologically, but they possess the same type of biological activity. The lysozyme content varies not only between species, but also between strains of the same species. On a dry-weight basis, chicken egg white contains about 3.5 percent lysozyme. This is the most readily available source, and hence, the most studied of the lysozymes.

Bacteria display various sensitivities to the lysis of the cell wall by lysozyme. Certain micrococci are rapidly lysed by 1 μg/ml of lysozyme. *Bacillus megaterium* requires 50 μg/ml and *B. cereus* barely is attacked by 50 μg/ml of lysozyme.

Gram-negative bacteria are rather insensitive to lysozyme because of their lipoprotein-lipopolysaccharide layer, which acts as a barrier protecting the sensitive mucopolysaccharides of the cell. If the barrier is damaged by certain treatments, the enzyme can penetrate to the mucopeptide layer and cause lysis, or partial lysis, of the cell wall. Starving Gram-negative cells at an abnormal pH, treatment with polymyxin B, or ethylenediamine tetraacetic acid, as well as reacting with complement, osmotic injury or aerosolization can render these cells susceptible to lysozyme action.

Enzyme Inhibitors. Enzyme inhibitors found in egg white are the ovomucoids, ovoinhibitors, and a ficin-papain inhibitor.

Rhodes, Bennett, and Feeney (1960) divided ovomucoids into four classes according to their inhibitory activities. Some ovomucoids, such as the chicken and goose, inhibit trypsin. Another group inhibits chymotrypsin (golden pheasant). A third group (turkey) inhibits approximately equal amounts of trypsin and chymotrypsin, and the fourth group included those from duck and emu, which inhibit twice as much trypsin as chymotrypsin. A duck ovomucoid inhibited a commercial bacterial proteolytic enzyme.

Matsushima (1958) discovered an inhibitor in egg white which he termed *ovoinhibitor.* This substance inhibited fungal proteinase of an *Aspergillus* and a bacterial proteinase prepared from a strain of *Bacillus subtilis.*

There is no evidence that these enzyme inhibitors have any significant effect on microbial growth in egg white.

Avidin. Each mole of avidin combines with two moles of biotin. Thus, organisms having a strict requirement for biotin as a nutrient are inhibited by this compound. The binding of avidin to bacterial cells has been discussed in detail (Korpela et al. 1984).

Conalbumin. This protein comprises about 12 percent of the total solids of egg white. It forms a stable complex with two ferric or ferrous ions per molecule, and by this action it inhibits organisms that require iron

for growth. This includes Gram-negative and Gram-positive bacteria as well as yeasts. It is evident that conalbumin does not destroy the microorganisms but merely inhibits growth by extending the lag phase. Feeney and Nagy (1952) postulated that dissociation of the complex would always result in traces of free iron which, when used by the organism, would allow the release of further iron from the complex. This results in slower-than-normal growth of the organism.

pH. The high pH established when carbon dioxide leaves egg albumen inhibits the growth of many microorganisms.

It might be postulated that because of lysozyme, and inhibition of Gram-positive bacteria, spoilage of eggs is primarily due to Gram-negative types. In the egg, conalbumin is the main inhibitor of Gram-negative bacteria. Although it does not prevent growth, it does inhibit or delay it. The albumen was more inhibitory at 39.5°C than at 30°C (Tranter and Board 1984). It is important that microbial growth be inhibited, because it is necessary to have shell eggs in commercial channels for the time needed for grading, packing, distribution, and retailing.

DAIRY PRODUCTS. Several substances may be present in milk that inhibit microorganisms. These include antibiotics, pesticides, bacterial viruses, sanitizing and cleaning compounds, and various natural inhibitors.

The naturally occurring microbial inhibitors in raw milk include lysozyme, cationic proteins, agglutinins or antibodies, leucocytes, lactenins, lactoperoxidase, lactoferrin, and perhaps, fatty acids. Many of the natural inhibitors in milk are heat labile and not found in pasteurized milk.

Of thirty-seven samples of cow milk tested, only one contained lysozyme at a level of 5 mg/kg (Carroll 1979). Only 10 percent of 200 samples of camel milk inhibited one or more of six test organisms (Barbour et al. 1984). The lysozyme of these samples averaged 6.48 mg/kg.

The antibiotic nisin is produced by certain strains of *Streptococcus lactis,* a common dairy organism, and may be present in small amounts in raw milk and certain milk products.

Cationic Proteins. These proteins carry a positive charge at physiological pH values and react with the anionic binding sites of cell walls and membranes of bacteria (MacMillan and Hibbitt 1973). This causes an increased permeability of the bacterial membranes, resulting in leakage from the cells. Hibitt, Brownlie, and Cole (1971) isolated cationic proteins from bulk milk samples. The antimicrobial activity was not destroyed by heating to 70°C for 30 min but was almost completely destroyed at 100°C. These proteins have activity to both Gram-positive and Gram-negative bacteria.

Lactoperoxidase. Lactoperoxidase is the peroxidase enzyme in milk. Peroxidases catalyze the oxidation of molecules, with peroxide (usually hydrogen peroxide) as the oxygen source. For antibacterial effects, besides peroxidase and H_2O_2, the presence of thiocyanate (SCN^-) is required. Reiter and Härnulv (1984) suggested adding minute amounts of H_2O_2 and SCN^- to milk to enhance lactoperoxidase effect and the shelf life of milk. The action on cells is greatest at pH 5.0 or less (Reiter et al. 1976).

According to the review of Reiter and Härnulv (1984), this system may be bacteriostatic or bactericidal. Bacterial enzymes are inhibited, the cytoplasmic membrane is damaged, and synthesis of protein, DNA, and RNA are inhibited.

Lactoferrin. This is an iron-binding protein in milk that acts on bacteria similar to conalbumin in egg white. Normal milk contains 0.1 to 0.3 mg/ml. This is increased from four to ten times in mastitic milk and in colostrum (Harmon et al. 1975). Antibodies in milk enhance the activity of lactoferrin (Dolby and Honour 1979; Spik et al. 1978). The addition of citrate reduces the inhibitory effect of lactoferrin (Bishop et al. 1976).

Fatty Acids. Fatty acids may be inhibitory or stimulatory, or they may have no effect on microorganisms. The action depends upon the type of organism, the type and concentration of the fatty acid, the pH of the growth medium, or the presence of certain other compounds. Short-chain fatty acids are about equally effective toward either Gram-negative or Gram-positive cells. Fatty acids with six or more carbon atoms are more inhibitory to Gram-positive cells than to Gram-negative cells. Long-chain fatty acids cannot penetrate the lipopolysaccharide layer of Gram-negative cells.

The mode of action and the type of inhibition depend upon the fatty acid and the concentration. At low concentration, some fatty acids are stimulatory, while at higher concentrations they are inhibitory to the same organism. The toxic effect is, in general, bacteriostatic, but for certain organisms the fatty acids may be bactericidal at high concentrations. The bacteriostatic effect may be due to the blockage of adsorption of essential nutrients. With extended incubation, the cells seem to overcome this difficulty, and have, at times, shown total cell growth greater than when no fatty acids were present. Besides affecting the permeability of the cell membranes and inhibiting transfer mechanisms, fatty acids inhibit the action of enzyme systems.

Although fatty acids are found in various foods containing fats, they have been of special concern to the dairy industry because of their prevalence in butter and cheese. The amount of free fatty acids increases during the aging of cheese. Some common fatty acids are listed in Table 4.9.

TABLE 4.9. SOME COMMON FATTY ACIDS

Common Name	Systematic Name	Chemical Formula
	Saturated Acids	
Formic acid[1]	Methanoic acid	$HCOOH$
Acetic acid	Ethanoic acid	CH_3COOH
Propionic acid	Propanoic acid	CH_3CH_2COOH
Butyric acid	Butanoic acid	$CH_3(CH_2)_2COOH$
Caproic acid	Hexanoic acid	$CH_3(CH_2)_4COOH$
Caprylic acid	Octanoic acid	$CH_3(CH_2)_6COOH$
Capric acid	Decanoic acid	$CH_3(CH_2)_8COOH$
Lauric acid	Dodecanoic acid	$CH_3(CH_2)_{10}COOH$
Myristic acid	Tetradecanoic acid	$CH_3(CH_2)_{12}COOH$
Palmitic acid	Hexadecanoic acid	$CH_3(CH_2)_{14}COOH$
Stearic acid	Octadecanoic acid	$CH_3(CH_2)_{16}COOH$
Arachidic acid	Eicosanoic acid	$CH_3(CH_2)_{18}COOH$
	Unsaturated Acids	
Crotonic acid	*trans*-2-Butenoic acid	$CH_3CH=CHCOOH$
Palmitoleic acid	9-Hexadecenoic acid	$CH_3(CH_2)_5CH=CH(CH_2)_7COOH$
Oleic acid	*cis*-9-Octadecenoic acid	$CH_3(CH_2)_7CH=CH(CH_2)_7COOH$
Linoleic acid	*cis*-9, *cis*-12-Octadecadienoic acid	$CH_3(CH_2)_3(CH_2CH=CH)_2(CH_2)_7COOH$
Linolenic acid	9,12,15-Octadecatrienoic acid	$CH_3(CH_2CH=CH)_3(CH_2)_7COOH$
Arachidonic acid	5,8,11,14-Eicosatetraenoic acid	$CH_3(CH_2)_3(CH_2CH=CH)_4(CH_2)_3COOH$

[1]Formic, acetic, and propionic acids are included because they are listed as fatty acids in some publications, although others consider butyric acid to be the simplest fatty acid.

TISSUE FOODS. Various natural microbial inhibitors have been found in tissue foods (meat, poultry, fish). Lysozyme, discussed in relation to inhibitors in egg white, is found in various tissues. Tolerances for antibiotic residues in meats have been established by the World Health Organization. Even though these inhibitors are at low levels, some microorganisms may be inhibited.

A living animal has defense mechanisms to inhibit the invasion of microorganisms. These include various antibacterial substances as well as natural and immune antibodies. Although, after slaughter, the mechanism for producing these materials is lost, those substances already produced in the tissue will remain. The extent to which they have an antibacterial function depends primarily upon their stability.

Tissues from cattle (brain, spleen, heart, kidney, and liver) possess antistaphylococcal substances. Basic polypeptides with antibacterial properties have been extracted from tissues (thymus, spleen, and thyroid). Synthetic basic polyamino acids are active against both bacteria and viruses. Spermine and spermidine are polyamines found widely distributed in animal tissues and are inhibitors of many types of microorga-

nisms (Bachrach and Weinstein 1970; Fair and Wehner 1971). The antibacterial action of basic polyamino acids is thought to be disruption of normal cell functions due to their combining with components of the cell wall.

Some hormones are bacteriostatic. Progesterone (progesterol) is antibacterial to Gram-positive organisms, and diethylstilbestrol (DES) has a bactericidal effect on *Staphylococcus aureus* (Yotis and Baman 1969) and *Candida albicans* (Yotis and Haan 1978). Deoxycorticosterone inhibits Gram-positive bacteria, yeasts, and molds. but not Gram-negative bacteria.

Plant Products

Since the discovery of antibiotics, there have been several surveys in which crude plant extracts or juices were tested for antibacterial activity. The extracts of several thousand plant species have been tested and, depending upon the plants surveyed and the test organisms used, from 30 to 50 percent of the extracts showed some antimicrobial activity. Juices and extracts from cabbage, carrot, green beans, celery, chicory, cucumber, chard, okra, rhubarb, corn, and turnip have shown antimicrobial activity to one or more microorganisms.

To be fair, it should be mentioned that some extracts have shown stimulatory effects. Pea medium is used to detect heat-stressed anaerobes in canned foods.

Several microbial inhibitors (flavanols, tannins, enzyme inhibitors, lectins, cyclopropene fatty acids, phytic acid) found to be naturally present in certain seeds, are discussed in a review by Halloin (1983). Although they do have inhibitory properties, these compounds either impart unsatisfactory properties to foods, are toxic, or may enhance carcinogenic properties of other compounds. These and some other selected microbial inhibitors are listed in Table 4.10.

Besides those inhibitors, lysozyme is found in some plant products, but at low levels. Enzyme inhibitors are found in various plant products. Gossypol, found in cotton seeds, inhibits Gram-positive bacteria, yeasts, and molds but has little or no effect on Gram-negative bacteria. The tomato contains tomatin or α-tomatine, which inhibits various fungi and bacteria. Anthocyanin pigments in cocoa possess antibacterial properties. Tannins comprise a group of natural phenolic compounds. English walnut meats, carob pods, sorghum grains, malt, mint, and peanuts inhibit microorganisms due to the presence of tannins. There is evidence that tannins interfere with enzyme activity, as well as absorbing on the cell surface and altering the permeability of the cell wall.

Benzoic acid and its salts are used as food preservatives and are found

TABLE 4.10. MICROBIAL INHIBITORS NATURALLY PRESENT IN PLANTS OR FOOD FROM PLANTS

Inhibitory Agent	Plant or Food	Organism Inhibited
Anthocyanin pigment	Fruits	Bacteria (enzymes)
Acetaldehyde	Fruit tissue	Yeast
Dimethoxyisoflavone	Peanut	*Aspergillus flavus*
Purothionins	Wheat endosperm	*Pseudomonas, Xanthomonas, Erwinia amylovora, Corynebacterium*
Juglone	Pecan	*Fusicladium effusum*
Protease inhibitor	Barley	*Aspergillus*
Trypsin inhibitor	Corn	Fungi
Enzyme inhibitor	Bean	Fungal cellulase
Lectins	"Seeds"	Virus
	Wheat germ	*Trichoderma viride*
Lipids	Peas	Fungi
Free fatty acids	Cocoa	*Aspergillus parasiticus*
Tannin(s)	Peanuts	*Aspergillus parasiticus*
	Mint	Antiviral
Phenolic	White potato	*A. parasiticus*
Lupulone Humulone Isohumulone	Hops	Gram-positive bacteria
Methylxanthines	Cocoa beans	*Aspergillus parasiticus*
Cinnamaldehyde Citral Perillaldehyde Citronellal	Oils (essential)	Fungi
Extract	Carrot root	*Aspergillus parasiticus*
Phytic acid (binds zinc)	Soybeans	*Aspergillus parasiticus*
Hydroxycinnamates	Fruits, vegetables	
p-coumaric acid	Grapes	*Saccharomyces cerevisiae, Escherichia coli, Staphylococcus aureus, Bacillus cereus*
ferulic acid		*S. cerevisiae*
p-methoxy cinnamate	*Curcurma zedoaria*	*Trichophyton rubrum* *Aspergillus niger, S. cerevisiae*
Caffeic acid		*Fusarium nivale*
Chlorogenic acid	Chicory	Bacteria
Antiviral	Grape juice	Human enterovirus "inactivation" Enteric virus
	Grape juice, apple juice, tea	Poliovirus

naturally in many berries. Cranberries are especially prominent for containing benzoic acid. Other compounds in cranberries with antimicrobial properties were described by Chu, Clydesdale, and Francis (1973).

Dallyn and Everton (1970) found that some vegetable oils contain a sporistatic or sporicidal agent. They believe that a peroxide precursor of a carbonyl is involved in the antimicrobial activity.

Pectin is a component of plant products. Although pectin shows no

inhibitory properties to microorganisms, a product of hydrolysis of pectin, α methyl D-galacturonate, inhibits the growth of many Gram-negative dysentery bacteria.

Olives contain oleuropein, a phenolic compound. Juven, Henis, and Jacoby (1972) reported that oleuropein was inhibitory to species of *Lactobacillus, Leuconostoc,* yeasts, and molds. An ethyl acetate extract of oleuropein reveals two components (glycoside oleuropein and its phenolic aglycone), which have antibacterial activity. Fleming, Walter, and Etchells (1973) found that oleuropein was not inhibitory to lactic acid bacteria, but two hydrolysis products, the aglycone and elenolic acid, were inhibitory for four species of lactic acid bacteria. The acid hydrolysate of oleuropein showed inhibitory properties to species of *Lactobacillus, Pediococcus, Leuconostoc, Staphylococcus, Bacillus, Salmonella, Pseudomonas, Erwinia, and Xanthomonas,* but no inhibition of the yeasts that were used. The inhibition of growth of *Lactobacillus plantarum* by oleuropein aglycone and elenolic acid also was detected by Federici and Bongi (1983). The inhibition is due to membrane damage and leakage of cellular constituents.

Homogenates and extracts of onions and garlic have antimicrobial activity. It is thought that, when the bulbs are crushed, enzymes react on compounds containing cysteine sulfoxide and form inhibitory thiosulfinates. One of the active compounds is allicin (thio-2-propene-1-sulfinic acid-5-allyl ester). It is not known whether its action is due to enzyme inhibition or to membrane disruption (Bogin and Abrams 1976). Of the factors in onion, the lachrymatory factor (thiopropanal-S-oxide) had the maximum inhibitory activity to spores of *Aspergillus parasiticus* (Sharma et al. 1981). The inhibitory action was lost by heating, drying, aeration, agitation, or prolonged storage. Neither garlic nor onion oil inhibited toxin production of *Clostridium botulinum* types B and E (DeWitt et al. 1979). Garlic extracts reportedly inhibit the growth of *Cryptococcus neoformans* and *Candida albicans.* Aqueous extracts inhibited the growth of *Bacillus cereus* (Saleem and Al-Delaimy 1982). Higher microbial counts of onion and garlic products are obtained when 0.05 percent K_2SO_3 is added to the dilution water.

Fruits contain various organic acids that inhibit the growth of microorganisms. A natural isoflavone found in some fruits has antifungal activity (Fukui et al. 1973). The essential oils of citrus fruits are found in the peel and can be recovered from this waste product. The disinfectant power of lemon and orange oil against spore-bearing organisms is reportedly greater than that of phenol. Orange oil at 0.1 percent and 0.2 percent was found to inhibit both Gram-positive and Gram-negative bacteria (Subba, Soumithri, and Rao 1967). At 0.2 to 0.3 percent orange and lemon oil suppressed growth of *Aspergillus parasiticus* and aflatoxin production (Alderman and Marth 1976).

Several spices or herbs and especially their essential oils inhibit various microorganisms (Shelef 1983). Mycotoxigenic molds were inhibited by cloves, allspice, anise seed, eugenol extracted from cloves, thymol from thyme, and anethol from anise seed (Hitokoto et al. 1980), by cloves, cinnamon, mustard, allspice, and oregano (Azzouz and Bullerman 1982), by o-methoxycinnamaldehyde from cinnamon (Morozumi 1978), and by various essential oils (Benjilali et al. 1984).

Low levels of oregano or cloves stimulated acid production of *Lactobacillus plantarum* and *Pediococcus cerevisiae,* while higher concentrations inhibited acid production (Zaika and Kissinger 1979, 1981). Lactic acid bacteria developed a resistance to the herbs oregano, rosemary, sage, and thyme (Zaika, Kissinger, and Wasserman 1983). Gram-positive bacteria were more sensitive than Gram-negative bacteria to the spices and herbs tested (Shelef, Jyothi, and Bulgarelli 1984).

Eugenol from oil of cloves was more effective than vanillin, an alcoholic extract of the vanilla bean, in inhibiting medically important yeasts (Boonchird and Flegel 1982). According to Davis and Cooks (1982), eugenol is also a constituent of nutmeg.

Apparently there are inducible resistance responses in plants to pathogenic bacteria (Caruso and Kuć 1979; Goodman 1978), which might carry over to the fruit.

Summary

There are many natural microbial inhibitors in foods. These inhibitors give the living animal or plant some protection from invading parasites. The food products, being an extension of the animal or plant, have a residual amount of these agents. Most of these natural compounds inhibit Gram-positive more than Gram-negative organisms. Perhaps this is one reason that spoilage is more often associated with Gram-negative organisms.

The microbial inhibitors naturally present in foods may maintain the food in a satisfactory state for transportation, processing, and distribution.

PROTECTIVE BARRIERS

Some foods are protected from direct contact with microorganisms by a natural cover or barrier. Examples of barriers are the shell and shell membrane of eggs, the testa of seeds, and the cuticle of intact plant organs.

Egg Shell and Membranes

An invisible, natural, proteinlike film on the egg shell, called the cuticle, or bloom, is considered by some to be the first line of defense against microbial penetration of the egg. The second physical barrier is the egg shell. Inside the shell are two membranes, the outer and inner shell membranes, which are the third and fourth barriers to microbial penetration. The structure of the egg and egg shell is shown in Figures 4.5 and 4.6.

CUTICLE. Washing of eggs and the use of abrasives to remove dirt disrupt the cuticle layer and allow easier penetration by microorganisms. There is a diversity of opinion on the effect of washing eggs on the resultant contamination of the interior. The cuticle tends to crack and deteriorate with time. Hence, it is not much of a barrier for stored eggs.

EGG SHELL. The egg shell is not a homogeneous structure. It consists of an organic framework of fibers and an interstitial substance of inorganic material. The two main layers of the shell are the outer or spongy layer and the inner or mammillary layer. The outer layer is thicker and contains most of the minerals.

The shell contains from 6,000 to 8,000 microscopic pores. These pores allow the exchange of water vapor and gases between the contents of the shell and the outer atmosphere. The pores vary in size, with extremes being reported as 1.6 to 7.47 μm. The average pore size is prob-

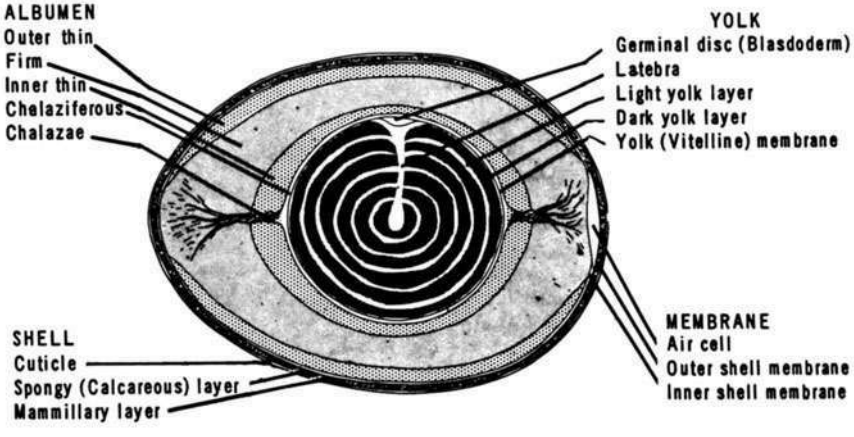

Figure 4.5. The parts of an egg.
Courtesy of USDA.

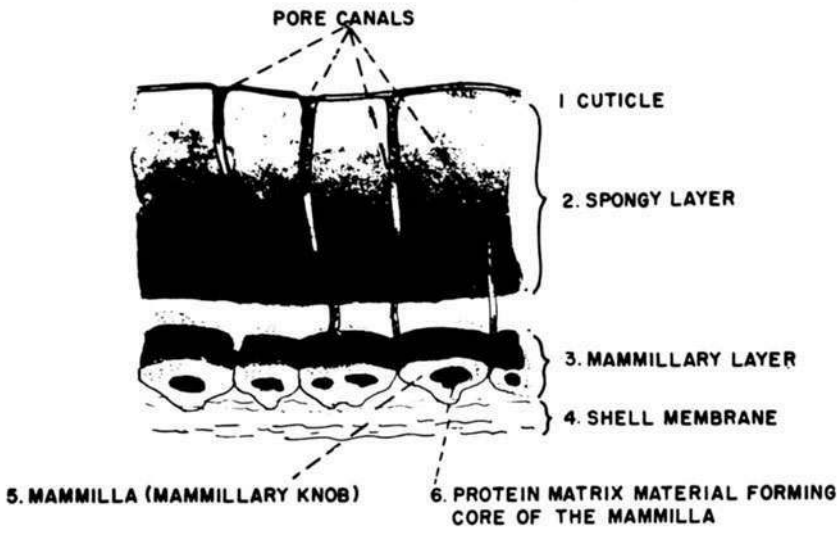

Figure 4.6. Magnified radial section through an egg shell.
Courtesy of USDA.

ably between 20 and 45 μm. The size of most pores will allow the passage of many microorganisms. Even yeast cells can be forced through pores, and mold mycelia grow through the pores. There is a linear relationship between shell porosity and infection of the eggs.

Shell weight and shell thickness do not correlate with penetration of microorganisms. However, thin shells have a tendency to crack more readily than thicker shells. Eggs with damaged shells are more subject to penetration of bacteria and egg spoilage.

If the temperature of the egg is higher than the surroundings, a condition that exists when the egg is laid, there is a greater possibility of penetration and, as the temperature differential increases, the potential for infectivity increases. As the egg contents cool, they contract, which results in a higher pressure outside the shell than inside. This pressure differential will force microorganisms through the pores of the shell. For further information about the egg shell, refer to Board (1980) and Tung, Garland, and Gill (1979).

Moisture on the egg shell, such as occurs during washing of eggs or when a cold egg is brought into a warm and humid room, increases the potential for microbial infection of the egg.

SHELL MEMBRANES. The outer shell membrane is in contact with the egg shell, and the inner shell membrane, sometimes called the egg membrane, is in contact with the albumen. They are composed mainly

of protein fibers (fibrin and mucin) strengthened with an albuminous cementing material. The membranes have been considered effective bacterial filters. The inner membrane is more effective than the outer membrane because of its closely knit fibers. Organisms inoculated onto the shell surface are found on the membranes almost instantly, but micoorganisms inoculated into the air cell and onto the inner membranes are not recoverable from the albumen for several days. This apparent retention of bacteria on the inner membrane may be partly due to conalbumin in the albumen, since adding iron to the inoculum or to the interior albumen of the egg hastens the penetration of the bacteria through the inner membrane. Wedral, Vadehra, and Baker (1971) found no difference in the permeability of the inner shell membrane before and after passage of either *Pseudomonas aeruginosa* or *Salmonella typhimurium,* indicating that no enzymatic activity is needed for penetration. With the conditions of their experiment, both bacterial species were able to penetrate from the shell surface through the inner membrane within two hours. By 36 hr after shell inoculation, all except one egg showed contaminated albumen. Although there are no pores, per se, in the membranes, there are apparently openings between the intermeshed fibers similar to a maze, which allow passage of bacteria through the membranes.

Even if bacteria can penetrate these barriers, they are confronted with the antimicrobial mechanisms of the albumen before they can attack the nutrients in the yolk.

Barriers in Plant Products

Foods such as fruits, vegetables, grains, and nuts contain a peel, skin, or hard surface that covers the interior food and protects it from microbial attack (Halloin 1983). The effectiveness of these barriers is evidenced by the growth of mold primarily on the stem end of fruit, where injury to the cover may occur during picking. Also, when injury occurs to the barrier by insects, birds, or bruising, the microbial population in that area is high and rotting of the fruit occurs there, before spreading to the remainder of the fruit.

In the section on inhibition, the presence of essential oils in the peel of citrus fruit is discussed. There are inhibitors in the waxlike coating on many fruits. Cabbage leaves contain wax esters on their surface. The waxlike coating of apples increases during storage, the extent of the increase depending upon the cultivar.

The protective coverings are not only microbial barriers, but also help control diffusion of materials into and out of the fruits or nuts. The sale of nuts in the shell has remained fairly constant, while purchases of shelled nuts have increased. Removing the nuts from the shell subjects

them not only to microbial contamination but also to oxidative rancidity of the fatty material and other deteriorative changes.

The seed coat and pericarp of grains, which allow the long-term storage of these foods, were important in establishing human civilization. According to Halloin (1983), the seed coat is the most important component in the resistance of seeds to deterioration.

TEMPERATURE

Since water, as the solvent, is necessary for microbial growth, the temperatures that will support microorganisms are limited to those at which water is a liquid. Pure water freezes at 0°C, but the addition of solutes lowers the freezing point. The exact minimum temperature for microbial growth is difficult to determine, because the solutes that are added to prevent freezing of water below −10°C may inhibit the microorganisms.

Water boils at 100°C and is no longer acceptable for growth at that temperature. The boiling point of water is increased by adding solutes or by increasing the pressure above atmospheric. By increasing the pressure in a chemostat, a culture of *Bacillus caldolyticus* grew at 105°C (Heinen and Lauwers 1981). With further increased pressure and higher boiling points, bacteria might grow at much higher temperatures.

The absolute highest and lowest temperatures for growth remain unknown, primarily because of problems of incubation at extremely high or low temperatures.

Temperature is one of the most important environmental factors that regulates the growth of microorganisms. Temperature is related to the ability of an organism to grow and to survive. The environmental temperature also has an effect on cell size, metabolic products such as pigments and toxins, nutritional requirements, enzymatic reactions, and the chemical composition of cells.

Ranges for Growth

Each organism has a minimum, optimum, and maximum temperature for growth. The minimum and maximum temperatures are those beyond which the organism ceases to grow. The optimum temperature is more difficult to describe, since it may be the optimum for total cell yield, rate of growth, rate of metabolism, respiration, or the production of some metabolic product. Usually the optimum temperature is based on the rate of growth.

In general microbiology, the microorganisms are usually divided into three arbitrary classes: psychrophilic (low temperature), mesophilic (me-

dium or middle temperature), and thermophilic (high temperature). Psychrophiles and thermophiles are divided into facultative and obligate types.

The temperature ranges for the various types are listed in Table 4.11. Microbiologists tend to disagree on the exact ranges of growth for the different types. The fact that some organisms grow at low, some at intermediate, and others at high temperatures is more important than the exact temperature for these ranges. The approximate temperature ranges for the growth of selected microorganisms are listed in Table 4.12. One should remember that the growth temperature depends upon the strains of a species and on the chemical and physical properties of the substrate.

In general, organisms can survive and show some growth at temperatures considerably lower than the optimum temperature. When the temperature is increased above the optimum, there is usually a very rapid decline in the growth rate. The maximum temperature is usually only a few degrees (5°–12°C) above the optimum. Above the maximum temperature, growth ceases and the life of the cell is in jeopardy.

PSYCHROPHILES AND PSYCHROTROPHS. There are many definitions of psychrophilic organisms. One of the simplest definitions is that psychrophiles grow well at 0°C. They should produce a visible colony at that temperature in seven, ten, or fourteen days. This definition does not consider optimum, minimum, or maximum temperatures. Perhaps some distinction should be made between those organisms with a maximum temperature of 20°C or less and those that can grow at a higher maximum. Morita (1975) suggested that psychrophiles have an optimal temperature for growth at about 15°C or lower, a maximum at about 20°C, and a minimum at 0°C or lower. Organisms that grow at low temperatures but do not meet the definition are called psychrotrophs. The term

TABLE 4.11. APPROXIMATE TEMPERATURE RANGES OF GROWTH FOR ARBITRARY CLASSES OF MICROORGANISMS

	Temperatures (°C)		
	Minimum	Optimum	Maximum
Psychrophilic	−15–5	10–30	20–40
Obligate	−15–0	10–20	20–22
Facultative	−5–5	20–30	30–40
Psychrotrophic	−5–5	25–30	30–40
Mesophilic	5–25	25–40	40–50
Thermophilic	35–45	45–65	60–90
Obligate	40–45	55–65	70–90
Facultative	35–40	45–55	60–80

TABLE 4.12. TEMPERATURE RANGES OF SELECTED MICROORGANISMS

Microorganism	Temperature (0°C)		
	Minimum	Optimum	Maximum
Bacteria			
Acetobacter	5	—	42
Aeromonas	0–5	25–30	38–41
Bacillus cereus	10	—	47–50
Brevibacterium	5	—	42
Clostridium	0–45	—	60
C. botulinum	3–10	30–40	42–45
C. perfringens	15–20	30–40	45–51
C. putrefaciens	0	20–25	30
C. thermosaccharolyticum	43–45	55–62	70–71
Escherichia coli	3–10	37–41	48–50
Lactobacillus	5	30–40	53
Leuconostoc	10	20–30	40
Micrococcus	10	25–30	45
Moraxella	−1–2	30	41–42
Propionibacterium	2–3	30–37	45
Proteus	10	37	43–45
Pseudomonas	−7–4	20–30	31–43
P. aeruginosa	8	—	42
P. fluorescens	0–6	20–25	40
Salmonella	5–10	35–37	46–49
Staphylococcus	5–10	35–40	46–48
S. aureus	5–10	35–39	44–48
Streptococcus cremoris	—	25–30	—
S. faecalis	5–10	37	49–51
S. lactis	10–15	25–30	40
S. thermophilus	20	40–45	52
Vibrio	—	10–37	—
V. parahaemolyticus	3–13	35–37	42–44
Xanthomonas	0–5	25–31	40
Yersinia enterocolitica	0–4	—	37
Molds			
Aspergillus fumigatus	—	30–40	—
Botrytis cinerea	−1	20	30
Cladosporium	−5–−8	—	—
Penicillium rubrum	—	25–28	—
Rhizopus stolonifer	5	—	25
Yeasts			
Candida	0	—	29–48
C. lipolytica	5	25	35–40
Hansenula	—	37–42	50
Saccharomyces	0–7	20–35	40
Torulopsis	0	17–25	30–35

psychrotroph was suggested by Eddy (1960) to apply to microorganisms capable of multiplying at 5°C and below, regardless of the optimum temperature. This definition does not include the minimum or maximum temperature. Eddy's definition of psychrotroph is essentially that of a psychrophile, except that the temperature has been increased from 0° to

5°C. There are few, if any, obligate psychrophiles in foods. Most microorganisms that grow in food at low temperature are psychrotrophic.

The spoilage of refrigerated food (meat, milk, eggs) is generally due to psychrotrophic growth. Psychrotrophs are found in many genera and include aerobes and anaerobes.

The pseudomonads are an important group of psychrotrophs. Some members of the genus can grow at 4°C, while others grow at 43°C. Overall, the optimum temperature is somewhere between 20° and 30°C. *P. fluorescens* grows at 4°C or less, but not at 41°C. The usual range for *P. aeruginosa* is 8° to 42°C. The optimum temperature for pigment production of *P. fluorescens* is 20°C. The diameter of colonies of *P. fluorescens* increases at a more rapid rate at 30°C than at 25°C or less (Johnson et al. 1970).

MESOPHILES. The mesophiles are organisms that grow in the middle temperature range. We can define mesophiles as those organisms with an optimum temperature of 25° to 45°C. There are two groups of organisms in the mesophilic range. The saprophitic organisms have an optimum temperature of 25° to 30°C, and potential pathogens have an optimum between 35° and 45°C.

THERMOPHILES. These organisms have been defined as growing at high temperatures, with an optimum of 45°C or higher. The information in Table 4.11 shows the minimum temperature for thermophiles somewhere between 35° and 45°C, the optimum between 45° and 65°C, and the maximum between 60° and 90°C. Since the minimum temperatures for thermophiles overlap the higher mesophilic range, the thermophiles have been divided into facultative and obligate types. The facultative thermophiles can grow in the mesophilic range of 37°C or less, but the obligate thermophiles cannot grow at 37°C.

ENUMERATION. If the arbitrary temperature classes have any meaning, methods for determining the organisms should follow the definitions. The incubation condition for psychrotrophs is listed as 7°C for ten days (APHA 1984).

Juffs (1972) compared the extremes of the British Standards Institute method of incubating at 5° to 7°C for seven to ten days to determine psychrotrophs. The high extreme (7°C for ten days) is the same as the APHA (1984) method. Juffs found that 7°C for seven days or 5°C for ten days gave counts that were not significantly different. However, counts from plates incubated at 7°C for ten days were significantly higher, and those using 5°C for seven days were significantly lower.

The incubation period of seven or ten days becomes impractical as

a control measure for perishable refrigerated foods such as milk. By the time the count is obtained, the product is consumed or may have developed defects. As a continuing control procedure, it is of value by revealing potential problems. To obtain a psychrotrophic count sooner, incubation at 21°C for 25 hr yields counts similar to the standard psychrotrophic method (Oliveria and Parmellee 1976).

Comparing the incubation temperature of 7°C to the definitions of classes, it fits neither the 0°C given for psychrophiles nor the 5°C given for psychrotrophs. Since the term *psychrotroph* has been substituted for *psychrophile,* and using the APHA plating system, we might define psychrotrophs as those organisms that grow at 7°C and produce a visible colony in ten days.

The mesophilic count can be related to the standard plate count, or aerobic plate count. Incubation temperatures of 25° to 37°C have been used. The 37°C temperature indicates the potential pathogen population better than the lower temperature; however, since spoilage organisms are a problem and higher counts are obtained, a temperature of 32° or 35°C is used (APHA 1984).

Thermophiles are enumerated by incubating the culture at 55°C ±2°C for 48 to 72 hr (APHA 1984).

RELATIVE IMPORTANCE. The relative importance of each of the temperature classes (psychrophiles, psychrotrophs, mesophiles, and thermophiles) depends upon the field of microbiology. Much of our food supply is held in cold storage, which makes psychrotrophs important as potential spoilage organisms.

Psychrotrophic microorganisms are ubiquitous in nature. Although they are favored by cold, they are not confined to temperatures of 5°C or less. Psychrotrophs are in fresh and salt water, in soil, and in or on nearly all raw food. Stokes (1968) reported that psychrophiles constituted 35 to 95 percent of the bacterial population on meats and 67 percent of the population on fresh chicken. At low temperatures (0°–5°C), carbohydrate fermenters generally are inhibited and acids are not formed. Proteolytic and lipolytic organisms grow at low temperatures, resulting in food defects.

Mesophilic species can be found in nearly every genus of organisms important in foods. Although spoilage of foods by mesophiles is significant, there is more concern with the mesophilic organisms that cause foodborne illness. These include species of *Salmonella, Staphylococcus, Clostridium, Shigella,* and *Bacillus.*

Since the rate of growth of microorganisms and enzyme activity is higher in the mesophilic range than in the psychrophilic range, spoilage of food occurs sooner.

Thermophiles can cause spoilage whenever foods are held at 50° to 70°C. This may occur during heating of the food while cooking, processing, or pasteurizing. Their growth in hot syrup may cause undesirable effects in sugar-processing plants. In some tropical countries, thermophiles—not psychrotrophs—are the important spoilage organisms.

The generation time is shorter for thermophiles than for either psychrophiles or mesophiles when each is grown at its optimum temperature. This results in a more rapid spoilage of food by thermophiles.

The thermophiles may be more important than is presently realized. Since food is not normally held at temperatures conducive for growth of thermophiles, they have not been given the attention that has been allocated to the other classes. Due to the rapid growth and metabolism at high temperatures, the new biotechnology is examining thermophiles for direct use in fermentations or as a source of enzymes (see Chapter 9).

Effect of Organisms on Temperature

Just as the environmental temperature alters the growth of microorganisms, the growth of microorganisms alters the temperature. When the organisms metabolize organic compounds, they not only grow and multiply, they also produce heat as a by-product. In a compost pile, the temperature may rise to a level at which only thermophilic bacteria can grow. In the production of compounds by microbial fermentation, it is often necessary to use a cooling system in the fermenter to maintain the temperature so that the maximum amount of desired product is developed.

GASEOUS ATMOSPHERE

The type of gas in the atmosphere surrounding the food may determine the types of organisms that become dominant. Oxygen in the atmosphere will favor the growth of aerobic types. The lack of oxygen, or a vacuum, will allow facultative anaerobes to become dominant.

Microorganisms vary widely in their tolerance to carbon dioxide. In a CO_2 atmosphere, the growth of some microorganisms is completely suppressed, while others are less affected. The lag period for growth of spoilage organisms on beef is increased in a 10 percent CO_2 atmosphere. Also, these organisms need a higher minimum water activity for growth in 10 percent CO_2 as compared to the normal atmosphere (0.033 percent CO_2). On the other hand, it is well documented that the presence of low concentrations (1 percent or less) of CO_2 stimulates oxygen uptake of bacteria. The amount of stimulation varies with bacterial species.

According to King and Nagel (1975), CO_2 inhibits the synthetic process of *Pseudomonas aeruginosa,* since resting cell metabolism is not affected.

MICROBIAL INTERACTIONS

Microorganisms can inhibit or stimulate the growth of one another. Although pure cultures are usually used in laboratory studies, mixed cultures are found in nature and on food products. Since the main goals of all living forms are self-preservation and perpetuation of the species, the microorganisms have to compete for food and the other necessities. As a result, there are various actions and reactions (or coactions) that may be harmful or beneficial. Burkholder (1952) listed nine possible coactions when organisms are in a mixed population:

Predation. In this relationship, the strong predators damage the weak prey.
Parasitism. The weak parasite benefits at the expense of the strong host.
Commensalism. The coaction results in the weak benefiting, and the strong is unaffected.
Amensalism. The opposite of commensalism. The strong benefits and the weak is unaffected.
Allotrophy. In this relationship, the strong feeds the weak.
Allolimy. The strong starves the weak.
Symbiosis. In symbiosis, there is mutual aid, with both organisms benefiting from the relationship.
Synnecrosis. There is mutual conflict resulting in the death of both organisms.
Neutrality. Neither organism benefits or loses from the relationship. There is no coaction between them.

Systems for determining interactions of microorganisms have been reported (MacFarlane and Makrides 1982; Nordbring-Hertz 1983).

Metabolism

The growth of microorganisms depends on the chromosomally determined primary metabolism. The loss of primary metabolism results in the death of the cell. Secondary metabolism is not essential for the growth and multiplication of microorganisms. Both primary and secondary metabolism produce metabolic products. Quite often the production

of secondary metabolites is induced by the presence of plasmids. The plasmids are composed of extrachromosomal nucleic acids that can be eliminated from the cell by various treatments and, under acceptable conditions, can be transferred from one cell to another. Simple and rapid methods for isolation and preparation of bacterial plasmids have been described by Anderson and McKay (1983) and Summerton, Atkins, and Bestwick (1983). By using plasmids to transfer genetic information, we may be able to develop more useful organisms than we have now.

The products of metabolism may stimulate the growth of other organisms. A strain of *Staphylococcus aureus* that required thymine and tryptophan was able to grow in mixed cultures or with culture filtrates of *Pseudomonas aeruginosa* (Gadbois, De Repentigny, and Mathieu 1973). The *Pseudomonas* supplied the thymine and tryptophan essential for the growth of the *S. aureus.*

Using a chemically defined medium that supported the growth of the yeast *Saccharomyces cerevisiae,* but not *Proteus vulgaris,* researchers found that both organisms grew in mixed culture (Shindala et al. 1965). A niacin-like factor, produced by the yeast, made it possible for the bacterium to grow. When niacin was added, the *Proteus* was no longer dependent upon the yeast and promptly outgrew it.

Several bacteria isolated from bananas stimulated the germination of the conidia of *Colletotrichum musae* on this fruit (McCracken and Swinburne 1980). On an agar medium deficient in L-cysteine, *Legionella pneumophila* did not grow, but when *Flavobacterium breve* was present, the *Legionella* grew as satellites around the *Flavobacterium* colonies (Wadowsky and Yee 1983).

Although these citations show one organism aiding another organism, the dominance of one species can result in the utilization of the nutrients so there is little or none left for other species.

pH. Microorganisms can alter the pH upward or downward, and thereby influence the growth of other microorganisms. The lactic acid bacteria (lactobacilli, streptococci, pediococci, leuconostoc) inhibit several bacteria (both spoilage types and foodborne pathogens) and viruses partially by acid formation with low pH and also by some undetermined factors (Abdel-Bar and Harris 1984; Collins-Thompson, Wood, and Beveridge 1983; Gilbert et al. 1983; Moon et al. 1982; Rao and Pulusani 1981; Reddy and Ranganathan 1983; Ross 1981; Talon, Labadie, and Larpent 1980). Montville (1982) reported that, in a model system at pH 4.4, growth of *Bacillus licheniformis* raised the pH sufficiently to allow growth and toxin production by *C. botulinum.* However, he found no toxin in canned tomatoes with elevated pH values.

INHIBITORS. The production of microbial inhibitors by microorganisms is well known. Besides the chemicals called antibiotics, there are other metabolic compounds that have antibioticlike or inhibitory characteristics. Several enzyme inhibitors are produced by microorganisms, but, according to Umezawa (1982), most of them have no significant antimicrobial activity. Some microbially produced inhibitors are listed in Table 4.13.

The pseudomonads produce several antibiotic substances, including pyocyanine and pyrrolnitrin. Species of *Pseudomonas* influence the growth of strains of *Vibrio parahaemolyticus, S. aureus, Bacillus,* as well as some molds and yeasts (Collins-Thompson, Aris, and Hurst 1973; Goatcher and Westhoff 1975; Hemming et al. 1982; Scannell et al. 1972; Vandenbergh et al. 1983).

Strains of species vary in their interactions with other organisms. In comparing the interactions of four strains of *Streptococcus cremoris* and four strains of *S. lactis,* it was found that two strains of *S. cremoris* dominated the four *S. lactis,* one strain of *S. cremoris* was compatible with the *S. lactis,* and one strain of *S. cremoris* was dominated by the four *S. lactis* strains (Reddy et al. 1971).

Bacillus species interact with other microorganisms. *B. subtilis* produces an antibiotic called subtilin. This polypeptide has a marked action against a wide range of Gram-positive, acid-fast, and certain Gram-negative bacteria. Depending on the concentration, subtilin is bacteriostatic or bactericidal. Barr (1975) found ten antimicrobial metabolites produced by a strain of *B. subtilis.* Some of these were antibacterial, while others were antifungal. *Bacillus* species isolated from mud inhibited the growth of *C. botulinum* type C (Graham 1978). A thermophilic species of

TABLE 4.13. SOME TYPICAL MICROBIALLY PRODUCED MICROBIAL INHIBITORS

Inhibitory Organism	Substance Produced	Organism Inhibited	Reference
Serratia marcescens	Prodigiosin	*E. coli, B. subtilis, Enterobacter aerogenes, S. aureus, P. aeruginosa*	Kalesperis, Prahlad, and Lynch (1975)
P. aeruginosa	Amino acid antimetabolite	*Bacillus* sp.	Scannell et al. (1972)
Saccharomyces cerevisiae	Glycoprotein	*Torulopsis glabrata*	Bussey and Skipper (1976)
Lactobacilli	Hydrogen peroxide	*Pseudomonas* sp. *Bacillus* sp. *Proteus* sp.	Price and Lee (1970)
C. perfringens	Unknown	*C. botulinum*	Smith (1975)
S. lactis	Nisin	*C. botulinum*	Scott and Taylor (1981)

Bacillus not only provided some protection of mushroom beds from the mold *Chaetomium olivaceum* but also increased the yield of *Agaricus bisporus* (Tautorus and Townsley 1983). A *B. subtilis* controlled the onion white rot organism, *Sclerotium cepivorum* (Utkhede and Rahe 1983).

Volatile metabolites of certain strains of bacteria were able to inhibit the growth, sporulation, and mycotoxin production of *Penicillium* and *Aspergillus* species (Barr 1976).

Aflatoxin, a substance produced by the mold *Aspergillus flavus,* inhibits various *Bacillus* species (Lillehoj and Ciegler 1970). Also, fusariotoxin T-2 produced by *Fusarium* inhibits the growth of *Saccharomyces cerevisiae* and *S. carlbergensis* (Schappert and Khachatourians 1983).

Certain strains of *Saccharomyces cerevisiae* produce a toxin that kills sensitive strains of this yeast (Wickner 1983; Young 1981).

Antagonism among microorganisms may be due simply to competition for space or nutrients. One organism may gain an advantage over another because of its motility (Lauffenburger and Calcagno 1983) or to environmental effects (temperature, moisture) favoring one organism over the others.

Fatty Acids. These substances are discussed as inhibitors naturally present in foods. They also are formed by microorganisms during metabolism.

The growth of *Clostridium botulinum* is inhibited on surface-ripened cheese due to the formation of fatty acids by *Brevibacterium linens. Salmonella gallinarum* is inhibited by *Leuconostoc citrovorum* due to the acidic pH and production of acetic acid (Sorrells and Speck 1970).

Hydrogen Peroxide. The mechanism of hydrogen peroxide (H_2O_2) formation by streptococci was described by Anders, Hogg, and Jago (1970). The amount of H_2O_2 that accumulates varies among strains. The addition of catalase or ferrous sulfate to milk prevents peroxide accumulation and results in an increased rate of acid production by the lactic streptococci (Gilliland and Speck 1969).

Organisms such as the micrococci produce catalase. It was reported by Nath and Wagner (1973) that in the presence of a *Micrococcus* species, the growth of, and acid production by, six cultures of lactic acid bacteria is stimulated. The stimulation is greater than that observed with the addition of catalase, indicating that factors besides hydrogen peroxide removal are involved.

Hydrogen peroxide produced by *Streptococcus mitis* and *Lactobacillus acidophilus,* in combination with a peroxidase and a halide, has viricidal activity to polio virus (Klebanoff and Belding 1974).

BACTERIOCINS. The bacteriocins are bactericidal substances produced by various species of bacteria. Although considered antibiotics,

they are not like the classical antibiotics since they are macromolecular, including or consisting of polypeptide or protein, and they generally act on strains of the same, or closely related, species (Vidaver 1983).

The colicins have been studied the most extensively. They are found in *E. coli* and other Enterobacteriaceae. There are over twenty types of colicins. In some respects, they act similarly to phage. They can be induced by UV light, similarly to prophage. In some cases, they are held intracellularly, and when released, the cell is lysed.

When bacteriocins attach to the receptor sites of a sensitive cell, one or more things may occur, depending upon the bacteriocin and the cell. One of the effects is the inhibition of macromolecular synthesis, such as proteins, RNA, and DNA (Konisky 1982). Some colicins inhibit only protein synthesis, while others inhibit all three. Inhibition of respiration may occur with some colicins, while others may block the function of permeases. Interference with the formation of ATP, or enhancement of its breakdown, has been reported. Leakage of cellular constituents has been reported but this may be due to secondary reactions and not caused directly by the bacteriocin. Colicins disrupt transport mechanisms of amino acids, which results in a lack of protein synthesis.

Bacteriocins of Gram-positive organisms generally have a wider spectrum of activity than those produced by Gram-negative cells.

Parasites

Two of the parasites of bacteria are the bacterial viruses or bacteriophages and the bacterial genus, *Bdellovibrio.*

Some phages have a limited number of microbial species that are susceptible hosts. In a species of bacteria there may be susceptible and resistant strains of bacteria. For other phages, there may be several species and even different genera of bacteria that contain susceptible cells. For instance, some pasteurella phages also attack strains of *Salmonella* and *Shigella.* The limited hosts for phages makes it possible to use phage typing to differentiate bacteria.

The *Bdellovibrio* resemble the virulent bacteriophages in their ability to lyse bacterial cells. However, unlike the phages, these microorganisms are actively motile (a single flagellum), small (0.25–0.4 μm wide and 0.8–1.2 μm long), vibroid, Gram-negative bacteria. The formation of plaques (holes in a host lawn due to lysis of cells) is usually developed in 12 to 24 hr by phage, but *Bdellovibrio* plaques are visible only after two to four days, and they enlarge up to six days of incubation. Host-independent *Bdellovibrio* populations have been grown, which has not been possible with phage. About one in a million cells is host independent, and these

host-independent cultures can revert to host-dependent varieties at about the same rate.

SUBLETHAL STRESS

During the processing of foods, microorganisms can be subjected to various physical or chemical stresses. These include treatments such as heat, freezing, drying, osmotic effects, irradiation, and various chemicals. To maintain the quality attributes of the food, the treatments may be minimal and the effect on the microorganisms is of secondary consideration. In other cases, the treatment may be an attempt to control certain organisms, but others may be affected. These treatments that do not kill, but damage or injure the cell, are called sublethal.

The damages or injuries to the organism due to sublethal treatments are called lesions. As a result of the injuries, the growth capabilities of the organisms may be altered, both in the food and on media during enumeration. The stressed cells may have an extension of the lag phase, more exacting growth requirements, or increased sensitivity to inhibitors or to selective agents in media. The longer-than-normal lag phase can be called the recovery period, during which the damage to the cell is repaired. Although the cells usually can recover in a noninhibitory growth medium, they can repair malfunctions in media in which growth is not evident. In other words, growth is not needed for repair of the lesion.

It has been stated that injured organisms that normally cause foodborne illness or food spoilage, when in a substrate conducive for repair, can, after repair, be involved in these unacceptable activities. However, in most populations with injured cells there are also some cells with no injury. Hence, given sufficient time for multiplication, the uninjured cells can cause foodborne illness or spoilage, without any assistance from the repaired cells.

Heating

In several processes (scalding, blanching, pasteurization, cooking, and spray drying), the food and associated microbial flora are subjected to heat. Since none of these processes are designed to obtain a sterile product, some organisms may be killed, many damaged, and others unaffected by the treatment.

Heat-induced lesions include damage of the cytoplasmic membrane, with leakage of cellular components, alteration of metabolic capabilities

of the cell, impairment of enzyme activity, and degradation of ribosomes and ribonucleic acid.

Heat-induced damage results in the loss of ability to grow in conditions in which normal cells can grow. Normal *S. aureus* cells can grow in media with 7.5 percent NaCl, but heat-damaged cells cannot tolerate 4 percent salt (Smolka, Nelson, and Kelley 1974). Repair, with the return of salt tolerance, occurs without growth in either a dilute synthetic or a rich complex medium (Hurst and Hughes 1981). The pH range for optimum growth is more narrow for heat-stressed cells than for normal cells. The addition of catalase or superoxide dismutase to tryptic soy agar protected heat-stressed cells from hydrogen peroxide (Bucker and Martin 1982). This results in higher counts of sublethally injured cells (Egan 1979). For increased recovery of injured *E. coli* cells in food samples, McDonald, Hackney, and Ray (1983) suggested adding 3, 3′-thio-dipropionic acid to supplement media for the repair process.

The temperature of incubation before and after heating influences the viability of *E. coli* (Katsui et al. 1982). They suggested that the fluidity of the cell membrane was involved in this reaction. The repair of the nucleoid structure of *E. coli* cells after heat shock (50°C for 5 min) is different than that after more severe heat treatments (Pellon 1983).

All solid media were more inhibitory to heat-stressed cells of *Salmonella typhimurium* than were the corresponding liquid media (Mackey and Derrick 1982). However, the addition of either catalase or pyruvate to nutrient agar resulted in increased recovery and comparable growth to that in nutrient broth.

Yeasts heated in water near the maximum temperature for growth are essentially undamaged, but the presence of glucose induces leakage of cell contents (Hagler and Lewis 1974). Calcium ions temporarily protect the yeasts against glucose-induced leakage. This indicates damage to the cytoplasmic membrane. If heated in the presence of glucose, injury affecting endogenous respiration is irreversible.

Yeasts have been enumerated on acidified (to pH 3.5) media; however, when twelve species of yeast were heat stressed, the lowest optimum pH for any of these was pH 6.8 and the maximum optimum was as high as pH 10 for two yeasts (Nelson 1972). Koburger (1972) found that most retail food samples yielded maximum counts of yeasts and molds when the medium was adjusted to pH 8. For enumeration of fungi, the addition of antibiotics to media, rather than acidification, to control bacteria, is now the usual procedure. The effects of various chemicals in media on recovery or inhibition of heat-stressed *S. cerevisiae* were discussed by Beuchat (1982). Conner and Beuchat (1984) found that various species of heat-stressed yeasts could not repair their injuries when exposed to rela-

tively low levels of essential oils of plants (allspice, cinnamon, clove, garlic, onion, oregano, savory, thyme).

Heat injury can occur in the spores of *Bacillus* and *Clostridium.* The germination system of *B. subtilis* is impaired by heating. Primary activation of spores at 90°C for 60 min, reduction of the incubation temperature by 10° to 15°C or addition of $CaCl_2$ and sodium dipicolinate, increases germination and colony formation of heat-damaged spores. The addition of lysozyme to enumeration media improves the recovery of severely heated *Clostridium* spores. Bacterial spore injury and recovery were reviewed by Foegeding and Busta (1981). Michels and Kagei (1983) recommended using trypticase soy agar supplemented with cysteine, methylene blue, and egg yolk emulsion for the enumeration of heat-damaged spores of *Clostridium sporogenes.*

Cold Effects

Freezing of cells can cause an apparent increase not only in nutritional requirements but also to sensitivity to selective agents used in media. Cells in the logarithmic growth phase are highly susceptible to cold shock and show an apparent loss in viability, which can be recovered by incubation with magnesium ions at 30°C. They gradually lose their capacity to recover if kept in cold magnesium-free buffer. Loss of viability due to cold shock is apparently due to damage of DNA (Sato and Takahashi 1970). The lethal effect of cold shock depends on the exposure time to low temperature, the growth phase, and physiological condition of the cells, the substrate, and the type of cell.

Drying

It is well established that organisms in dried foods may have metabolic injuries that impair the proliferation of the cells in selective media containing inhibitors in a concentration well tolerated by normal cells of the same species. These stressed cells can recover their tolerance to inhibitors if allowed to recuperate or repair in a nonselective medium.

Dried organisms are stressed by freezing (freeze-drying or lyophilization), aerosolization (spray-drying), or heat (roller, drum, tunnel or spray-drying), as well as existing in an environment of very low water activity.

The initial physiological state, the composition of the food, the location of the organisms in the food, the processing history, storage conditions, and the method of rehydration, will influence the survival and recovery of organisms from dried food.

Freeze-dried cells have altered permeability. Freeze-dried *E. coli* are

susceptible to antibiotics that do not affect normal cells. Also, there is leakage of RNA from cells stressed by freeze-drying. The alterations in the stressed cells are reversible, since during or immediately after recovery of cellular permeability, the damaged RNA, as well as metabolic damage, is repaired (Sinskey and Silverman 1970). After freeze-drying, resynthesis of cell wall and membranes and the reestablishment of transport mechanisms are needed before cellular growth occurs. Many freeze-dried cells require pyruvate, hematin, and menadione for recovery. The effectiveness of the three compounds during repair of the cells is additive. It is believed these compounds may aid in reestablishing the permeability barrier of the cell.

Besides the need for nonselective nutrients in the recovery medium, due to the need for rehydration, the temperature at which this is performed is important. Ray, Jezeski, and Busta (1971a) recovered a higher number of cells at 15° to 25°C, but there was an earlier initiation of growth and more rapid growth if rehydration was done at 35°C. The rate of repair is reduced by lowering the temperature of the recovery medium from 35°C to 10°C and is extremely low at 1°C (Ray, Jezeski, and Busta 1971b).

Usually dried food is prepared for the analysis of microorganisms by rapidly blending the food with a sterile diluent. However, a more gradual system of rehydration such as soaking the food in diluent at a 1:1 ratio for 30 to 60 min allows the microorganisms to adapt to a normal moisture level. Rapid rehydration can cause lethal or sublethal injury to the dried cells. This results in a lower count of cells when the sample is plated onto a selection medium (Ting and Banwart 1985).

Chemical Inhibitors

Since preservation processes cause injuries to microorganisms, perhaps chemical inhibitors may do likewise. Besides chemical preservatives, organisms on equipment surfaces are exposed to cleaning and sanitizing agents. If these organisms remain on the equipment, they may contaminate the food; if injured, they may be difficult to enumerate.

The action of physical factors can be readily altered and the microorganisms tested for injury, but it is more difficult to eliminate chemicals if they are sorbed to the cell or are inside the cell. In these cases, even dilution may not eliminate the chemical effect on the cell, and death can result.

Injury by acid and recovery of *Salmonella bareilly* was reported by Blankenship (1981). For recovery, it was indicated that synthesis of protein and RNA and electron transport are needed. The exposure of *E. coli* to

hydrogen peroxide or to low levels of cadmium causes single-strand breakage in DNA (Ananthaswamy and Eisenstark 1977; Mitra and Bernstein 1978). The transfer of injured *E. coli* to a cadmium-free liquid medium resulted in recovery without significant synthesis of DNA (Mitra 1984).

Chlorine injury to coliforms (*E. coli*) was manifested as inhibition of uptake of metabolites, reduced adhesion and ability to colonize the small intestine, and inhibition in selective media for enumeration (Camper and McFeters 1979; Graham and Brenniman 1983; Walsh and Bissonnette 1983).

Ultraviolet Light

The ultraviolet portion of the spectrum includes radiations from 150 to 390 nm. When living cells are irradiated with this light, some may be "killed" and others may be mutated. The UV wavelengths (around 260 nm) most active in producing either of these effects are those most readily absorbed by the nucleic acids. Both lethal and mutagenic effects of ultraviolet light can be partially reversed by repair or reactivation.

Although the ultraviolet light may be absorbed by many cellular components, absorption by the nucleic acids results in damage to cellular mechanisms for division. Repair of DNA in the light is called photoreactivation. Reactivation of UV-irradiated cells also can occur in the dark (dark repair, dark reactivation, or excision repair).

Various compounds have been given credit for activation or protection of irradiated cells. One such, the enzyme catalase, breaks down the microbial inhibitor, peroxide, that is formed during irradiation. Another action of ultraviolet light is the inactivation of disulfide enzymes due to disruption of cystine. Inactivation of enzymes can affect microbial growth but not necessarily cause death.

Nalidixic acid (Eberle and Masker 1971), coumarin, pyronin Y, 6, 9-dimethyl 2-methylthiopurine and caffeine (Grigg 1972; Koukalová and Reich 1981) inhibit or block repair of UV-treated cells.

Other Repair Systems

Microoganisms can repair damage by gamma rays (Foegeding and Busta 1981; Yatvin, Wood, and Brown 1972), fluorescent and photo light (Eisenstark and Ruff 1970), as well as water stress (McFeters, Cameron, and LeChevallier 1982; Rhodes, Anderson, and Kator 1983). The injury and repair of these sublethal treatments are similar to those discussed for other treatments.

Summary

Various treatments as discussed in this section can cause injury to microorganisms. Often these injuries can be repaired, but the treatment rendered to the organism can be lethal depending upon the extent or amount of treatment. A truly lethal injury cannot be repaired, but microorganisms given a sublethal treatment, under certain conditions, can recover, grow, and multiply. In some instances, the treatment appears to kill the cells, since they are not able to multiply. However, they may still be alive, since they are metabolizing and even growing, such as the filamentous forms of *E. coli.*

The sublethal injuries are important since, without repair, the organisms are not able to grow on selective media used in enumeration of a particular type. Microorganisms in a stressed condition are more susceptible to inhibition and death by other stress factors.

OTHER FACTORS

Environmental conditions, such as pressure, surface tension, surfaces, light and photosensitizers, and pesticides, as well as colloidal characteristics, emulsion structure, and concentration of nutrients, might help determine the microorganisms that are able to metabolize, grow, reproduce, and spoil food products or result in a public health hazard. Compared to the other factors listed, these are of minor consequence in foods.

INTERACTIONS OF FACTORS

In a few instances, only one factor causes stresses on microorganisms. The interations of these factors are important in foods.

Water Activity and Nutrition

The minimum a_w for growth of a microorganism depends upon the nutrients present in the growth medium. Christian (1955) showed that the range of a_w for growth of *Salmonella oranienburg* was appreciably smaller in a chemically defined glucose-salts medium than in a nutrient broth. The addition of small amounts of proline and four other amino acids to the glucose-salts medium stimulated the growth of *S. oranienburg* at 0.97 a_w and permitted growth down to 0.96 a_w. Addition of vitamins showed further stimulation, and growth occurred at 0.95 a_w. Thus, the minimum a_w levels for microorganisms should be determined in a medium with ample nutrients for the growth of the organism tested.

Since foods differ in their nutrient content, microorganisms may be able to grow in some foods at lower a_w than required in other foods.

Water Activity and Temperature

The greatest tolerance to low a_w occurs at the optimum temperature. As the temperature is changed from optimum, the range of a_w is reduced in which spore germination and growth occur. Studying the growth of fungi on jam, Horner and Anagnostopoulos (1973) found the interaction of a_w and temperature had a significant effect.

For *Salmonella* survival and multiplication, there was a close correlation between a_w and storage temperatures of meat and bone meal (Liu, Snoeyenbos, and Carlson 1969). The relationships of a_w and temperature as well as pH and antifungal agents on growth of *Aspergillus flavus* and *A. parasiticus* were determined by Holmquist, Walker, and Stahr (1983).

Water Activity and pH

As the pH is increased or decreased from the optimum, the minimum a_w needed for growth is increased. Decreasing the pH of the growth medium increases the minimum a_w at which *Clostridium botulinum* spores will germinate and initiate growth (Baird-Parker and Freame 1967).

pH and Temperature

Outgrowth of spores of *Clostridium botulinum* type E occurred at 15.6°C and a pH of 5.4 to 5.6. When the temperature was lowered to 5°C, a pH of 6.2 to 6.4 was needed for outgrowth to occur (Emodi and Lechowich 1969).

The optimum pH for *B. subtilis* spore germination is markedly changed by altering the incubation temperature (Ishida, Ishido, and Kadota 1976). The optimum pH was 7.4 at 37°C and 5.4 at 10°C.

Eh and pH

There is a definite relationship between the pH and the functional OR systems in bacterial cultures. Staphylococci are more acid tolerant aerobically than anaerobically (Barber and Deibel 1972). Most staphylococcal strains initiate growth and produce detectable enterotoxin at pH 5.1 when grown aerobically. In anaerobic conditions, most strains fail to produce enterotoxin below pH 5.7. When held at 5°C, *Brochothrix* grew aerobically at pH as low as 5.4, but anaerobically no growth was evident

below pH 5.8. It grew readily at pH 6.0 or higher (Campbell et al. 1979).

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5

Sources of Microorganisms

The microbial flora of a food consists of the microorganisms associated with the raw material, those acquired during handling and processing, and those surviving any preservation treatment and storage.

Since these microorganisms do not arise by spontaneous generation, they must contaminate the food at some stage of production, harvesting, handling, processing, storage, distribution, or preparation for consumption. Most foods are subjected to many potential sources of microorganisms.

Why should we be concerned with sources of contamination? Primarily so that we can control contamination and keep the microbial load on or in the food as low as possible. By doing this, we obtain a longer shelf life for the food; we hope that this reduces the chance of microbial food-borne illness when the food is ingested. By keeping the contamination low, we can more easily control or eliminate the microorganisms with food-preservation techniques.

The potential sources of contamination are soil, water, air, plants, feed or fertilizer, animals, human beings, sewage, processing equipment, ingredients, product to product, and packaging materials.

Microorganisms can be exchanged between these sources. For example, animals contaminate the soil with fecal material. Then rain washes the microorganisms into the creeks and rivers. This water may be used for irrigation and contaminate plants used for food. Thus, although water is the carrier, the microorganisms originally come from animals.

For some foods, it is difficult to determine how many of the organisms in the flora are contaminants and how many are the result of multiplication on or in the food.

SOIL

Soil is the natural habitat of many microorganisms which, at times, may be found there in high numbers. The microbial density is greater nearer the soil surface than at deeper levels. The types and numbers of microorganisms vary with the type of soil (sandy, clay), as well as with

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environmental conditions. The environment is changing constantly, especially moisture and temperature.

Growth of Microorganisms

The microbial growth in soil is limited to areas of organic material. These areas include the roots of plants, plant debris falling onto the soil, animal carcasses, fecal deposition by animals, and dead microorganisms. Besides the animal carcasses on the soil surface, dead earthworms, insects, and other small animals are in the soil.

The growth of microorganisms is influenced by the chemical composition of the materials undergoing decomposition, the rate of decomposition of the chemical constituents of these substances, and the environmental conditions. Fats, waxes, and lignins may gradually accumulate in the soil because of their relative resistance to decomposition. This residual, called humus, is subject to slow and gradual attack by a variety of microorganisms. The main factor affecting multiplication is nutrient depletion. In general, the soil environment is not a good medium for the growth of microorganisms. However, they tend to survive.

Number and Types of Microorganisms

The microbial numbers can vary from a few organisms in sandy and desert soil to as many as 10^{10}/g in fertile soil. Due to the usually poor environmental conditions, microorganisms in soil are in some form of resting stage or in a reduced state of metabolic activity. Spores of *Bacillus* and *Clostridium* are quite prevalent in some soils. Generally, bacteria outnumber other types of microorganisms. Those that are common in food and found in soil include *Acinetobacter, Alcaligenes, Arthrobacter, Bacillus, Clostridium, Corynebacterium, Flavobacterium, Micrococcus, Pseudomonas,* and *Streptomyces.* Yeasts are prevalent in the soil of orchards and vineyards. Molds, especially spores, are found in some soils (Christensen, Raper, and States 1978; McKenzie and Taylor 1983).

The natural habitat of *Clostridium botulinum* types A and B is soil. *C. botulinum* type E was not found in cultivated soil or woodlands (Huss 1980). Thermophilic sporeformers that can spoil canned foods are found in soil, usually in low numbers.

Enteroviruses survive sewage treatment and can be found in soil on which treated wastewater is discharged. The viruses are adsorped to soil particles, which makes them relatively resistant to inactivation. Another important factor that influences inactivation is temperature. At 4°C, poliovirus persisted at least 180 days in saturated soil (Yeager and

O'Brien 1979a). No infectivity was found in dried soil, regardless of temperature. The loss of infectivity was due to irreversible damage including dissociation of viral genomes and capsids as well as degradation of viral RNA (Yeager and O'Brien 1979b). Poliovirus survives in soil longer in winter than in summer (Tierney, Sullivan, and Larkin 1977).

Other microorganisms become inactivated in soil by predators (Casida 1980, 1983), bacteriolytic enzymes (Hemelt, Mares, and Upadhyay 1979), or toxins (Lynch 1982). In dry organic soil *Ustilago* spores remain viable for one year, but in wet organic soil they should remain viable for less than one week (Andreis 1980). St. John and Matches (1982) reported survival of *C. perfringens* for at least twenty-four weeks in tilled soil.

Contamination of Food

Microorganisms in the soil can contaminate tubers or root crops by direct contact. Also, dirt is blown by the wind or is splashed by rain falling onto the soil, so that the dirt can contaminate crops such as strawberries, beans, cabbage, or peas that grow near the ground level. The microbial numbers and types on crops are influenced by the degree of contamination of the soil in which they are grown.

Mechanical harvesting has increased the amount of soil contamination as well as breakage of fruits and vegetables. Cereal crops are contaminated mainly during harvesting.

Marine sediments have microbial counts in the range of 10^4 to 10^9/g. The numbers are usually higher near the shore than at deeper levels. The bacteria found in these sediments include *Aeromonas, Bacillus, Chromobacterium, Citrobacter, Escherichia, Pseudomonas,* and *Vibrio.* These sediments serve as a source of microorganisms for water as well as for fish and shellfish. During trawling along the bottom with nets, the sediment is disturbed so that it contaminates the fish or shellfish that are caught in the nets.

WATER

Water is a potential source of microbial contamination of food. Rain contains microorganisms that are washed from the air. As the water lands on the ground, it is further contaminated by soil microorganisms. In the ocean, organisms are in greater abundance near the shore than in regions distant from land. The dumping of wastes such as sewage and the runoff from animal feedlots result in considerable microbial contamination of waterways with enteric types of bacteria.

Types of Microorganisms

The genera of bacteria that tend to be part of the "normal" flora of water include *Pseudomonas, Flavobacterium, Cytophaga, Acinetobacter, Moraxella, Aeromonas, Corynebacterium, Streptococcus, Klebsiella, Alcaligenes, Bacillus, and Micrococcus.* Also, enteroviruses are quite resistant to the treatment given to sewage and therefore can be found in water supplies.

Escherichia coli is used as an indicator of fecal pollution of water in temperate climates. This organism supposedly dies rapidly in the relatively hostile environment of rivers due to low temperature, sunlight, toxic chemicals, and a lack of nutrients. However, some reports indicate that the coliforms and *E. coli* seem to maintain their population (Coleman et al. 1974; Goodrich et al. 1970). Water is still the main carrier of organisms that cause gastroenteritis in humans. From 1971 to 1982, there were 387 outbreaks of gastroenteritis caused by consumption of contaminated water in the United States. Of these, 150 outbreaks were caused by municipal water supplies (CDC 1983).

Food Contamination

Water contacts food during production, harvesting, and processing. If the water used for irrigation of various crops is contaminated or is sewage effluent, the fruits and vegetables can be potential health hazards (Rosas, Báez, and Coutiño 1984; Tamminga, Beumer, and Kampelmacher 1978; Tierney, Sullivan, and Larkin 1977; Van Donsel and Larkin 1977).

Seafoods are harvested from water. The microorganisms in the water contaminate the surface, gills, and intestinal tract of fish and shellfish. The occurrence of fecal coliforms in fish is a reflection of the pollution level of their water environment. The skin of Atlantic salmon was analyzed by Horsley (1973). The predominant genera were *Cytophaga, Flavobacterium, Moraxella,* and *Pseudomonas.* Other organisms found were *Acinetobacter, Bacillus, Aeromonas, Vibrio,* coryneforms, members of the families Enterobacteriaceae and Micrococcaceae.

When bivalve mollusks feed, they filter large quantities of water and concentrate bacteria and viruses that are present in the water environment. Shellfish are normally found in water near the shore. This water is subject to contamination of runoff water carrying soil microorganisms and sewage outfalls. The accumulation and concentration of these microorganisms from the water is of particular concern when they are potential pathogens such as salmonellae and human enteric viruses.

Potential pathogens in the drinking water of animals can be a health hazard to the animals and to people who work with the animals. These

organisms from infected animals also can contaminate the carcass during the slaughter operations.

During harvesting, water may be used for hydrocooling of vegetables. Irrigation or surface water that is polluted with sewage or animal wastes can serve as a source of inoculation with various organisms, including potential pathogens. The water can redistribute organisms already on the vegetables from high levels of contamination or spoilage to other areas so that all of the vegetables are contaminated. Since many vegetables and fruits are eaten raw, the use of untreated water to wash these foods can serve as a vehicle of transmission of pathogenic organisms.

Ice is used to cool and maintain coolness in a variety of fresh foods. As the ice melts, the organisms associated with it are passed on to the foods and vice versa. The reuse of ice is neither a recommended nor an accepted procedure, since this ice is contaminated. Chen and Chai (1982) reported that ice-melt drainage in fishholds of fishing vessels contained bacteria at levels of 10^7 to 10^9/ml.

The use of water in a food-processing plant can be a source of microorganisms for contaminating food. Water is the food industry's number-one raw ingredient. Water enters into the processing of most foods for cleaning equipment and processing areas, for washing food, for conveying (fluming) food, or as an ingredient.

The safety of water that is used in foods is of major importance. However, although considered to be safe for drinking (potable), municipal water supplies are not always acceptable for food processing. This is because of the presence of food-spoilage microorganisms as well as chemicals that may produce objectionable odors and flavors in food. The organisms that tend to be present in municipal water supplies are psychrotrophs such as pseudomonads which can grow in water with a limited nutrient content. Hence, when a food processor stores water in small tanks, the growth of the pseudomonads may result in microbial levels of 10^5 to 10^6 per milliliter.

In the dairy industry, these psychrotrophs are especially troublesome. The wash water may contain up to 10^3 psychrotrophs per milliliter and be one of the most important sources of microbial contamination. Psychrotrophs cause spoilage of refrigerated milk and milk products.

In poultry-processing plants, water is used in the scald tank for washing of the carcass and for cleanup of the building and equipment. There has been much speculation that water from the scald tank is a source of contamination of the carcass. This is due to the fact that the dirt from the head, feet, and feathers, as well as feces (due to defecation at death) contaminate the scald water. The temperature of the water (53° to 61°C) can kill many pathogenic and spoilage types but is not sufficient to kill

all of the microorganisms that are present, and thermophilic types can grow. Bacterial counts of the scald water range from 10^2 to 10^6 per milliliter. When the birds enter the scald tank, they may be taking their last breath and the heart may be pumping. The contaminated water then can contaminate the internal portions of the bird through the lungs and vascular system. The addition of NaOH to raise the scald water pH to 9.0 effectively reduced the level of bacterial population (Humphrey, Lanning, and Beresford 1981).

Ice water used to cool the carcasses after processing can be a source of microorganisms that can penetrate deep into the skin surface (Thomas and McMeekin 1982).

In food-canning operations, water is used to cool the cans of food after heat processing. Due to the heat and expansion of metal, the seams and seals on the cans are under stress, and leakage can occur. Hence, it is important that the cooling water be maintained with as few microorganisms as possible. Methods of monitoring this water for microorganisms have been described (Rey et al. 1982). Thompson and Griffith (1983) found clostridial spores at low levels in about 10 percent of their samples of cooling water. Since no further processing occurs after cooling, any contamination of the food by microorganisms that can grow in the canned product may cause spoilage or, if pathogenic, foodborne illness.

Water is used as an ingredient in many foods. In these cases, the water is a direct source of microbial contamination.

With the present shortages and high cost of water, there is an effort to reduce water consumption and to recycle water in the food-processing plant. Caution must be used so that contamination or high microbial loads on foods do not result from these pratices.

AIR

Today, the main concern with air pollution is with chemicals (carbon monoxide, hydrocarbons, soot, fly ash) rather than with biological agents. Nature is the major contributor of not only the chemical pollutants but of biological agents such as plant cells, animal hair, pollen, algae, protozoa, bacteria, yeasts, mold spores, and viruses. Food is subjected to airborne contamination until it is sealed in a package.

Types and Numbers of Microorganisms

There is no natural or normal microflora of air, which is contaminated from various sources. Generally, mold spores are more prevalent than are other microorganisms. The main source of microorganisms, es-

pecially mold spores, is decaying plant materials near the ground surface. Contamination of the air is caused by gusts of wind picking up the microorganisms or spores. Compared to mold spores, there are relatively few yeast cells in the atmosphere, and these are mainly near ground level. One would expect yeasts to be prominent in the air of orchards and vineyards. However, even here, they may comprise only 25 percent of the fungal population of air.

There are many sources of microorganisms for contamination of the air. Liquids are aerosolized by spraying, splashing, the bursting of bubbles, by forcing through a small orifice, or by vibrations. Trickling filters used in sewage-treatment plants produce aerosols by the spraying action and by the splashing of liquid sewage. The application of wastewater to land by spraying under pressure causes aerosolization of organisms. Teltsch et al. (1980) devised equations to predict aerosolized pathogenic microorganisms downwind from a wastewater spray or aeration site.

The types of organisms in air are often associated with the type of activity in the area. Downwind from a sewage-treatment plant, Pereira and Benjaminson (1975) found species of *Klebsiella, Bacillus, Flavobacterium, Streptococcus,* and *Micrococcus.* Streptocci are found near dairies, and yeasts are found near bakeries and breweries. The microbial flora of the air in a food-processing plant reflects the sanitary condition of the plant unless the air is purified. Incinerators, if not operated properly, can be a source of aerosols that may contain infectious microorganisms.

Human beings shed organisms and also produce aerosols during talking, coughing, and sneezing. The microbial load of the air is proportional to the numbers of persons in an enclosed space, their activity, and the rate of air circulation. Although humans might not contribute significantly to the microorganisms in outdoor air, they can be a significant source for air in a processing room.

Gerba, Wallis, and Melnick (1975) reported that bacteria and viruses become airborne when a toilet is flushed. These organisms settle out on surfaces throughout the bathroom.

Thus, air is subjected to a number of sources of microorganisms (decaying matter, water, soil, human beings, animals, sewage). The numbers and types of organisms present in the air or atmosphere depend upon many factors, such as tendency to settle out and maintenance of viability.

There is a considerable variation of the microbial load of air. It has been suggested that country air in the summer contains an average of 10,000 fungal spores/m^3. Lacey (1973) suggested the level may reach $10^8/m^3$ during haymaking or harvesting. The handling of moldy grain or hay indoors can result in a level of 10^9 spores/m^3. One study found more bacteria over an alfalfa field than over bare soil (Lindeman et al. 1982).

At 46 m downwind from a wastewater spray irrigation site there were

10^4 bacteria/m^3 (Bausum et al. 1982). Clark, Rylander, and Larsson (1983) reported a level of 10^5 bacteria/m^3 in both poultry- and swine-confinement buildings. Lenhart, Olenchock, and Cole (1982) stated that the atmosphere in a poultry-processing plant contained bacteria that might be a potential health hazard to the workers.

Lacey (1973) listed several genera of fungi as being present in rural air, with *Cladosporium* predominating. This dominance was also reported by Ogunlana (1975). Other researchers have reported that species of yeasts, *Aspergillus, Penicillium, Cladosporium, Alternaria, Helminthosporium,* and *Fusarium* are the most dominant fungi in corn dust (Hill et al. 1984).

The air near the earth is more contaminated than at higher altitudes, although the microbial population at 2,000 to 3,000 meters is more stable than that at lower altitudes. Air over land is more contaminated than air over the ocean, although wind currents can carry micoorganisms for several hundred miles out to sea. The upper atmosphere over the ocean is more contaminated than air near the ocean surface.

Clouds often contain high levels of bacteria and fungal spores. Turbulent air tends to carry high numbers of cells aloft. Temperature inversions can affect the microbial load at various altitudes. Rain and snow tend to wash the microflora from the air. The atmosphere is more contaminated in the summer than in the winter.

Survival

Although Dimmick, Wolochow, and Chatigny (1979) reported that, with special conditions of growth and aerosolization, bacterial cells in air can divide, this has not yet been shown for naturally occurring microorganisms in air.

The stability of microorganisms in air is affected by the relative humidity, temperature, oxygen, solar factors including ultraviolet radiation, and chemical components. Spores of molds and bacteria retain their viability better than do vegetative cells. Also, capsules help protect cells in the atmosphere.

Aerosols of microorganisms have been produced in the laboratory and the viability determined. Aerosolization damages the cytoplasmic membrane and at least some of the associated transport mechanism (Benbough et al. 1972).

The stability of bacteria in aerosols is similar to that of viruses. There is rapid death during aerosolization, followed by a less rapid death rate. The method of recovery and enumeration influence the apparent effect of relative humidity and oxygen on survival of bacteria in aerosols.

When air is sampled downwind from a source of contamination, there is a significant decrease in the microbial load as the distance from

the source increases. This is because the organisms settle out onto various surfaces, the organisms lose viability, and the air is contaminated by less contaminated air. However, bacterial spores were found to travel in air from near the Black Sea to Sweden (Bovallius et al. 1978).

Sunlight reduces the number of viable organisms in air. By analysis of air near a spray irrigation system, Teltsch and Katzenelson (1978) found higher counts at night than during the day.

Food-Processing Operations

The contamination of air in food-processing plants was reviewed by Heldman (1974). The effect that air has on the microflora of food depends upon the level of contamination of air and the time of contact of air with food.

Aerosols are produced in food-processing plants by spray washing or spray cooling of food, by high-pressure sprays used in cleaning, by flooding of floor drains, by mixers and motors, and by the operation of various other equipment. Workers in the area produce aerosols. The movement of equipment, supplies, and people in a food plant causes turbulent wind currents that increase the microbial load of the air.

There is considerable variation in the microbial load of air in various areas of a processing plant. In clean areas there are very few organisms in the air. In areas in which live animals are handled or raw products are brought into the processing operation, the microbial load can be quite high.

One method that is used to control the microbial load in the air of a processing plant is to move air from clean areas to dirty areas or by using positive air pressure in clean areas. With positive air pressure, if a door is opened, air flows out of a room and outside air does not come in. Fresh air entering the clean areas is filtered to remove dirt as well as some microorganisms.

ANIMAL AND PLANT FOOD

Animal and plant food can serve as a source of potential chemical and biological contamination of food.

Microorganisms in animal feed can contaminate the feet, hide, hair, and feathers of animals. Consumption of the feed adds organisms to the digestive tract. Those organisms that survive the rigors of digestion, upon elimination, can contaminate exterior portions of the animal. When the feed contains potential pathogens such as salmonellae, these may cause illness in the animal, invade the body and locate in lymph nodes, contam-

inate the carcass during slaughter, as well as contaminate external portions of the animal (MacKenzie and Bains 1976).

Plant food may contain organisms that can contaminate the surfaces of plants and associated human foods. When animal or human wastes are used as fertilizer, pathogens from these sources can contaminate the plant and associated foods. Since fruits and vegetables are eaten raw, they can be a source of foodborne illness. In countries that use night soil as a fertilizer for food crops, washing of the food is suggested. The decomposition of plant tissue can cause an increase of potential toxin-forming molds in fields (Griffin and Garren 1976).

Although pathogens may be present in manure or sewage used to fertilize crops, sunlight tends to eliminate microorganisms from the plant parts (Bell, Wilson, and Dew 1976; Jones 1976).

PLANTS

Plants are contaminated by microorganisms from several sources (dirt, water, air, fertilizer, animals, and humans). Once contaminated, certain microorganisms can grow on the plant surfaces and plant pathogens can attack their host plants. The microbial flora on plant surfaces varies with the kind of plant.

One study found a total count of 10^9/g including 10^7 coliforms, 10^8 psychrotrophs, 10^7 streptococci, 10^5 yeasts, and 10^6 molds per gram on wild rice (Goel et al. 1970).

Pseudomonas species are quite prevalent on vegetables. Kominos and colleagues (1972) isolated *P. aeruginosa* from tomatoes, radishes, celery, carrots, endive, cabbage, cucumbers, onions, and lettuce. They believe that raw vegetables are a source of this organism contaminating patients in hospitals. Various bacteria, including pseudomonads, can adhere to and colonize the surface of leaves (Ercolani 1978; Hildebrand, Alosi, and Schroth 1980; Leben and Whitmoyer 1979).

The flowers of fruits are inhabited by many genera of yeasts, including the ascospore formers *Saccharomyces* and *Hansenula,* as well as the imperfect *Torulopsis, Candida, Rhodotorula,* and *Kloeckera.* In some fruits such as grapes, the yeasts might not be present on the flowers but are found on the ripe fruit.

A *Pseudomonas* and an *Arthrobacter* introduced into open flowers of soybean plants were isolated from 24 of 177 resultant beans pods (Leben 1976).

There is evidence that not only the surface but also the interior tissue of plants can be contaminated. Vegetables can harbor fecal streptococci within unopened pods, heads, and other structures. Meneley and Stanghellini (1974) found that 44 percent of apparently healthy cucumbers

contained bacteria internally. Enterobacteria (*Proteus, Citrobacter,* and *Enterobacter*) were present in 10 percent of the cucumbers.

Thus, live plants can serve as a source of microorganisms. When the plant dies, the decaying vegetation becomes an important source of airborne, waterborne, and soil microorganisms, which then contaminate plants the following year.

ANIMALS

Animals have a normal or natural microflora that is established very early in life. Besides this microflora, animals tend to harbor the types of organisms found in their environment, since they are contaminated by soil, water, air, feed, and excreta.

Animals harbor organisms that can cause food spoilage or foodborne disease. Organisms are found in animals' gastrointestinal tract, nasal passages, cutaneous lesions, and on the skin, feet, hair, or feathers on the outer surface. These organisms are readily transferred to the edible portions of the animal during processing.

Wild animals contaminate growing crops and stored products. One of the means by which viruses are transmitted to plants is by insects (Coudriet, Kishaba, and Carroll 1979). Insects, birds, and rodents destroy the protective covering on foods, so that not only do they contaminate the food, they also make the food more susceptible to microorganisms that cause spoilage. These animals also carry potential human pathogens that may be transferred to the foods they contaminate (Kapperud 1975; Stek 1982).

Flies, much more so than many other types of common food plant insects, are a serious potential carrier of disease-producing organisms. Flies pick up organisms on their hairy legs and feet. They then contaminate human food by walking or leaving their excreta on it. Flies have a part in the spreading of *Salmonella, Shigella, Vibrio, Escherichia,* and other disease-causing organisms as well as food-spoilage types.

The muscle tissue of most live, healthy animals is essentially sterile. In a diseased state or after slaughter (or after harvesting in the case of fish), bacterial invasion of the muscle tissue can occur. Most of the microorganisms in tissue are thought to result from surface or intestinal contamination or from processing operations. Vanderzant and Nickelson (1969) removed muscles from slaughtered animals aseptically. The maximum number found in any sample was 1,400 per gram in beef. Of sixty-five samples analyzed, forty-four did not yield isolates when incubated at 37°C. Hence, it is evident that even after slaughter, a majority of muscle tissues are sterile.

One of the sources of microorganisms in the interior portion of ani-

mals is the lymph nodes. These nodes tend to filter out bacteria from the lymphatic system. Microbial counts of 10^5/g have been obtained from the lymph nodes, with many types of organisms being present. Both cattle (61 percent) and sheep (14 percent) at slaughter contained *Salmonella* in lymph nodes (Samuel et al. 1981). Of forty-two hog carcasses passed by federal meat inspectors, Brown and Neuman (1979) found mycobacterial infection of the lymph nodes.

The microbial flora that grows on refrigerated beef is composed mainly of soil organisms derived from the hide, hair, and hooves of the animals. The hides of animals may contain from 10^3 to over 10^9 microorganisms per square centimeter. Although they do not penetrate the intact skin, these microorganisms are a source of contamination of the edible portions of the carcass during processing.

The predominant organisms in the intestinal flora of both animals and humans are obligate anaerobes such as species of *Bacteroides* and *Peptostreptococcus*. They may reach levels of 10^{10} to 10^{11} per gram. The facultative anaerobes may reach levels of 10^7 to 10^9 per gram of intestinal contents. These include coliforms, enterococci, and lactobacilli. The natural habitat of the human pathogen *Salmonella* is the intestinal tract of animals as well as of humans. Other organisms, such as fungi and viruses, can be found in the intestinal tract. There is a possibility that microorganisms can pass through the intestinal wall into other organs and tissues. Viruses were found in chicken tissues that appeared normal (Robertson, Wilcox, and Kibenge 1984).

During life, the intestinal wastes can contaminate the outer surface of the animal. During slaughter and processing, the sphincter muscle becomes nonfunctional, and also the intestines can be punctured accidentally. In either case, the contents and associated organisms can contaminate the carcass.

The circulatory system is sterile except for the occasional invader or during an infection. Although a majority of samples of hemolymph from the horseshoe crab contained no detectable microorganisms, 27 percent did reveal low numbers of bacteria (Brandin and Pistole 1985).

The respiratory system acquires microorganisms from the air during breathing. The nose filters out many microorganisms, and it tends to be the most contaminated part of the system. The respiratory system may serve as a passageway for microorganisms to contaminate internal tissues.

The surface of fish may contain 10^2 to 10^5 bacteria per square centimeter and the intestinal contents vary from 10^4 to 10^7 organisms per gram. These microorganisms reflect the condition of the environment from which the fish are obtained (Austin 1982). This is especially evident for shellfish (Ellender et al. 1980; Larkin and Hunt 1982; Metcalf et al. 1980; Perkins et al. 1980).

Milk in a healthy udder has few microorganisms, if any. However, aseptically drawn milk usually ranges from less than 500 to 5,000 and occasionally, up to 10,000 organisms per milliliter (Thomas, Druce, and Jones 1971). The flora is predominantly coagulase-negative staphylococci, micrococci, and corynebacteria. Infection of the bovine mammary glands is called mastitis. With this condition, counts of 10^5 or more per milliliter of milk have been found. A variety of organisms can cause mastitis, and organisms such as coagulase-positive staphylococci, coliforms, and streptococci, will dominate in the milk. Aseptically drawn milk contains mesophiles, but few psychrotrophs or thermophiles. When milk is obtained normally, it is contaminated by bacteria in the teat canal and on the surface of the teat. The surfaces of the four teats, even after washing with disinfectant and drying, had a mean colony count of 10^6 bacteria (Thomas, Druce, and Jones 1971). Perhaps this is due to the strong attachment of bacteria to the teat surfaces (Notermans, Firstenberg-Eden, and Van Schothorst 1979). Hibbitt, Benians, and Rowlands (1972) found an average of 22,500 viable microorganisms per teat canal. McKinnon, Cousins, and Fulford (1973) analyzed milk obtained normally from twenty-five cows and found bacterial counts from 44 to 11,400 per milliliter, which is similar to the so-called aseptically drawn milk referred to by Thomas, Druce, and Jones (1971). Due to microorganisms normally present in the teat canal, the first drawn milk contains more organisms than the last portion from any particular quarter of the udder.

When an egg is laid, it is essentially sterile. Some bacteria (salmonellae) and avian viruses may invade the ovaries of hens and contaminate the yolk before the egg is formed. Even for those few eggs that might be contaminated, the bacterial number is relatively low. Most of the contaminants on the egg shells are of intestinal origin. Other sources of organisms are nesting material, feather dust, the feet and body of the bird, as well as subsequent handling and storage. Eggs obtained from poultry flocks on wire have less bacterial contamination than do eggs from flocks in houses with floor litter. This is because nesting material, including feces, has less chance of coming into contact with the shell.

HUMANS

A human embryo develops in a relatively sterile environment. At birth the baby is subjected to a massive invasion of microorganisms from the mother, other people, bedding, air, food, water, and other materials.

The skin is the most available part of the body for colonization after birth. Staphylococci are predominant on normal infant skin (Carr and Kloos 1977). The colonization is followed by microbial invasion of the nose, oral cavity, throat, and the respiratory, digestive, and urogenital

tracts. Many microorganisms are merely transients, while others become established as a normal, perhaps permanent, flora. Besides the microbial population on or in a human, clothing can be contaminated by external sources or by the person, and then serve to pass the microorganisms on to the person or to food products.

The skin is never free of bacteria, and the dirtier the skin, the greater the contamination. Normal human skin contains a relatively stable microflora. However, the numbers and types of microorganisms vary from person to person, from site to site, and from day to day. There are differences in the environment at various places on the body. Some areas are too dry and, in other areas, the pH is not satisfactory for bacterial growth. The palm of the hand has primarily surface organisms, whereas the forehead has both surface and subsurface flora, but mainly subsurface. Surface organisms consist of transient and normal resident types, while subsurface organisms are primarily normal resident types.

Washing the skin removes most of the transient microorganisms, but it is practically impossible to remove all of the normal flora. Some microorganisms associate with the sweat glands, sebaceous glands, and hair follicles, where they not only can grow but also are difficult to remove. The predominant bacteria on the skin are staphylococci, corynebacteria, and propionibacteria. Also, species of *Micrococcus, Bacillus, Alcaligenes, Pseudomonas, Enterobacter, Klebsiella, Proteus, Escherichia,* and *Citrobacter* are quite prevalent (Adams and Marrie 1982; Sunga, Heldman, and Hedrick 1970). *S. aureus* is found more often on the hands and face than on other parts of the body. The organism is associated with the nose and is spread when people handle their faces and noses. *S. aureus* is associated with infections such as acne, pimples, and boils.

Humans are a source of airborne microorganisms as well as an important source of food contamination through handling of food. The hands are subject to contamination by considerable numbers of bacteria, most of which are unable to multiply on the hands and usually die. However, these transient bacteria are passed on to food products when they are handled. Many of the transient bacteria result from the handling of foods (DeWit and Kampelmacher 1982; Seligmann and Rosenbluth 1975). Workers in meat- and chicken-processing plants had rather high levels of *E. coli,* and several had salmonellae on their hands.

The hair covering the skin is a potential source of microorganisms. There is no normal flora of the hair, but hair acts as a carrier to retain and shed organisms. Beards, sideburns, and mustaches may be especially bad if worn in a food-handling area. Barbeito, Mathews, and Taylor (1967) found that, although washing with soap and water reduced the level of contamination of beards, it did not eliminate microorganisms and toxins from them. If a person grows a beard to hide acne or pimples,

or a nasal carrier has a mustache, these hairy growths will merely serve for continued dissemination of *S. aureus*. Besides the microorganisms that may be shed into foods, the hairs and hair clippings may contaminate the food.

Clothing is directly associated with humans and can serve as a carrier to contaminate foods with human microflora or can contaminate people with environmental microorganisms. Laundering does not necessarily remove all of the microorganisms from clothing. The effectiveness of laundering depends upon the type of organism, fabric, temperature, detergent, antibacterial agents, bleaches, flushes and rinses, drying, ironing, and final packaging. Viruses tend to adsorb to the fabric and are difficult to remove. Microorganisms tend to remain viable in wool longer than in cotton. Synthetic fabrics require less drastic laundering, allowing greater microbial survival. Microorganisms tend to survive storage for a shorter time on these fabrics than on either cotton or wool.

During laundering, microorganisms are transferred among the clothing in the same wash load, and it is conceivable that microorganisms can carry over from one load to the next. Even dry cleaning does not assure the sterility of clothes.

The gastrointestinal tract may be considered a hollow tube within the body, but the internal portion is outside the body. The gastrointestinal tract is important as a potential source of organisms, as well as the primary site of action of foodborne organisms causing illness (gastroenteritis). Factors that can influence the flora of the gastrointestinal tract include the diet, antibiotics or other chemotherapeutic agents, and certain diseases.

Although enormous numbers of microorganisms are ingested with food and with mucus from the respiratory tract, most do not survive in the digestive system.

In the normal stomach, the microbial flora increases and decreases. The ingestion of food or mucus causes a rise in numbers. The flora is reduced by acid and digestive enzymes and by emptying of the stomach into the duodenum. There are usually low numbers of organisms in the normal stomach, duodenum, jejunum, and upper ileum. The pH of the stomach varies among individuals. Higher pH values will allow the survival of more organisms. Foods tend to react with stomach acids and raise the pH. Also, some microbial cells are imbedded in food particles, which protects them from the acid.

Normal peristalsis, bile, and immunoglobulins tend to remove organisms from the upper small intestine. However, drugs such as morphine slow peristalsis; this and several other conditions can result in colonization (Gracey 1979). Some organisms are able to adhere to the epithelial cells of the small intestine by at least two systems (Kleeman and Klaen-

hammer 1982). Most of the organisms that cause foodborne illness must attach to the epithelium and become established in the intestine, where they produce toxins (see Chapter 6).

As the cells move down the intestines, the conditions improve for microbial growth. Active multiplication begins in the lower small intestine and continues through the large intestine. Bacterial levels of more than 10^{10}/g of contents are attained in the colon and excreta. The redox potential varies from -150 mv in the duodenum to -250 mv in the colon (Holdeman and Moore 1972). This low level in the colon favors anaerobic types of organisms.

The fecal flora of infants is composed primarily of species of *Bifidobacterium.* Other organisms include the anaerobic *Bacteroides* as well as coliforms, enterococci, lactobacilli, and staphylococci (Mitsuoka and Keneuchi 1977). In the adult, *Bacteroides* surpasses the anaerobic bifidobacteria. These two groups range from 10^9 to 10^{10} cells per gram of feces. The eubacteria and anaerobic lactobacilli are also at levels of 10^9 per gram of feces. The remaining organisms consist of enterobacteria, enterococci, lactobacilli, clostridia, bacilli, veillonellae, pseudomonads, yeasts, molds, and viruses (Haenel 1970). The relationship of viruses to the fecal microflora has not been examined thoroughly. Enteric viruses are present in the intestines and feces of apparently healthy people.

Speck, Calloway, and Hadley (1970) tested the effect of diets on the fecal flora. Their results indicated that a change in diet influences the fecal microflora. The type of diet affected the aerobic population more than the anaerobic microflora. A protein diet supported a higher aerobic population than did either a carbohydrate or fat diet. Haenel (1970) found no dramatic differences in the intestinal microbial flora with meat-egg, milk-vegetable, vegetarian, or raw vegetarian diets. Even the consumption of 1 kg of Bulgarian yogurt per day failed to dramatically alter the microbial flora in most individuals. Conn and Floch (1970) fed eight to twenty capsules, each containing five (10^{10}) viable *Lactobacillus acidophilus* for seven to ten days. They found no significant increase in fecal lactobacilli, but they did notice a decrease in total anaerobic bacteria. Similar results were reported by Paul and Hoskins (1972). A comparison of Seventh Day Adventists revealed few statistically significant differences in the fecal flora of vegetarians and nonvegetarians (Finegold et al. 1977). Although some reports indicate that diet does influence the fecal flora, in a review, Hentges (1980) stated that there are minor and inconsistent changes in fecal flora composition in response to dietary alterations.

People with acute diarrhea may show the following features of intestinal dysfunction: (1) bacterial colonization of the small intestine with defective microbial clearing systems; (2) malabsorption of fat, carbohydrate, and vitamin B_{12}; (3) fluid accumulation due to alterations in electrolytes and water transport; and (4) alteration of the mucosal morphology. Those

abnormalities are not necessarily manifested in all persons with acute diarrhea or in all cases of diarrhea.

Mendes and Lynch (1976) found fecal organisms on various surfaces in restrooms (door handles, flush handles, tap handles, toilet seat, floor). The tap handles were more contaminated than the door handles. The wash basin overflow was more contaminated than the floor and the area under the rim of the toilet. They believed that their findings showed that a person with *Salmonella* or *Shigella* infection could contaminate various surfaces in a washroom or toilet. These areas could conceivably serve as a source of contamination for other people.

Since people are carriers of pathogenic types of microorganisms, they are especially hazardous when handling processed (cooked or pasteurized) products that may be held for short periods and eaten with no further treatment.

The hands of workers have been cited as increasing the microbial load of many products. Poultry products show increased counts when the carcass is hand transferred from the picking to the eviscerating line, during eviscerating, cutting up the carcass, and deboning by hand. In the shellfish industry, the hand shucking of oysters and clams, shelling of shrimp, and picking of crabmeat from the shell are associated with increased bacterial counts. Slicing, weighing, and hand packaging of meat products increase the contamination through human handling of the product. The unpacking, trimming, sorting, and repackaging of fresh produce (fruit and vegetables) provide an opportunity for contamination of the food with human pathogens. Fresh produce may be eaten raw, which makes contamination a potential hazard.

The carelessness of humans is an important cause of microbial contamination of foods. Failure to properly clean and sanitize equipment, failure to wash one's hands, working and handling food with an infection, poor personal hygiene, lack of care in handling of food, making unsatisfactory alterations of equipment, and failure to keep foods at the proper temperature are some of the careless things that people do that can increase the microfloral load of foods.

Some shoppers have been seen opening containers of foods. This is a deplorable practice that can cause contamination of the food with resultant spoilage or a health hazard.

During the analysis of foods, microbiologists can be a source of contamination (Denny 1972).

SEWAGE

Animal manure or, in some cases, human wastes are used as a fertilizer on crops. These biologically produced substances contain microorgan-

isms, including human pathogens. When added to soil, these pathogens may survive for periods sufficient to contaminate the harvested crop.

In rural areas, septic tanks are used. Quite often these are not properly installed and do not operate effectively, and raw sewage leaks into the soil. Salmonellae are quite prevalent in raw sewage. Processing in sewage-treatment plants lowers the incidence, but even at 99 percent reduction, if the original level of salmonellae is high, they remain in the effluent and sewage sludge (Farrah and Bitton 1984; Langeland 1982; Nabbut, Barbour, and Al-Nakhli 1982; Yaziz and Lloyd 1982).

Enteric viruses survive the treatment process, may remain infective when bound to solids, and can contaminate soil and groundwater when sewage sludge is applied to agricultural land (Goyal, Keswick, and Gerba 1984; Larkin et al. 1978; Ward and Ashley 1980).

EQUIPMENT

During the Industrial Revolution, machines were developed to do most of the work of humans. Hence, there is less contact with food by people and more contact by machines and equipment.

Thousands of small pieces of equipment, such as knives, cutting boards, and bins, are used in food-processing or -handling establishments. Although most equipment is metal, some parts may be made of rubber or plastic. In some cases, cardboard is used in boxes used to hold harvested fruits or vegetables. Cardboard or paper is used as packaging material for bulk shipment and storage of various ingredients.

Metal processing equipment does not support the growth of microorganisms. It has no natural or normal microbial flora. Yet, food-processing equipment is one of the major sources of contamination of foods.

Equipment may be cleaned and sanitized, but this does not mean that it is sterile. Even on washed, visibly clean surfaces, the survival and growth of bacteria are possible. Even visibly clean surfaces may have food deposits or films that provide microenvironments acceptable for survival and growth of microorganisms. The potential for microbial buildup on equipment is enhanced when equipment is improperly cleaned and sanitized. If food residues are visible on cleaned equipment, one can be assured of a potentially high microbial level.

Pitted surfaces or poorly soldered joints are places in which foods can lodge on the equipment. During the course of the daily operation, bacterial growth will occur in these food films and deposits. This then serves as a source of contamination when food contacts these surfaces.

It is well established that microorganisms adhere to surfaces. *Streptococcus thermophilus* adheres to stainless steel during the pasteurization

process (Driessen, DeVries, and Kingma 1984). However, it is easier to remove soil and bacteria from stainless steel than from rubber or plastics. Although stainless steel is more expensive than some other materials, it pays for itself through savings in cleaning, as well as durability and product quality. When flexibility is needed, such as in milking machines or in chicken-picking machines, rubber or plastic becomes an essential part of the equipment.

Gilbert and Maurer (1968) surveyed equipment in shops and cafes. Swabbings from a meat-slicing machine revealed bacterial loads of 10^6 to $10^7/cm^2$. The cleaning methods used were ineffective in reducing the bacterial loads to an acceptable level. When we consider that most outbreaks of foodborne illness are due to contamination of the food in foodservice units or in the home, it should be obvious that the equipment in the home and foodservice institutions needs more attention.

Meat

The contamination of meat by equipment begins with slaughtering the animal. Contaminants from the knife used to cut the artery may circulate through the body during the bleeding process. Humane slaughter by immobilizing the animal with CO_2 or electric stunning before bleeding helps reduce the heart action and hence, contamination.

During the slaughtering operation, the carcass contacts various equipment surfaces. Various pieces of equipment and humans can cause the spread of *Salmonella* throughout the slaughter operation (Smeltzer, Thomas, and Collins 1980a, b; Stolle 1981). When the carcass is cut up and packaged for retail, further transfer of organisms is made from the carcass surface by means of saws, knives, slicers, and cutting blocks. These increased surfaces with their associated juices support large populations of bacteria.

The grinding of meat further increases the surface area, releases juices, and distributes the original surface bacteria throughout the meat. Equipment may be involved in the distribution and increased counts of the product. Meat grinders are highly contaminated with millions of bacteria (Dempster 1973).

Poultry

During processing there may be a tenfold increase in the microbial count on the skin of poultry carcasses. Some of the increase is due to contaminated scald water and to human factors. The equipment used in processing the birds plays a role in contaminating the carcasses. At every stage of the processing operation (Fig. 5.1) there is ample opportunity

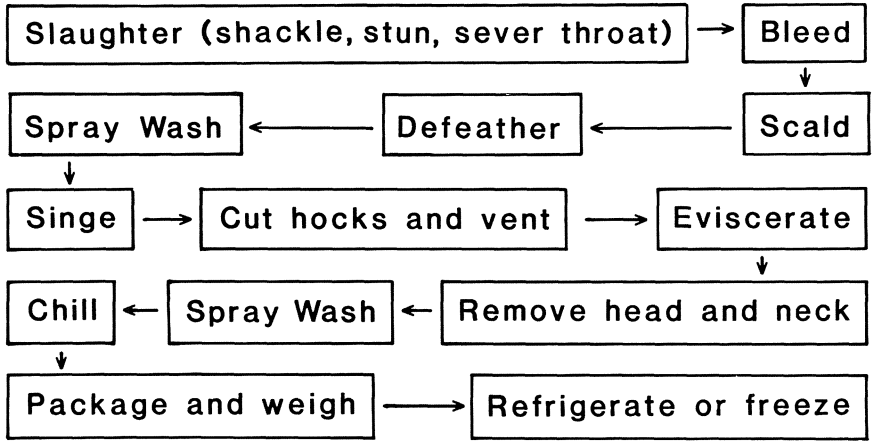

Figure 5.1. Poultry-processing operation.

for microbial contamination (McMeekin and Thomas 1979; Thomas and McMeekin 1980).

Schuler and Badenhop (1972) sampled seventeen equipment sites on two sampling days in thirteen poultry-processing plants. On only five of the pieces of equipment was the average bacterial count less than 1,000/cm^2 on both sampling days prior to start of the operation. Sites of contamination were the flexible fingers on the picking machine (11,000 to 99,000/cm^2), the hock cutter (7.000 to 97,000/cm^2), the transfer belt between dressing and eviscerating lines (5,000 to 56,000/cm^2), lung gun (3,500 to 73,000/cm^2), and cut-up belt (2,000 to 35,000/cm^2).

The picker fingers are difficult to clean properly, and the contamination of these fingers increases significantly during plant operation. There is increased contamination of poultry carcasses during feather removal, which can be due to leakage from the vent and the bacteria from the feathers contaminating the skin via the flexible rubber fingers.

Further processing (cutting up the carcass and deboning) can increase the microbial level of poultry meat (Denton and Gardner 1981). Sources of organisms include both equipment and humans.

Milk

Since milk is a liquid, it is in contact with some type of equipment or surface from the time it is removed from the cow until it is packaged. Unsatisfactorily washed and sanitized milking and milk-handling equipment constitute the main source of bacteria in milk at the farm level.

The bacteria in farm milk tanks include many psychrotrophs and few thermodurics (MacKenzie 1973; Thomas, Druce, and Jones 1971). Thermodurics are more prevalent than psychrotrophs in pipeline milking plants (MacKenzie 1973).

Psychrotrophs can grow and cause spoilage of refrigerated milk, but few thermodurics grow below 10°C. Pasteurization of milk destroys most of the psychrotrophs. Contamination of pasteurized milk by psychrotrophs from poorly cleaned pipes and filling machines can result in spoilage of the refrigerated milk.

Seafood

When fish are caught they are subjected to contamination from fish boxes (plastic, wood, or metal) and other fish-holding systems on the ship. However, there are few, if any, bacteria in the flesh of fish before filleting and subsequent handling.

Filleting may be done manually or by filleting and skinning machines. When fish are filleted by hand, the knives and cutting boards provide a source of microorganisms, and filleting machines become contaminated by the surface microflora on the fish, which is transferred to the flesh of the fillet. The bacterial count on the filleting equipment varies from 10^3 to 10^8 bacteria/cm^2. When the bacterial load on the surface of the fish is high, there is a higher bacterial count on the filleting boards. Gloves worn by the filleters often have a higher bacterial load than the incoming fish.

Fruits and Vegetables

The microbial load of fruits and vegetables increases during harvesting and processing. Peas aseptically removed from the pods are essentially sterile. When the peas are separated by a viner, the bacterial count may reach 10^6/g. The microbial load of most vegetables is reduced by washing and by blanching. After blanching, the bacterial load increases due to contamination by equipment and may attain levels of 10^4 to 10^5/g. The equipment causing contamination includes conveyor belts, hoppers, and filling machines.

Splittstoesser (1973) stated that the source of most organisms on frozen vegetables is contaminated equipment surfaces. His data show that corn cutters, French bean slicers, spinach choppers, and inspection belts for peas significantly increased the microbial level of these vegetables. In one instance, he found that moving peas to the freezer by an airlift (to

conserve water) resulted in significantly more bacterial contamination of the peas than when they were flumed.

INGREDIENTS

The quality of a processed food is influenced by the quality of the ingredients. Although ingredients may constitute a small part of the total food, they may add a substantial number of microorganisms. Hence, specifications, including acceptable microbiological levels, are needed for the purchase or production of ingredients.

Spices or seasonings are often the source of high microbial numbers. Spices are parts of plants such as dried seeds, buds, fruit or flower parts, bark, or roots, usually of tropical origin. These seasonings and flavors may contain over 10^8 aerobic bacteria per gram. Also, they contain aerobic and anaerobic spores. A sample of ginger contained 48 million total bacteria, 12 million yeasts and molds, 26 million aerobic sporeformers, and 0.72 million anaerobic sporeformers (Tjaberg, Underdal, and Lunde 1972). Pepper is usually highly contaminated with bacteria and fungi. Flannigan and Hui (1976) reported that fourteen of twenty spices contained *Aspergillus flavus* and four of these spices supported growth of this mold and the production of aflatoxin. Mean standard plate counts of over 10^6 per gram were obtained for black pepper, ginger, and paprika (Julseth and Deibel 1974). These spices also contained over 10^6 bacterial spores per gram. Psychrotrophic spores were absent in herbs and spices analyzed by Michels and Visser (1976). A nationwide survey of ten spices and herbs revealed less contamination at retail than had been reported for these ingredients at import (Schwab et al. 1982). Some samples of all ten spices and herbs had an aerobic plate count over 10^6/g. However, many samples had less than 100/g. Thus, the geometric means of individual spices and herbs ranged from 1,400 for cloves to 820,000 for pepper.

Thermophilic bacteria, usually as spores, are added to foods with ingredients such as spices, starch, flour, and sugar. Thermophilic spores are important when added to canned foods. The higher the level of heat-resistant spores, the greater the chance that some may survive the heat treatment and be a potential spoilage problem.

The total aerobic count of flour sampled at the mills is generally in the range of 10^2 to 10^3/g (Graves et al. 1967). They reported levels of total aerobic thermophilic spores from 0 to 1,200 per 10 g of flour, and although thermophilic flat-sour spores were not present in 10 g of most flour samples, one 10-g sample contained 730 flat-sour spores.

Since sugar is a potential source of spores, the National Canner's As-

sociation has set limits of aerobic thermophilic flat-sour and anaerobic thermophilic spores when this ingredient is to be added to canned foods.

Eskin, Henderson, and Townsend (1971) suggested that infections of confectionery items by osmophilic yeasts are due to contaminated nuts, fruits, chocolate, flour, or flavoring syrup. Bakery products can be contaminated by the ingredients in the doughs or by adding materials to the baked food. Icing and nuts are added to already baked sweet rolls. The toasting of almonds and pecans is not sufficient to assure sterility. These products may contain 10^4 to 10^5 bacteria and 10^4 yeasts and molds per gram.

Batters used in the breading of seafoods, onion rings, or other products are a potential source of high levels of microorganisms. Batters used in the manufacture of breaded onion rings had a total bacterial count of 2,500 to 8,500,000/g (Maxcy and Arnold 1972). There were 110 to 32,000 coliforms per gram of the batters they examined.

Since ingredients are an important source of microorganisms, food processors must establish acceptable microbiological limits for these substances. Then sufficient testing is needed to assure the processor that the ingredients are acceptable and, if they are used in the manufacture of a food, adjustments of the process are needed to take care of any excessive bacterial load.

PRODUCT TO PRODUCT

There is an old saying that one bad apple can spoil the barrel of apples. This is very true, because the spoilage organisms from the bad apple can contaminate and cause spoilage of surrounding apples until all of the apples are spoiled.

In this example, the transfer of the organisms can be disastrous because they are known to cause spoilage of apples. Although the transfer of microorganisms from one food to another can occur directly if the foods contact each other, more often it occurs through contamination of wash water, equipment, or handling.

Researchers have reported that the shelf life of vacuum-packed sliced cooked meats would be improved if the meats were handled and sliced in an area strictly separated from raw cured meats (Mol et al. 1971).

People handling both cooked and raw foods can transfer microorganisms from the raw to the cooked product. With no further treatment, this is a potential health hazard. The cook who cuts up a raw chicken on a cutting board that is then used for cutting up raw vegetables for a salad may transfer *Salmonella* from the raw chicken to the raw vegetables. The

meat department in a retail store may use the same knife and block to cut fish, cold meat, chicken, beef, and other types of food. Some of these foods may be eaten with no further treatment or only a minimal heating. A potential health hazard can result from this. Cross-contamination of food is one of the ten main factors that contribute to foodborne illnesses.

PACKAGING

Packaging is a potential source of contamination. There are many studies concerning the effect of different types of packaging materials on the shelf life of food. The possibility that the packaging material might contribute to the flora is not mentioned. This is probably because the number of flora on most foods is much greater than that on packaging material. In their survey of microorganisms on poultry-processing equipment, Schuler and Badenhop (1972) reported the total bacterial count of bulk containers for ice-packed poultry. The counts ranged from 290/4 in.^{2} (11/cm^2) to 9,770/4 in.^{2} (376/cm^2). They stated that the holding of boxes overnight led to contamination by dust, rodents, and birds.

Korab (1963) tested one-trip glass bottles used for carbonated beverages. He reported that a majority of bottles had no bacteria, yeasts, or molds when they were received at the beverage company, but about 6.5 percent had more than fifty bacteria per bottle. Storage of uncovered bottles for one week increased the microbial contamination. Storage outside resulted in more contamination than storage inside, and the contamination increased with storage time. Compared to returnable and reused bottles, the one-trip bottles have very low microbial counts.

Plastic materials are used for packaging many food products. During manufacture and handling, electrostatic charges can occur on the plastic. These charges attract airborne material such as dust and microorganisms (Baribo, Avens, and O'Neill 1966). Although plastics are essentially sterile when made into packaging material, they may become contaminated if not handled in an acceptable manner. The static charges add to the problem of maintaining plastic free of microbial contamination.

There was a time when containers used for shipping poultry were reused to pack fresh vegetables for shipment. Due to the potential for contamination of the containers by salmonellae from the poultry and then the transfer of these pathogens from the container to the vegetables, this practice was halted.

Packages serve as a protective covering to limit or prevent microbial contamination; however, they do not prevent microbial growth. It is important that preparation of the food for packaging is designed to limit

or prevent microbial contamination prior to packaging. Also, the package must be durable enough to maintain its integrity during storage and distribution.

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6

Foodborne Agents Causing Illness

Throughout our lifetimes we are subjected to risks and hazards of all kinds. The food supply in the United States is one of the most abundant, nutritious, and safest on earth. However, there is no absolute degree of safety, not even for the food we consume.

Foods should be safer today than in the "good old days," due to the knowledge we have gained of bacteria and sanitation, as well as to increased regulations. However, due to large-scale, high-speed food processing, alteration of traditional processing methods resulting in less control of microorganisms, proliferation of heat-and-eat convenience foods, and nationwide distribution with increased potential for mishandling, it is possible for outbreaks of foodborne illness to occur that involve many people. This is evident by the five-state outbreak of salmonellosis that occurred in 1985. The number of reported outbreaks and cases fluctuates from year to year, but since 1967 the number of cases per 100,000 people has tended to increase. Part of this increase may be due to more complete reporting of foodborne illness, and part may be due to an actual increase in the number of cases of such illness.

TYPES OF FOOD HAZARDS

The agents that cause human illness and can be transmitted by foods are bacteria, viruses, fungi, parasites, chemicals, and toxins naturally present in plants and animals. Bryan (1973) listed approximately 200 etiologic agents of foodborne illness.

In 1961, the Communicable Disease Center (since renamed the Centers for Disease Control, or CDC), became responsible for maintaining records and reporting foodborne illnesses in the United States. It is well recognized that all outbreaks or cases of foodborne illness are not reported to the CDC. However, the data collected by the CDC are the best that we have. Annual summaries of foodborne illnesses have been published since 1966. The confirmed etiologies of foodborne outbreaks for 1981 (CDC 1983b) are listed in Table 6.1. In 1981, as well as in other years, bacteria were involved in most of the outbreaks and cases.

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TABLE 6.1. ETIOLOGY OF CONFIRMED FOODBORNE DISEASE OUTBREAKS, CASES, AND DEATHS IN THE UNITED STATES IN 1981

	Outbreaks		Cases		Deaths	
Etiology	No.	%	No.	%	No.	%
Bacterial						
Bacillus cereus	8	(3.2)	74	(0.9)	0	(0.0)
Campylobacter jejuni	10	(4.0)	487	(5.6)	0	(0.0)
Clostridium botulinum	11	(4.4)	22	(0.3)	1	(3.1)
Clostridium perfringens	28	(11.2)	1,162	(13.4)	2	(6.3)
Salmonella	66	(26.4)	2,456	(26.8)	21	(65.6)
Shigella	9	(3.6)	351	(4.1)	0	(0.0)
Staphylococcus aureus	44	(17.6)	2,934	(33.9)	1	(3.1)
Streptococcus Group A	2	(0.8)	307	(3.5)	0	(0.0)
Streptococcus Group D	1	(0.4)	24	(0.3)	0	(0.0)
Vibrio cholerae non-O1	1	(0.4)	4	(<0.1)	0	(0.0)
Vibrio parahaemolyticus	2	(0.8)	13	(0.2)	0	(0.0)
Yersinia enterocolitica	2	(0.8)	326	(3.8)	0	(0.0)
Other	1	(0.4)	48	(0.6)	0	(0.0)
TOTAL	185	(74.0)	8,208	(93.2)	25	(78.1)
Chemical						
Ciguatoxin	15	(6.0)	152	(1.8)	3	(9.4)
Heavy metals	2	(0.8)	4	(<0.1)	0	(0.0)
Monosodium glutamate	2	(0.8)	4	(<0.1)	0	(0.0)
Mushroom poisoning	11	(4.4)	25	(0.3)	3	(9.4)
Scombrotoxin	7	(2.8)	67	(0.8)	0	(0.0)
Other	14	(5.6)	75	(0.9)	0	(0.0)
TOTAL	51	(20.4)	327	(3.8)	6	(18.8)
Parasitic						
Giardia lamblia	1	(0.4)	61	(0.7)	0	(0.0)
Trichinella spiralis	7	(2.8)	70	(0.8)	1	(3.1)
TOTAL	8	(3.2)	131	(1.5)	1	(3.1)
Viral						
Hepatitis A	6	(2.4)	128	(1.5)	0	(0.0)
TOTAL	6	(2.4)	128	(1.5)	0	(0.0)
Confirmed Total	250	(100.0)	8,794	(100.0)	32	(100.0)

SOURCE: CDC (1983).

Not all of the diseases that may be transmitted by foods can be described thoroughly in this text. Therefore, those organisms and illnesses that have been reported most frequently are stressed.

Definition of an Outbreak

A foodborne disease outbreak is defined by the CDC as an incident in which two or more persons experience a similar illness, usually gastrointestinal, after ingesting a common food, and epidemiological analysis implicates the food as the source of the illness. For botulism or chemical poisoning, one case constitutes an outbreak (see Figure 6.1).

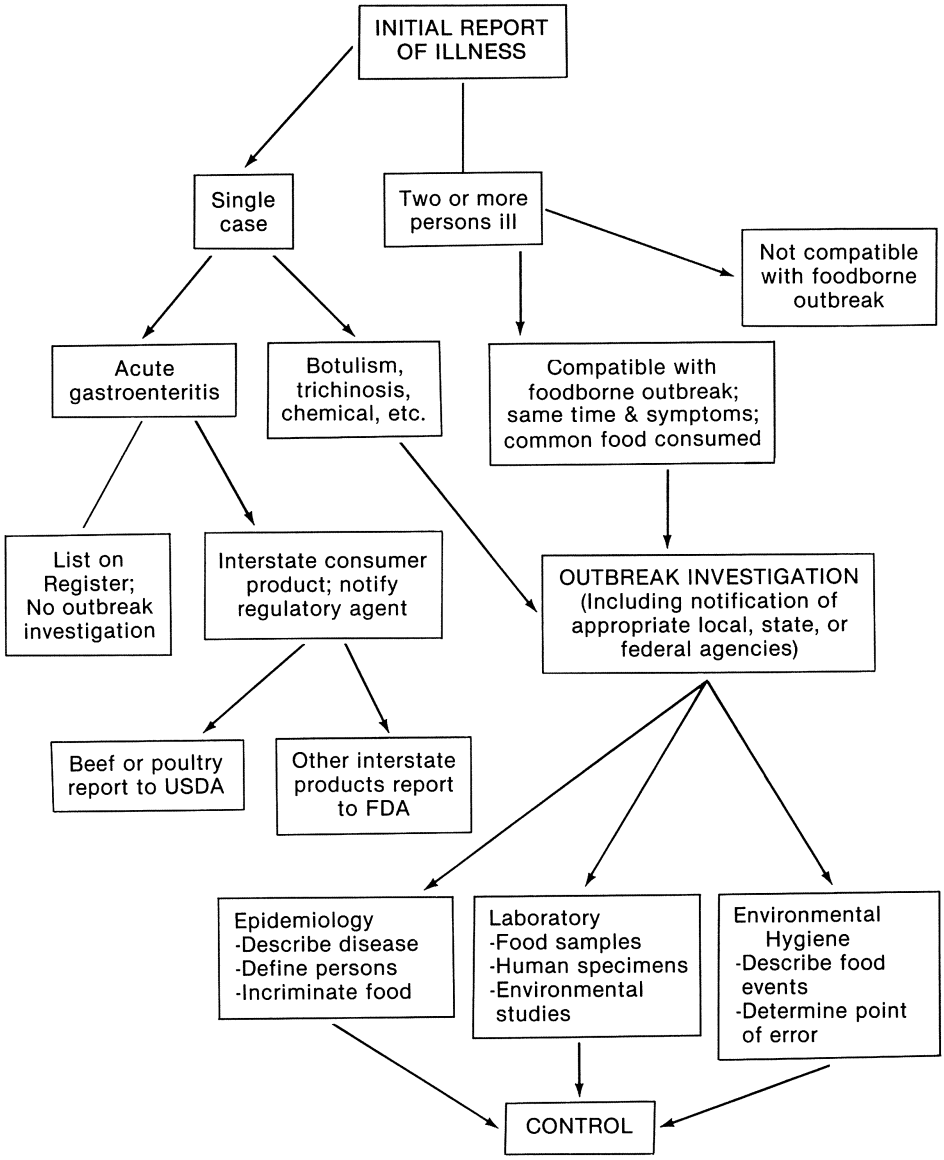

Figure 6.1. A scheme for handling foodborne disease complaints, to be implemented by state and local health departments.
Courtesy of CDC.

A microbial foodborne illness may result from ingesting a food containing either pathogenic microorganisms or a toxin or poison. When a pathogenic microorganism is the etiologic agent, the illness is called an infection. If a toxin or poison is the causative agent, the illness is called a food intoxication or food poisoning.

Epidemiology

Epidemiology attempts to identify the cause and the mode of transmission of infections and to suggest and evaluate methods for control.

The diagnosis of the specific disease is important for treatment and control. With a known etiology, acceptable therapy can be prescribed, dangers from handling patients with infections can be avoided, and the patient can be informed of the possible course of the illness.

Confirmed etiologies are those in which laboratory evidence is obtained and fulfill specific criteria of the CDC. The present reporting system involves many people and agencies. If the affected people do not seek medical help, if the doctor does not report the illness, or if there is no further investigation to confirm the cause of the illness, it is not recorded.

The data recorded by the CDC showed that bacterial agents accounted for 92 outbreaks and 3,270 cases in 1976. This increased to 185 outbreaks and 8,208 cases in 1981 (Table 6.1). It has been estimated that less than 10 percent of the actual outbreaks and cases are reported.

For the United States, estimates as high as 10 or 20 million cases a year have been made. Of the reported outbreaks, only about 50 percent have a confirmed etiology.

The data in Table 6.1 show that two bacterial agents, staphylococci and salmonellae, account for 44 percent of the outbreaks and more than 60 percent of the cases. The salmonellae accounted for more than 65 percent of the deaths due to foodborne agents. The three most prominent agents are salmonellae, *S. aureus*, and *C. perfringens*, not only in the United States but also in Canada (Todd 1983a), the Netherlands (Beckers 1982), England and Wales (Roberts 1982), and Europe in general. The most commonly reported cause of foodborne illness in India is staphylococcal enterotoxin (Hobbs 1982).

Other important bacteria have been *C. botulinum* and *Shigella*, with *B. cereus* showing an increase. Recent additions are *Vibrio parahaemolyticus*, *Yersinia enterocolitica,* and *Campylobacter jejuni.* Pasteurized milk was the vehicle for *Listeria monocytogenes* (Fleming et al. 1985).

Cholera is an important illness worldwide, spread by poor sanitation and contaminated food. There are illnesses, such as traveler's diarrhea, in which symptoms resemble those of several foodborne diseases. The CDC defines traveler's diarrhea as an acute intestinal illness that devel-

ops one or more days after arrival in a foreign country. Enterotoxigenic *E. coli* is the major agent, but other bacteria (*Shigella*, salmonellae), parasites (*Giardia, Entamoeba*), and viruses (rotavirus, reovirus, parvovirus) may be involved (Dupont 1981; Morgan et al. 1984).

Diarrheal syndromes are much more prevalent in countries in which sanitation is lacking and health resources are less available than in the industrialized nations. Nichols and Soriano (1977) estimated that 2.7 million children under five years of age die each year from diarrhea. Of these deaths, 76 percent occur in Asia, and 13 percent in Africa.

Foods Involved

Various foods are involved in foodborne illness (Table 6.2), but meat and meat products are the vehicles for transmission of a major share of foodborne illnesses. Precooked roast beef and ham are frequently re-

TABLE 6.2. FOODBORNE DISEASE OUTBREAKS BY VEHICLE OF TRANSMISSION, 1977–1981

	Year						
Vehicle	1977	1978	1979	1980	1981	Total	%
Beef	27	14	20	17	34	112	4.4
Pork	8	10	10	5	7	40	1.6
Ham	10	12	10	8	9	49	1.9
Sausage	10	2	0	5	6	23	0.9
Other meat	10	8	8	12	3	41	1.6
TOTAL MEAT	65	46	48	47	59	265	10.4
Poultry products	14	7	13	24	34	92	3.6
Shellfish	12	16	9	17	9	63	2.5
Other fish	22	29	31	53	28	163	6.4
TOTAL SEAFOOD	34	45	40	70	37	226	8.9
Milk	0	2	0	2	9	13	0.5
Ice cream	4	2	2	2	5	15	0.6
Other dairy	4	4	3	5	4	20	0.8
TOTAL DAIRY	8	8	5	9	18	48	1.9
TOTAL ANIMAL FOODS	121	106	106	150	148	631	24.8
Baked foods	12	5	4	9	12	42	1,6
Fruits and vegetables	9	6	9	15	11	50	2.0
Salads, miscellaneous	23	20	19	19	21	102	4.0
Mushrooms	32	1	4	0	11	48	1.9
Chinese food	17	7	3	3	1	31	1.2
Mexican food	12	9	6	10	5	42	1.6
Other food	41	23	21	28	35	148	5.8
Unknown	169	304	289	371	324	1457	57.1
TOTAL	436	481	461	605	568	2,551	

SOURCE: Data from CDC *Annual Summaries.*

ported as vehicles of salmonellae and *S. aureus*, respectively. Poultry and poultry products are a source of salmonellae, *C. perfringens, S. aureus*, as well as other organisms.

Before pasteurization, milk was involved in outbreaks of many diseases. Unfortunately, milk may be contaminated after pasteurization or may be used raw. Many of the illnesses caused by fishery products are due to ciguatoxin, scombrotoxin, or paralytic shellfish poisoning. Foods that contain these toxins are from an unsafe source. Fishery products also are vehicles for clostridia and *Vibrio* species as well as parasites.

Canned foods are vehicles for botulinum toxin. If the container leaks, the foods may be the vehicle for other agents that cause illness. Chinese foods, another common source, have been studied as well (Bryan et al 1982).

Place of Mishandling

When a processor of foods is involved in a foodborne disease outbreak, there is a great potential for many cases because of the widespread distribution of the defective product. Fortunately, the processor is involved in relatively few outbreaks (Table 6.3). Most outbreaks are caused by the mishandling of foods in foodservice establishments (Bryan 1982). The home is also an important place for such mishandling to occur. One can assume that there are very few cases per outbreak for those occurring in the home. The CDC no longer determines the place of mishandling, but rather the place of consumption (CDC 1983b).

Contributing Factors

Various factors contribute to outbreaks of foodborne illness. The main ones are improper holding temperatures (failing to properly refrigerate food), inadequate cooking, contaminated equipment (failure to clean and disinfect kitchen or processing plant equipment), and poor

TABLE 6.3. PLACES IN WHICH FOODS WERE MISHANDLED, 1968–1978

Place	No. of Outbreaks	%
Food-processing establishment	171	4.1
Food-service establishment	1,707	40.4
Home	639	15.1
Unknown, unspecified, or not applicable	1,704	40.4
TOTAL	4,221	

SOURCE: Data from CDC *Annual Summaries.*

TABLE 6.4. CONTRIBUTING FACTORS CAUSING FOODBORNE ILLNESS, 1977–1981

Factor	Year					Total	%
	1977	1978	1979	1980	1981		
Improper holding temperature	168	150	115	184	181	798	39.5
Inadequate cooking	67	53	53	41	62	276	13.7
Contaminated equipment	58	45	51	47	70	271	13.4
Food from unsafe source	23	16	7	23	40	109	5.4
Poor personal hygiene	87	63	70	80	110	410	20.3
Other	46	32	24	20	34	156	7.7
TOTAL						2,020	

SOURCE: Data from CDC *Annual Summaries.*

personal hygiene (Table 6.4). Other factors that contribute to foodborne illness include preparing food a day or more before serving, with improper holding and reheating, cross-contamination (from raw to cooked products) and adding contaminated ingredients to previously cooked food without reheating (Bryan 1982; Roberts 1982: Todd 1983b). After foods have been contaminated, the main danger is allowing them to remain at a temperature that permits the growth of potentially hazardous microorganisms.

Most of these problems could be controlled with only a little effort on the part of food handlers, whether in a processing plant, a restaurant, a cafeteria, or a home. The tremendous turnover of food workers makes effective training difficult. However, since many outbreaks occur because of carelessness at home, everyone might benefit from training in food handling and personal hygiene. A course in the sanitation of food handling could be offered in high school. According to Feachem (1984), the incidence of diarrhea can be reduced by 14 to 48 percent by simply washing contaminated hands with soap and water. Food microbiologists should set a good example for others to follow.

Symptoms and Severity

The symptoms of the illnesses are variable. However, diarrhea, nausea, vomiting, and abdominal cramps are evident in most foodborne illnesses.

Usually, each day some 8 to 10 liters of water enter the intestines of adults. The source of this fluid includes ingested liquid (food and water), swallowed saliva, and secretions from the stomach, pancreas, liver, and intestinal glands. Normally, about 90 percent of this liquid is resorbed from the small intestine, and of the remaining liquid, about 90 percent,

is absorbed from the colon, so that only about 100 to 200 ml of fluid passes out of the colon. Most of the excess water in the body is removed by the kidneys. Perspiration and respiration account for the remaining loss of water from the body.

Either malabsorption or excessive secretion of water and electrolytes into the intestines can result in diarrhea. Researchers have defined diarrhea as the occurrence of four or more unformed stools within one day, together with one or more other symptoms (abdominal cramps, nausea, vomiting, fever) (Dupont et al. 1980). Mild forms of diarrhea may result in only a few hundred milliliters of extra fluid per day passing out of the colon. However, in severe cases, up to 1 liter per hour may be expelled. This excessive loss of water causes dehydration, which can result in death. Other effects, especially in very young children, include malnutrition and growth retardation.

Fortunately, in most cases, the illness is not severe and the patient recovers in one or a few days. However, twenty-five deaths from diarrhea occurred in the United States in 1981 (Table 6.1). Although salmonellae were involved in most of the deaths, the mortality rate was higher for botulism (4.5 percent) than for salmonellosis (0.9 percent).

Diarrhea has various causes. These include the ingestion of certain drugs such as antibiotics and antimetabolites, excessive laxatives, herbal teas, sugar alcohols in dietetic foods and beverages, various "health" foods, certain carcinomas, and from modification of a diet or from anxiety caused by traveling. Additional factors are listed in Table 6.1.

BACTERIAL DISEASES

Bacterial foodborne illness can result from toxins produced in the food before consumption (*S. aureus, C. botulinum*) or by the organism either infecting cells or producing toxins in the intestinal tract after ingestion. Once in the intestinal tract, the organism must contact and adhere to the cells of the epithelium (Archer 1984; Hill 1985; Smith 1984). Otherwise, they will be removed by peristalsis and the movement of the intestinal contents, as well as by mucociliary action.

Once attached to the cells, some organisms may invade the cells and others produce enterotoxins; in either case, the water secretion into or adsorption from the intestines may be affected.

For epidemiology of an outbreak and the prevention of foodborne illness, both microorganisms and their toxins must be detected. Although there are basic similarities in the various methods, there are also specific differences.

Staphylococcal Intoxication

This illness accounted for over 17 percent of the outbreaks and almost 34 percent of the cases of reported foodborne illnesses in the United States in 1981 (Table 6.1). However, the actual extent of this illness is not known.

CHARACTERISTICS OF THE INTOXICATION. The characteristics of an illness can aid in determining the causative agent in an outbreak of foodborne illness.

Incubation Period. The incubation period of a typical outbreak usually ranges from 30 min to 8 hr, with most illnesses occurring 2 to 4 hr after ingestion of the suspect food (Fig. 6.2).

Symptoms. Not all of the people eating a suspect meal become ill, and not all ill people experience the same symptoms. The severity of the symp-

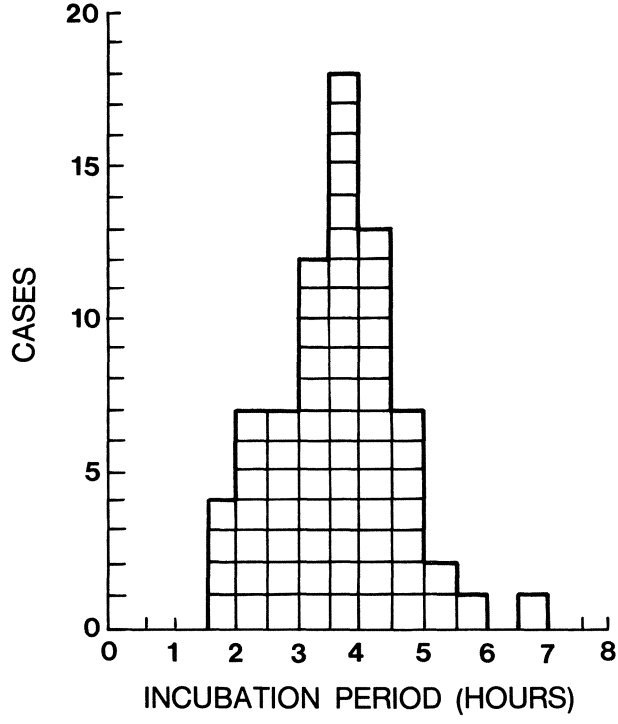

Figure 6.2. Depiction of an outbreak of staphylococcal intoxication. Incubation period.
Courtesy of CDC.

TABLE 6.5. PERCENTAGE OF ILL PERSONS EXPERIENCING SPECIFIC SYMPTOMS DUE TO STAPHYLOCOCCAL INTOXICATION

	Outbreaks				
Symptoms	1	2	3	4	5
Nausea	—	76	100	50	41
Vomiting	70	44	100	77	76
Diarrhea	19	67	100	82	73
Abdominal cramps	71	71	58	15	16
Chills	—	25	46	—	—
Headache	42	—	6	41	35
Prostration	—	—	63	—	—
Weakness	—	—	—	68	73
Leg cramps	—	—	—	—	11
Muscle soreness	—	—	5	—	—
Collapse	—	9	—	—	—
Hypotension	—	1	—	—	—
Fever	—	25	7	21	24

— = Not reported.
SOURCE: Data from *CDC Morbidity and Mortality Weekly Reports.*

toms varies with the concentration of enterotoxin in the food, the amount of food consumed, and the susceptibility of the individual. The principal symptoms listed in Table 6.5 are nausea, vomiting, abdominal cramps, and diarrhea.

Duration and Therapy. Symptoms of the illness usually subside after one or two days. The illness is rarely fatal. Due to the sudden onset and short duration of the illness, treatment usually is not needed. However, hospitalization is required in cases in which shock, dehydration, and extensive vomiting have occurred. In these cases, therapy includes replacing lost fluids and electrolytes. According to Holmberg and Blake (1984), 10 percent of the victims seek hospital care.

ETIOLOGIC AGENT. This illness is called an intoxication because the etiologic agent is an enterotoxin. Payne and Wood (1974) listed six known staphylococcal enterotoxins (A, B, C, D, E, F), (SEA, SEB, SEC, SED, SEE, SEF) based on serological reactions. Although SEC_1 and SEC_2 react with the same antibody and are often lumped together as SEC, they also react with distinct minor antibodies. The presence of a third SEC, enterotoxin C_3, has also been reported (Reiser et al. 1984).

Enterotoxin A is the most frequently encountered enterotoxin in food poisoning outbreaks in the United States. In New Zealand, Jarvis and Harding (1972) found enterotoxins C and D to be more prevalent than SEA. SEB rarely is involved in food poisoning outbreaks in the United States.

Properties of the Enterotoxins. Staphylococcal enterotoxins are simple proteins with a molecular weight between 25,000 and 35,000. They are readily soluble in water and salt solutions.

The heat stability is an important characteristic of staphylococcal enterotoxins. Enterotoxin B is more heat resistant than A or D. The heat resistance is influenced by the medium in which it is heated (composition, pH). In foods, the enterotoxins are not completely inactivated by normal cooking, pasteurization, or other usual heat treatments. Tatini (1976) stated that thermal processing cannot be relied upon to inactivate these toxins. Further, he found that heated toxins had greater biological activity than unheated toxin when tested at the same dose level.

Action of Enterotoxins. Researchers used monkeys to study enterotoxin-induced emesis (vomiting) (Elwell et al. 1975). Their results indicate that enterotoxin is not absorbed from the intestine. Orally ingested enterotoxin is thought to mediate emesis by acting on sites in the intestine. This stimulus apparently is transferred by the vagus and sympathetic nerves to the vomiting center, which is part of the central nervous system. The vomiting center somehow induces retroperistalsis of the stomach and small intestine, resulting in emesis. The exact systems involved in the vomiting syndrome have not been determined. The known facts and hypotheses have been discussed and described elsewhere (Robins-Browne 1980; Thompson and Malagelada 1982; Van Miert et al. 1983). Since the nervous system is involved, it has been suggested that the enterotoxins should be called neurotoxins. However, neuronal binding of SEA in the intestinal tract was not demonstrable (Beery et al. 1984).

The action of staphylococcal enterotoxins in the diarrheal syndrome is not known. Staphylococcal enterotoxin shows an affinity to the walls of the stomach and the small and large intestines. If sufficient enterotoxin is present in consumed food, it causes inflammation and irritation of the lining in the stomach and intestinal tract. Working with flounder intestine in vitro, the data of Huang, Chen, and Rout (1974) suggested that enterotoxin stimulates active sodium and chloride secretion. Others found that enterotoxin B did not interfere with water absorption in the guinea pig ileum, but that staphylococcal delta toxin did inhibit absorption in the jejunum and ileum (Kapral et al. 1974).

Amount of Enterotoxin Needed for Illness. There are no definite data concerning the minimum amount of enterotoxin needed to cause symptoms in a human. Gilbert (1974) listed estimates that ranged from 0.015 to 0.357 μg of enterotoxin per kilogram of body weight. Besides body weight, individuals vary in their sensitivity to enterotoxins.

FOODS INVOLVED. Various foods have been involved in staphylococcal intoxication since 1974 (Table 6.6). More than 35 percent of the out-

TABLE 6.6. FOODS INVOLVED IN STAPHYLOCOCCAL INTOXICATIONS 1976–1981

Food	Years		Total	%
	1976–1978	1979–1981		
Beef	3	5	8	4.5
Ham	22	19	41	23.3
Pork	1	1	2	1.1
Other meat	5	7	12	6.8
Poultry	7	16	23	13.1
Shellfish	1	0	1	0.6
Other fish	0	1	1	0.6
Milk	0	1	1	0.6
Other dairy	1	2	3	1.7
Bakery products	3	10	13	7.4
Fruits and vegetables	1	1	2	1.1
Salads	13	13	26	14.8
Other foods	6	6	12	6.8
Unknown	11	20	31	17.6
TOTAL	74	102	176	

SOURCE: Data from CDC *Annual Summaries.*

breaks involved red meats, with 23 percent being caused by contaminated ham. Fresh meats are involved less often than cured meats since, being perishable, they are refrigerated. Cured meats are not as perishable and may be mishandled by being allowed to remain at room temperature. Also, it is difficult for *S. aureus* to compete with the microbial flora on fresh meat as compared to that on cured meat.

The poultry products involved are usually either barbecued or in salads. Other salads (potato, macaroni, and tuna fish) have been implicated in more than 14 percent of the outbreaks. Bakery products containing custard or cream are important offenders.

In many outbreaks, the food is cooked, then contaminated with *S. aureus* during handling and held at a temperature for growth and production of enterotoxin.

THE ORGANISM (*S. AUREUS*). *S. aureus* is described briefly in Chapter 3. These Gram-positive cocci produce colonies that may be white or have pigment ranging from yellow to orange. The color is influenced by growth conditions as well as strain variation.

In both aerobic and anaerobic conditions, *S. aureus* produces acid from mannitol, glucose, lactose, and maltose. Aerobically, many carbohydrates are metabolized with the production of acid, but acid is not produced from arabinose, cellobiose, dextrin, inositol, raffinose, rhamnose, or xylose.

Typical *S. aureus* strains produce α-toxin, coagulase, and a heat-stable nuclease.

SOURCES. Some reports call *S. aureus* ubiquitous because it is so widespread (air, dust, clothing, floors, water, sewage, and insects). The principal source of *S. aureus* is the human nose, although it is found on the skin, especially on the hands, in infected wounds, burns, boils, pimples, acne, in nose and throat discharges, and in feces. The primary site on the hands is the fingertips, which relates to the habit of handling one's nose with the fingers. The extent of nasal carriers is difficult to determine, but surveys have shown the carrier rate to vary from 6 percent to over 60 percent of the population. People associated with hospitals tend to have a higher carrier rate than the normal population.

Animals are a source of *S. aureus*. Most of the strains of *S. aureus* isolated from animals tend to have characteristics different from those associated with people (Devriese 1980; Kibenge, Wilcox, and Perret 1982). The organism is found in food-processing operations (Harvey, Patterson, and Gibbs 1982; Notermans, Dufrenne, and Van Schothorst 1982), and at a high level on and in healthy food handlers (Holmberg and Blake 1984).

GROWTH. Researchers believe that 10^5 to 10^6 cells of *S. aureus* per gram of food must generally be present before the production of enterotoxin reaches a level that can cause intoxication. Due to the normally low numbers in food, multiplication must occur. By knowing and understanding the factors affecting the growth of *S. aureus*, we can control the growth, enterotoxin production, and outbreaks of staphylococcal intoxication. The general factors affecting the growth of *S. aureus* are described in other sections of this text. *S. aureus* is a relatively poor competitor, and various bacteria can inhibit or outgrow it. This inhibitory interrelationship of other bacteria with *S. aureus* is important in preventing toxin production in foods. It may be a primary reason for certain foods to be less involved than others in outbreaks of staphylococcal intoxication. In foods with a_w of 0.90 to 0.95 or with 5 to 10 percent salt, *S. aureus* can dominate because most other bacteria cannot grow.

Strains of lactic streptococci (*S. lactis* and *S. cremoris*) inhibited *S. aureus* inoculated into milk prior to cheese making (Ibrahim 1978). Since cheese has been a vehicle of staphylococcal enterotoxin, it is evident that microbial competition cannot be relied upon to always inhibit growth and toxin production of *S. aureus*. Also, it has been suggested that some competitive organisms may degrade the enterotoxins.

Obviously, *S. aureus* is able to grow on or in foods that have been involved in staphylococcal intoxication. Some outbreaks have occurred by holding food at room temperature for less than 4 hr. Longer incubation times increase the risk. The behavior (growth or inhibition) of *S. aureus* has been reported for various cheeses (Koenig and Marth 1982;

Magrini, Chirife, and Parada 1983), pumpkin pie (Wyatt and Guy 1981), potatoes (Tamminga et al. 1978), and meat (Luiten, Marchello, and Dryden 1982). The maximum temperature for growth of *S. aureus* was increased by the addition of salt, monosodium glutamate, or soy sauce (Hurst and Hughes 1983).

Sublethal treatments and other factors (pH, a_w, Eh, temperature) discussed in Chapter 4 act to control growth and enterotoxin production of *S. aureus*. Growth was not observed at a_w 0.85 or below, pH 4.3 or less, or at 8°C or less (Notermans and Heuvelman 1983). Enterotoxin production required higher a_w, pH, or temperature than that needed for growth but was mainly inhibited by the effect of a_w.

When warm or hot foods are refrigerated, a long time may be needed to cool the food sufficiently to prevent growth of *S. aureus*. Foods with gravies or sauces cool slower than do those without, and the substrate is more readily available for growth of organisms.

TOXIN PRODUCTION. *S. aureus* produces several toxins, including those listed in Table 6.7. The main difference between exotoxins and endotoxins is not whether they are outside or inside the cell, but rather their structure. Exotoxins are proteins with little or no nonprotein residues. Endotoxins are primarily polysaccharide and lipid complexes (lipopolysaccharides). Food microbiologists are concerned with the enterotoxins, which, by definition, are exotoxins, since they are protein as well as being found free from the cell.

Not all strains of *S. aureus* are enterotoxigenic. Schroeder (1967) estimated that only 4 percent of the staphylococcal strains in milk were capable of producing enterotoxin. In a survey of *S. aureus* in meat, dairy products, and other foods, Payne and Wood (1974) found that 125 of 200 strains (62.5 percent) produced enterotoxins. Wieneke (1974) found a somewhat higher ratio of enterotoxigenic strains in cooked food than in

TABLE 6.7. SOME TOXINS OF *STAPHYLOCOCCUS AUREUS*

Toxin	Action
α-toxin	Hemolytic, dermonecrotic, lethal
β-toxin (phospholipase C)	Hemolytic
γ-toxin	Hemolytic
δ-toxin	Hemolytic, enteric
Hyaluronidase	Spreading factor
Staphylococcal coagulase	Coagulates plasma
Staphylokinase	Fibrinolytic
Leukocidin	Kills leucocytes
Epidermolytic	Exfoliation
Enterotoxins (A, B, C, D, E, F)	Emetic
Toxic shock toxin	Lymphoid hyperplasia

raw food. The higher ratio may be due to people contaminating cooked foods with human strains. Such strains are more frequently enterotoxigenic than are strains from animal or other sources.

The production of SEA results from a chromosomal gene (Mallonee, Glatz, and Pattee 1982; Pattee and Glatz 1980; Shafer and Iandolo 1978b). Shafer and Iandolo (1978a) reported that SEB production of strains S6 and 277 is determined by a chromosomal gene or genes and suggested that this might be the case for many enterotoxigenic strains of *S. aureus*. Dyer and Iandolo (1981) proposed that SEB production is dependent on at least two unlinked genes. Altboum, Hertman, and Sarid (1985) reported that elimination of plasmid pZA10 from strain 6344 caused the loss of SEB and SEC_1 production. According to these researchers, the plasmid can be transferred and confer toxigenicity to other *S. aureus* strains, and chromosomal DNA can integrate the plasmid.

The conditions necessary for growth and enterotoxin production have been reviewed (Smith, Buchanan, and Palumbo 1983). In general, toxin production occurs in a more narrow range of environmental characteristics than those discussed for growth in Chapter 4.

The relationship of growth of *S. aureus* and SEB production was reported by Markus and Silverman (1969) and is shown in Figure 6.3. The rate of synthesis of SEB is greater than that for SEA, so higher concentrations of SEB are obtained. Tweten and Iandolo (1983) suggested that a precursor of SEB is bound to the cell membrane. At some stage, mature SEB is formed and released by the membrane into a specialized compartment in the cell wall. From there, it is released to the exterior environment.

A constant dissolved oxygen (DO) level of 100 percent stimulated growth, but enterotoxin production was not observed (Carpenter and Silverman 1974). A DO of 10 percent yielded a higher level of enterotoxin than did a DO of either 100 percent or 50 percent.

Aerobically, certain strains of *S. aureus* produce enterotoxin at a pH of 4.8, but anaerobically, no enterotoxin is found at pH 5.4 (Barber and Deibel 1972). Their results indicate that biological acid production cannot be relied upon to inhibit *S. aureus* in fermented sausage, and they recommended chemical acidulation. Since the toxin is produced at a higher level aerobically than anaerobically, they suggested sampling aerobic portions of foods for the toxin. Mixing and cross-sectional sampling merely dilute the toxin with anaerobic portions of the food, which contain little or no toxin.

S. aureus (A100) grew and produced enterotoxin in precooked bacon stored aerobically either at 37°C and a minimum a_w of 0.84, or at 20°C and an a_w of 0.89 (Lee, Silverman, and Munsey 1981). Anaerobically, the organism required a minimum a_w of 0.90 at 37°C, and 0.94 at 20°C.

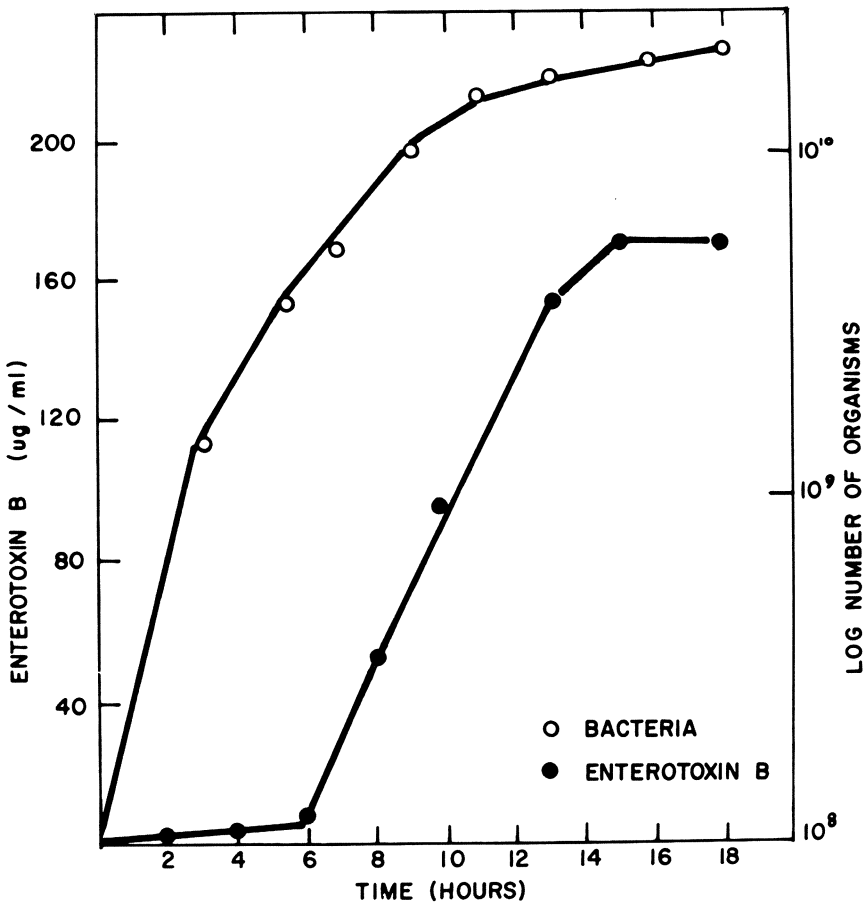

Figure 6.3. Growth and Enterotoxin B production of *S. aureus*. *Courtesy of Markus and Silverman (1969).*

The effect of sodium chloride on growth and enterotoxin production was reported by McLean, Lilly, and Alford (1968) and is shown in Figure 6.4.

METHODOLOGY. Foods or other samples are examined for *S. aureus* and the enterotoxins to demonstrate postprocessing contamination, to determine a potentially hazardous product, or to confirm a causative agent in a foodborne illness. Since *S. aureus* is sensitive to heat and sanitizing agents, the presence of the organism or its toxins in processed food or on equipment generally indicates poor sanitation or handling. Post-

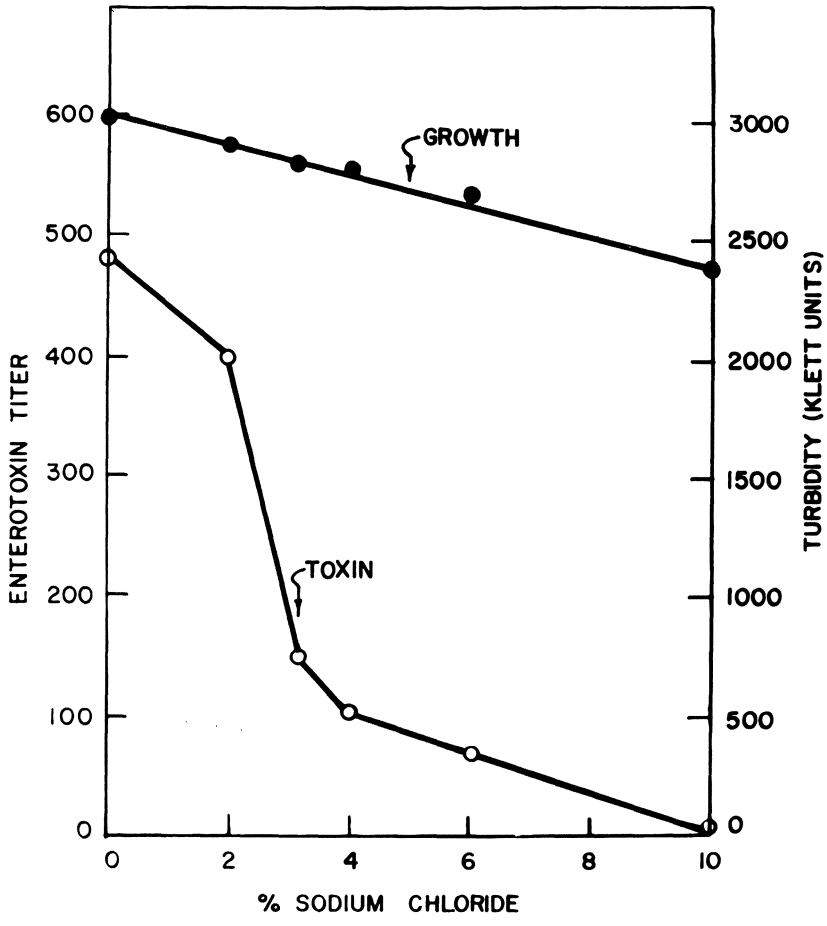

Figure 6.4. Effect of salt concentration on the production of enterotoxin by *S. aureus*.
Courtesy of McLean, Lilly, and Alford (1968).

processing contamination usually is due to human contact or to improperly sanitized food contact surfaces.

The methodology involved in conjunction with *S. aureus* includes the detection and enumeration of the organism; the characterization of the organism by testing for coagulase, nuclease, and other enzymes; serological reactions; phage typing and the qualitative and quantitative analyses for the enterotoxins.

Detection and Enumeration of the Organism. Many media have been developed and suggested for the direct plating and enumeration of *S. aureus*.

The FDA (1978) method uses Baird-Parker agar. This medium contains potassium tellurite as a selective agent and egg yolk and tellurite as differential agents. On this medium, *S. aureus* colonies appear circular, smooth, convex, moist, gray to jet-black, frequently with an off-white margin, due to reduction of the tellurite to elemental tellurium. They are surrounded by an opaque zone and an outer clear zone due to the reaction on the emulsified egg yolk. When touched with an inoculating needle, the colonies have a buttery to gummy consistency. There may be variations to this description. Typical colonies are selected for further study.

A collaborative study revealed that Baird-Parker (BP) agar is satisfactory for recovery of cells stressed or injured by heat or other processing conditions (Baer et al. 1975).

Baird-Parker agar, modified by substituting pig plasma for egg yolk, uses the coagulase reaction for differentiation and is as effective as the original agar for growing stressed *S. aureus* cells (Becker, Terplan, and Zaadhof 1983; Idziak and Mossel 1980). The addition of bovine fibrinogen to plasma-modified BP agar was found to be effective in detecting *S. aureus* and also determines the nuclease reaction (Beckers et al. 1984; Chopin et al. 1985).

Characterization. Colonies typical of *S. aureus* on an agar surface are selected for further testing and characterization. These tests may include various fermentations of carbohydrates, the presence of coagulase, heat-stable nuclease or lysozyme, and determining the resistance to chemical inhibitors or antibiotics (Baker 1984). None of these tests, or a combination of them is an absolutely reliable indicator of enterotoxin formation by the organism.

The coagulase test is considered the most reliable single test for differentiating potentially pathogenic *S. aureus*. However, not all coagulase-positive strains produce enterotoxins. On the other hand, there are reports that coagulase-negative strains produce enterotoxin or have been involved in cases of staphylococcal intoxication (Lotter and Genigeorgis 1975). Some enterotoxigenic strains of staphylococci found to be coagulase negative on first isolation, become coagulase positive after subculturing several times, and some strains may lose the ability to produce coagulase but remain enterotoxigenic.

The coagulase reaction in clotting blood serum was reviewed by Tager (1974). According to this review, there are three main reactions, ending with the conversion of fibrinogen to fibrin and formation of a clot. According to Baird-Parker (1972), over 90 percent of the strains of *S. aureus* produce a coagulase. Several tests have been suggested and evaluated to detect the presence of coagulase. A tube test is used to determine free coagulase, and a slide test detects a clumping factor. These are distinct

entities. A latex slide agglutination test to determine clumping factor and protein A simultaneously was described by Essers and Radebold (1980) and reported to be as accurate as the tube test and more accurate than the slide test by Doern (1982).

Although other organisms may produce nucleases, the heat stability of *S. aureus* nuclease is unique. It is generally agreed that the enzyme loses but little activity by boiling for 30 min. It retained 10 percent of its activity after heating at 100°C for 180 min, 120°C for 34 min, or 130°C for 16 min (Erickson and Deibel 1973).

The basic test for staphylococcal nuclease was described by Lachica, Genigeorgis, and Hoeprich (1971). Since then, several suggestions for improvement have been made (Ibrahim 1981; Pham and Davis 1979; Stadhouders, Hassing, and Galesloot 1980). However, other staphylococci produce thermostable nuclease (Gudding 1983). These nucleases can be distinguished from each other by seroinhibition, the inhibition of nuclease activity by specific antibodies (Gudding 1983; Lachica, Genigeorgis, and Hoeprich 1979).

The thermostable nuclease test might be used as a screening method for foods to detect the possible presence of high populations of *S. aureus* or enterotoxin (Tatini et al. 1975; Tatini, Cords, and Gramoli 1976). This system seemed to be applicable to certain foods, but was of questionable value for others. Ibrahim and Baldock (1981) always detected nuclease in cheese that contained enterotoxin.

Medwid and Grant (1980) questioned the value of the thermonuclease test for foods containing high numbers of *Streptococcus faecalis* subsp. *liquefaciens*, since this organism inactivated the enzyme.

Phage Typing. Some bacteriophages are very specific for the cells they will lyse, while other phages have a wider spectrum of susceptible cells. Obviously, for phage typing of *S. aureus*, those phages that have a limited spectrum of susceptible cells are desirable.

The International Committee on Nomenclature of Bacteria established a Subcommittee on Phage Typing of Staphylococci in 1953. The Staphylococcus Reference Laboratory of the British Public Health Laboratory Service in London became the International Reference Center for the subcommittee. This subcommittee established a set of phages for routine testing of *S. aureus*. In 1961, it became the World Health Organization Centre for Staphylococcus Phage Typing. A description of the system used for phage typing was reported by CDC (1976b) and Parker (1972).

Phage typing is of particular help in the epidemiological study of staphylococcal intoxications or infections. Various strains isolated during an outbreak from patients, foods, and food handlers can be tested for

their phage patterns. This information makes it possible to differentiate between a strain responsible for the outbreak and unrelated strains.

Smith (1972) stated that there are only two good reasons for typing *S. aureus*, either by phages or antisera. The main reason is for epidemiological investigations, and the second reason is to give a label to the organism for research work.

Enterotoxin Detection. The detection of enterotoxin is important. If an outbreak occurs and isolates of *S. aureus* are obtained from a food, the enterotoxins produced can be compared to the enterotoxin causing the intoxication. In one outbreak involving dried milk, no *S. aureus* isolates were detected. In this case, the organisms multiplied, produced the toxin, and then were killed by heat during the processing. In cases such as this, the food can be analyzed for the enterotoxin.

The principal methods for detecting enterotoxins can be separated into biological and serological systems.

A biological system can be some living entity or part thereof. The best living entity to determine enterotoxin activity is a human. However, humans are not always readily available. Also, the amount of enterotoxin cannot be determined, because people vary in their sensitivity to enterotoxin.

Animals such as monkeys, chimpanzees, dogs, pigs, cats, and kittens have been tested as biological systems. Monkeys and chimpanzees can be given the enterotoxin orally, whereas the other animals are injected either intraperitoneally or intravenously. Because vomiting is the first symptom to occur and is the most readily observable reaction to enterotoxin, animals such as rodents that do not have a vomiting mechanism are not used. Pigeons, frogs, tropical fish, nematodes, or various protozoa or bacteria show no obvious reaction with enterotoxins. Tissues including human intestinal cells, rabbit gut segments, chicken embryos, tissue cultures (HEp-2 and HeLa cells) and isolated enzyme systems have been tested for assay of enterotoxin. Although crude enterotoxins have shown some effect on some of these systems, the purified toxins have revealed little or no effect.

Since monkeys or chimpanzees can be fed the enterotoxin orally, they are the preferred test animal. The monkey is not as sensitive to enterotoxin as humans are and the sensitivity of chimpanzees is somewhere between the sensitivities of humans and of monkeys. Another disadvantage is that these animals are expensive to maintain and, as they are used to assay enterotoxin, they tend to develop a resistance (a limited immunity) to the toxins. Thus, their usefulness for enterotoxin detection is limited. A direct skin test on sensitized guinea pigs reportedly has a sensitivity of 10 to 100 pg of SEB per milliliter of prepared food sample (Scheuber et al. 1983).

Tests using animals were necessary before the enterotoxins were purified and serological tests could be developed. It is evident that the animal tests are still important for the detection of any new types of enterotoxins, or testing the toxicity of isolated compounds, or a particular chemical portion of a toxin.

Serological systems that have been suggested for assaying enterotoxins include Ouchterlony double-diffusion plate (Ouchterlony 1968), Oudin single-gel diffusion tube, Oakley double-gel diffusion tube, microslide (Casman and Bennett 1965), polyvalent single radial immunodiffusion (Meyer and Palmieri 1980), hemagglutination inhibition, reversed passive hemagglutination (Yamada, Igarishi, and Terayama 1977), immunofluorescence, radioimmunoassay and enzyme-linked immunosorbent assay (ELISA) tests.

The Ouchterlony system was adapted for use on a glass slide by Wadsworth. A modification of this microslide method was used by Casman and Bennett (1965) to detect enterotoxin in foods. Bennett (1971) stated that 0.005 μg of enterotoxin A per gram of cheese was detectable. In order to determine low levels of enterotoxin, procedures of extraction, separation, and concentration are needed (Bennett and McClure 1980).

According to a collaborative study reported by Bennett and McClure (1976), the microslide gel double-diffusion test has a high degree of specificity, is simple, and has good reproducibility in the identification of enterotoxins. This system has been adopted as official by the AOAC.

The various diffusion tests have been used with varying results. The radioimmunoassay (RIA) technique reportedly can detect as little as 0.1 ng/ml of SEA and 0.5 ng/ml of SEB (Areson, Charm, and Wong 1980). However, the radioactive label deteriorates and should be replaced after about two months. Radioactive waste is created. The system also requires expensive equipment and scarce and expensive purified toxins.

To overcome some of the problems of the RIA, Saunders and Bartlett (1977) suggested an enzyme immunoassay (EIA). Using an EIA, Kauffman (1980) stated that it was sensitive to 2 ng SEA/ml. He considered this method to be sensitive and precise enough to serve as an alternative to the RIA.The FDA guidelines for acceptable enterotoxin detection is 1 to 2 ng/g of food.

Procedures using ELISA now have the greatest potential for the detection of recognized enterotoxins. First, a measured amount (100 μl) of purified enterotoxin is added to wells in a polystyrene plate. After sufficient time at 4°C to allow the toxin to bind to the plastic, the liquid is removed and the wells filled with serum albumin to block any residual binding sites on the plastic. Then the wells are emptied, washed with buffer, and known antitoxins, to which an enzyme such as peroxidase or alkaline phosphatase is conjugated, are added. After about 2 hr, the wells are aspirated to remove the liquid and washed to remove excess conju-

gated antitoxin. Then a substrate is added which the conjugated enzyme can change to a colored compound. If there is no enterotoxin, there is nothing to which the antitoxin can react. It, along with the enzyme, are removed by washing, and there will be no color reaction when the substrate is added. It is evident that the amount of enterotoxin is related to the amount of enzyme-conjugated antitoxin that remains, and that is related to the intensity of color that is formed by the enzyme acting on the substrate for a specified time. The testing of known amounts of enterotoxin is used to establish the color intensities produced in a specified time, so that the ELISA is a quantitative procedure. Extensive purification of the toxin in the sample is not needed for the ELISA system. There are several variations to this basic system.

Using a competitive ELISA, SEA, SEB, or SEC concentrations of 0.1 ng or less were measured (Stiffler-Rosenberg and Fey 1978). Four versions of the ELISA were compared by Fey, Pfister, and Rüegg (1984). They stated that the sandwich with labeled antibody was the best, with a sensitivity of 0.1 ng of enterotoxin per milliliter. To detect enterotoxin in food, the ELISA is as sensitive as the RIA (Freed et al. 1982; Kuo and Silverman 1980) and more sensitive than the microslide technique (Lenz et al. 1983).

Injection of an antigen into an animal results in the production of antibodies. A single B cell produces a single antibody specific for a single antigenic determinant. However, the B-cell population is polyclonal in nature and produces heterogeneous, polyclonal antibodies even when a highly purified antigen is injected. If the B cells were a monoclonal population, a large amount of single antibody would be produced. It is difficult to culture individual B cells. However, when B cells are fused with myeloma (tumor-producing) cells, the resultant hybridoma combines the antibody production of the B cell and the reproduction of the myeloma cell. The hybridoma can be cultured and antibody harvested from the culture supernatant, or it can be obtained from the ascites fluid of a host animal bearing a hybridoma tumor.

The monoclonal antibodies produced by the hybridoma are very specific for a particular antigenic determinant. However, monoclonal antibodies do not readily cross-link when binding to antigen. Cross-linking is important in diagnostic tests, which depend on precipitation reactions, such as the gel-diffusion tests. Also, if two or more antigens share a common determinant recognized by the monoclonal antibody, cross-reactions can occur.

The above is a rather simplistic description of monoclonal antibody production. More complete descriptions of production or use can be found in many publications, including articles by Løvborg (1984), Sherman (1984), and van Hell and Helmich (1984). Monoclonal antibodies

have been produced for staphylococcal enterotoxins (Edwin et al. 1984; Meyer et al. 1984; Thompson, Ketterhagen, and Bergdoll 1984).

Although the biological activity of the enterotoxins is not measured by these tests, Bergdoll (1970) stated that correlation between the two is adequate to justify the use of serological tests to analyze for enterotoxins. Chang and Dickie (1971) interpreted their observations with enterotoxin B to mean that antigenic sites and toxin sites are probably not the same. They did not speculate on whether this would invalidate the serological tests as methods for the analysis of the enterotoxins. Other researchers reported that heating in gelatin caused both immunological inactivation and biological inactivation of SEB, indicating a possible relationship (Notermans et al. 1983).

CONTROL OF STAPHYLOCOCCAL INTOXICATION. It should be relatively simple and easy to prevent staphylococcal intoxication. The obvious control measure would be to keep the organisms out of foods. For those *S. aureus* that do invade food, methods to destroy them or to prevent their growth and enterotoxin production can be employed. If toxin is produced, then destruction of the toxin is needed.

Prevent Contamination. It is impossible to keep all foods free from *S. aureus* due to the ubiquitous character of the organisms. However, it would seem possible to use common sense measures to keep the contamination low.

Humans are the main reservoir of the organism. The health, hygiene, and work habits of food handlers can influence the level of contamination of foods. People who are sick should not be allowed to handle food. They contaminate the food not only with *S. aureus*, but also with other disease organisms. People with colds, sinusitis, or other respiratory disturbances are good sources of *S. aureus*, as are boils, pimples, acne, and infected cuts. It is not possible to segregate all carriers of *S. aureus*, since a sizable portion of the population may be involved. However, those people who are obviously contaminators of food should be removed from food-handling areas.

S. aureus has been isolated from knives and slicers during investigations of outbreaks of food poisoning. Food-handling equipment should be designed so that it can be properly and effectively cleaned and sanitized. The cleaning and sanitizing of equipment are discussed in Chapter 9.

Prevent Growth and Toxin Production. The main control of *S. aureus* is to hold the food at a temperature unsatisfactory for growth. There is little or no growth below 4°C or above 46°C.

Toxin production occurs in a narrower temperature range than cellular growth. Troller (1976) suggested a range of 10° to 45°C. Greater

amounts of enterotoxin are produced in the range of 33° to 38°C than at either higher or lower temperatures.

Obviously, to warm the product for serving, or to cool leftover heated food, the temperature of the food must go through the growth range. This range must be traversed as quickly as possible. Three hours has been suggested as the maximum time allowed. To facilitate cooling of leftover food, the food should be placed in shallow layers in shallow pans. The deeper the food, the longer it will take to cool. The warm food should not be held at room temperature to cool, but should be placed in the refrigerator. This means that the refrigeration system must be adequate to cool these potential extra loads. The refrigerator cannot be jammed full of hot food, since no refrigerator can handle such a condition.

In fermented sausage products, *S. aureus* can be controlled by lowering the pH with starter cultures or chemical acidulation (Daly et al. 1973; Raccach 1981). A combination of a culture and acidulation or chemical inhibitor is more effective than either treatment alone. Adding solutes or drying of foods to lower the a_w below 0.85 should prevent growth and enterotoxin production of *S. aureus*.

Dipping poultry carcasses in a 5 percent potassium sorbate solution reduced the growth of inoculated *S. aureus* (To and Robach 1980). In model systems, concentrations above 3 percent were needed to inhibit 50 percent growth of *S. aureus* (Lahellec, Fung, and Cunningham 1981). Butylated hydroxyanisole (BHA) was reported to be lethal to *S. aureus* at 100 μg/ml (Degré and Sylvestre 1983). Although Parada, Chirife, and Magrini (1982) found some growth inhibition of *S. aureus* in a model system by 100 μg/ml of either BHA or butylated hydroxytoluene, neither was effective in delaying growth in a cheese spread (a_w of 0.976).

Destroy S. aureus, *Inactivate Toxin.* Heating is the principal method used to kill *S. aureus* cells as well as other organisms. Although heat may kill the cells, if enterotoxin has been produced, the toxin may persist, because, in a food, the enterotoxin is more heat stable than the organism.

Not only is the temperature important, the time of exposure is also important. The temperature and time needed to kill an organism depend on factors such as the heat resistance of the organism, the number of cells present, the age of the cells, the type of food or suspending medium, the ingredients added to the food, the pH, a_w, and previous treatment of the food or organism (stress conditions). Thermal processing is described elsewhere in the text, so only a brief preview is given here.

To help describe the heat resistance of an organism, D and z values are used. The D value is used to describe the time-temperature relationship to the killing of an organism. A 1-D is the time needed at a specified temperature to reduce the number of cells by 90 percent, or, conversely,

10 percent of the cells survive this treatment. The z value is a measure of the effect of a change in temperature on the resistance of an organism. It is the °F or °C required for the thermal resistance to change by a factor of 10. Most of the thermal evaluations are reported in °F; however, when possible, the °F have been converted to °C.

Angelotti, Foter, and Lewis (1961) determined the *D* values of various organisms in foods. In custard heated at 60°C, the *D* value for *S. aureus* 196E was 7.82 min, while for *S. aureus* Ms 149 it was 7.68 min. This means that if 100 cells of *S. aureus* Ms 149 were present per gram of custard, heating for 7.68 min at 60°C would reduce this number to 10. In another 7.68 min, there would be only 1 cell per 10 gram and after another 7.68 min, only 0.1 of a cell per gram, or 1 cell per 10 grams of custard would be viable. The importance of aseptic methods to keep the original population as low as possible should be quite evident. In chicken à la king, the respective *D* values for *S. aureus* 196E and Ms 149 were 5.37 and 5.17 min. This indicates that the organisms are more susceptible to heat in chicken à la king than in custard, since the *D* values are lower for the organisms in chicken à la king. Other *D* values for *S. aureus* are listed in Chapter 12.

It is relatively easy to kill *S. aureus* by normal pasteurization or cooking procedures. The problem lies in recontamination of the heated food with *S. aureus*. Deboning poultry or slicing of ham after cooking furnishes opportunities for recontamination from the hands of food handlers. With the destruction or reduction of the indigenous flora during heating, the recontaminating cells of *S. aureus* are not competitively inhibited. If this recontaminated food is allowed to remain at temperatures for growth of *S. aureus*, enterotoxin production may occur, resulting in a potential outbreak of staphylococcal intoxication. To reduce the occurrence of staphylococcal intoxications, it is important that strict sanitation and hygiene be followed anywhere that food is handled.

Heating is not a practical means to eliminate enterotoxins from food, because of the heat stability of these agents (Fig. 6.5). Calcium hypochlorite at 50 to 200 μg/ml detoxified SEB-contaminated water (Meyer, Hinterberger, and Korte 1977). Using chlorine to detoxify food might not be acceptable, but it may be effective for surface decontamination.

Botulism

From 1899 through 1976, there were 1,875 reported cases of botulism in the United States. Since 1976, cases of botulism have been classified into four categories (foodborne, infant, wound, and undetermined). From 1977 through 1980, there were 170 cases of foodborne botulism, 171 cases of infant botulism, and 5 cases of wound botulism reported in

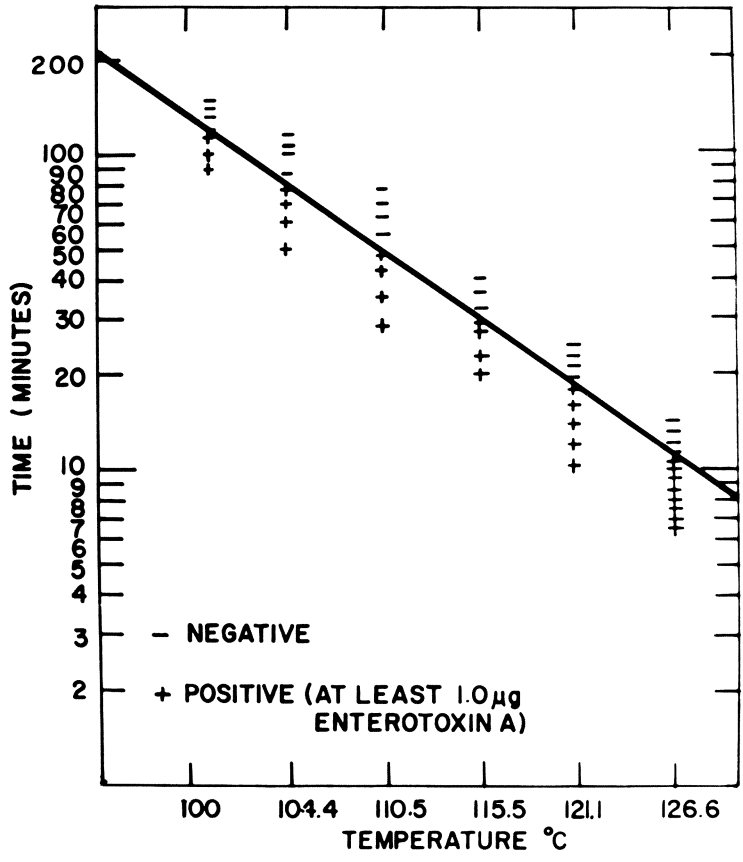

Figure 6.5. Thermal inactivation of staphylococcal enterotoxin. *Courtesy of Hilker et al. (1968).*

the United States (CDC 1979, 1981; Gunn and Terranova 1979; Morris and Hatheway 1980).

Although the number of cases of botulism is comparatively low, from 1899 to 1976 there were 992 deaths, for a case fatality rate of 52.9 percent. The case fatality rate for 1977 was reported as 6.3 percent (CDC 1979a) or 8.2 percent (Gunn and Terranova 1979). The case fatality rate was 5.2 percent, 0, and 16.7 percent for 1978, 1979, and 1980, respectively. This rate has decreased considerably since 1950 (Fig. 6.6). The average outbreak involves only two or three people. The largest outbreak reported in the United States occurred in Michigan in 1977. This outbreak involved 59 people, but there were no deaths. Home-processed food was the source of the toxin.

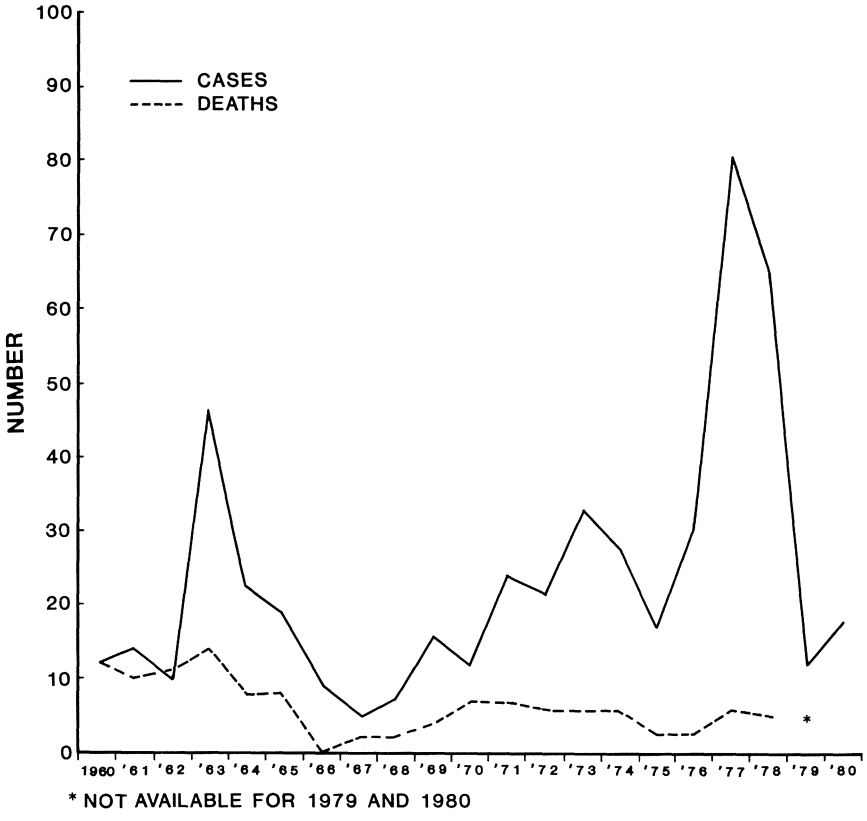

Figure 6.6 Foodborne botulism—death-to-case ratios, by 10-year periods, 1899–1973.
Courtesy of CDC.

ETIOLOGIC AGENT. Botulism is a neuroparalytic intoxication caused by neurotoxins produced by toxigenic strains of *Clostridium botulinum*. Eight antigenically distinct toxin types (A, B, C_1, C_2, D, E, F, and G) are recognized. Sometimes types C_1 and C_2 are grouped as C, but they are serologically distinct. Types A, B, E, and F cause botulism in humans. Although types C and D have been reported as causing illness in humans, the involvement is rare and uneventful. Types C and D affect animals. Type C is especially involved with birds and type D with cattle. Swine are moderately to highly resistant to most of the toxins.

The number of outbreaks caused by types A, B, E, and F in the United States from 1899 to 1976 is shown in Figure 6.7. Type A toxin is the main cause of botulism in the United States, followed by types B and E. Only one outbreak of foodborne illness has been attributed to type F toxin.

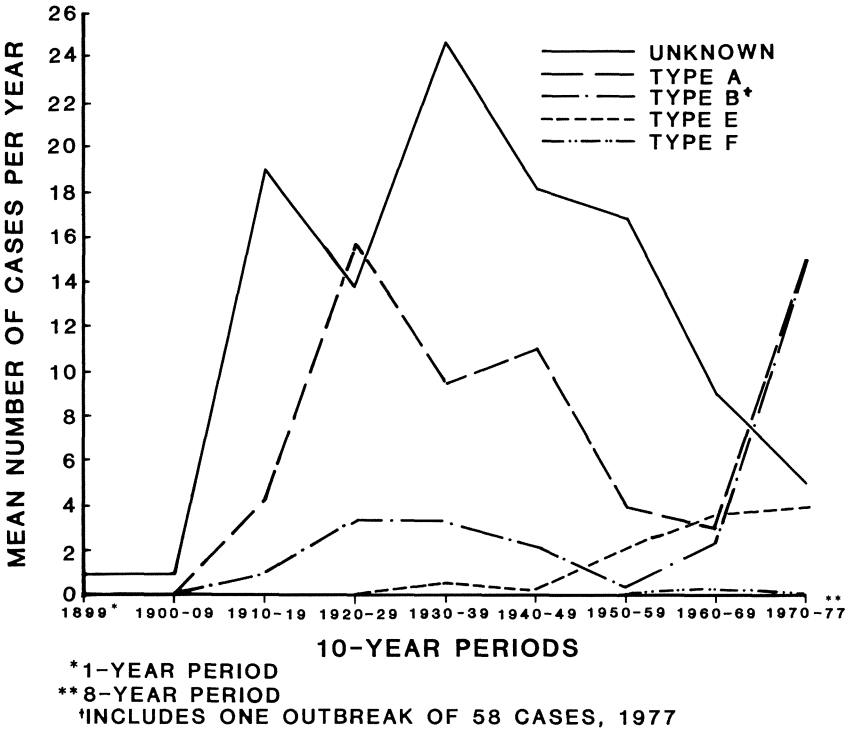

Figure 6.7. Number of cases of botulism, by type of toxin.
Courtesy of CDC.

No human cases of type G have been reported in the United States, but the organism has been isolated from necropsy samples of four adults and one infant in Switzerland (Sonnabend et al. 1981). Type G toxin was demonstrated in the serum of three of these cases.

Most of the type A outbreaks occur in the western states, while type B is predominant in the east and central United States. This correlates with the finding that type A spores predominate in the soil in the west, while type B spores predominate in the east and central areas. In Europe, type B botulism is the predominant type, which correlates with the occurrence of type B spores in the soil. Although not prominent in the contiguous forty-eight states of the United States, type E botulism has accounted for more than 40 percent of the outbreaks in Japan, Canada, and the Scandanavian countries, and for almost all of the cases in Alaska.

It is generally agreed that for foodborne botulism, the *C. botulinum* grows in the food and produces the neurotoxin, which is then ingested orally. However, for wound botulism, the organism produces the toxin

in the infected wound, in vivo. Infant botulism apparently results from the ingestion of the organism with subsequent intraintestinal production of the toxin. Hence, there may be an unusual circumstance in which the toxin might be formed in the intestinal tract of adults after ingestion of the organism or its spores.

During the logarithmic growth phase, the toxin accumulates intracellularly, with small amounts found extracellularly. Beyond the logarithmic phase of growth, by cell lysis, the toxin is released to the extracellular medium.

Properties of the Toxins. The toxins are simple proteins that are water soluble, heat sensitive, and acid stable. On a molar basis, the toxins produced by *C. botulinum* are the most lethal natural products known. Since the toxins are protein, they are antigenic. Various molecular weights ranging up to 900,000 have been reported. According to Sugiyama (1980), these large compounds are complexes of toxins and nontoxic material. The molecular weights range from 128,000 to 170,000 depending upon the type of toxin and culture. The toxins are dichains composed of two single-chain polypeptides of different length combined by a sulfide linkage. Proteolytic organisms form these dichains naturally. For nonproteolytic strains, trypsinization of the toxin components produces similarly linked dichains of the peptides. This has been called "activation." The presence of two dissimilar components was shown for type C_2 toxin (Iwasaki, Ohishi, and Sakaguchi 1980; Ohishi 1983b). Neither component manifested the original toxicity, but when mixed together and trypsinized, the original toxicity was restored.

Action of the Toxins. Except for wound or infant botulism, the preformed toxin is considered to be ingested orally. To react with and affect the nerves, the toxin must traverse the barrier of the gastrointestinal tract and be transported to the susceptible nerves. The action on the nerves causes neuromuscular blockage, paralyzing the muscles.

The type of food in the GI tract can affect the toxin and its action. The food might protect the toxins from the enzymes or other destructive actions, such as by stomach acids. Food increases or decreases the secretion of digestive juices. The food might combine with the toxin, forming larger particles less able to penetrate the intestinal wall. Foods may affect the rate of peristalsis, which increases or decreases the time the toxin is in an area of the intestine affording the greatest opportunity for penetration of the wall.

The major site for absorption of the toxin is the small intestine (Bonventre 1979; Ohishi 1983a). Perhaps only a small amount of the ingested toxin is absorbed, but Bonventre (1979) estimated that about 10^{11} molecules of toxin reaching the peripheral nerve endings is sufficient to cause

symptoms of botulism in adults. Also, the large intestine may be an absorption site, since, according to Lamanna (1968), intrarectal instillation of toxin into monkeys and rabbits caused death, although the animals lived longer than when the toxin was given orally. The different toxins are absorbed at different rates (Sugii, Ohishi, and Sakaguchi 1977).

That the toxin is "wasted" in the alimentary tract is evident from work with animals. When type C toxin is administered to mink intraperitoneally, only 50 MLD (mouse lethal dose) causes death. Oral administration of 4,000 MLD causes death, but with 2,500 MLD, the mink survive. With pigs, the oral to intravenous ratio of amount of toxin needed to cause death is 16,700:1 (Smith, Davis, and Libke 1971). Data from various outbreaks indicate that not all of the toxin penetrates the intestinal wall, and that it can be present in the intestines for a prolonged period.

The toxin passes with the lymph through the thoracic duct and is dumped into the bloodstream. It is carried by the vascular system to the nerves. Lamanna (1968) reported that when the lymph system was prevented from emptying into the bloodstream, animals fed botulinum toxin did not develop symptoms of botulism.

At the nerves, the toxin attaches to the presynaptic terminals of cholinergic nerves, and interferes with the release of acetylcholine at the myoneural junctions (Sellin 1981; Sugiyama 1980). The failure of impulses to be transmitted across the nerve fiber junctions results in paralysis of the muscles, which are controlled by the nerves.

CHARACTERISTICS OF THE INTOXICATION. Since the toxins affect the nerves, this illness differs from most other foodborne illnesses.

Incubation Period. After food contaminated with botulinum toxin is ingested, the usual time for symptoms or signs of botulism to appear is 12 to 48 hr (CDC 1983b). However, this time may vary from 2 hr to 8 days, depending upon the amount of toxin ingested, the type of toxin, the resistance of the individual, and perhaps even the type of food. Generally, if the incubation period is less than 24 hr, the person will be more severely affected, and death is more likely than if the incubation period is longer.

Attack Rates. Although botulinum toxin is the most potent natural poison known, not everyone who consumes contaminated food will acquire symptoms or succumb to the toxin. This has been related to the amount of contaminated food ingested. However, there have been reports that merely tasting a food and spitting it out resulted in intoxication and death. This led to the supposition that toxin could be absorbed directly from buccal exposure. The contention of Lamanna, Hillowalla, and Al-

ling (1967) is that the normal swallowing reflex transfers the toxin from the mouth to the GI tract.

Symptoms and Signs. The cardinal features of botulism are as follows:

1. Fever is absent but may develop if a complicating infection occurs.
2. Mental status is normal. Patients may be anxious or agitated and some are unusually drowsy; however, in the absence of secondary complications, patients are responsive.
3. The pulse rate is normal or slow, but tachycardia may occur if hypotension develops.
4. Although vision may be blurred, numbness and paresthesias are absent, and a sensory deficit does not occur.
5. Neurological manifestations are symmetrical.

The symptoms and signs from cases reported to the Centers for Disease Control are listed in Table 6.8. The first symptoms of illness are gastrointestinal. Nausea or vomiting, substernal burning or pain, abdominal distension, and decreased bowel sounds may occur. Some cases have initial transitory diarrhea but subsequently become constipated. These symptoms and signs may mislead physicians to diagnose the illness as appendicitis, bowel obstruction, or even diaphragmatic myocardial infarction. The mucous membranes of the mouth, tongue, and pharynx may be red, dry, and painful, which might be diagnosed as pharyngitis.

Sometimes other illnesses are diagnosed as botulism. Investigation of 438 suspect botulism outbreaks by CDC (1974a) revealed that only 75 were actually botulism, and the rest were due to other disorders.

The neural symptoms usually begin with the eyes and face, and paralysis progresses downward to the throat, chest, and extremities. When the diaphragm and chest muscles become fully involved, respiration is not possible. Death is due to asphyxiation. This usually occurs in three to six days. In nonfatal cases, complete recovery may take several months. Mental processes are usually clear during the illness.

Therapy and Duration. Prompt clinical, epidemiological, and laboratory efforts are required because each hour is critical to the survival of a patient with botulism. The patient should be hospitalized for treatment.

The treatment of botulism can be separated into three parts: removing unabsorbed toxin from the alimentary tract, neutralizing toxin with antitoxin, and treating the symptoms, such as respiratory distress.

The sooner the antitoxin is administered, the better the chances for recovery. The reduction of the mortality rate in recent years has been attributed to the prompt treatment of the patients and administration of antitoxin. In some cases, guanidine hydrochloride is used as an adjunct.

TABLE 6.8. SYMPTOMS OR SIGNS OF BOTULISM

	Type A	Type B	Type E	Type F	Und[a]	Total	
Outbreaks	34	15	10	1	44	104	
Cases	97	46	36	3	90	272	
	Number of Outbreaks						% with Symptom or Sign
Symptoms							
1. Blurred vision, diplopia, photophobia (double vision)	31	13	9	1	40	94	90.4
2. Dysphagia (difficulty in swallowing)	27	14	3		35	79	76.0
3. General weakness	22	12	4		22	60	57.7
4. Nausea and/or vomiting	15	13	10	1	19	58	55.8
5. Dysphonia (confused speech)	25	8	5		19	57	54.8
6. Dizziness or vertigo	8	4	5		15	32	30.8
7. Abdominal pain, cramps, fullness	5	6	3		7	21	20.2
8. Diarrhea	5	6			5	16	15.4
9. Urinary retention or incontinence	2	2	1		2	7	6.7
10. Sore throat	4	2	1			7	6.7
11. Constipation		2		1	3	6	5.8
12. Paresthesia	1					1	1.0
Signs							
1. Respiratory impairment	32	7	7		30	76	73.1
2. Specific muscle weakness or paralysis	23	9	3		13	48	46.2
3. Eye muscle involvement, including ptosis	16	9	3	1	17	46	44.2
4. Dry throat, mouth or tongue	7	6	2		7	22	21.2
5. Dilated, fixed pupils	3	4	2		8	16	15.4
6. Ataxia	3	1		1	4	9	8.7
7. Postural hypotension			1		2	3	2.9
8. Nystagmus	1		1		1	3	1.0
9. Somnolence		1				1	1.0

[a]Und = toxin type undetermined or unspecified.
SOURCE: Data from CDC (1974a).

This chemical seems to compensate for neural effects of the toxin. The use of germine in combination with guanidine is beneficial in the treatment (Cherington, Soyer, and Greenberg 1972).

The treatment of symptoms is especially important with respiratory difficulty or failure. Tracheotomy of the patient is used to assist in breathing. In a severe case, a mechanical respirator may be needed to maintain breathing.

With very little exposure to the toxin, the patient may remain asymptomatic or develop symptoms with little distress that pass in a few days. With increased amounts of toxin and development of respiratory failure, so that a respirator is needed, it may take several months for the patient to recover. Partial paralysis may persist for six to eight months (Bryan 1973).

One reason given for the lower mortality rate in Europe compared to the United States is that type B is prevalent in Europe and type A in the United States. Type A toxin is known to bind rapidly to tissue. In one case, 24 hr between ingestion and treatment was sufficient for the type A toxin to adhere irreversibly to the patient's myoneural tissue (Dillon et al. 1969). The antitoxin reacts with free toxin. When the toxin is irreversibly bound, administration of antitoxin has little effect on the recovery of the patient. Muscles injected with type A toxin were paralyzed up to seven days, while type B caused paralysis for less than five days (Sellin, Thesleff, and Dasgupta 1983).

Although immunization of people could be accomplished, the rare occurrence of botulism makes widespread immunization impractical. Immunization is recommended for laboratory or other personnel who are exposed to the toxin.

FOODS AS VEHICLES OF THE TOXIN. Usually botulism is associated with foods that have been given a preservation treatment, stored for some time, and consumed without appropriate heating. The preservation treatment in these foods is inadequate to destroy the spores that were present in the food.

The foods involved in botulism outbreaks in which the toxin type was determined are listed in Table 6.9. In more than 54 percent of the outbreaks, vegetables were the vehicle of the toxin. Fish, fruit, and condiments were other important vehicles, whereas meat, poultry, and dairy products have been involved rarely in botulism. Home-processed foods accounted for the majority of the outbreaks (72 percent), while commercially processed foods were involved in less than 10 percent of the outbreaks. Unknown vehicles accounted for slightly less than 20 percent of the outbreaks (Table 6.10). Although commercially processed foods have been involved in fewer outbreaks than home-processed foods, the inci-

TABLE 6.9. FOODS INVOLVED IN BOTULISM OUTBREAKS IN WHICH THE TYPE OF TOXIN WAS DETERMINED, 1899–1977

	Botulinum Toxin Type						
Food Group	A	B	E	F	A + B	Total	%
Vegetables	115	31	1	—	2	149	54.4
Fishery products	11	4	25	—	—	40	14.6
Fruit	22	7	—	—	—	29	10.6
Condiments	17	5	—	—	—	22	8.0
Beef	6	1	—	1	—	8	2.9
Dairy products	3	2	—	—	—	5	1.8
Pork	2	1	—	—	—	3	1.1
Poultry	2	2	—	—	—	4	1.5
Other	8	3	3	—	—	14	5.1
TOTAL	186	56	29	1	2		
%	67.9	20.4	10.6	0.3	0.7		

SOURCE: CDC (1979b).

dents which involve commercial foods are publicized, while those caused by home-packed foods rarely are mentioned. The reason is the widespread distribution of the commercial packs. If 500 cans of home-packed corn are contaminated, chances are only one of these will be involved in botulism in one family. If 500 cans of commercially canned corn are contaminated, they could cause botulism in 500 families. So far, this has not happened in the United States.

The outbreaks of botulism due to commercially packed foods since 1960 caused some health "experts" to suggest that families should process food at home. Data certainly do not support the theory that home canning reduces the number of botulism outbreaks.

THE ORGANISM. The genus *Clostridium* is divided into four groups according to spore formation and gelatin liquefaction. *Clostridium botulinum* is in Group II. In this group, the spores are terminal and gelatin is hydrolyzed.

The species includes a heterogeneous group of strains that are divided into types A through G based on the antigenic neurotoxins that are produced. These seven types are divided into four groups according to their deoxyribonucleic acid homologies and biochemical, physiological, and serological characteristics. The members of these groups are as follows: group 1, type A and proteolytic strains of types B, C, D, and F; group II, type E and nonproteolytic strains of types B and F; group III, nonproteolytic strains of types C and D; and group IV, type G (Buchanan and Gibbons 1974).

The optimum temperatures for growth are 30° to 40°C for group 1, 25° to 37°C for group II and 30° to 37°C for groups III and IV

TABLE 6.10. OUTBREAKS OF FOODBORNE BOTULISM ATTRIBUTED TO COMMERCIALLY PROCESSED OR HOME-PROCESSED FOODS, 1899–1977

Source of Food	1899	1900–1909	1910–1919	1920–1929	1930–1939	1940–1949	1950–1959	1960–1969	1970–1977	Total	%
Home processed	1	1	48	77	135	120	50	42	74	548	71.6
Commercially processed	0	1	14	26	6	1	2	10	6	66	8.6
Unknown	0	0	8	13	13	13	51	26	28	152	19.8
TOTAL	1	2	70	116	154	134	103	78	108	766	

SOURCE: CDC (1979b).

organisms. Group I organisms digest milk, group II coagulate milk with a soft curd but do not digest it, group III organisms do not change milk, and those in group IV digest milk slowly. Other differences and similarities were described by Lynt, Kautter, and Solomon (1982).

Toxigenic Strains. Apparently most strains of *C. botulinum* are capable of producing a toxin; however, this ability may be lost due to laboratory cultivation, and nontoxigenic clostridia have been isolated from both marine and terrestrial environments.

Inoue and Iida (1970) suggested that production of toxin by *C. botulinum* is influenced by prophages in the cells. By curing the cells of the phage, some toxigenic strains lose the ability to produce toxin. With reinfection by phage, the cells regain toxigenicity. It has been established that the phage type can determine the toxin type that is produced and not all lysogenic phages will confer toxigenicity on the infected cells (Hariharan and Mitchell 1976; Oguma, Iida, and Inoue 1975). Most of the nontoxigenic strains that were converted to the toxigenic state reportedly were not stable to serial transfer through cooked meat medium (Oguma 1976).

Some strains of *C. botulinum* cured of their prophages by researchers remained toxigenic (Eklund et al. 1970). The researchers suggested that the organisms might carry more than one prophage, or alternatively that not all *C. botulinum* toxins are induced by bacteriophage.

The studies of phage inducement of toxin formation primarily have used type C and D strains. The relationship of phage to toxigenicity of strains causing human botulism has not been evaluated sufficiently.

Sources. The organism is widely distributed in nature. It occurs in soil apparently throughout the world. The type of *C. botulinum* depends upon the locality. The factors that affect the distribution of the different toxin types are not known. Healthy animals may be carriers of types of organisms to which they are immune, thus serving as reservoirs for replenishing the organisms in the soil and spreading the organisms to various places.

Bottom sediments of marshes, lakes, and coastal ocean waters contain *C. botulinum*. The primary type in marine environments in northern areas appears to be type E; however, other toxin types of the organisms also are found. Type E is common in the intestine of fish taken from certain waters. The organism has been isolated from the environment of trout farms and from trout grown in these establishments (Cann and Taylor 1982) and in a salmon hatchery (Eklund et al. 1982). The organism does not multiply in living fish, but it does multiply in aquatic vegetation and bottom deposits. In marshes, the growth of algae reduces the oxygen content of the water to a level that allows *C. botulinum* to multiply.

Since the organism is widely distributed in nature, it is evident that food, especially vegetables and fish, can be contaminated during production, harvesting, or processing. *C. botulinum* has been reported in commercial honey (Huhtanen et al. 1981; Kautter et al. 1982; Midura et al. 1979), corn syrup (Kautter et al. 1982) and liver sausage (Hauschild and Hilsheimer 1983). Fortunately, if present, the spores are usually in low numbers, and they can germinate only if the conditions of microenvironments of the food are favorable. After germination, sufficient time is needed for growth and toxin production.

GROWTH AND TOXIN PRODUCTION. The foods involved in outbreaks of botulism obviously present an environment or microenvironments that allow the growth of *C. botulinum*.

Various minimum and maximum temperatures have been reported for growth and toxin production. Eklund, Wieler, and Poysky (1967) reported growth and toxin production of a strain of type B *C. botulinum* at 3.3°C. The maximum temperature of growth for types A and B is about 48°C. Type E can grow in a range from 3.3° to 45°C. Growth and toxin production of type E is less pronounced at 37°C than at either 30° or 22°C (Ajmal 1968). A higher number of cells developed at 30° than at 22°C, but no difference was found for toxin production, indicating that the amount of toxin produced per cell was greater at 22° than at 30°C. When incubated at 4°C, growth and toxin production occurred in four to five weeks.

Type A (14 strains) and proteolytic type B (13 strains) did not multiply, but decreased in number during holding in cooked meat medium at 10°C for eighteen weeks (Smelt and Haas 1978). At 12°C some strains formed toxin in three to four weeks. At 4°C, nonproteolytic type B and type F cultures grew in tryptone glucose yeast extract broth, but not in crabmeat (Solomon, Kautter, and Lynt 1982). For growth in crabmeat, a temperature of 12°C was needed.

At pH 4.8, the growth of *C. botulinum* was detected in only one food (pineapple rice pudding) of the many tested (Townsend, Yee, and Mercer 1954). No growth was found at any pH below 4.8. Several other reports have confirmed that *C. botulinum* fails to grow below pH 4.8 (Baird-Parker and Freame 1967; Huhtanen et al. 1976; Ito et al. 1976; Segner, Schmidt, and Boltz 1971). These and similar reports have been the basis for the FDA setting an upper limit of pH 4.6 for acid foods. These require less heat treatment in the canning process than do those foods over pH 4.6. However, growth and toxin production have been detected at pH 4.3 to 4.6 when media with high amounts of protein in suspension were used (Smelt et al. 1982; Tanaka 1982).

In some outbreaks of botulism, the pH values of the foods involved

were below that regarded as necessary for growth and toxin production. One explanation is that there are pH gradients in some foods so that some microenvironments are acceptable for growth and toxin production. Another possibility is that the growth of other organisms can result in acceptable microenvironments (Anderson 1984; Montville 1982; Odlaug and Pflug 1979).

The effect of salt on the growth and toxin production varies with the types and strains of *C. botulinum*. An aqueous phase level of 5.5 percent salt in homogenized cod was sporostatic to two strains of *C. botulinum* type E (Boyd and Southcott 1971). In lumpfish caviar, no toxin was formed at water phase salt concentrations of 5.56 percent or greater (Hauschild and Hilsheimer 1979).

C. botulinum is an anaerobic organism. The clostridia lack cytochromes, cytochrome oxidase, catalase, and peridoxase, so the oxidation-reduction potential (Eh) must be low for growth to occur (Lund and Wyatt 1984). *C. botulinum* type E inoculated into meat and fish produced toxin under both aerobic and anaerobic conditions on incubation (Ajmal 1968). There are microenvironments in these foods which have low oxidation-reduction potentials favorable for the growth of this organism.

Sugiyama and Yang (1975) inoculated spores of *C. botulinum* into fresh mushrooms, which were then packaged and stored at 20°C. The respiration of the mushrooms removed the free oxygen and allowed the spores to germinate and the cells to produce toxin in three to four days. The mushrooms appeared to be edible. No toxin was detected in mushrooms held at 4°C.

Since vacuum packaging in gas-impermeable plastic produces anaerobic conditions, it was reasoned that this practice might encourage the growth of *C. botulinum* and result in toxic foods. The rate of toxin production is higher in vacuum-packaged fish than in fish packaged without vacuum (Pace and Krumbiegel 1983). The slowest rate of toxin production is in unpackaged products. Research on vacuum packaging of raw fish has revealed that when the fish are stored at 10°C or less, spoilage is evident before toxin is detected (Eyles and Warth 1981). This safety margin is reduced if storage is at 20°C. Flushing the package with CO_2 may increase the risk of botulism, since CO_2 tends to inhibit spoilage organisms but enhance the germination of *C. botulinum* spores and cellular growth. However, 100 percent CO_2 delays toxin production (Doyle 1983). Vacuum-packaged and modified atmosphere-packaged foods must be stored below 4°C to assure safety (Notermans, Dufrenne, and Keijbets 1981; Stier et al. 1981).

The interaction of microorganisms is both stimulatory and inhibitory to growth of *C. botulinum*. The growth of other organisms tends to reduce the Eh of the growth medium, which may then allow growth of *C. botuli-*

num. Some organisms, such as yeasts, may produce growth factors favorable for *C. botulinum*.

The growth of *C. botulinum* in foods can cause a foul, putrid odor which should serve as a warning to the consumer. In many of the outbreaks of botulism, either a patient or an asymptomatic participant stated that the food had an off-odor or flavor, but these warning signs were ignored. Some people have a high tolerance to off-flavors or off-odors. In some foods (smoked, spiced, fermented), slight off-flavors or off-odors are difficult to detect.

There are reports that toxin is present in spores (spore-bound toxin) of *C. botulinum* (Booth et al. 1972; Yamakawa, Nishida, and Nakamura 1983). Phagocytosis releases the spore toxin, which has the same lethal effect as other free toxins.

METHODOLOGY. The methodology of *C. botulinum* involves the detection and enumeration of the organism, the characterization of the organism, or qualitative or quantitative determinations of the neurotoxins.

Detection and Enumeration. Due to the hazards involved, before working with *C. botulinum*, a person should be protected by suitable toxoids.

The demonstration of botulinum toxin in a food implicated in a botulism outbreak is often all that is done. However, the detection and isolation of *C. botulinum* from the food furnishes confirmatory evidence. Also, methods of detection, isolation, identification, and enumeration are basic in determining the distribution of the orgamism in nature.

A system for detecting the organism is described in the *Bacteriological Analytical Manual* (FDA 1978). The medium for detection of *C. botulinum* is anaerobic egg yolk agar or liver veal-egg yolk agar.

Characterization. There is a widely held view that no organism isolated from a suspected case of botulism should be identified as *C. botulinum* unless it is toxigenic. Therefore, the outstanding characteristic of a strain of *C. botulinum* is the toxin it produces.

Once an organism has been isolated and found to be a Gram-positive rod that forms spores and is anaerobic, other tests can be made. These include digestion of gelatin, milk, and meat; fermentation of various carbohydrates; indole production; hemolysis; and lecithinase and lipase activity. Various biochemical reactions of *C. botulinum* types are listed in Table 6.11. However, there may be disagreement on some of these reactions. Others are listed by Dezfulian and Dowell (1980).

It is evident from this information that biochemical tests can be considered reliable only to a limited extent in determining or distinguishing *C. botulinum*. This is because different toxin types of *C. botulinum*, or even strains within toxin types, represent quite distinct metabolic groups.

TABLE 6.11. REACTIONS OF *CLOSTRIDIUM BOTULINUM*

	Proteolytic Types A, B, C, D, F	Non-Proteolytic Types B, C, D, E, F	G
Digestion of			
Gelatin	+[a]	+	+
Casein	+	−	+
Meat	+	−	N
Acids produced from			
Glucose	+	+	−
Fructose	+	+	−
Mannose	−	+	−
Lactose	−	−	−
Salicin	V	−	−
Other tests			
Indole	−	−	−
Urease	−	−	−
Nitrates reduced	−	−	−
Lipase	+	+	−
Hemolysis	+	+	+

[a] + = positive reaction; − = no reaction; V = variable reactions; N = insufficient data.

Serological Reactions. Besides the toxins, the organism has three sources of antigenic substances: the flagellar antigens (H), the somatic antigens (O), and the spore antigens. The flagellar antigens show a narrow range of specificity, and the somatic antigens are often common to all members of a species or type. The spore antigens are type specific and are dissimilar to either the flagellar or somatic antigens of the vegetative cell (Solomon et al. 1969).

Toxin Detection. The *Bacteriological Analytical Manual* (FDA 1978) describes a procedure for detection of preformed toxin in food and for detection of toxins produced by organisms in the food with an enrichment procedure.

An extract is analyzed for toxin by intraperitoneal injection of mice. The injected mice are observed over a period of 92 hr. Death indicates the presence of toxin. The mouse neutralization test for toxin has been accepted as official by the AOAC.

Serological techniques, as described for detection of staphylococcal enterotoxin, have been tested for the analysis of botulinum toxins. The ELISA system has been adapted for type A (Notermans, Dufrenne, and Van Schothorst 1978), type B (Kozaki et al. 1979), type E (Notermans, Dufrenne, and Kozaki 1979) and type G toxin (Lewis et al. 1981). Cross-reactivity and nonspecific reactivity are problems not only for the ELISA, but also for other serological systems for analyzing for botulinum toxins.

Dezfulian and Bartlett (1984) produced specific type A antitoxin by selective suppression of the immunological response of inoculated rabbits to unwanted antigens and subsequent immunization with a toxoid. Even then, there was some cross-reaction of the antitoxin with culture filtrates of type B organisms. Monoclonal antibodies to type C_1 and D toxins were produced and used in the ELISA to detect these toxins (Oguma et al. 1984).

CONTROL OF BOTULISM. For botulism to occur, the spores or cells of toxigenic *C. botulinum* must be present in the environment and gain access to the food; viable cells or spores must remain in the food after processing; and the food must have an environment favorable for germination and outgrowth of the spores and growth of the vegetative cells to produce the toxin; then the food must be eaten cold or with insufficient heating to destroy the toxin.

Therefore, the principal methods of controlling staphylococcal intoxication also are important in the control of botulism. Quite simply, we can prevent contamination of the food, prevent growth and toxin production, destroy the organisms or toxin, and not eat suspect foods.

Perhaps the best method of controlling botulism is to heat the food during processing to a temperature that will destroy the spores of toxigenic *C. botulinum*. However, there are many foods, such as cured meat products, that would suffer organoleptically if these heat treatments were used. For these foods, additives are needed that will inhibit toxin production by *C. botulinum* contaminants.

Prevent Contamination. Generally, *C. botulinum* gains access to food as a soilborne or dustborne contaminant. Vegetables are harvested from the soil or in close proximity to the soil. Washing of vegetables will remove some soil and associated organisms, but it is difficult to remove all organisms by this method.

Fish and marine animals feeding in waters that contain *C. botulinum* can be contaminated. Most of the contamination is in the intestines of fish. Proper cleaning and eviscerating procedures result in minimal contamination of the fish flesh.

High levels of contamination are more difficult to destroy than are low levels or, ideally, no spores. Therefore, sanitary practices are very important.

Prevent Growth and Toxin Production. Foods that do not have an environment conducive to the growth and toxin production of *C. botulinum* will not become toxic.

One of the simplest methods to inhibit the growth of microorganisms

is to hold the food at temperatures below which growth will occur. Since certain strains of *C. botulinum* can grow and produce toxins at 5°C or less, normal refrigerator temperatures cannot be relied upon to control toxin formation in foods. Freezing does not destroy the toxin or the spores of *C. botulinum*, but as long as the food is frozen, germination and growth of the vegetative cells with toxin production does not occur. Frozen foods should not be allowed to thaw and remain in that condition for extended periods. *C. botulinum* does not grow at an a_w of 0.93 or less. Foods dried below this level should not become toxic (there is the possibility that toxin could be formed prior to drying).

A chemical added to cured meats is sodium nitrite. This chemical is important in color fixation of cured meat, as well as in inhibiting growth and toxin formation by *C. botulinum*. Various other chemicals have been tested for inhibition of the organism. These chemicals and nitrites are discussed in Chapter 11.

During processing, the destruction of microbial competitors, the removal of air by heating (lowering the redox potential), and the destruction of cellular tissues with the release of cellular fluids, favors the growth of *C. botulinum*.

Destruction of C. botulinum. Since this organism produces spores, it is more difficult to destroy than is *S. aureus*. The method used to destroy the spores is thermal processing.

Most of the outbreaks of botulism have been caused by consumption of toxic heat-processed food. Thus, it may seem odd that heating is the only presently acceptable method of destroying the spores of *C. botulinum*. The spores of *C. botulinum* vary in their heat resistance. Type E spores are the least resistant and thus are most easily destroyed by heat.

Due to outbreaks of botulism from eating contaminated fish, it was recommended that during smoking, the fish be held at 82.2°C for 30 min. Pace and Krumbiegel (1973) examined commercially smoked fish and found that 0.9 percent to 2.0 percent contained C. *botulinum* type E spores, although the fish had been subjected to 82.2°C for 30 min.

Various results of heat processing are obtained undoubtedly because of the difference in heat resistance of various strains of type E spores. One group of researchers determined the thermal-death-time of five strains of type E spores in fish paste (Crisley et al. 1968). The *D* values in this medium at 80°C varied from an average of 1.6 min to 4.3 min for the five strains.

Boiling (100°C) is not recommended to destroy spores of type A or B, since several hours would be required. Processing of low-acid foods is accomplished at a temperature of 121.1°C. A 12*D* thermal process (see Chapter 12) is considered essential for low-acid canned foods. With this

treatment, the probability of survival of *C. botulinum* spores is very remote.

Of the clostridial spores, those of *C. botulinum* are included in the most radiation-resistant group. The spores vary in their resistance, with type F generally considered to be the most resistant and type E spores the least resistant, with types A and B intermediate.

A combination of radiation and heat has a synergistic effect on destruction of spores of *C. botulinum*. Spores that are exposed to low-level radiation are more susceptible to heat inactivation than untreated spores. The heat is also of value in halting enzyme action and inactivating any toxin that might be present. Thus, pasteurization levels of radiation followed by heating at values less than those presently required may be beneficial in controlling *C. botulinum*. Heating before radiation does not lower the radiation resistance, but radiation lowers the heat resistance of the spores.

Inactivation of Toxin. Radiation pasteurization does not inactivate preformed botulinum toxin, whereas heat pasteurization does. The heat stability of botulinum toxin is considerably less than that of staphylococcal enterotoxin. The types of botulinum toxin vary in their heat resistance. The time necessary to inactivate the toxins depends upon the amount of toxin present, the temperature, and the substrate in which they are heated. The toxins are more stable at acid pH levels and are most stable about pH 5.0. Various times and temperatures have been recommended for inactivating botulinum toxins (Bradshaw, Peeler, and Twedt 1979, 1981; Woodburn et al 1979).

Although the toxins are relatively heat labile, being inactivated in 2 min or less at 90°C, this heat treatment usually is not suggested for destroying toxins formed in canned foods. Bryan (1973) recommended that before serving, canned food should be heated to boiling (about 100°C) and held for 5 min to 15 min.

The resistance to chemicals depends upon the type of toxin, temperature, pH, and other substances in the medium. In water, the purified toxin is sensitive to chlorine, bromine, or iodine. Free available chlorine with a residual of 1 mg/L (1 ppm) will destroy at least 99.9 percent of all the types of botulinum toxin in 5 min or less. Type E toxin is the most resistant to destruction by chlorine.

Canned food that is not normal should not be consumed. In most outbreaks of botulism involving canned food, one or more people noted swelling of the can, bubbles in the food, off-odors, or turbid liquid. If the food is suspicious, it should be returned to the store, if purchased, or disposed of in a manner so that it is not consumed by animals or by other people.

Clostridium perfringens Foodborne Illness

This illness has been called a food poisoning, an intoxication, a foodborne illness, an infection, and an infective food poisoning. These different designations are undoubtedly due to the fact that the release of the toxin is different from that of *S. aureus* or *C. botulinum.* Large numbers of the organism are associated with the illness. This is true for infections; however, in this illness, a toxin is the causative agent.

The actual number of outbreaks and cases in the United States is not known. Due to the rather mild symptoms and short duration of the illness, medical help usually is not needed. Hence, most cases probably are not reported to the CDC. Table 6.12 shows the number of confirmed outbreaks and cases reported in recent years. The majority of the reported outbreaks are caused by food consumed in mass-feeding establishments. Outbreaks that occur in the home might not be reported because of the type of illness and the small number of people involved.

CHARACTERISTICS OF THE ILLNESS. The illness is the result of a sequence of events. The food is contaminated by the organism. During cooking, the vegetative cells and the heat-sensitive spores may be killed, but heat-resistant spores will survive. The heat experienced during cooking may activate the spores to germinate.

If the food is held at a temperature allowing growth (10° to 50°C), the vegetative cells will multiply. A mean generation time of 10 to 12 min was reported for the organism in cooked poultry at 37°C (Mead 1969). With this short generation time, the organism can increase 1,000-fold in 2 hr.

The exact number of *C. perfringens* needed to cause the illness has not been determined. However, over 10^6 to 10^8 cells per gram of food can be a potential health hazard. The organism must first pass through the low pH of the stomach to reach the intestines. Organisms in the early lag phase are least resistant, and resistance increases to a maximum at the end of the growth phase. It is well documented that the acid in the stom-

TABLE 6.12. OUTBREAKS AND CASES CAUSED BY *CLOSTRIDIUM PERFRINGENS*

Year	Outbreaks	Cases	Cases per Outbreak
1977	6	568	94
1978	9	617	68
1979	16	936	58
1980	25	1,463	58
1981	28	1,162	41
TOTAL	84	4,746	56

SOURCE: Data from CDC *Annual Summaries.*

ach combines with consumed proteins, causing an increase in the overall pH. This effect of protein on the stomach acidity, and resultant protection of the bacterial cells, may be the factor in the involvement of protein foods in outbreaks of the illness.

When the remaining organisms reach the small intestine, they find an environment acceptable for multiplication and sporulation. The sporulating cells produce the enterotoxin, which is released during cell lysis. The organisms may sporulate in some foods, but usually, by the time the food is toxic, it is not palatable (Craven, Blankenship, and McDonel 1981).

Incubation Period. The interval of time between the meal and onset of illness for an outbreak of *C. perfringens* food poisoning is shown in Figure 6.8. In this example, the range is 2 to 29 hr, with a median of 13 hr. CDC (1983b) listed the incubation period as 9 to 15 hr.

Attack Rate. Not everyone who eats the contaminated food becomes ill. The reasons for the escape of some people are similar to those described for staphylococcal intoxication and botulism. People with hyperacid stomachs may be protected, due to the death of *C. perfringens* cells. Previous contact or any immunity apparently has little effect, if any, on acquiring the illness when contaminated food is ingested.

The disease is not prevented in parenterally immunized animals that possess neutralizing serum antibody against the toxin (Niilo 1971). In a survey of human sera, a majority contained antibody against *C. perfringens* enterotoxin (Uemura et al. 1974). They speculated that antibody could be induced by the acute *C. perfringens* food poisoning or be due to the continuous presence of the organism and small amounts of toxin in asymptomatic people.

Symptoms. The symptoms reported in four outbreaks of *C. perfringens* illness are listed in Table 6.13. The two prominent symptoms of the illness are diarrhea and abdominal cramps. Nausea and headache were fairly important in two outbreaks but were of minor occurrence in the others. Fever, vomiting, dizziness, and bloody stools may occur, but they are rare.

In young people, the symptoms are usually mild, but in elderly, ill, or debilitated people, the illness can be severe. Although rare, death has occurred in cases complicated by other illness.

Duration and Therapy. The illness is usually of short duration, from less than 12 hr up to 24 hr. A few cases may persist for 48 hr. The illness is followed by a complete and uneventful recovery. Although weakness may be evident the day after symptoms begin, a person usually can return to work or normal activity. Due to the rather mild symptoms and short duration, usually no therapy is needed.

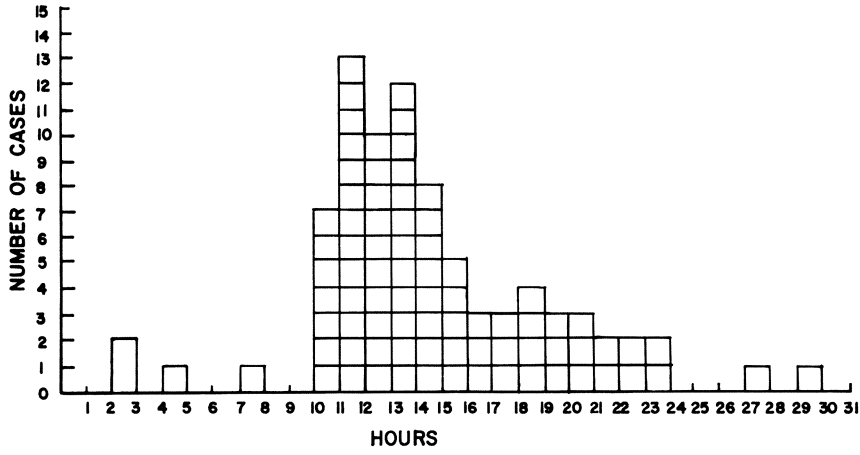

Figure 6.8. Typical outbreak of *Clostridium perfringens* foodborne illness—incubation period.
Courtesy of CDC.

Foods Involved. The foods involved in outbreaks of *C. perfringens* illness are generally protein-type foods that have been boiled or lightly roasted or meat and poultry stews, sauces, gravies, pies, casseroles, salads, and dressings (Table 6.14). Usually the incriminated food is cooked one or two days in advance, then refrigerated until reheated for serving.

When meat is cooked in bulk, the heat gain is slow, and subsequent cooling is slow. Heating lowers the oxygen content of the food, providing a more anaerobic environment for the clostridia to grow.

TABLE 6.13. *CLOSTRIDIUM PERFRINGENS* FOODBORNE ILLNESS: PERCENTAGE OF PEOPLE REPORTING SYMPTOMS IN FOUR OUTBREAKS

	Outbreak			
Symptom	1	2	3	4
Diarrhea	82	85	89	91
Abdominal cramps	75	70	86	72
Headache	40	4	—	44
Nausea	33	13	48	42
Fever	8	—	—	8
Bloody stools	7	—	—	1
Vomiting	6	—	16	11
Dizziness	—	2	—	—
Prostration	—	—	—	39
Chills	—	—	—	29

SOURCE: Data from CDC *Morbidity and Mortality Weekly Reports.*

TABLE 6.14. FOODS INVOLVED IN OUTBREAKS OF *CLOSTRIDIUM PERFRINGENS* FOODBORNE ILLNESS

Food	Number	%
Beef	19	41.3
Pork	2	4.3
Other meat	2	4.3
Poultry	8	17.4
Fishery products	2	4.3
Other, multiple, unknown	13	28.3
TOTAL	46	

SOURCE: Data from CDC *Annual Summaries.*

ETIOLOGIC AGENT. The etiologic agent for the illness is an enterotoxin. The enterotoxin protein is a structural part of the spore coat that is overproduced in some strains of *C. perfringens* (Frieben and Duncan 1973). Synthesis of the spore coat is an early event in sporulation (Labbe and Duncan 1977).

Two enterotoxigenic strains followed different patterns of production: one strain produced a high amount of toxin during 2.5 to 4 hr of growth, with the level stabilized at 5 hr; the other strain revealed a rapid increase of toxin after 4 hr, with a maximum level after 25 hr. Using the ELISA, researchers detected levels 1 ng/ml to 10 ng/ml of enterotoxin during vegetative growth even with an enterotoxin negative strain (Granum et al. 1984). They assumed that 1 ng/ml of the compound was needed for sporulation. The 12 μg and 18 μg/ml produced by the enterotoxigenic strains during sporulation was excess compound which happens to be an enterotoxin. Most methods will not detect 1 ng/ml of enterotoxin, but the ELISA is a very sensitive system (Bartholomew and Stringer 1984). This low level of enterotoxin (1 ng/ml) does not noticeably affect biological systems. Hence, it is necessary for sporulation of the cells to produce sufficient levels of toxin to cause enteric distress.

Properties of the Enterotoxin. The purified toxin is a protein. It is essentially free of nucleic acids, lipids, and reducing sugars (Hauschild 1971). The molecular weight is between 33,000 and 35,000, with an isoelectric point of pH 4.3. Being protein, the toxin is antigenic, and only one type has been detected, regardless of the producing strain of organism.

The heat needed to inactivate the enterotoxin is influenced by the substrate (type, pH). According to Naik and Duncan (1978), the biological activity is destroyed within 5 min at 60°C. However, other researchers reported that inactivating 12.5 μg/ml enterotoxin at 61°C required 23.8 min for gravy and 25.4 min for buffer, both at pH 6.1 (Bradshaw et al. 1982).

The toxin is inactivated by pronase, but not by trypsin, lipase, chymotrypsin, or papain (Duncan and Strong 1969; Hauschild and Hilsheimer 1971).

Enterotoxigenic Strains. On the basis of the production of four major exotoxins (alpha, beta, epsilon, and iota), strains of *C. perfringens* are divided into five groups, A to E (Table 6.15). This grouping makes it possible to correlate strains with illnesses of people and animals.

The main food-poisoning strains are included in type A, although type C strains were involved with a severe gastroenteritis in Germany shortly after World War II, and in other countries since then (Hauschild 1973). Skjelkvålé and Duncan (1975) isolated enterotoxin from type C strains and Uemura and Skjelkvålé (1976) reportedly isolated enterotoxin from a strain of type D. These enterotoxins are identical serologically, whether from type A, C, or D.

Not all strains of type A produce enterotoxin. Even organisms that have been isolated from foodborne outbreaks seem to lose the ability to produce the enterotoxin after transfer in laboratory media. Some organisms do not sporulate, and hence fail to produce the enterotoxin.

Strains of *C. perfringens* that fail to produce fluid accumulation in ligated rabbit ileum or overt diarrhea when injected into the rabbit ileum, do not cause illnesses when fed to human subjects (Strong, Duncan, and Perna 1971).

Action of the Enterotoxin. Hauschild (1971) stated that the enterotoxin causes increased capillary permeability, vasodilation, and excess fluid movement into the intestinal lumen, resulting in diarrhea. Also, there is an increase in intestinal motility. McDonel and Duncan (1977) reported that ileal loops responded to purified enterotoxin, with net secretion of fluid, sodium, and chloride, inhibition of glucose uptake, and substantial sloughing of epithelial cells. Damage to the epithelial layer of villus tips was evident, and even complete destruction of epithelial cells was observed (McDonel et al. 1978). The primary site of action of the enterotoxin is the brush border membrane of the villus tip epithelial cells (McDonel 1980). The enterotoxin is absorbed from the small intestine and can be detected in lymph or blood (Ohishi, Yamamoto, and Saka-

TABLE 6.15. TYPES OF *CLOSTRIDIUM PERFRINGENS*

Types	Major Toxins Produced
A	alpha
B	alpha, beta, epsilon
C	alpha, beta
D	alpha, epsilon
E	alpha, iota

guchi 1981). The action is greatest in the ileum, less in the jejunum and least in the duodenum.

Tests with ligated loops indicate that the toxin acts directly on the intestine, since fluid accumulates only in inoculated loops and not in adjacent control loops (see Fig. 6.9). There does not seem to be any invasion of the intestinal mucosa by the cells of *C. perfringens*, which is suggestive of a toxic action on the intestine.

THE ORGANISM. The cells in this species are anaerobic, Gram-positive, sporeforming, straight rods. They occur singly, in pairs and, rarely, in short chains, are not motile, produce capsules, and liquify gelatin. This species ferments many of the common carbohydrates. Acid is produced in litmus milk (fermentation of lactose), but the clot is broken up by the production of large amounts of gas, resulting in a stormy fermentation.

Sources. C. perfringens has been called ubiquitous, due to its widespread distribution in nature. It is found in soil, dust, air, water, sewage, human and animal feces, and on many food products.

The presence of *C. perfringens* in the intestines of humans and animals is well established. People who practice poor hygiene or partake of communal meals tend to have a higher incidence of *C. perfringens* than do people using good hygienic practices. The level of *C. perfringens* contamination normally is 10^2 to 10^4 organisms per gram of fecal material. However, Nakagawa and Nishida (1969) reported that about one-third of the positive samples had 10^3 or less, one-third had 10^4 to 10^5, and one-third

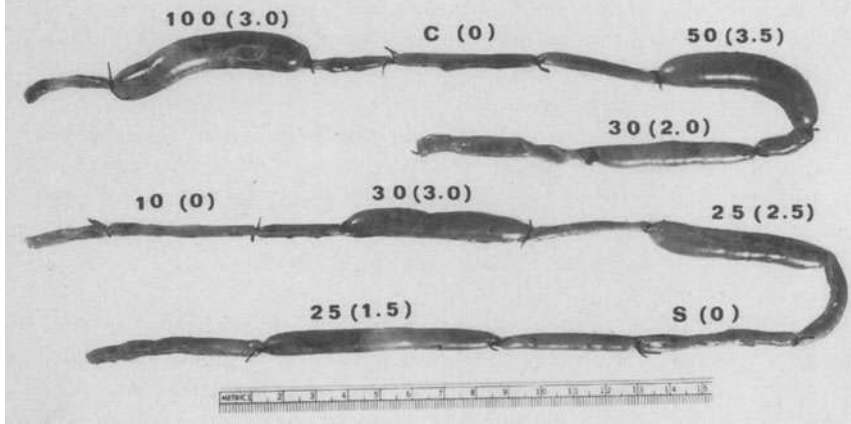

Figure 6.9. Reactions in ligated intestinal loops of two chickens. Numbers in parentheses = micrograms of enterotoxin injected per fluid volume, in milliliters; c = enterotoxin-free cell extract; s = saline.
Courtesy of Niilo (1974).

had 10^6 or more *C. perfringens* per gram from normal human intestines. Others found that some healthy aged adults persistently excreted from 10^7 to 10^9 *C. perfringens* per gram of feces (Yamagishi et al. 1976). However, these may not be enterotoxigenic types.

With the widespread distribution of *C. perfringens* in the environment, and the fact that the organism can produce spores that are resistant to adverse conditions, the organism is a logical common contaminant of food.

Growth. Although *C. perfringens* is an anaerobe, it is aerotolerant. Therefore, strict anaerobic conditions are not needed for growth. The redox potentials of meat products are favorable for growth of the microorganism.

Good growth occurs between pH 5.5 and 8.0, and no growth occurs below pH 5.0 or above pH 9.0. At pH 5.5, vegetative growth occurred, but not sporulation and enterotoxin production (Labbe and Duncan 1974). They reported the optimum range for enterotoxin production as pH 6.5 to 7.3.

C. perfringens grows readily at temperatures between 20° and 50°C, with maximum growth between 37° and 47°C. Mead (1969) found no growth at either 15° or 52°C. Reportedly, some strains show limited growth at 15°C. A long lag phase is a characteristic of growth at low temperatures. Sporulation and toxin production were optimum at 37°C (Labbe and Duncan 1974). No sporulation or toxin production was noted at 46°C.

At a_w values less than 0.995, the rate of growth is reduced. The minimum a_w for growth depends upon the solute, temperature, pH, and other factors. The absolute minimum a_w for growth appears to be 0.93 (Kang et al. 1969). In this case, glycerol was used to lower the a_w.

C. perfringens can grow in concentrations of curing salts (sodium nitrite, and nitrate) considerably higher than those used in normal curing operations (Gough and Alford 1965). Ando (1975) reported that with 2 percent NaCl, a level of 0.05 percent (500 ppm) $NaNO_2$ was needed to completely prevent outgrowth of germinated spores of *C. perfringens*. Lower concentrations permitted some outgrowth.

METHODOLOGY. The clinical and epidemiological pattern of *C. perfringens* foodborne illness is so characteristic that it is almost diagnostic. However, it is necessary to analyze appropriate samples in order to confirm the presence of the organism. Because the enterotoxin is released primarily during sporulation, and most of the sporulation occurs in vivo, it is assumed that the analysis of food for the enterotoxin may be of little value. In one study, however, researchers inoculated ground and auto-

claved chicken meat with *C. perfringens* and analyzed it for enterotoxin (Al-Obaidy et al. 1985). They reported a level of 0.76 μg/g of enterotoxin after incubation for 14 hr, indicating that the toxin might be detected in foods involved in gastroenteritis due to *C. perfringens*.

To help establish *C. perfringens* as the cause of a foodborne outbreak, three laboratory tests are suggested: (1) organisms of same serotype in epidemiologically incriminated food and stool of ill individuals; (2) isolation of organisms with same serotype in stool of most ill individuals and not in stool of controls; or (3) demonstration of greater than 10^5 *C. perfringens* cells per gram of incriminated food, provided the food sample was properly handled.

Detection and Enumeration of C. perfringens. Food should always be analyzed promptly. This is especially true for *C. perfringens* detection because unprotected cells are sensitive to low temperatures. A buffered glycerol-sodium chloride solution mixed in equal portions with a food and stored in dry ice at −56°C protected cells of *C. perfringens* (Harmon and Placencia 1978). The number of *C. perfringens* was only slightly lowered by this treatment.

For the enumeration of *C. perfringens*, tryptose sulfite cycloserine (TSC) agar without egg yolk was effective (Harmon, Kautter, and Peeler 1971). This medium with or without added egg yolk emulsion is recommended for enumeration of *C. perfringens* (AOAC 1985; APHA 1984; FDA 1978). The medium is differential because of the presence of sodium bisulfite and ferric ammonium citrate. Bacteria that reduce sulfite to sulfide produce black colonies due to the formation of iron sulfide. Since many organisms can reduce sulfite to sulfide, D-cycloserine is added to select for clostridia.

For enumeration of this organism in foods, the MPN technique using thioglycollate-cycloserine (Debevere 1979), iron milk (Abeyta 1983; Abeyta, Michalovskis, and Wekell 1985; St. John, Matches, and Wekell 1982), or rapid perfringens medium (DeBoer and Boot 1983; Erickson and Deibel 1978) reportedly is simpler and more effective than direct plating systems.

Confirmational Tests. C. perfringens is the only clostridial species that reduces sulfite, is nonmotile, reduces nitrate, and produces a stormy fermentation in milk (Buchanan and Gibbons 1974).

Nitrate reduction and motility are determined by stab inoculation of nitrate motility medium. It is recognized that all strains do not reduce nitrate in this medium. However, when the medium is modified by adding 0.5 percent galactose and glycerol, then good nitrate reduction is obtained (Hauschild and Hilsheimer 1974).

Serology. Not all of the strains are typable with the presently available antisera. The many problems associated with serotyping *C. perfringens* were reviewed by Stringer, Turnbull, and Gilbert (1980).

Bacteriocins. The specificity of bacteriocins can be used to type microorganisms. Watson (1985) used fifty bacteriocins to type 802 isolates of the organism. A greater proportion of those strains involved in foodborne illness were bacteriocinogenic than were other isolates.

Toxin Detection—Enterotoxin. Although the determination of enterotoxin may not be important epidemiologically, it is important when studying the role of toxin in pathogenesis and the mechanisms of enterotoxic action. Various systems have been used for the assay of the enterotoxin.

Ligated intestinal loops have been used to assay for the enterotoxin (Hauschild, Hilsheimer, and Rogers 1971; Niilo 1974). When toxin is injected into a ligated loop, it causes fluid accumulation (Fig. 6.9). Measurement of this fluid as compared to control loops is an indication of the amount of toxin that is injected. Since it is inconvenient to assay toxin in this manner, other methods have been devised.

Intradermal injection of the enterotoxin into rabbits and guinea pigs causes erythema around the injection site in 1 to 2 hr, and reaches a maximum in 18 to 24 hr. The diameter of this reddened area is related to the concentration of enterotoxin (Niilo 1975). The reaction is distinctive for the enterotoxins. According to Hauschild (1970), this system is 1,000 times as sensitive as the ligated intestinal loop, as well as being more rapid and accurate.

Various mammalian cell lines have been tested. African green monkey kidney (Vero) and dog kidney (MDCK) cells are sufficiently sensitive to assay for the enterotoxin (Giugliano, Stringer, and Drasar 1983; McDonel and McClane 1981).

The immunodiffusion methods listed for assaying *S. aureus* enterotoxin can be used to assay for *C. perfringens* enterotoxin. According to Skjelkvålé and Uemura (1977), both the reversed passive hemagglutination and counterimmunoelectrophoresis assays are rapid and reliable systems. The radioimmunoassay detects enterotoxin at a level of 1 ng/ml (Stelma et al. 1983).

Various ELISA systems are specific and sensitive for *C. perfringens* enterotoxin. In one study, as little as 25 ng/ml of enterotoxin was detected with an indirect ELISA, while 1 ng/ml was detected with a four-layer sandwich ELISA (McClane and Strouse 1984). Other researchers have detected 5 ng/g of enterotoxin in feces with a sandwich ELISA (Bartholomew et al. 1985).

CONTROL OF *C. PERFRINGENS* FOOD POISONING. The three general methods for control of this foodborne illness are (1) limit or prevent contamination of the food; (2) prevent or inhibit growth of the organism; or (3) destroy the organism. Any combination of these methods also may be used. Since most, if not all, of the enterotoxin is produced in the intestinal tract, systems to prevent production or to destroy the toxin are not applicable.

Limit or Prevent Contamination. Due to the ubiquitous character of the organism, preventing contamination of food cannot be relied upon as a means of control. However, this does not mean that we should ignore good sanitary practices.

Prevent Growth. The presence of a few *C. perfringens* cells in a food product is not considered to be a health hazard. The hazardous level is some 10^6 to 10^8 cells per gram of food. The prevention of germination of spores with outgrowth and multiplication of the vegetative cells is the only practical method to control the illness caused by *C. perfringens*.

One simple method of control is to cook and serve the food without an extended holding period. If this is not possible, refrigeration of the food in small quantities for quick cooling will inhibit growth. The minimum temperature of growth is about 15°C. Any acceptable refrigerator should be able to maintain this temperature unless it is overloaded. The lower the temperature of the refrigerator, the sooner the food will be cooled. As the temperature approaches 0°C, the population tends to decline, since the vegetative cells are not stable at such low temperatures. Even the spores appear to be damaged at low refrigerator or freezer temperatures.

The maximum temperature for growth is 50° to 52°C. Above 52°C, no growth should occur, and the vegetative cells may die. There was greater than a 99 percent reduction in the number of *C. perfringens* on cooked beef cubes when held at 53.3°C (Brown and Twedt 1972).

Destroy the Organism. Vegetative cells of *C. perfringens* are destroyed by thorough cooking, but heat-resistant spores can survive. Subjecting the spores to sublethal temperatures stimulates germination. The heat resistance of the spores varies from strain to strain; hence, heating to a certain temperature is not required, since the killing effect is dependent upon the strain of organism that is present.

If heat-resistant spores are present, it is not possible to heat foods sufficiently to inactivate all of them without damaging the organoleptic properties of the food. It must be assumed that there are surviving spores in the cooked food, and the food must be kept hot until it is served.

If any food is cooked one day for serving the next, and even if it is refrigerated, it should be reheated prior to serving to kill the vegetative cells resulting from germination of spores and possible further multiplication. To be safe, foods such as gravies should be reheated by boiling for 10 to 15 min. and roasts or poultry should be reheated to a temperature sufficient for inactivation of *C. perfringens* (Roy, Busta, and Thompson 1981).

Gamma radiation will destroy spores of *C. perfringens*. Clifford and Anellis (1975) calculated 12*D* values for eight strains of *C. perfringens* spores. These values, divided by 12, revealed *D* values ranging from 130 to 350 Krad.

Salmonellosis

All members of the genus *Salmonella* are potentially pathogenic for humans as well as for vertebrate animals. The transmission of the disease is usually from animals to humans by the ingestion of food of animal origin. Direct transmission is also possible from human to human, from human to animal, and from animal to human (Fig. 6. 10). Diseases or infections naturally transmitted between vertebrate animals and humans are called zoonoses.

The illness caused by salmonellae can be divided into four syndromes which may occur individually, simultaneously, or consecutively in the course of an infection. These syndromes are the carrier state (convalescent or asymptomatic), enteric fever (typhoid or paratyphoid fever), gastroenteritis (food infection), and septicemia (characterized by a brief febrile illness or a prolonged or relapsing illness with localized lesions).

Acute gastroenteritis, caused by nearly all serotypes of *Salmonella*, is the most frequent syndrome encountered. It is of primary importance to food microbiologists.

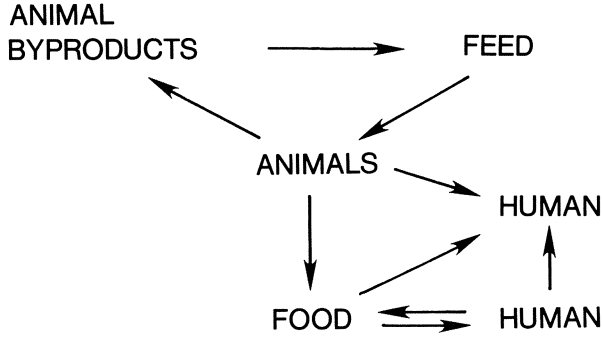

Figure 6.10. Cycles of infection for salmonellosis.

According to CDC (1983b), salmonellosis accounted for 26.4 percent of the reported outbreaks, 26.8 percent of the cases, and 65.6 percent of the deaths due to foodborne illness in 1981. Only the number of cases of *S. aureus* intoxication surpassed salmonellosis in these categories. The reported isolations of *Salmonella* from humans (1968–1980) is shown in Figure 6.11. Most isolations occur during the warm months. The number of reported isolations each year appears to increase.

The increase in the number of cases has been attributed to better diagnosis, laboratory procedures, and reporting. However, in those areas in which diagnosis and reporting have always been adequate, increases have occurred in the number of outbreaks and cases of salmonellosis. The prevalence of salmonellosis is increasing, although much energy has been expended to control this illness. This increase of *Salmonella* in both human and animal sources is associated with national and international commerce in animals, feeds, and foods; with large-scale, intensive animal raising; and with increased use of ready-to-serve or heat-and-eat foods.

It has been estimated that only 1 to 10 percent of the actual number of cases are reported. There may be over 2,000,000 cases of salmonellosis each year in the United States. Many cases are mild and no professional care is solicited. Salmonellae may be the major cause of bacterial gastroenteritis and foodborne illness in developed countries.

The economic loss due to *Salmonella* was estimated to be at least $300 million annually in the United States (Foster et al. 1970). The cost includes medical care, hospitalization, lost time and income through absence from work, death of animals, decreased production of animals, loss or reduced value of contaminated products, testing and control proce-

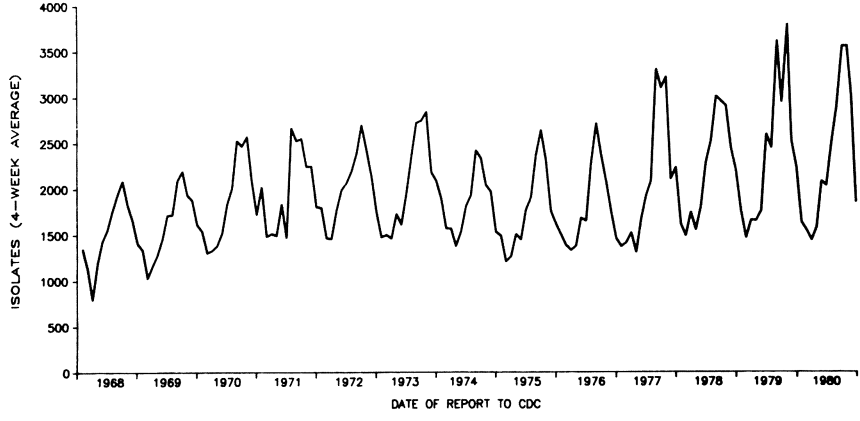

Figure 6.11. Reported number of isolations of salmonellae from humans by month, 1968–1980.

dures, and recall of products from market channels. A more recent estimate (CDC 1978b) is that the annual cost for salmonellosis is $1.5 billion for medical expenses alone. Depending on the circumstances, the average cost per case ranged from $277 to $4,548 for 12 outbreaks (Todd 1985). The median cost per case was $769. At this rate, 2 million cases cost more than $1.5 billion per year in the United States.

More deaths are caused by salmonellae and salmonellosis than by botulism. The death-to-case ratio is rather low, with a case fatality ratio of 0.26 percent from 1962 to 1968 (CDC 1971) and 0.15 percent for 1974 (CDC 1975). In 1981, the number of deaths jumped to 21 for a case-to-fatality ratio of 0.86 percent (CDC 1983b). Most deaths occur in very young or old people.

CHARACTERISTICS OF THE ILLNESS. Salmonellosis has been considered to be an infection caused by the action of the organism in the intestine. However, there is evidence that a toxin or toxins may be involved.

Incubation Period. In thirty-four random outbreaks reported by the CDC, the range for the incubation period varied from 1 hr to eight days. The usual incubation period is reported as 6 to 48 hr (CDC 1983b).

Symptoms. The symptoms of nine random outbreaks reported by CDC are listed in Table 6.16. The symptoms and their severity depend on the number of organisms and the serotype of *Salmonella*, as well as the resistance of the host. The most reported symptom is diarrhea, followed by abdominal cramps, fever, nausea, vomiting, chills, and headache.

TABLE 6.16. SALMONELLOSIS: PERCENTAGE OF PEOPLE EXPERIENCING SYMPTOMS IN RANDOM OUTBREAKS

	Outbreak								
Symptom	1	2	3	4	5	6	7	8	9
Diarrhea	87	100	100	96	93	75	93	96	95
Diarrhea (bloody)	4	—[a]	—	—	5	—	5	—	—
Abdominal cramps	70	47	70	81	86	82	76	66	57
Fever	68	82	—	85	62	80	48	97	43
Nausea	53	69	80	62	69	70	52	—	38
Vomiting	53	82	80	40	40	62	26	54	19
Chills	38	54	70	79	—	—	52	86	—
Headache	36	66	60	—	65	—	63	—	29
Dizziness	—	—	—	—	42	—	—	—	—
Muscle aches	—	—	70	—	—	—	—	95	—

[a]— = Not reported; however, the symptom might have existed.
SOURCE: Data from CDC *Morbidity and Mortality Weekly Reports.*

Duration. In normal, healthy adults, the gastroenteritis usually lasts for only two to three days, but if further infection occurs, the illness may persist for months or years, and it may result in death. Thomas and Mogford (1970) stated that the duration of infection was longer than two months in nearly 25 percent of the cases, and intermittent excretion of the organisms was observed in 20 percent of the cases. Children are more susceptible than adults to prolonged excretion of the organism.

Therapy. The majority of the cases need no therapy. When hospitalization is required, an attempt is made to stabilize loss of fluids and prevent dehydration, as well as to maintain the electrolyte balance.

The administration of broad-spectrum antibiotics usually gives no appreciable benefit; the salmonellae become resistant to antibiotics. Treatment with antibiotics can cause the asymptomatic carrier state to become an active case (Saroglou and Bisno 1978).

Eating yogurt was suggested as part of the therapy for the salmonellosis outbreak involving 2 percent milk in 1985. Although yogurt did not prevent salmonellosis in rats, it did significantly reduce the mortality as well as other effects (Hitchins et al. 1985a, 1985b).

FOODS INVOLVED. Various foods have been the vehicle for transmission of salmonellae. Most of these foods are of animal origin or contaminated by foods of animal origin, as shown in Table 6.17.

The involvement of meat and meat products continues to increase and accounted for more than 50 percent of the known foods from 1977 to 1981. During 1977, there were several outbreaks and more than 180

TABLE 6.17. FOODS INVOLVED IN SALMONELLOSIS, 1977–1981, NUMBER OF OUTBREAKS

	Year						
Food	1977	1978	1979	1980	1981	Total	%
Beef	11	3	4	4	13	35	14.9
Pork	2	7	2	2	1	14	6.0
Other meat	1	3	1	0	0	5	2.1
Poultry	6	4	3	7	9	29	12.3
Eggs	0	0	2	1	0	3	1.3
Fishery products	0	0	0	0	2	2	0.9
Dairy products	4	4	2	3	5	18	7.7
Bakery products	1	0	1	1	1	4	1.7
Salads	3	2	2	3	2	12	5.1
Other, unknown	13	22	27	18	33	113	48.1
TOTAL	41	45	44	39	66	235	

SOURCE: CDC *Annual Reports.*

cases of salmonellosis related to the consumption of precooked roast beef. These, as well as outbreaks in 1975 and 1976, stimulated action by the USDA to require specific times and temperatures for heating beef roasts (USDA 1983).

Poultry and poultry products have been implicated in many outbreaks of salmonellosis. Since the USDA required the pasteurization of egg products, these have been eliminated as a significant vehicle for salmonella organisms. The failure to cook chicken and turkey sufficiently to eliminate the natural contamination is the usual contributing factor causing salmonellosis.

Raw milk has been the vehicle for salmonellae causing salmonellosis throughout the world (Bryan 1983; CDC 1984a; Reilly et al. 1983; Tacket et al. 1985a). All milk should be pasteurized before human consumption, but even pasteurized milk has been implicated in outbreaks of salmonellosis (Bryan 1983; CDC 1984b). A widespread outbreak in 1985, affecting over 16,000 people in several states, involved pasteurized milk. This incident should make us aware that pasteurization systems and proper procedures for handling pasteurized milk are needed to prevent salmonellosis and perhaps other illnesses.

Apple cider involved in an outbreak was produced from apples that had fallen onto the ground on which animal waste had been used as fertilizer. Although the apples were washed, sufficient salmonellae remained to cause illness. *S. eastbourne* was isolated from the environment in which cocoa beans were processed (CDC 1974b), apparently causing the contamination of chocolate candy which was involved in a widespread outbreak. No source was determined for *S. napoli*, which was isolated from chocolate bars involved in several cases of illness (Greenwood and Hooper 1983).

ETIOLOGIC AGENT. The mechanism by which the salmonellae cause gastroenteritis has not been fully explained. Perhaps more than one system is involved. The heating of food to destroy salmonellae results in a safe product. Hence, it was believed that salmonellosis was an infection that required high numbers of the organism to cause the illness. However, the heating needed to kill the organism also might inactivate any heat-labile toxins produced by the organisms.

The number of ingested *Salmonella* needed to cause an infection varies with the strain of the organism and the characteristics of the individual ingesting the organisms. Healthy adult males can ingest 10^5 to 10^7 cells (depending upon the strain of the organism) before symptoms of the illness occur. However, person-to-person transmission in hospitals indicates that only a few organisms are needed to cause illness. Blaser and Newman (1982) reviewed eleven outbreaks and calculated that, in six of

these, less than 1,000 ingested organisms caused the illness. In a series of cases involving chocolate bars, it was estimated that an infective dose was roughly fifty organisms (Greenwood and Hooper 1983). The very young, aged, debilitated, and undernourished, and individuals with other illness are more susceptible than normal healthy individuals to infection by *Salmonella.*

The pH of the gastric juice can have a marked effect on salmonellae (Giannella, Broitman, and Zamcheck 1971). Healthy gastric juice with low pH (1–2) kills small numbers of salmonellae, but when it is deficient (pH 3.0 or higher), an otherwise harmless number of cells may survive.

When salmonellae reach the small intestine, they can live and multiply. The extent of multiplication depends on factors such as the peristaltic rate, attachment to the mucosa, the composition of the intestinal flora, the ingestion of various therapeutic agents (such as antibiotics), exposure to radiation, and presence of iron-deficiency diseases.

Attachment of the salmonellae to the intestinal mucosa is considered to be essential for the organisms to remain in the intestine and cause illness. Although invasion of the intestinal epithelium is needed for excess secretions (Giannella 1979), invasion, per se, is not sufficient to cause salmonellosis (Formal, Hale, and Sansonetti 1983). The results of rectal biopsies of twenty-two patients varied from normal mucosa to edema, hemorrhages, and sloughing (Day, Mandal, and Morson 1978). Some strains of salmonellae are known to cause mucosal damage and increased fluid output, some cause minimal mucosal damage and increased fluid, while others cause damage and no increase in fluid output. *Salmonella* infection causes lesions in the ilea of pigs (Wilcock 1979) and calves (Wray and Sojka 1978).

Researchers in one study suggested that *S. typhimurium* causes fluid secretion by altering the sodium and chloride transport systems (Giannella et al. 1975). When infected, the intestinal mucosa showed an increase in adenyl cyclase activity, which results in ileal secretion of fluids. In a review, Smith (1977) suggested that an extracellular product is involved, because the microvilli degenerate before coming into contact with the bacteria.

Only four of thirteen strains of *Salmonella* tested produced a positive reaction in the rabbit ligated gut loop when living organisms were used (Sakazaki et al. 1974). When culture filtrates were tested in the ligated loops, eleven of thirteen strains of *Salmonella* gave a positive reaction. The researchers contended that the culture filtrates contained enterotoxic activities.

The rabbit ileal loop model was used to detect enterotoxin produced by salmonellae (Sedlock and Deibel 1978). The serotypes that were tested varied in their ability to produce the toxin, but even old stock cultures

maintained this trait. This indicates that the action is not due to a plasmid.

A factor isolated from *S. typhimurium* caused cells of Chinese hamster ovary (CHO) to elongate (Sandefur and Peterson 1977). This action is indicative of an enterotoxin. They found that this effect was blocked by antisera for cholera toxin and the B fragment of cholera toxin.

It is quite evident that most, if not all, strains of *Salmonella* produce enterotoxins that elongate CHO cells and cause secretion in gut loop tests (Baloda et al. 1983; Sharma, Singh, and Singh 1983). Jiwa and Månsson (1983) and Alouf (1982) reported that both heat-labile and heat-stable enterotoxins are produced by some strains of salmonellae. Although the enterotoxin usually reacts with antitoxin to cholera enterotoxin, one study reported a salmonella enterotoxin that did not cross-react with antisera to either cholera or heat-labile *E. coli* enterotoxins (Baloda et al. 1983).

Other researchers reported that, in salmonellosis, the cyclic adenosine monophosphate (cAMP) is increased in epithelial cells in a manner similar to that of cholera (Peterson et al. 1983). They stated that it was not clear whether the increase in cAMP was due to enterotoxins, to prostaglandins formed during the inflammatory response to the bacteria, or to both systems.

THE ORGANISMS. The organisms in this genus are separated on the basis of their somatic (O) and flagellar (H) antigens. The antigenic classification of *Salmonella* is known as the Kauffmann-White scheme. New species or serotypes are continually being found and added to the many already classified. In 1964, there were about 900 known serotypes. Now there are more than 1,700 serotypes (CDC 1982a). The Arizona group has been incorporated into the Kauffmann-White scheme for *Salmonella* by Rohde (1979).

Although there are many serotypes, only about fifty cause nearly all of the outbreaks and cases of salmonellosis. Each year, some ten serotypes account for more than 60 percent of the isolations of *Salmonella*. The ten most prominent serotypes isolated from humans for the years 1977 through 1980 are listed in Table 6.18. *S. typhimurium* is the most frequently reported serotype isolated from both human and nonhuman sources (Table 6.19). Also, *S. typhimurium* is the most frequently isolated serotype in many other countries.

Growth Factors. Most strains of salmonellae can grow in a simple medium consisting of ammonium nitrogen, mineral salts, and glucose. A few strains need essential growth factors, primarily vitamins. The minimum water activity for growth is between 0.93 and 0.96. The salt concentration

TABLE 6.18. MOST FREQUENTLY REPORTED *SALMONELLA* SEROTYPES ISOLATED FROM HUMAN SOURCES: RANKING (1-10) FOR YEARS 1977-1980

Serotype	Year			
	1977	1978	1979	1980
S. typhimurium[a]	1	1	1	1
S. heidelberg	3	2	3	2
S. enteritidis	4	3	2	3
S. newport	2	4	4	4
S. infantis	5	6	5	5
S. agona	6	5	6	6
S. saintpaul	7	9	7	7
S. montevideo	9	7	9	8
S. typhi	8	8	8	9
S. oranienburg	10	—	10	10
S. javiana	—	10	—	—

[a]Includes *S. typhimurium* var *copenhagen.*

needed to inhibit growth of salmonellae depends on the temperature and other factors (Alford and Palumbo 1969). In ground pork, at pH 5.0, stored at 10°C, a salmonella grew with 3.5 percent salt, but did not grow at 5 percent salt. At 20°C or 30°C, salmonellae grew at the 5 percent salt level. With ground pork at pH 6.5, growth occurred with 8 percent salt added at 20°C, or 30°C, but no growth occurred at 10°C. The pH values for growth of salmonellae are listed in Table 4.6.

TABLE 6.19. MOST FREQUENTLY REPORTED *SALMONELLA* SEROTYPES ISOLATED FROM NONHUMAN SOURCES: RANKING (1-10) FOR YEARS 1977-1980

Serotype	Year			
	1977	1978	1979	1980
S. typhimurium[a]	2	1	1	1
S. derby	1	2	3	2
S. panama	3	5	5	3
S. agona	4	3	2	4
S. manhattan	—	4	—	5
S. infantis	8	10	4	6
S. weltevreden	6	—	10	7
S. heidelberg	10	6	6	8
S. oranienburg	—	—	7	9
S. habana	—	—	—	10
S. montevideo	—	—	8	—
S. cholerasuis	—	—	9	—
S. worthington	—	—	10	—
S. tennessee	—	7	—	—
S. london	—	8	—	—
S. meleagridis	9	9	—	—
S. anatum	5	—	—	—
S. newington	7	—	—	—

[a]Includes *S. typhimurium* var *copenhagen.*

Since salmonellae are pathogenic, they generally are considered to be mesophilic. However, some strains appear to be psychrotrophic (see Table 4.12). In general, these organisms require a temperature of 2° to 4°C above the minimum reported in order to grow in food products. Even then, at low temperatures, the more psychrophilic organisms tend to outgrow the salmonellae. According to Tesone, Hughes, and Hurst (1981), the addition of 0.2 molar NaCl raises the maximum temperature of growth for some strains.

The interaction of other microorganisms naturally present in the food with salmonellae can result in the inhibition of these potential pathogens. There are numerous reports on the observed inhibition of *Salmonella* by lactic acid bacteria. Gilliland and Speck (1972) believed the antagonistic action was caused by factors in addition to the acidic environment created by the bacterial fermentation.

Biochemical Reactions. The main biochemical reactions of the salmonellae are listed in Table 6.20. Even though a "+" or "−" reaction is listed, this does not mean that all of the many serotypes and strains, types or variants show that reaction. Subgenus I includes the more prominent salmo-

TABLE 6.20. MAIN BIOCHEMICAL REACTIONS OF THE SALMONELLAE

	Subgenus[a]	
Test or Substrate	I	III
Gas from glucose	+	+
Methyl red test	+	+
Indole production	−	−
Voges-Proskauer test	−	−
H_2S production	+	+
Growth in Simmons citrate	+	+
Urease production	−	−
Gelatin liquefaction	−	(+)
Phenylalanine deamination	−	−
Growth in KCN medium (D)	−	−
Growth in malonate	−	+
Motility	+	+
Reduction of nitrate	+	+
Fermentation of:		
Adonitol	−	−
Dulcitol	d	−
Inositol	d	−
Lactose	−	+ or X
Maltose	+	+
Mannitol	+	+
Salicin	−	−
Sucrose	−	−

[a] + = prompt, positive; X = late and irregularly positive; (+) = delayed positive; d = different biochemical types; − = negative.

nellae serotypes, and subgenus III consists of the organisms commonly referred to as the Arizona group, now designated as *S. arizona.*

Motility. Although the organisms are said to be motile, there are mutants that lose this ability, and two serotypes, *S. pullorum* and *S. gallinarum* are not motile and do not possess flagella.

Sources. As a genus, the salmonellae are said to be ubiquitous. They are worldwide and found in or on soil, air, water, sewage, animals, humans, food, feed, processing equipment, and some plant products. Some serotypes tend to be localized in a region or a country, but with national and international travel and trade, the organisms are readily disseminated.

The natural habitat of the organisms is the intestinal tract of humans and animals. Thus, it is logical that humans, animals, and their environments are the primary sources of *Salmonella.*

Few animals, if any, are born with intestinal contamination by *Salmonella.* Just as human infants are highly susceptible to salmonellosis, so are very young animals. Some young animals may survive salmonellosis and become temporary or long-term shedders of the organisms. However, it is more likely that the animal is continually infected by some source of *Salmonella.*

The sources of salmonellae for infection of domestic animals include the parent stock, feed, water, rodents, wildlife, pets, humans, insects, arachnids, soil, and vegetation. The salmonellae can spread from animal to animal during production and processing. They spread from animals to people by direct interactions of humans and livestock, wild animals and pets, as well as through poor food-handling practices and the consumption of raw or undercooked foods of animal origin.

The relationship of *Salmonella* in feed to human salmonellosis is apparent with *S. agona.* Prior to 1971, the serotype was reported from human hosts in the United States only six times. It was isolated from imported fish-meal on several occasions from 1970 to 1972. Since these isolations, it has been found in domestic animals, and since 1973 it has ranked as one of the ten most prevalent salmonellae from humans (see Table 6.18). This suggests the involvement of feed in a chain or cycle of infection (see Fig. 6.10).

The role of free-living (wild) animals as a source of *Salmonella* was thought to be quite important. Now it is thought that rodents, birds, and other wild animals may be victims of their environment rather than the source of the organisms. Even so, these free-living animals can transfer the organism from infected areas to clean areas.

Since the waste materials from infected farms contain salmonellae, the use of this material as fertilizer for crops or pastures can spread the potentially infectious material to contaminate future animals. Rain fall-

ing on the farms, feed lots, or fields containing contaminated wastes can wash the organism into streams and lakes. Besides the runoff from animal quarters, sewage sludge and effluents from abattoirs and poultry-processing plants contain salmonellae. These polluted waters can cause infection of farm animals and wild animals that drink from them.

Various pets are potential sources of salmonellae. They may contaminate food or directly transmit these organisms to humans. Pet turtles are an important source of *Salmonella* for transmission primarily to children. An estimated 280,000 cases of turtle-associated salmonellosis occurred in the United States in 1970 and 1971 (Lamm et al 1972). The interstate shipment of turtles less than 4 inches long was banned by the FDA in 1975, because studies showed that 14 percent of the reported salmonellosis in the United States was linked to pet turtles. However, some turtles are still being sold in the United States, and some 3 or 4 million are raised each year for export.

The direct transmission of *Salmonella* from animals to humans poses a problem in zoos that have children's petting areas or other open areas such as the aviary (Komorowski and Hensley 1973).

Although the cooking of raw meat and poultry will destroy salmonellae, the raw foods are a source of contamination to the food preparer and the kitchen equipment and can cross-contaminate cooked foods and other foods that are eaten raw.

Fish and shellfish are contaminated by water that contains waste products from humans and animals. Salmonellae were recovered from oysters, clams, and crabs harvested from both the east and west coasts of Florida (Fraiser and Koburger 1984). The east coast site was approved for shellfish harvest, but 25 percent of the trapped crabs and more than 13 percent of the clams contained salmonellae. The west coast site was an unclassified harvesting area, and the incidence of salmonellae in clams was higher than 43 percent. The highest level of contamination reported was 2.2 salmonellae per 100 grams of shellfish.

Fruits and vegetables are rarely involved in outbreaks of salmonellosis. However, vegetables can be contaminated by organisms present in animal wastes, or polluted irrigation water used on crops. Velaudapillai, Niles, and Nagaratnam (1969) found that 1.3 percent of the vegetables sampled were infected with *Salmonella, Shigella*, or enteropathogenic *E. coli*. *Salmonella* serotypes were found on celery, leeks, spinach, green beans, and carrots. Wild olives contained *S. sandiego*. Samples of lettuce (68.3 percent) and fennel (71.9 percent) yielded one or more serotypes of salmonellae (Ercolani 1976). Salmonellae were isolated from over 22 percent of the vegetable samples analyzed by Tamminga, Beumer, and Kampelmacher (1978a). Other researchers found that four samples of vegetables (radish, carrot, celery and spinach) of fifty examined contained

salmonellae (Rude et al. 1984). They suggested that vegetables eaten raw should be considered a source of salmonellae in outbreaks of salmonellosis. Further, raw vegetables may be a source of human parasites. Coconut products are believed to be contaminated by salmonellae in the soil. This contamination is transferred to the inner, edible portion during processing.

Humans may be one of the main reservoirs of *Salmonella*. If not a source, humans can act as the vehicle for transfer of the organisms from one area to another.

Food handlers or workers in animal by-product plants often have a high rate of carriage of *Salmonella*. Although they might not have an active infection, the children of these workers have been observed to have higher rates of gastroenteritis than children of other workers. This indicates that a worker who handles products that contain *Salmonella* can bring the organisms home to the other members of the family.

Survival in Nature. The natural habitat of *Salmonella* is the intestinal tract of animals. Excreted organisms are found in fecal material, food, and various parts of the environment. To cause salmonellosis, they must be able to survive in these unnatural surroundings.

Salmonellae in slurry (liquid manure) sprayed onto grass may survive from eighteen days (Taylor and Burrows 1971) to thirty-three weeks (Findlay 1972). Jones, Hagler, and Hamilton (1977) found that the survival of salmonellae in cattle slurry was influenced by the normal microbial flora. In sterilized slurry, *S. dublin* multiplied and survived for more than 370 days, while in natural slurry, there was no multiplication and shorter survival times. Survival time in a pasture is influenced by grass cover, sunlight, temperature, and rainfall.

Effluent from apparently healthy pigs was applied to land (Chandler and Craven 1981). The salmonellae were detected for eight months from soil and nearly two months from pasture. *S. typhimurium* survived in soil for seven to forty-two days, depending upon soil moisture and temperature (Zibilske and Weaver 1978).

Liu, Snoeyenbos, and Carlson (1969) found that the survival of salmonellae in meat and bone meal was related to water activity and holding temperature. As water activity and temperature increased from low levels, the survival decreased. At an a_w of 0.82 and temperature of 50°C, a 5-log reduction occurred in 72 hr. However, at 10 percent moisture (a_w about 0.55) at 4°C, there was essentially no reduction in viable cells of *S. senftenberg* 775W for 20 days. Elevated temperatures, oxygen, and unsaturated fatty acids accelerate the death of *S. oranienburg* in fish meal (Lamprecht and Elliot 1974).

When fingertips were contaminated with 500 to 2,000 cells of *S. an-*

atum, the cells could be recovered 3 hr later. Contamination, followed by washing, eliminated levels of less than 1,000, but when 6,400 were inoculated, they could be recovered 10 min after washing. Even fewer than 100 cells of *S. anatum* inoculated onto fingertips infected samples of corned beef and ham handled 10 min after exposure (Pether and Gilbert 1971). This shows how *Salmonella* can be spread by workers handling first contaminated products and then clean products.

Survival in Food. There is less destruction of inoculated salmonellae during pan drying than during spray drying of egg white. Pan drying is more similar to the natural drying of materials. Once dried, the organisms can survive for long periods at room temperature or below. Salmonellae were isolated from dried eggs after forty-one weeks at 20°C, and at sixty-five weeks when held at 2°C.

Salmonellae can survive up to eleven weeks on vegetables held at 2° to 4°C, and seven weeks at room temperature. In acid foods, survival is relatively short, but if salmonellae survive freezing, such as in certain fruits, they may remain viable for considerable periods. *Salmonella* can survive at least thirteen months on frozen poultry held at −21°C. In ice cream, the organisms can survive for several years.

Salmonellae inoculated into cold-pack cheese food survived in excess of twenty-seven weeks at 4.4°C, when no preservative (potassium sorbate) or acids were added (Park, Marth, and Olson 1970). Six *Salmonella* strains were inoculated into sheep milk at 10^4 cells/ml and processed into Manchego cheese (Medina, Gaya, and Nuñez 1982). The organisms were absent from all lots of eight-week Manchego cheese.

The cells of *S. typhimurium* tended to die when present in yogurt (Rubin 1985). The presence of casein and an increased pH of yogurt protected the cells from the inhibitory agent, lactic acid.

When inoculated into a beef-pork mixture at a level of 10^4 cells per gram, *S. dublin* survived pepperoni processes and persisted after forty-two to forty-three days of drying (Smith et al. 1975). *S. typhimurium* was more sensitive than *S. dublin* to the acid condition of Lebanon bologna during processing and aging (Smith et al. 1975b).

Salmonellae inoculated into chocolate bars at a level of about 10^6 cells/g were viable after storage for nine months (Tamminga et al. 1976). The reduction was greater in bitter chocolate than in milk chocolate. *S. typhimurium* inoculated at a level of about 1,000/g of chocolate was not detected in 55 g after fifteen months (Tamminga et al. 1977). *S. eastbourne* survived better than *S. typhimurium*.

In a review, Bryan (1968) summarized the survival of salmonellae. These organisms can survive in the low range of temperature and relative humidity. According to references cited by him, they can survive in con-

taminated earth and pasture for more than 200 days, cloth for 228 days, plastic cover slips for 93 days, sweeper dust for 10 months, rodent feces for 148 days, roach pellets for 199 days, poultry feces for more than 9 days, dried cattle feces for more than 1,000 days, on egg shells from 21 to 350 days, in dried whole eggs for more than 4 years and in meat salad for 77 days.

BASIC METHODOLOGY. It is not the intent of this text to precisely describe the procedures needed to obtain samples, or to detect or enumerate salmonellae in the samples. For the procedures, the reader is referred to FDA (1978), AOAC (1985), APHA (1984), and USDA (1971). Sampling and testing plans for salmonellae were reviewed by Olson (1975).

We are concerned primarily with the analysis of food samples, swabs from food surfaces and water, and environmental swabs, although the analysis of other samples may be required, even in a food microbiology laboratory.

In many foods, the organisms have been subjected to debilitating or sublethal processes such as freezing, desiccation, extremes of pH, heat, osmotic pressures, or curing of ingredients during the manufacture or storage of the product. When present, salmonellae usually comprise a very small component of the total population. For most foods, there is no tolerance for *Salmonella*. Even one *Salmonella* detected in a 25-g sample is considered to be adulteration of the food (exceptions are raw chicken and meat). Hence, it is necessary to detect very low levels of these organisms. This small number of salmonellae, especially in comparison to the larger number of other organisms, means that the presence of salmonellae is unlikely to be demonstrated by direct plating on selective media, and that an enrichment procedure in broth is needed.

The basic procedure for the examination of foods for *Salmonella* consists of preenrichment and enrichment in broths, streaking and detection on selective-differential agars, and then characterizing typical colonies to confirm that they are salmonellae. Confirmation is done by means of biochemical tests, serological typing and, for a few serotypes, phage typing. It is evident that a long time period is needed before results of any kind are obtained. Hence, many attempts have been made to develop rapid procedures. Besides the basic procedure, some of these rapid methods are discussed.

Preenrichment. The preenrichment phase is accomplished by incubating the sample in a noninhibitory broth (usually in the ratio of one gram of sample to nine of broth) for 18 to 24 hr at 37°C. Shorter preenrichment times (4 to 6 hr) have been suggested, but longer preenrichment times (18 to 24, or 48 hr) have been more effective (D'Aoust 1981; Kafel 1981).

The recovery of salmonellae from dried food products may be improved by reduced hydration systems (Andrews, Wilson, and Poelma 1983; Poelma, Andrews, and Wilson 1984; Wilson et al. 1985).

Media that have been used for preenrichment include lactose broth, trypticase soy broth, reconstituted nonfat dry milk, nutrient broth, and buffered peptone water.

Generally, processed foods that may contain sublethally injured salmonellae are preenriched. Raw or heavily contaminated products are put directly into a selective enrichment broth. The review by Andrews (1985) pointed out that preenrichment increased the recovery of salmonellae from these foods. However, for the examination of oysters, Miller and Koburger (1984) reported greater recovery by direct enrichment in selenite cystine broth than by using a preenrichment broth.

Enrichment. The enrichment process is the creation of a special environment which permits the growth or selection of the desired species or group of microorganisms from a mixture of microorganisms with which they are likely to be found. The procedure usually functions by furnishing conditions for the desired organisms to outgrow all others. Ideally, an enrichment system for salmonellae should allow these organisms to grow and inhibit the growth of all other organisms.

It has been difficult to develop the ideal enrichment medium. Since there are some 1,700 serotypes of *Salmonella*, any selective agent is likely to be inhibitory to one or more serotypes or to be ineffective in inhibiting the nonsalmonellae that are present.

Foods vary in their nutrient content, pH, inhibitory substances, and other factors that may influence the growth of *Salmonella*. Therefore, when a food is added to an enrichment medium, usually in a ratio of 1:9, the characteristics of the medium are changed. Each food will alter the medium in its own peculiar way. Therefore, to be effective, a different enrichment medium should be designed for each food.

The enrichment broths commonly used in the United States are selenite cystine (SC), tetrathionate (T), tetrathionate brilliant green (TBG), and TT. Rappaport's medium (with modifications) reportedly is an effective broth for salmonellae enrichment (Fricker and Girwood 1985; Tongpim et al. 1984; Vassiliadis et al. 1984). However, it has never received much attention in the United States.

The enrichment broths are used as secondary enrichment of samples that have been preenriched (usually 1 ml of sample from the preenriched broth to 10 ml of enrichment broth), and as direct enrichment for raw or heavily contaminated food. Most analytical procedures suggest adding 25 g of food sample to 225 ml of broth. For feed samples, 30 g is added to 100 ml of broth. Due to multiple sampling of a food lot to assure the

absence of *Salmonella*, the analysis of larger quantities of pooled material from these samples has been suggested.

Some foods have certain peculiarities that call for special procedures for better detection of *Salmonella*. When gelatin is added in enrichment media, the viscous suspension that results can interfere with *Salmonella* analysis. Rose (1972) suggested adding gelatinase to the enrichment medium. The enzyme was not inhibitory to salmonellae, and better recovery was made from gelatin artificially inoculated with *S. typhimurium*.

When milk is analyzed, the acid produced causes formation of curds. *Salmonella* trapped inside these clots of milk are not detected. The addition of trypsin aids detection by digesting the protein of the milk, but apparently it does not injure the salmonellae.

The inoculated enrichment broth is incubated at either 37° or 43°C for 18 to 24 hr, and then a loopful is streaked onto an agar surface that is selective and differential.

Isolation. Many more *Salmonella*-positive samples are found when two or three agars are used. Due to the results of collaborative studies, researchers recommended retaining the use of bismuth sulfite (BS) agar, but recommended that salmonella-shigella agar and brilliant green (BG) agar be replaced by Hektoen enteric (HE) agar and xylose lysine desoxycholate (XLD) agar for detection and isolation of salmonellae (Sanders 1981). Bailey, Cox, and Thomson (1983) reported that BG and HE were superior to BS agar for recovery of *Salmonella*. Hawa, Morrison, and Fleet (1984) used a dulcitol bile novobiocin agar for isolating salmonellae from chicken carcasses.

It is recommended that BS plates be observed after 24 hr and, if no typical salmonellae colonies are observed, the plates should be incubated for another 24 hr before they are called negative.

With preenrichment for 24 hr, enrichment for 24 hr and BS incubation for 48 hr, a total of 96 hr (four days) is needed even for samples with no *Salmonella*, before any decision can be made.

The procedure to be used (type of preenrichment medium or no preenrichment, types of enrichment and plating media, as well as incubation temperatures) is dictated somewhat by the type of sample being analyzed, as well as the types of organisms and serotypes of *Salmonella* that might be present.

It is difficult to compare media for the preenrichment, enrichment, and isolation for *Salmonella*. If salmonellae are present in the sample at a high level, almost any of the typical media can be used to detect them. If the organisms are at a very low level, some of the samples for preenrichment or direct enrichment may not contain any *Salmonella*. To overcome these circumstances, many samples need to be tested and the results analyzed by an acceptable statistical system.

In all of the research that has been done, it is evident that the more different types of broths and agars that are used, the greater is the chance for recovery of *Salmonella*, especially if sublethally injured and at a very low level, with various other organisms at high levels.

Biochemical Characterization. If no typical colonies are found on the isolation medium, this is satisfactory evidence that the sample is *Salmonella*-negative. If typical colonies are detected, this is presumptive evidence that the sample may be *Salmonella*-positive. For further characterization of the organisms in these typical colonies, both biochemical and serological tests are needed. Neither the biochemical nor the serological tests are specific for *Salmonella*, and neither system will detect all the serotypes of the genus.

Rapid and easily run methods have long been the desire of microbiologists. One method employed to get a quick reaction is to use a large inoculum in a small amount of medium.

With the number of biochemical tests that need to be run, considerable time and material are required to prepare the media, inoculate, incubate, determine the reactions, and classify the cultures. To cut down on the time and cost, rapid methods, composite media, and simplified test kits have been developed. Commercial preparations have the potential benefit of providing many laboratories with standard materials for use in assessing bacterial reactions. These commercial systems include the API 20E, Enteric-Tek, Enterotube II, Micro-ID, Minitek, Repliscan II, and Sensititre systems.

These systems have certain things in common. They are all miniaturized and rapid tests based primarily on reports in the literature by various workers. None of the systems agrees 100 percent with the conventional tube method for determining reactions. The common opinion is that these tests are acceptable for general use.

Serology. Serological considerations are used in the immunization of humans and animals for certain diseases caused by salmonellae, for testing the serum of humans or animals for potential carriers of the organisms, and for determining the serotype of an isolated test culture. The serological reactions of the test culture are needed to supplement the information obtained from biochemical tests.

The salmonellae possess O, H, and K antigens (Kauffman 1972). The O (somatic) antigens are considered to be a constitutive part of the cell wall, but they may extend beyond this barrier. They are a complex of lipopolysaccharide (LPS) and protein. The detailed structure of the O antigen varies from strain to strain. This variation forms the basis for the serological differences of the organisms. The O antigens are considered to be heat stable, resisting boiling for 2½ hr.

Salmonella can change in colony morphology form smooth to rough

(S→R variation). These rough forms are caused by mutations that block the synthesis of polysaccharides. Thus, they have no O side chains, lose O agglutinating ability, and have a reduced virulence.

The H antigens, or flagellar antigens, are found only in motile cultures. They are destroyed at 100°C (heat labile), and by dilute alcohol or acid. Flagella are composed of the protein flagellin, with a molecular weight of about 40,000. Many species of salmonellae have two types of flagellar antigens (phase 1 and phase 2). A single bacterium manifests only one of these.

Phase 1 antigens are specific and are shared by only a few serotypes of salmonellae, while phase 2 antigens are nonspecific and are common to many serotypes.

Usually, only the O and H antigens are determined when typing a culture, although the Vi antigen is sometimes of interest. The cultures are typed according to the Kauffmann-White scheme. This system is outlined by Kauffmann (1972) and Krieg and Holt (1984).

The specific H or O antigens are determined by cell genetics. Due to mutations or phage conversions, the antigens can be altered. This may account for the multiplicity of serotypes and the overlapping patterns of antigenicity.

If there is an agglutination with the O antisera, it is not proof that the culture is a salmonella. There are cross-reactions with the *Salmonella* O antisera and nonsalmonellae. Some strains of closely related organisms, especially strains of *E. coli, Citrobacter, Shigella*, and *Enterobacter*, either share certain somatic antigens or have antigens similar to those of salmonellae (Refai and Rohde 1975). Other unrelated microorganisms reportedly react with *Salmonella* O antisera (Aksoycan and Sağanak 1977; Corbel 1975).

The H antigens are determined by a tube test. A motile organism is necessary for reaction with H antigens. Since *S. pullorum* and *S. gallinarum* are not motile, they will not react. For the exact procedure for detecting H antigens, the reader is referred to AOAC (1985) or FDA (1978).

The medium in which the organism is grown can influence the reaction with H antiserum (Banwart and Kreitzer 1972; Stamper and Banwart 1974).

Besides biochemical and serological tests, for epidemiological purposes phage typing, antimicrobial resistance patterns (resistotyping), and plasmid typing may be used for characterizing salmonellae (Brunner et al. 1983; Farrar 1983; Riley et al. 1983; Somerville, Nhlapo, and Alberts 1983).

OTHER METHODS. The conventional procedure requires considerable material and time to detect *Salmonella*. Other tests have been devised to obtain results more quickly and easily. Usually rapid tests result in a

loss of accuracy or precision. However, even with the conventional test, not all of the *Salmonella* serotypes are detected in all cases. The results obtained with a rapid test should at least compare favorably with those obtained with the conventional test (FDA 1978).

To detect the low number of salmonellae in most foods, the sample must be incubated in a growth medium to enrich the salmonellae prior to analysis, even by the "rapid" methods.

Hydrophobic-Grid-Membrane Filtration (HGMF). After preenrichment and enrichment, a portion of enrichment broth is passed through an HGM filter (Entis et al. 1982). The filter is transferred to a selective agar surface and, after incubation, observed for salmonellae. Presumptive colonies must be evaluated biochemically and serologically as in the conventional test. The HGMF system has been adopted by the AOAC.

Phage Systems. The use of bacteriophage has been suggested for detection, characterization, and the typing of certain serotypes of salmonellae.

To detect *S. typhosa* (*S. typhi*) in soil samples, Kandelaki (1964) added typhoid indicator phage to the sample. An increase in the phage titer during incubation was evidence that *S. typhi* was present in the sample. The phage titer increase detected more positive samples than did the bacteriological plating system Kandelaki used.

For determining the character of colonies on selective agars, one team of researchers (Cherry et al. 1954) suggested using bacteriophages. According to them, the method is rapid, simple, and quite specific. They used the O-1 phage of Felix and Callow. This system was reevaluated by Welkos, Schreiber, and Baer (1974). They tested 652 strains of *Salmonella* and found that the phage (O-1) reacted with 640 (98.2 percent). Of 1,463 nonsalmonellae strains, only *E. coli* strains were lysed. Of 239 strains of *E. coli* tested with 10^{12} plaque-forming units/ml (PFU/ml), 14 (5.9 percent) were susceptible to lysis. Gunnarsson, Hurvell, and Thal (1977) found that 98 percent of 5,287 strains of salmonellae were lysed by O-1 phage. They warned that monophasic strains of subgenus III and strains of subgenus IV usually are not sensitive to the O-1 phage.

The increase in O-1 phage was determined with high performance liquid chromatography to detect salmonellae in milk and feces (Hirsch and Martin 1983; Crane, Martin, and Hirsch 1984). A plaque system using O-1 phage reportedly can detect salmonellae in milk in 24 hr (Hirsch and Martin 1984). Since not all strains of salmonellae are affected by the O-1 phage, Güdel and Fey (1981) suggested adding a G_{47} phage to the O-1 phage to increase the number of strains that are lysed.

Immunoassays. There are many immunoassay systems, some of which are discussed for the detection of *S. aureus* enterotoxin. The immunoassays

of interest in salmonellae detection are those using a labeled reactant (usually the antibody).

For the radioimmunoassay (RIA), a radioactive substance is conjugated onto one of the reactants. Although the RIA is a sensitive system, there are problems (see *S. aureus* enterotoxin detection, page 215).

When a fluorescent compound is conjugated onto a reactant in salmonellae determinations, it is called the fluorescent antibody (FA) technique. This test has been adopted by the AOAC (Sanders 1975). After an enrichment procedure, a direct or indirect FA test can be used.

For the direct test, the specific antibody is conjugated with a fluorescent dye. When this conjugate is added to a solution containing homologous antigen, they will react, and the product will be fluorescent in ultraviolet light.

For the indirect method, if the antibody is produced in a rabbit, rabbit serum is injected into another animal, such as a goat, to produce antirabbit serum. This goat antirabbit serum is conjugated with the fluorescent dye. For the test, the antigen is reacted with the specific antiserum produced in the rabbit. Then the conjugated goat antirabbit serum is added, which will react with the antiserum produced in the rabbit.

The advantage of the indirect method is that, as long as the antibodies are produced in a rabbit, only one conjugated antiserum is needed (goat antirabbit). This conjugate will work in FA tests for clostridia, staphylococci, salmonellae, and any organisms, as long as the specific antiserum to the antigens of the organism are produced in a rabbit.

Besides the requirement for a good fluorescent (ultraviolet) microscope and well-trained personnel, the diagnostic sera used to detect salmonellae by the FA is expensive. The serum should react with all of the serotypes of *Salmonellae* and not cross-react with the nonsalmonellae. This type of serum is not yet available.

In many surveys, the FA procedure reveals more *Salmonella*-positive samples than the conventional culture test. Various reasons for this have been suggested. One proposal is that the FA is more sensitive than culturing. A more logical explanation is that cross-reactions with nonsalmonellae by the FA antisera result in more apparently *Salmonella*-positive samples. Thus, the samples that appear positive with the FA technique must be run through the conventional test for confirmation.

When an enzyme is conjugated onto a reactant, it is called an enzyme immunoassay (EIA) (see *S. aureus* enterotoxin detection, page 215). For salmonellae testing, this and the other immunoassays might be used to detect the organisms as well as their enterotoxins.

For their EIA, Mattingly and Gehle (1984) stated that 10^6 *Salmonella*/ml were needed for detection. This means that the salmonellae in the samples must be enriched prior to analysis. However, the enrichment

period for the EIA has been shortened from that suggested for the conventional test (FDA 1978). Of these immunoassays, the EIA is the newest and most promising (Mattingly et al. 1985; Swaminathan, Aleixo, and Minnich 1985). The ELISA has been used to detect salmonellae enterotoxins (Richter 1983).

For all of these immunoassays, specific antibodies are needed that will react with all of the salmonellae and not cross-react with nonsalmonellae. Although many claims have been made regarding various antibodies, no such antibodies are presently available and might never be found. Even the FDA (1978) test yields a few false negative samples, however, and sometimes even a false positive sample.

Although these systems have some advantages over the conventional method (FDA 1978), to study the organisms in an epidemiology of an outbreak of salmonellosis, they must be isolated.

Deoxyribonucleic Acid (DNA) Hybridization Assay. DNA is composed of two parallel, complementary strands that are uniquely matched and held together by chemical bonds. These strands can be separated. The two complementary strands, under specific conditions, can find each other and rejoin or hybridize, even in the presence of other molecules. Specific strands of DNA, called probes, that are unique to salmonellae and reportedly do not hybridize with other organisms are used in this assay. These probes make it possible to detect the presence of salmonellae in a complex mixture of organisms that are present in a food sample. It is necessary to enrich the sample to obtain sufficient salmonellae for the assay to work properly.

In the past, radioactive phosphorus (p^{32}) was used as a label on the probes to detect hybridization. However, enzyme labels have been developed for this purpose (Olsiewski, Thalenfeld, and Engelhardt 1985). Using enzyme labels (rather than radioactive phosphorus) makes the assay safer and more acceptable.

Reportedly, a properly formulated DNA probe will neither cross-react with other organisms nor give a false positive result. Other DNA probes can be used with the basic system for the detection of various organisms (Fitts 1985; Sayler et al. 1985).

Comments. At the present time, *Salmonella* methodology is confusing to some. There is more than one so-called official method, as well as many other proposed tests, several of which are not listed or discussed in this text.

However, considering the many types of samples and the various compositions of foods, as well as the bacterial populations, and considering that salmonellae, if present, are usually in very low numbers, perhaps many different methods are needed. Perhaps we should evaluate all of

the potential media and systems with each food to develop the best system for each food.

The results of research have shown that, with the FDA (1978) method, the more types of media that are used, the greater the likelihood that all *Salmonella*-positive samples will be detected. With nearly all of the so-called rapid methods, only one medium is used. Are these systems that superior to using a selective agar isolation system? It seems doubtful. Obtaining a false positive sample may increase the cost of food due to re-examination, reprocessing or discarding the suspect food. However, missing a *Salmonella*-positive sample with a resultant salmonellosis outbreak can result in an even higher cost to the processor.

CONTROL OF *SALMONELLA*. The measures needed to control *Salmonella* and salmonellosis are essentially the same as those needed for the control of *S. aureus* and the clostridia (prevent contamination, prevent growth, destroy the organisms). The number of cases of typhoid fever due to *S. typhi* has been drastically reduced. Proper sewage disposal, protection and chlorination of communal water supplies, restricting the harvesting of aquatic food from contaminated water, pasteurization of certain products, sanitary control of food processing and sales establishments, exclusion of typhoid carriers from food-handling occupations, and immunization programs, have reduced the incidence of typhoid fever in the United States.

Since the prevalence of salmonellosis is increasing, it is evident that these procedures are not effective in controlling this illness; other measures need to be incorporated.

There is an incentive for the food industry to market products with no salmonellae. Section 402 of the Federal Food, Drug and Cosmetic Act defines a food to be adulterated if it bears or contains any poisonous or deleterious substance that may render it injurious to health, and if it has been prepared, packed, or held under unsanitary conditions whereby it may become contaminated with filth or whereby it may be rendered injurious to health. Foods containing *Salmonella* or other pathogens fall within this definition.

The control of salmonellae in foods includes the acquisition of *Salmonella*-free raw materials, with processing, storage, and distribution under conditions that prevent the increase of *Salmonella* and, ideally, a terminal treatment of the food to destroy any salmonellae that may be present.

Prevent Contamination. Because salmonellae are ubiquitous, keeping them out of food might seem to be a hopeless task. However, it is generally believed that, with a sincere effort by all factions of feed and food pro-

duction, handling, processing, and preparation, the contamination of food by salmonellae can be prevented or reduced.

A major cycle of infection is animal by-product to feed, feed to animals, and animals to food to humans (Fig. 6. 10). There are other cycles, such as human to food to human or even human to human. If the cycle of feed to animals to humans can be broken, the incidence of salmonellosis might be reduced significantly.

There should be *Salmonella*-free breeding stock, feed, and water, strict sanitary practices, segregation of sick animals, and sale of only healthy animals. Sanitation includes many facets, such as pest control, cleaning and disinfecting housing, removal and disposal of waste, storing feed so that it is not contaminated, preventing workers from carrying diseases with them from one farm area to another, and keeping out visitors. There are sanitary requirements for dairy farms, but other farming practices are usually left to the farmer. However, Oosterom, Van Erne, and Van Schothorst (1982) reported that using practical farming situations, even with sanitary precautions, it was impossible to produce *Salmonella*-free pigs.

To reduce or eliminate *Salmonella* from chickens, a system called competitive exclusion shows promise (Blanchfield et al. 1984; Goren et al. 1984; Pivnick et al. 1982; Schneitz, Seuna, and Rizzo 1981). It is based on colonization of the gastrointestinal epithelial surface by indigenous organisms. Feeding or otherwise inoculating fecal, cecal, or anaerobic cultures from adult fowl into the intestines of day-old chicks will allow these bacteria to occupy the attachment sites. This deters or prevents salmonellae from colonizing the intestines of the chicks. Competitive exclusion also protected turkeys from salmonellae (Reid and Barnum 1983). The feeding of antibiotics to animals can alter the indigenous flora, allowing salmonellae to colonize the intestinal tract (Barrow, Smith, and Tucker 1984; Que and Hentges 1985).

For seafoods, the habitat in which such animals grow must be free of *Salmonella.* The first step is to prevent untreated sewage or water from entering streams, lakes, or oceans. Shellfish, such as oysters, are believed to cleanse themselves of salmonellae and *E. coli* in 48 to 72 hr if placed in clean water. At an initial level of 400 salmonellae per gram, oysters contained no detectable cells after 2.5 days, whereas 900 cells per gram required 3.5 days (Son and Fleet 1980). However, clean beds are needed for the production and harvesting of shellfish, since the depuration process cannot be relied upon for the elimination of all pathogens.

In the preparation of food, one of the important factors for spreading *Salmonella* is the cross-contamination from raw to cooked food. Until such time as raw poultry or meat products can be produced with no *Salmonella*, we must assume that the organisms will be present. Handling of these

products and then handling ready-to-eat products transfers the bacteria from the raw to the ready-to-eat products.

Thorough daily cleaning and sanitizing of the processing plant and equipment can reduce the dissemination of salmonellae and other organisms into the environment and onto foods. Testing for salmonellae should be part of the control effort. If the organisms are in the plant environment, there is a good chance they will be in the finished product. Cleaning and sanitizing are discussed further in Chapter 10.

Foster (1969) pointed out that elimination of salmonellae from foods would not necessarily eliminate human salmonellosis. We also need to consider the human carrier as well as household pets. The human carrier is important in the processing plant, in the food-preparation and serving industry, and in the home.

Foods have been placed into five categories in terms of hazard to human health (Foster 1971). No safe level of salmonellae has been established, but it is evident that illness is more likely when the number ingested is increased. Three hazard characteristics were established for a particular food. These are (1) the food or an ingredient is a significant source of *Salmonella*; (2) there is no control step (heating) in the process to destroy the organism; and (3) if the product is mishandled, growth can occur, resulting in an increased hazard. The higher the number of hazard characteristics of a food, the more potentially dangerous it is. A food with no hazard characteristics is not considered dangerous, foods with one or two are intermediate, and foods with all three are hazardous. The susceptibility of people varies, with infants, the aged, and the infirm the most susceptible to *Salmonella* infection. With this aspect included, the five categories are established:

Category I foods possess any of the hazard characteristics and are intended for use by infants, the aged, or the infirm.
Categories II, III, IV, and V foods are for general uses.
Category II foods possess all three hazard characteristics.
Category III foods have two of the hazard characteristics.
Category IV foods have one hazard characteristic.
Category V foods have no hazard characteristics.

With these hazard categories, it is evident that fewer samples of foods in category V need to be tested (if negative results are obtained), than those in category I or II. Considering 25-g samples, it was proposed that the food should be accepted if there are no positive samples in fifteen samples (Category III, IV, or V), thirty samples (Category II), and sixty samples (Category I) or, if no more than one positive sample is detected in twenty-four samples (III, IV, V), forty-eight samples (Category II) or

ninety-five samples (Category I). Thus, the lower the risk category number and the greater the hazard, the greater the number of samples that need to be analyzed and the lower the percentage of positive samples that are allowed.

Prevent Growth. Although refrigeration at 4° to 5°C will control the multiplication of *Salmonella*, it will not improve the hazard characteristic, since the food can be abused in market channels or during preparation and serving. Foods should not be held between 10° and 50°C. These temperatures should be traversed as rapidly as possible during heating and cooling.

The salmonellae do not grow in foods with naturally low pH (below pH 4.0). Acids can be added to some foods or feeds to inhibit growth of the organisms. Although salmonellae might not grow, they may survive in some foods at a low pH.

Potassium sorbate reportedly inhibits the growth of salmonellae (Rice and Pierson 1982; To and Robach 1980). Sodium sulfite and metabisulfite inhibit growth of *Enterobacteriaceae* in ground meat (Banks and Board 1982). Although sulfites are allowed in some foods in the United States, they are not allowed in fresh red meats.

Destroy the Organisms. The destruction of the organisms is the best way to make certain that they do not present a health hazard. It is preferable if the destruction can be accomplished in the final package, so that recontamination does not occur during further handling.

Liquid propylene oxide was effective in destroying salmonellae in meat and bone meal (Tompkin and Stozek 1974). The concentration of propylene oxide needed for destruction of salmonellae depends on the relative humidity, level of contamination, and time of exposure. There is a residual limit of 300 μg/g in certain food ingredients.

Fumigation of feed (fish, meat, or bone meal) with formaldehyde gas is an effective means of destroying salmonellae (Duncan and Adams 1972). The gas eliminates salmonellae to a depth of 1.91 cm of feed.

The chlorination of water is used to destroy *S. typhi*. This treatment can aid in the control of salmonellae in food-processing plants whether chlorine is used in wash water, equipment rinse water, or cooling water.

The heating of foods is the principal system to eliminate salmonellae. Pasteurization is used to control salmonellae in liquid egg products. This is discussed in Chapter 11. Pasteurization of coconut at 80°C for 8 to 10 min effectively killed salmonellae without affecting the product quality (Schaffner et al. 1967). Another report stated that the present milk pasteurization process (63.33°C for 30 min or 71.67°C for 16 sec) will inactivate salmonellae, as long as the number does not exceed 3×10^{12} salmonellae per milliliter of milk (Read et al. 1968). It would be extremely

unlikely that this level of salmonellae would be attained without spoilage, due to other bacteria.

Although bacteria have an increased heat resistance in dried products, it is not impossible to eliminate salmonellae from dry animal feed. Carrol and Ward (1967) found a combination of 87.8°C for 10 min consistently reduced salmonellae to a nondetectable level in fish meal. Van Cauwenberge, Bothaste, and Kwolek (1981) stored salmonellae-inoculated corn flour (10 and 15 percent moisture) at 49°C. After 24 hr, 99.9 percent of the salmonellae cells were destroyed. However, heating salmonellae-inoculated poultry feed in a microwave oven (reaching a temperature of 186°C), in a hot air oven at 110°C for 1 hr, or in flowing steam for 10 min failed to consistently eliminate the salmonellae (Burdick et al. 1983).

The cooking of foods usually destroys any salmonellae that may be present. *S. senftenberg* 775W is killed during the normal processing of frankfurters in the smokehouse when the internal temperature of the product reaches 71.1°C (Palumbo, Huhtanen, and Smith 1974). Heating pepperoni to 60°C or Lebanon bologna to 51.7°C eliminated 10^4 inoculated cells of salmonellae (Smith et al. 1975a, 1975b). Goodfellow and Brown (1978) determined *D* values (time needed to destroy 90 percent of the population) for salmonellae in a ground beef system. The *D* values they reported were 51.6°C (61–62 min), 57.2°C (3.8–4.2 min), and 62.7°C (0.6–0.7 min). These *D* values were used to establish the times and temperatures needed to cook roast beef to assure the death of any salmonellae.

Salmonellae are only moderately resistant to radiation treatment. Although the needed radiation dose varies with the type of product, conditions of radiation, and level of contamination, Thornley (1963) suggested that 0.5 Mrad would reduce the number of salmonellae by a factor of 10^7. Ostovar, Pereira, and Gallop (1971) found that 0.5 Mrad was sufficient to destroy salmonellae in smoked whitefish. Another team of researchers reported that, for frozen meat, 0.6 Mrad would reduce *Salmonella* numbers by a factor of at least 10^5 (Ley et al. 1970). For decontamination of herring meal, radiation doses of 0.8 to 1.3 Mrad were recommended (Underdal and Rossebo 1972). Mulder, Notermans, and Kampelmacher (1977) suggested treating poultry carcasses with 0.25 Mrad to eliminate salmonellae. More recently it was reported that, after treatment with 0.25 Mrad at 5°C, the poultry might still contain salmonellae, but that, after storage of this treated poultry at −18°C for three months, no salmonellae were detected (Mulder 1982). Pearce (1979) found that salmonellae were quite stable in stored casein, but treatment with 0.5 Mrad eliminated these organisms from the protein.

The low doses of radiation needed to eliminate salmonellae have no

apparent adverse effect on the flavor or nutritional quality of the food or feed. It would seem to be a very useful process for animal feed, since gamma irradiation can be accomplished after the feed has been packed into impervious bags.

Shigellosis (Bacillary Dysentery)

This illness accounts for less than 10 percent of the reported outbreaks of foodborne illness in the United States. Since only a few states report this illness to CDC, it is probably more widespread than the data indicate. The predominant mode of transmission is human to human, but contaminated water and food also are important.

Sometimes the illness is called the filth disease, because it is associated with poor personal hygiene and sanitation. The illness is often prominent in areas where there are large groups of people. Summer camps, mental institutions, Indian reservations, and low socioeconomic urban communities are the high-risk population areas. The increase in infant day care centers has resulted in another high risk situation.

The mortality rate appears to be greatest in infants and in adults over fifty years of age. Most of the cases are children under ten years, and the highest attack rate is children in the one-to-four age group. This is probably due to their lack of training in personal hygiene at this young age.

In the twenty-to-twenty-nine age group, there is a higher rate of illness in females than in males. This is believed to be due to women's having closer contact with children, especially sick children, than do men of this age group.

Although considered a disease of the poor, the affluent have acquired the illness through travel to foreign areas, such as Mexico, Central America, and the Far East.

Shigellosis is primarily an illness of humans, but other primates, including monkeys and chimpanzees, can acquire the disease (Rout et al. 1975). By using procedures such as starvation, folic acid deficiency, control of normal intestinal flora, and administration of opium to slow peristaltic action, experimental shigellosis has been established in mice and guinea pigs (Nelson and Haltalin 1972; Maier and Hentges 1972). The ligated small intestine of rabbits serves as an animal model for studying shigellosis.

PATHOGENESIS. The pathogenesis of *Shigella* is rather complex. After reaching the small intestine, the first requirement is the attachment to and penetration of cells of the intestinal mucosa. If the cells cannot invade, they are avirulent. After invasion, shigellae multiply and cause the destruction of cells (Levine et al. 1973). This results in mucosal ulcera-

tion. Speelman, Kabir, and Islam (1984) studied colonic lesions in thirty-three patients with shigellosis. Their findings indicated that the rectosigmoid area is the most frequently and most severely affected, and that during the course of illness, the colonic lesions extend in a proximal direction. In some cases, lesions extended to the splenic flexure, the distal transverse colon, and the proximal transverse colon. In 15 percent of the patients, pancolitis was evident.

Shiga toxin, an enterotoxin produced by *S. dysenteriae* 1, the most virulent *Shigella*, was purified by one group of researchers (Brown et al. 1982). The toxin was cytotoxic to HeLa cells, caused secretion in rabbit ileal loops, and was lethal to mice. *S. sonnei* and *S. flexneri* also produce toxins that are antigenically and biologically related to Shiga toxin of *S. dysenteriae* (Keusch and Jacewicz 1977; O'Brien et al. 1977). These three species, as well as *S. boydii*, produce a toxin that causes morphological changes in Chinese hamster ovary cells (Takeda, Okamoto, and Miwatani 1979).

The exact role of the enterotoxins in shigellosis has not been established, but there are various speculations as to their action (Fernandez et al. 1984; Formal, Hale, and Sansonetti 1983; Mathias et al. 1980). The mechanism of the pathogenic action of shigellae and their toxins is not known exactly. The illness may be an infection caused by the organism, some form of intoxication, or a combination of these actions.

CHARACTERISTICS OF THE ILLNESS. The number of organisms needed to cause the infection may be rather low, since person-to-person transfer is the main mode of transmission. One study reported that as few as ten organisms of virulent strains produced illness in human volunteers (Levine et al. 1973).

In most outbreaks of shigellosis there is an index case. One person develops symptoms of the disease, and then one or two days later, others became ill. If the index case is a food handler, a foodborne outbreak can occur. An example of an outbreak is shown in Figure 6.12.

Incubation Period. The time from ingestion of the organisms until symptoms appear varies with the individual (health, age), number of organisms ingested, and virulence of the organism. CDC (1983b) listed the incubation period for shigellosis as 12 to 50 hr. Due to person-to-person spread, it sometimes is difficult to determine the exact times for the infections.

Symptoms. The symptoms of shigellosis as listed in several outbreaks are shown in Table 6.21. The main symptom is diarrhea. Infections associated with mucosal ulceration usually cause more fever, abdominal pain, and rectal bleeding than do toxin-induced diarrheas.

In the United States, the disease is usually self-limiting and rather

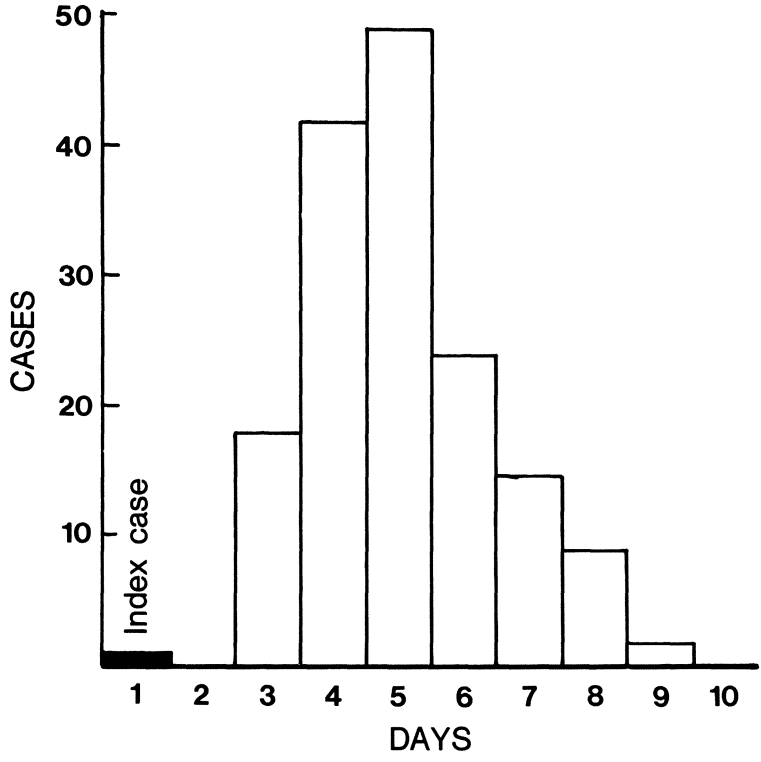

Figure 6.12. Progress of a typical outbreak of shigellosis.

mild. However, the diarrhea may be severe; the drastic loss of fluids with resulting dehydration is the major consideration. In young children the disease may be very severe, or even fatal, due to dehydration and extraintestinal manifestations.

Duration and Therapy. The illness may last for 12 hr to three weeks, with the average illness lasting from five to six days. The infection usually persists beyond clinical recovery, and the duration of the convalescent carrier state is commonly three to four times that of the duration of symptoms.

Since very young patients may not understand the need for strict personal hygiene, drug therapy is often warranted to prevent secondary spread. Food handlers should be considered potential spreaders of the organism and should be given therapy. The type of drug used is dependent on the sensitivity of the cultured strain and on the sensitivity of the patient to various antibiotics.

TABLE 6.21. SHIGELLOSIS: PERCENTAGE OF PEOPLE EXPERIENCING SYMPTOMS IN EIGHT OUTBREAKS

	Outbreak							
Symptom	1	2	3	4	5	6	7	8
Diarrhea	100	98	91	92	100	98	100	91
Bloody	6	6	11	9	23	6	5	24
Mucus	—	19	19	—	—	19	—	—
Tenesmus	—	31	54	17	—	31	—	54
Abdominal pain or cramps	94	85	64	76	79	85	16	69
Chills	56	54	57	27	51	54	—	—
Fever	53	47	57	78	95	47	27	76
Nausea	41	59	47	56	—	59	35	57
Vomiting	33	27	21	53	49	27	13	48
Headache	—	66	43	29	51	66	—	57
Muscle ache	—	55	—	—	—	55	—	—
Weakness	—	—	71	—	—	—	—	—

SOURCE: CDC *Reports.*

After recovery from shigellosis, an apparent state of immunity often develops, but repeated infections may occur. Immunization of a population would not be practical, due to the low incidence of shigellosis. In institutions or other areas of high incidence, immunization would be beneficial.

Foods Involved. As might be expected, the foods that have been involved in most outbreaks are those that are handled the most. These are salads (potato, tuna, shrimp, macaroni, and chicken). The ingredients may be clean, but during the preparation, the salad is contaminated by hand manipulation or mixing. The organisms can multiply readily in moist foods held at room temperature, so even a small inoculum can cause a large-scale outbreak of shigellosis.

THE ORGANISMS. The genus *Shigella* contains four species: *S. dysenteriae, S. flexneri, S. boydii,* and *S. sonnei.* They are separated mainly by serological tests and DNA relatedness. Each of the four species contains several biotypes or serotypes. In the United States more than 60 percent of the isolates are *S. sonnei,* and more than 25 percent are *S. flexneri.* On a relative basis, *S. sonnei* is decreasing and *S. flexneri* is increasing in the frequency of isolation. Typically, the shigellae are nonmotile, fail to ferment lactose, and are anaerogenic (do not produce visible gas from fermentation of carbohydrate).

Sources. The normal habitat is the intestinal tract of human beings and other primates. Isolation from other animals is rare.

In a survey of market vegetables in Ceylon, Velaudapillai, Niles, and Nagaratnam (1969) found cabbage, leeks, spinach, beans, carrots, and pumpkins contaminated with shigellae. They believed that the contamination may have originated from the water used for irrigation, fertilizer (manure), or from handling the vegetables at the market.

The main source of shigellae involved in outbreaks is people who are symptomless carriers, or ambulant cases. The organisms can be isolated for several weeks after recovery from the illness. Testing of healthy food handlers for these organisms would be meaningless; however, a person experiencing diarrhea should not be allowed to handle food.

METHODOLOGY. The methodology involved with the shigellae includes enrichment, plating, and differentiation to detect the organism. Serological reactions, colicin typing, and DNA evaluation are aids to further separate the shigellae. The use of colicin typing as an epidemiological tool was questioned by Vlajinac and Krajinovic (1983).

Virulent and avirulent strains can be differentiated by feeding human volunteers. A simpler method is based on the fact that keratoconjunctivitis develops in the eyes of rabbits and guinea pigs when they are inoculated with virulent strains of shigellae (Cross and Nakamura 1970). The cytotoxic activity of *Shigella* toxins can be assayed with HeLa cells (Gentry and Dalrymple 1980).

CONTROL. Foodborne outbreaks of shigellosis are caused by the mishandling of food. A high standard of personal hygiene by food handlers (washing hands after using the toilet, not handling food during illness or a diarrheal symptom), and sanitation of the premises, with proper cooking and refrigeration of foods, should help control shigellosis.

Escherichia coli Enteritis

Certain strains of *E. coli* cause enteric disease syndromes in people and other vertebrate animals. In countries lacking basic sanitation, *E. coli* is a prominent cause of enteritis. In most of the outbreaks in the United States, the organisms are transmitted by water. Some outbreaks are foodborne, although *E. coli* is not listed as a causative agent in Table 6.1.

In 1971, an outbreak due to enteropathogenic *E. coli* (EEC) in imported cheese resulted in 387 reported cases. This was the first well-documented United States outbreak in which food containing EEC affected adults.

In a survey of infants and children with diarrhea, salmonellae or shigellae were isolated from 42 percent of the patients, while enterotoxin-producing strains of *E. coli* were found in 86 percent of the diarrhea

group and 41 percent of the control group (Rudoy and Nelson 1975). This indicates that in enteric diseases of young people, *E. coli* may be as important as salmonellae or shigellae.

ETIOLOGIES. The role of *E. coli* in enteric disease, especially the diarrheal syndrome, is quite complex. Although all virulent strains can be called enteropathogenic, there are classical enteropathogenic strains, invasive strains that cause illness similar to shigellae, strains that produce enterotoxins, and a more recently recognized enterohemorrhagic *E. coli* (Levine and Edelman 1984).

Classical enteropathogenic strains of *E. coli* that are neither invasive nor produce enterotoxin (ST or LT) have been reported (Levine et al. 1978; Robins-Browne et al. 1982; Toledo et al. 1983). The colonization of the intestines by enteropathogenic strains can result in moderate to severe injury to the mucosa (Rothbaum et al. 1982; Ulshen and Rollo 1980). It has not been established that mucosal injury causes the gastroenteritis. The possibility that a toxin different from the conventional enterotoxins is involved in the diarrheal syndrome has been proposed (Klipstein et al. 1978). Other researchers have suggested that a shigalike cytotoxin may play a role in enteropathogenic *E. coli* gastroenteritis (Cleary et al. 1985). In a study of an enteropathogenic strain, Ryder, Kaslow, and Wells (1979) found that it produced a heat-stable enterotoxin.

The invasive strains of *E. coli* cause a dysentery (fever, cramps, and diarrhea containing blood and mucus) similar to that caused by shigellae (DuPont 1978; Sakazaki et al. 1974). The invasive strains can be distinguished by the development of keratoconjunctivitis when placed on the guinea pig eye, or by the penetration of HeLa or HEp-2 cells (Mehlman et al. 1982).

Enterotoxigenic *E. coli* (ETEC) may form a heat-labile (LT), a heat-stable (ST), or both LT and ST toxins. The heat-labile toxin of *E. coli* is readily destroyed at 60°C. It is similar to cholera enterotoxin (choleragen) immunologically, structurally, and functionally. Cholera antitoxin reacts not only with choleragen and *Salmonella* enterotoxins, but also with the LT of *E. coli*. Both choleragen and LT are composed of two subunits, designated as A and B.

Acting alone, neither A nor B is toxic. For toxicity, the B subunit binds with host cell membrane receptors and acts as the messenger for the A subunit, which activates adenyl cyclase (Gill and Richardson 1980; Moss et al. 1984). This alters the cyclic adenosine monophosphate metabolism in intestinal cells, which induces an enhanced secretion of fluid and electrolytes into the intestine, resulting in the diarrheal syndrome in the same manner as cholera toxin.

In one study, researchers separated the A and B subunits of LT and

choleragen (Y. Takeda et al. 1981). Then they recombined them to form hybrids of the two toxins. These hybrids revealed a toxicity similar to that of the parent toxins. According to Clements and Finkelstein (1979), one difference between LT and choleragen is that LT is plasmid mediated, while choleragen is chromosomally mediated.

The heat-stable (ST) enterotoxins of various *E. coli* strains have been purified, and their properties determined. Heating at 100°C for 1 hr has no effect on biological activity (Robertson, Dreyfus, and Frantz 1983), but the toxin is inactivated when heated at 121°C for 15 min (Lallier, Lariviere, and St. Pierre 1980). They reported that the maximum amount of enterotoxin was produced after 7 hr of growth.

There may be more than one ST (Burgess, Mullan, and Newsome 1980). It has been observed that not all ST shows activity in the suckling mouse assay. Those toxins showing activity in both the suckling mouse and piglet assays have been designated as ST_a. Other toxins having activity in the piglet assay but not the suckling mouse assay have been called ST_b.

The ST molecule is a polypeptide with a molecular weight of less than 5,000. This means that it is a rather poor antigen. By coupling with a protein carrier and using multiple booster doses, researchers have produced antiserum (Frantz and Robertson 1981). The antiserum reacted with ST_a from various strains of *E. coli* but not with ST_b. Frantz and Robertson stated that ST_a from various strains of *E. coli* share at least one antigen determinant.

In the lower small intestine, *E. coli* ST_a stimulates the enzyme guanylate cyclase, which increases cyclic guanosine monophosphate, including secretion of fluids into the intestine (Argenzio et al. 1984; Dreyfus, Jaso-Friedmann, and Robertson 1984; Gianella and Drake 1979; Rao et al. 1980). The mechanism of the activity of ST_b is unknown, but apparently the increase in cyclic nucleotides such as GMP or AMP is not involved (Dreyfus, Frantz, and Robertson 1983; Kennedy et al. 1984).

The ability to produce the enterotoxins is determined by plasmids (Echeverria et al. 1985; T. Takeda et al. 1981). Since the plasmids are transferable, theoretically any *E. coli* might become enterotoxigenic. However, only certain serotypes of *E. coli* tend to accept the plasmid (Echeverria et al. 1982; Reis 1980). The plasmids were successfully transferred to *Salmonella, Shigella, Citrobacter, Enterobacter, Klebsiella, Edwardsiella, Serratia*, and *Proteus* species (Neill, Twiddy, and Holmes 1983; Takeda et al 1983).

There appear to be several steps in the development of *E. coli* gastroenteritis. Normally, when a person ingests cells of *E. coli* they pass through the stomach and small intestine and may colonize in the large intestine. If the cells can adhere to the epithelial cells in the small intes-

tine, they may remain for a sufficient period to multiply, and, if toxigenic, to produce enterotoxin.

Reportedly, the enterotoxins are produced by the organisms in the small intestine. However, they have been produced in laboratory media, with the culture filtrates showing fluid accumulation in ileal loops. Hence, it is possible that some enterotoxin is preformed in foods. No one has reported finding *E. coli* enterotoxin in foods.

CHARACTERISTICS OF THE ILLNESS. Since there are enteropathogenic, enteroinvasive, enterotoxigenic, and enterohemorrhagic strains of *E. coli*, there is more than one type of illness.

Incubation Period. The incubation period is listed as 6 to 36 hr (CDC 1983b). In an outbreak of enteropathogenic organisms involving imported cheese, the mean incubation time was 18 hr, and the average duration of symptoms was two days (Marier et al. 1973). For an outbreak involving invasive strains, the incubation period ranged from less than 24 hr to more than 72 hr, with most cases between 24 and 47 hr (Tulloch et al. 1973). In an outbreak due to heat-labile enterotoxin, symptoms began by 12 hr and peaked at 36 to 48 hr (CDC 1976a). In an outbreak involving toxigenic *E. coli*, the incubation ranged from 3 to 166 hr, with a median of 44 hr (Taylor et al. 1982).

Symptoms. The main symptom is diarrhea. In a foodborne outbreak due to imported cheese (Marier et al. 1973), 88 percent of the people reported this symptom. Other symptoms (with the percentage of people experiencing the symptom) were fever (72 percent), nausea (71 percent), cramps (66 percent), chills (38 percent), vomiting (35 percent), malaise (34 percent), aches (28 percent), and headache (19 percent). With severe diarrhea, dehydration can occur. Newborn infants are especially susceptible to enteropathogenic *E. coli*. In an outbreak of invasive strains, 100 percent of the patients had diarrhea and 7 percent had bloody diarrhea, similar to that with *Shigella* (Tulloch et al. 1973). Tenesmus was a symptom in 96 percent of the patients. In another outbreak, the main symptoms were diarrhea, nausea, abdominal cramps, headache, chills, and dizziness (Snyder et al. 1984).

Duration and Therapy. The duration of the illness depends upon the severity of the disease and the type of individual. The average illness persists for three to four days, but in some cases, the diarrhea may persist for fourteen days. Usually, coliforms are removed from the upper intestine in seven to ten days, but in rare cases, they may persist for several weeks.

With severe watery diarrhea, fluids and electrolytes must be replaced. Glucose tends to induce absorption of sodium and water; thus, treatment

with oral glucose-electrolyte solutions has been suggested (Klipstein and Engert 1978).

Foods Involved. The only well-documented foodborne outbreaks in the United States involved soft cheese. In other countries, outbreaks have been associated with consumption of dairy products, salads containing raw vegetables, meat and meat products, poultry, fish, and baked products.

Number of Organisms Needed. Although newborn humans and animals are highly susceptible, adults are more resistant. In a survey by Rudoy and Nelson (1975), EEC was isolated from 41 percent of a control group of children without diarrhea. This indicates that, even if a few organisms are present, they do not cause illness if the numbers are kept under control by the body defenses or by interactions with other organisms in the intestine.

One study found that all of their human volunteers developed diarrhea 11 to 48 hr after ingestion of 10^6 to 10^{10} colony-forming units of *E. coli* (Donta et al. 1974). The type and number of cells of *E. coli* and the age and physical condition of the host determine the occurrence and the severity of the illness.

NATURE OF THE ORGANISM. *Escherichia coli* is an important organism in the microbiology of foods. Besides being involved in gastroenteritis, the organisms in this species are considered to be indicators of possible fecal contamination and can cause spoilage of some foods. There are more than 160 serotypes of *E. coli*.

Sources. The main habitat of *E. coli* is the intestinal tract of humans and animals. However, fecal contamination causes it to spread to other environments, especially soil and water.

Researchers analyzing various foods of animal origin found enterotoxigenic *E. coli* in each type (Sack et al. 1977). They believed that these foods are potentially important vehicles for these toxigenic organisms, especially if the food is not handled properly.

EEC was detected in about 10 percent of 2,000 samples of cheese (Fantasia et al. 1975). None of the foods examined from a village in the Philippines (Echeverria et al. 1978) or Ethiopia (Jiwa, Krovacek, and Wadström 1981) contained enterotoxigenic *E. coli*.

METHODOLOGY. There is no simple culture medium or system that will differentiate pathogenic from nonpathogenic *E. coli*. After enrichment and isolation of *E. coli*, it is necessary to characterize the organisms with biochemical, serological, and other tests.

Invasiveness and Enterotoxigenicity. Laboratory models are available to detect and assay invasiveness or enterotoxigenicity of *E. coli* (Table 6.22).

Invasive strains are detected by their ability to penetrate the superficial layers of the cornea and cause keratoconjunctivitis in the eye of the guinea pig. The invasion of rabbit intestinal mucosa can be tested if the culture causes keratoconjunctivitis.

The ligated intestinal loop test can be used to determine pathogenicity of *E. coli.* By using bacteria-free inocula, the reactions of enterotoxins, assay of enterotoxins, and the effect of antibodies on the reaction can be obtained. Generally, rabbit intestine is used, but other animal models are acceptable.

In rabbit ileal loops, the LT causes maximum fluid accumulation in about 18 hr, with the ratio of volume (ml) to length (cm) about 2.0:2.5. To detect ST, the loops are examined at 6 to 8 hr, although some effect of LT is evident at the time. To determine both LT and ST, the test material is introduced through the stomach with a tube or injected into the intestine of a seven-to-nine day-old rabbit. After 6 or 7 hr, the animal is sacrificed, the entire GI tract is removed and weighed, the fluid is removed and measured, and the ratio of volume to fluid is determined (Gorbach and Khurana 1972). The rabbit as an animal model is relatively expensive and the method is time consuming with both false positive and false negative reactions (Thorne and Gorbach 1978).

An infant (suckling) mouse assay has been used to detect ST. The one-to-four day old mice are inoculated intragastrically. After 4 hr, the mice

TABLE 6.22. DETECTION OF INVASIVENESS AND THE ASSAY OF ENTEROTOXINS OF *ESCHERICHIA COLI*

Property	Test
Invasiveness	Guinea pig eye-keratoconjunctivitis test (Sereney test)
	Invasiveness of rabbit (or other animal) intestinal mucosa
	Plasmid-140 megadalton
Heat-labile enterotoxin (LT)	Rabbit ileal loop (18 hr)
	Skin-permeability test (rabbit)
	Chinese hamster ovary tissue culture
	Y-1 mouse adrenal tumor cells
	Seological tests (ELISA, Biken, latex particle, RIA)
	Gene probe
	Hamster-uterine growth
Heat-stable enterotoxin (ST)	Suckling mouse
	Rabbit ileal loop (6 hr)
	Guanylate cyclase activation
	Gene probe
	ELISA
LT and ST	Suckling mouse test (6 hr)
	Gas-liquid chromatography

are sacrificed and the ratio of intestinal tract plus contents weight to the remaining body weight is determined. A ratio of greater than 0.83 indicates the presence of ST. Researchers reported, however, that the mouse test was not sufficiently sensitive (Merson et al. 1980). Frantz and Robertson (1981) found the assay to be impractical for screening large numbers of isolates. They stated that it was fairly expensive and did not detect all forms of ST. A dog assay detected more ST-producing strains than did the mouse assay (Nalin et al. 1978).

Rabbit skin tests have been used to detect LT. The preparation is injected intracutaneously, followed 24 hr later by an injection of Evans blue dye. Areas of induration and bluing are read 1 to 2 hr later. Young rabbits show a better reaction than do old rabbits.

Certain cell lines can be used to assay for these enterotoxins. The LT causes elongation of Chinese hamster ovary cells.

Originally flat Y-1 mouse adrenal tumor cells become rounded when exposed to the enterotoxins (Gurwith 1977). Vero (African green monkey) cells appear enlarged, thick-walled, and refractile, with several filamentous tendrils in response to *E. coli* LT (Speirs et al. 1977). The ST has no effect on these three cell lines (Giugliano, Stavric, and Konowalchuck 1982). The Y-1 test for LT and the suckling mouse test for ST were adopted as official first action by the AOAC (Lovett and Peeler 1984).

Another team of researchers described a gas-liquid chromatographic method that could detect and differentiate LT and ST (Brooks et al. 1984). The activation of guanylate cyclase was used to detect ST (Waldman et al. 1984). They claimed it was more sensitive than the suckling mouse test. The increase in uterine weight after inoculation i.p. into cycling hamsters was used to detect the presence of LT (Alleva and Lamanna 1984). Cholera toxin also gave this response.

Various serological tests similar to those used to detect staphylococcal enterotoxins have been used for *E. coli* LT (Honda et al. 1981; Tsukamoto et al. 1980; Yano et al. 1982), and ST (Giannella, Drake, and Luttrell 1981). A staphylococcal coagglutination test for LT was developed by Brill, Wasilavskas, and Richardson (1979) and modified by later researchers (Honda et al. 1983a). Monoclonal antibodies have been developed for both LT and ST (Belisle, Twiddy, and Holmes 1984; Brandwein et al. 1985).

Enzyme-linked immunosorbent assays (ELISA) have been developed to detect LT (Honda, Sato, and Miwatani 1984; Merson et al. 1980; Ristaino, Levine, and Young 1983; Sack et al. 1980; Svennerholm and Wiklund 1983) as well as ST (Klipstein et al. 1984; Thompson et al. 1984). At the present time, the ELISA systems seem to be the most promising for the detection of toxins as well as other antigenic materials.

Methods have been developed to detect enterotoxigenic *E. coli* (ETEC)

by means of DNA colony hybridization using DNA or gene probes (Moseley et al. 1980; Patamaroj, Seriwatana, and Echeverria 1983). According to these authors, this system can detect ETEC in food or stools without the need to enrich the sample. The use of this system for determining ETEC among isolated organisms was approved as official first action by the AOAC (Hill and Payne 1984).

CONTROL. The control of the illness is similar to that for other enteric diseases. As far as foodborne illness is concerned, the number of cases would not warrant the immunization of all humans in developed countries but might be worthwhile for people at risk in developing countries. Some data suggest that there is a natural development of immunity in an endemic area (Black et al. 1981).

To prevent contamination, Bryan (1973) stated simply that we should practice personal hygiene, prepare foods in a sanitary manner, protect and treat (chlorinate) water, and dispose of sewage in a sanitary manner.

The easiest way to prevent growth of the organism is to provide an unacceptable environment for *E. coli*.

Since apparently large numbers of cells are needed to cause illness in other than newborn babies, simply cooking the food sufficiently provides a safe product.

Bacillus cereus Gastroenteritis

Under normal circumstances, *B. cereus* is not considered to be a pathogen. Foodborne illness due to *B. cereus* rarely is reported in the United States. The number of reported outbreaks is much higher in European countries and is one of the major causes of foodborne illness in Hungary.

There are two distinct syndromes caused by enterotoxins produced by *B. cereus*. One enterotoxin causes an illness similar to that produced by the *S. aureus* enterotoxin. The primary symptoms are nausea, vomiting, and abdominal cramps. Diarrhea may occur in the later stages of some cases. The incubation period usually is less than three hr. A second enterotoxin results in an illness similar to that caused by the *C. perfringens* enterotoxin. In this case, the effects are primarily in the lower intestinal tract, and the primary symptom is diarrhea. The incubation period is about 10 to 12 hr.

In both syndromes, fever is absent, indicating that toxins are involved. Usually, the symptoms last about 12 to 24 hr, but some cases, especially of the emetic type, may be severe. Due to the usually mild illness, with short duration, people might not seek medical aid. Therefore, most cases probably are undetected and unreported.

High numbers of *B. cereus* (10^6 to 10^9/g) have been associated with

foods involved in the illness. Due to multiplication of the cells in the food before analysis, these numbers may be higher than when the food was ingested.

THE ORGANISM. *B. cereus* is an aerobic, Gram-positive, motile, spore-forming rod. It hydrolyzes esculin and starch, reduces litmus milk and nitrate, and is catalase positive.

Since many cells ($> 10^5$/g of food) are needed to cause foodborne illness, the food must be abused sometime during preparation. In outbreaks in England due to rice, the rice was boiled and then allowed to sit at room temperature for 12 hr to three days. The rice was either boiled or fried before being served. The original heating was not sufficient to kill the spores of *B. cereus* but could activate the spores to germinate. Holding the rice at room temperature allowed the vegetative cells to multiply to levels that caused illness.

Source. B. cereus is common in soil and dust, so it is logical that foods that are readily contaminated by soil and dust will contain the organism. Plant products (cereals, flour, starch, bakery products, spices), animal products, and mixtures of ingredients (spaghetti sauce, pudding, soup mixes, gravy mixes) can contain a few or many cells or spores of *B. cereus*.

There is a similarity between foods involved in illnesses due to *B. cereus* and to *C. perfringens*. In either case, the food is prepared ahead of time in large batches and is not properly cooled before reheating (if needed) and serving. The reheating is not sufficient to destroy the cells.

METHODOLOGY. To detect *B. cereus* a spread plate system using mannitol-egg yolk-polymyxin (MYP) agar is suggested (FDA 1978). On this medium, the organism produces pink colonies surrounded by a dense precipitate (lecithinase-positive reaction). *B. cereus* is mannitol negative. Harmon, Kautter, and McClure (1984) found MYP slightly superior to two other media (polymyxin pyruvate-egg yolk-mannitol-brom thymol blue and trypticase-soy-polymyxin blood agars), but all three were satisfactory for detection of *B. cereus*.

Selected colonies are isolated and the organisms tested for the Gram reaction and biochemical tests. A method for differentiating *B. cereus* was described by Harmon (1982) and was adopted by the AOAC as its interim official first action.

ENTEROTOXIN. Turnbull (1981) discussed the various toxins of *B. cereus*. Gilbert and Kramer (1984) compared the various properties of the diarrheal and emetic enterotoxins. The diarrheal toxin is a protein with a molecular weight of about 50,000, while the emetic toxin is a peptide (not antigenic) and has a molecular weight of less than 5,000.

The diarrheal toxin can be detected by the rabbit ileal loop test or skin-permeability system of either rabbits or guinea pigs. Monkey feeding with observation of vomiting is the system used to detect emetic enterotoxin.

CONTROL. The methods of control are the same as those for *C. perfringens*. Keeping *B. cereus* out of the food would appear to be a difficult task. However, proper holding temperatures (55°C or higher, or below 10°C) for the food would prevent growth of the organisms. Multiplication of the cells is needed to attain sufficient numbers to cause illness.

Streptococcal Infections

Group A (*Streptococcal pyogenes*) may be present in food. This organism causes scarlet fever or septic sore throat. Besides a sore throat, symptoms may include vomiting and diarrhea.

A beta-hemolytic group A streptococcus caused an outbreak of pharyngitis (CDC 1982b). The food involved was not determined. Outbreaks of pharyngitis believed due to contaminated salads were reported by CDC (1984c).

Although group A streptococci usually are involved, group G streptococci occasionally cause pharyngitis. Among the possible symptoms are nausea, vomiting, and diarrhea. An outbreak in 1979 involving chicken salad was reported by Stryker, Fraser, and Facklam (1982).

A group C streptococcus contaminating homemade cheese caused fever, chills, and vague constitutional symptoms (CDC 1983a).

When streptococcal foodborne infection is mentioned, the organisms in group D, often referred to as enterococci, or fecel streptococci, are considered to be the etiologic agent of the illness.

The reported number of outbreaks due to enterococci is not significant. In cases in which enterococci have been involved, the incubation period ranged from 2 to 30 hr. The symptoms included nausea, vomiting, and diarrhea. Usually, the symptoms were milder than those in other foodborne illnesses. The duration was usually only a few hours.

Some people believe that the enterococci do not cause foodborne illness, and some are convinced that these organisms are involved in outbreaks. Other workers are unsure, but they realize that more convincing proof is needed to relate enterococci to foodborne illness.

Illness Due to Vibrios

The genus *Vibrio* is discussed in Chapter 3. The two important species causing gastroenteritis are *V. cholerae* and *V. parahaemolyticus*. Other *Vibrio* species such as *V. fluvialis* (Tacket et al. 1982) and *V. mimicus* (Ciufecu,

Nacescu, and Israil 1983) have been involved in causing diarrhea in humans.

V. CHOLERAE. This organism causes cholera in humans. Generally, the ingestion of 10^6 to 10^{10} cells is needed to cause the illness. The normally low pH of the stomach will kill unprotected cells, but they can pass through the stomach if the pH is altered, such as by the ingestion of antacids. Those cells that pass into the small intestine must be able to adhere to the mucosa on which they grow and produce an enterotoxin. It is believed that the flagellum is associated with attachment, since nonmotile cells generally cannot adhere to the intestine. From 24 to 48 hr after ingestion, symptoms of cholera may occur. The symptoms include profuse watery diarrhea, abdominal cramps, nausea, and vomiting. In severe cases, a liter or more of water may be lost each hour, resulting in rapid dehydration, shock, and eventually, death.

The principal strains of *V. cholerae* that cause illness are in serogroup 0:1, although other strains have been involved. The 0:1 serogroup contains two biovars, the classical and the eltor (Krieg and Holt 1984). There are two subtypes, Ogawa and Inaba. A third subtype, Hikjima, has been proposed. Although the toxin is produced in response to a gene located on the chromosome of the cell, not all 0:1 strains are enterotoxigenic.

The cholera enterotoxin (CT) and *E. coli* heat-labile toxin (LT) share some antigenic properties. The CT has a molecular weight of about 84,000. It consists of five B subunits arranged in a ring in which there is an A subunit. The B subunits combine to the membrane receptor, GM1 ganglioside, in the intestine, which then allows the A subunit to traverse the cell membrane and activate the membrane-bound adenyl cyclase, thereby causing an increase in cyclic adenosine monophosphate (cAMP). This causes an inhibition of absorption of sodium and chloride at the villus tip, which is accompanied by water being dumped into the intestine. There is no structural damage to the intestinal mucosa. The presence of bile acids seems to enhance the excretion of water into the intestine (Eidels, Proia, and Hart 1983; Lange, Hansson, and Lönnroth 1983; Middlebrook and Dorland 1984).

Cholera toxin is determined by methods similar to those used for *E. coli* LT, such as the skin reaction (Kuroki 1981), CHO cells (Nozawa, Yokota, and Kuwahara 1978), animal models (Richardson, Giles, and Kruger 1984; Spira, Sack, and Froehlich 1981), and immunological reactions (Beutin et al. 1984; Holmes and Twiddy 1983; Ito, Kuwahara, and Yokota 1983; Shah, Kauffman, and Boutin 1982). A gene probe for detecting enterotoxigenic strains was described by Hanchalay et al. (1985).

Cholera is associated with poor sewage treatment and disposal, inadequate treatment of drinking water, and crowded conditions with poor

sanitation. Although it is important in Asia and Africa, it has been reported only rarely in countries with proper sanitation. Cholera was not reported in the United States from 1911 to 1973, when a single case occurred in Texas. In 1978, there were eight cases of cholera in Louisiana, apparently caused by eating inadequately cooked crabs (CDC 1978). Since 1978, cases involving *V. cholerae* have occurred in Florida and Texas due to consumption of raw oysters and water, as well as foods contaminated with the water (CDC 1981d; Johnston et al. 1983; Shandera et al. 1983). Non-0:1 and toxigenic 0:1 *V. cholerae* were involved in these cases.

The various cases of cholera have increased interest in the presence of *V. cholerae* in United States coastal waters. It has been reported in the gulf coast from Texas to Florida (De Paola et al. 1984; Hood et al. 1983; Roberts et al. 1982), the Chesapeake Bay (Colwell et al. 1981) and California waters (Kenyon et al. 1984). The contamination of coastal waters means that shellfish (oysters, clams) present the greatest risk when consumed. Also, fish from these waters may contain *V. cholerae*.

V. PARAHAEMOLYTICUS. Although perhaps not as important as cholera on a worldwide basis, the prevalence of this illness seems to have increased. In 1968, *V. parahaemolyticus* was not listed as an etiological agent of foodborne illnesses by CDC. In 1971, there were three outbreaks with 370 cases and in 1972, six outbreaks with 701 cases. From 1975 to 1981 there was an average of two outbreaks per year.

The organism was recognized as a cause of illness in Japan during the early 1950s. During the summer months, the organism accounts for a majority of the reported foodborne illnesses in Japan.

The illness was thought to be unique to Japan. However, due to a spreading epidemic of cholera in the early 1970s, techniques to isolate vibrios were improved, and diarrheal stools were examined for vibrios. It was discovered that *V. parahaemolyticus* was an important cause of foodborne illness, not only in Japan, but also in the United States and other countries.

The Illness. The main symptom is diarrhea. Other symptoms are listed in Table 6.23.

The incubation period usually is 12 to 24 hr with a range of 4 to 96 hr. The usual duration is two to three days, but it ranges from a few hours to ten days. If treatment is needed, fluid replacement and antibiotic medication may be used.

Foods that have been incriminated in outbreaks in the United States include steamed crabs, crab salad (made from canned crabmeat), raw crab, processed lobster, boiled shrimp, roasted oysters, and raw oysters.

TABLE 6.23. *VIBRIO PARAHAEMOLYTICUS* GASTROENTERITIS: PERCENTAGE OF PEOPLE REPORTING SYMPTOMS IN FOUR OUTBREAKS

	Outbreak			
Symptom	1	2	3	4
Diarrhea	100	100	100	98
Abdominal cramps	85	96	86	78
Nausea	46	63	51	76
Vomiting	33	59	38	74
Headache	33	46	32	25
Chills	45	71	37	10
Fever	28	34	17	26
Bloody diarrhea	3	5	1	—

SOURCE: CDC *Reports.*

The seafoods were inadequately cooked or refrigerated or there was cross-contamination between cooked and raw products.

Besides these foods, in other countries the consumption of raw fish, as well as such diverse foods as meat, eggs, cereal products, and vegetables have been involved in outbreaks.

Etiologic Agent. The exact mechanism whereby *V. parahaemolyticus* causes gastroenteritis has not been determined. When the organism is grown on Wagatsuma blood agar (Miyamoto et al. 1969), those strains isolated from human cases exhibit a beta hemolysis, while most of the strains from natural environments or seafoods are not hemolytic. This hemolytic activity is called the Kanagawa phenomenon, with hemolytic strains being Kanagawa positive (K^+) and nonhemolytic strains Kanagawa negative (K^-). There is a close correlation of hemolytic activity and human pathogenicity.

To remain in the intestinal tract, the organisms must be able to attach or adhere to the intestinal cells. The adherence to intestinal cells was greater for K^+ than for K^- strains (Carruthers 1977). In comparing K^+ and K^- cells, Gingras and Howard (1980) found no difference in adherence, while a second group of researchers (Reyes et al. 1983) found one strain, a K^-, which adhered to a significantly greater extent than the other strains.

Invasiveness is a pathogenic factor. Both K^+ and K^- strains penetrated into the lamina propria of the ileum (Boutin et al 1979). They suggested that the organism is capable of more than a superficial colonization of the gut.

Craig (1972) stated that it seemed unlikely that the same molecule would have hemolytic and enterotoxic properties. He suggested that an enterotoxin is elaborated by the organism in the intestine. Other re-

searchers found a pathogenic strain that was not hemolytic (Sakurai et al. 1974). They believed that a factor or factors other than a hemolysin was responsible for the illness. Yet another team reported that antisera to the hemolysin did not prevent accumulation of fluid in ileal loops when challenged with living cells of either K^+ or K^- strains (Honda et al. 1983b). This would indicate that a substance other than the hemolysin was responsible for this action.

A study by Bradshaw and others (1984) stated that there is no satisfactory evidence for a filterable enterotoxin, but that the invasive and adhesive traits substantiate the theory that the Kanagawa hemolysin is the primary virulence factor.

It is not known how many cells need to be ingested to cause the illness. Smith (1971) stated that 10^6 to 10^9 cells were needed, depending on the condition of the stomach, type and quantity of food eaten, and buffering capability of the ingested food. Accidental ingestion of 10^5 cells caused symptoms in a laboratory worker. Using rabbit ileal loops, Twedt, Peeler, and Spaulding (1980) found that concomitant inoculation of other nonvirulent vibrios can increase the number of cells of *V. parahaemolyticus* needed to cause dilation of the loops.

Nature of the Organism. V. parahaemolyticus is a Gram-negative, straight or curved, facultatively anaerobic rod. It is halophilic, requiring 1 to 3 percent salt for growth. The cells grow in broth with 8 percent, but not 10 percent salt (Kourany and Vasquez 1975). The minimum a_w is about 0.94. Usually the pH range for growth is 5 to 11 (Twedt, Spaulding, and Hall 1969). The optimum is pH 7.6 to 8.6.

The organism is sensitive to cold temperatures, so the minimum for growth is usually listed as 10° to 13°C. The upper temperature is about 42° or 43°C, with an optimum between 30° and 37°C. The effect of low temperature on the cells is related to the type of product and the temperature of storage. When exposed to 2°C, the membrane is damaged (Van den Broek and Mossel 1977). The presence of salt tends to protect the cells from low-temperature damage.

Smith (1971) stated that the organism has a very short generation time of 9 to 11 min. The average generation time of four strains was 13.6 min (Lee 1973).

The natural habitat is coastal and estuarine ocean water throughout the world. It lives in sediment during the cold winter months, and, as the temperature rises in late spring, the organism colonizes the water and animal life. Direct relationships have been demonstrated between the water temperature, abundance of the organism in the water, and outbreaks of vibrio foodborne illness. The organism has been isolated from a variety of seafoods.

Methodology. V. parahaemolyticus, if present in a food, is usually in low numbers. Hence, an enrichment procedure using a broth such as glucose salt teepol (GST) is used (FDA 1978). For chill-stressed cells, GST supplemented with magnesium or ferric iron salts increases recovery (Ma-Lin and Beuchat 1980). A loopful of the enriched sample is streaked onto thiosulfate citrate bile salts sucrose (TCBS) agar (FDA 1978). On this medium, the organism appears as round colonies 2 to 3 mm in diameter with green or blue centers. The salt concentration (1 percent) and pH (8.4), plus other inhibitors, suppress the growth of most organisms except halophiles, such as *V. parahaemolyticus*. Sucrose-fermenting organisms produce yellow colonies. These include *V. cholerae*, some enterics and *S. aureus*. Organisms besides *V. parahaemolyticus* that can form green colonies include some vibrios, pseudomonads, and enterics.

A hydrophobic grid membrane filtration system using *V. parahaemolyticus* sucrose agar was described by Entis and Boleszczuk (1983). They claimed that significantly higher counts were obtained with their system than with the FDA (1978) method.

Kourany (1983) found TCBS to be of dubious value in certain cases and suggested a specially supplemented trypticase soy agar for detection of *V. parahaemolyticus*.

Since organisms besides *V. parahaemolyticus* can grow on the isolation media, biochemical tests are needed to differentiate it from other vibrios as well as other Gram-negative organisms.

The organism has O, H, and K antigens. Serological typing schemes using the K and H antigens have been devised (Beuchat 1982; Shinoda et al. 1983).

Control. The control of *V. parahaemolyticus* foodborne illness should be relatively simple. The organism is readily killed by heat. When inoculated at a level of 10^5/ml in shrimp homogenate, no survivors were found after heating at 100°C for 1 min (Vanderzant and Nickelson 1972). According to CDC (1978a), for boiled seafoods, Louisiana law requires a minimum of 7 mın of boiling to ensure the destruction of pathogens. Delmore and Crisley (1979) reported D values at temperatures from 49° to 55°C for nine strains of *V. parahaemolyticus*. At 55°C, the D ranged from 0.03 to 0.24 min in sterile clam homogenate.

Although cooking seafoods will destroy the organisms, it is difficult to change the habit in some cultures of eating raw seafood. The use of radiation treatment of seafoods might be a solution.

Campylobacter

Prior to 1980, *Campylobacter* was not listed as a causative agent of foodborne gastroenteritis by CDC. *C. jejuni* was involved in five outbreaks in 1980 and ten in 1981 (CDC 1983b).

Analysis of 8,097 fecal samples obtained from eight hospitals in various areas of the United States yielded twice as many isolations of *C. jejuni* as salmonellae (4.6 percent to 2.3 percent) (Blaser et al. 1983). Similar findings were reported by other researchers (Drake et al. 1981; Svedhem and Kaijser 1980). Stool samples of patients admitted to an Australian hospital were examined for various potential pathogens (Cavanagh et al. 1980). The percent of stools containing these organisms was as follows: *C. jejuni*, 13 percent; salmonellae, 20 percent; shigellae, 4 percent; and parasites, 13 percent.

If *C. jejuni* is so prominent, why was it not listed as a cause of foodborne illness in the CDC reports prior to 1980? With the development of simplified and selective methods to isolate *C. jejuni* from stools (Skirrow 1977), the number of confirmed cases has increased, and only confirmed cases are listed by CDC. Although there are several species of *Campylobacter*, the one most often associated with human distress is *C. jejuni*.

The organism is a Gram-negative, slender, curved to spiral rod. The cells have a single polar flagellum and move with a corkscrewlike motion.

C. jejuni does not ferment or oxidize carbohydrates. Energy is obtained from amino acids or from intermediates of the tricarboxylic acid cycle. It does not hydrolyze gelatin, casein, deoxyribonucleic acid, or esculin. The organism is catalase and oxidase positive and reduces nitrate to nitrite; most strains hydrolyze hippurate.

It is microaerophilic with 5 percent O_2 and 10 percent CO_2 optimal for growth. The minimum temperature for growth is 32° to 35°C and the maximum is about 44° to 45°C. At 42°C it grows in 1.5 percent but not in 2 percent NaCl. At 25°C, 1.0 to 2.5 percent salt enhances the death of the organism, and it is sensitive to 1.0 percent salt at 4°C (Doyle and Roman 1982a).

The minimum pH for growth generally is 5.3, but some strains grow at pH 5.1 (Gill and Harris 1983) and one strain showed growth at pH 4.9 (Doyle and Roman 1981).

The illness is a typical gastroenteritis, with the dominant symptoms being diarrhea and abdominal pain, usually with fever. Other symptoms include nausea, vomiting, mucus or blood in the stool, headache, chills, fatigue, backache, and dehydration. The illness may mimic appendicitis (Chan, Stringel, and MacKenzie 1983; Pitkänen et al. 1983), and can cause abortion (Gilbert et al. 1981). The illness may be severe in some patients. Smith and Blaser (1985) reported on two fatalities associated with *C. jejuni*. The incubation period usually is 2 to 5 days but may range from 1 to 11 days. The duration of the illness usually is 1 to 3 days but may last for three weeks. The organism may be excreted for over a year (Richardson, Koornhof, and Bokkenheuser 1981). If the illness is severe, treatment with erythromycin and fluid replacement may be needed.

C. jejuni can invade the intestinal epithelium and cause damage to the mucosa. In calves, Al-Mashat and Taylor (1980) found a thickening of the wall of the ileum, inflammation of the jejunal and ileal mucosa, an enlargement of mesenteric lymph nodes, as well as dilated capillaries in the mucosa of the small intestine. Invasion and lesions in the intestinal tract also were noted with experimentally infected hamsters (Humphrey, Montag, and Pittman 1985) and newly-hatched chicks (Welkos 1984).

Mills and Bradbury (1984) stated that little is known of the mechanism of the pathogenicity of the organism. Since fever is a symptom, it would seem to be an infection. However, the researchers have detected toxin production in twenty-four of thirty-two strains of *C. jejuni* (Ruiz-Palacios et al. 1983). Culture supernatants caused fluid secretion in rat ileal loops, but not in rabbit loops or the infant mouse assay. It caused elongation of CHO cells. Cholera antitoxin prevented the toxic response, indicating a serological similarity to cholera enterotoxin, *E. coli* LT, and possibly the enterotoxins of salmonellae. These observations were confirmed by Klipstein and Engert (1984, 1985) and McCardell, Madden, and Lee (1984).

Domestic and wild animals are the main reservoirs of *C. jejuni*. Hence, foods derived from animal sources tend to be those involved as vehicles for infection of humans by the organism. Drinking raw (untreated) water is the highest risk factor (Hopkins, Olmsted, and Istre 1984). They listed further risks (in descending order) as drinking raw (unpasteurized) milk, having a cat in the home, and eating undercooked chicken. Besides the ingestion of contaminated water or food, the organism can be transferred directly from animals to humans or from humans to humans (Norkrans and Svedhem 1982).

The exact mechanism by which milk becomes contaminated is not known. However, milk from cows with *Campylobacter* mastitis might be used. Also, according to CDC (1981e), up to 60 percent of healthy cows shed *Campylobacter* in their feces. Without very careful handling, fecal contamination of milk might occur. The incidence of *C. jejuni* in farm bulk milk is about 1 to 2 percent (Lovett, Francis, and Hunt 1983).

Milk has been involved in several outbreaks of campylobacteriosis (CDC 1981a, 1984d; Hudson et al. 1984; Potter et al. 1983; Wright et al. 1983). In a survey of raw milk, deBoer, Hartog, and Borst (1984) concluded that there was no association between developing campylobacteriosis and consuming raw milk. They, as well as Doyle and Roman (1982c), reported that *C. jejuni* survived longer in sterile milk than in raw milk when held at 4°C.

C. jejuni is quite prevalent in poultry-processing plants (Luechtefeld and Wang 1981; Oosterom et al. 1983b; Wempe et al. 1983) and on the carcasses of the processed poultry (Rayes, Genigeorgis, and Farver 1983; Shanker et al. 1982). Fortunately, *C. jejuni* tends to die during refrigerated

holding so there is less contamination at the retail store than immediately after processing (Kinde, Genigeorgis, and Pappaioanou 1983). From a survey of chicken livers, Barot, Mosenthal, and Bokkenheuser (1983) concluded that most of the contamination with *C. jejuni* occurred at the processing plant.

The organism has been detected on red meats in processing plants and at retail stores (Stern et al. 1985; Turnbull and Rose 1982). Koidis and Doyle (1983) reported that *C. jejuni* survived well in refrigerated ground beef. The extent of survival depended upon the strain of the organism. Conversely, Bolton, Dawkins, and Robertson (1982) and Oosterom et al. (1983a) found that cold storage reduced the level of *C. jejuni* on carcasses below detectable levels.

When food is contaminated, it is usually at a low level. Due to temperature and oxygen requirements, the organism generally does not multiply in food. Hence, it is evident that low levels of contamination are sufficient to cause illness in humans, although the exact number of cells needed is not known. Only a low-level infection occurs during direct transfer from animals or people to the recipient. A level of 90 *C. jejuni* was the minimum infective dose causing diarrhea in 90 percent of inoculated chickens (Ruiz-Palacios, Escamilla, and Torres 1981).

Various methods have been used to detect the presence of *C. jejuni* in foods. Since a relatively low number of cells are in foods, an enrichment procedure usually is needed. Selective enrichment broths are generally a basal medium to which various supplements and antibiotics are added to stimulate the growth of *C. jejuni* and inhibit other organisms (Barot and Bokkenheuser 1984; Doyle and Roman 1982a; Rogol et al. 1985; Rothenberg, Stern, and Westhoff 1984). After incubation of the inoculated enrichment broth in a microaerobic environment at 42°C, a portion is surface plated onto a selective agar surface. In some cases, the sample may be plated directly without enrichment. The selective agars tend to be similar to the selective enrichment broths, with 1.5 to 2.0 percent agar added for solidification (Fricker 1985; Moskowitz and Chester 1982; Ng, Stiles, and Taylor 1985; Stern 1982). After incubation microaerobically at 42°C, the inoculated surfaces are observed for typical colonies of *C. jejuni*.

These can be screened by microscopic observation (phase-contrast) for motility and morphology, and then for oxidase and catalase-positive characteristics. After these simple tests, other biological and biochemical tests can be run (Lior 1984; Roop, Smibert, and Krieg 1984; Stern 1982). Serological and other tests may be performed to further confirm the presence of *C. jejuni* (Bradbury et al. 1984; Harvey and Greenwood 1983; Kosunen, Bång, Hurme 1984; Penner, Hennessy, and Congi 1983; Tenover et al. 1984, 1985; Wong et al. 1985).

The control of *C. jejuni* enteritis is similar to that for salmonellae.

Foods derived from animal sources should be handled as though they contain the organism. *C. jejuni* is not very heat resistant. In skim milk, the D values for *C. jejuni* ranged from 7.2 to 12.8 min at 48°C and 0.74 to 1.0 min at 55°C (Doyle and Roman 1981). In ground beef, these values were 5.9 to 6.3 min at 50°C and 0.2 to 0.35 min at 58°C (Koidis and Doyle 1983). In autoclaved ground chicken, the D values ranged from 20.5 min at 49°C to 0.79 min at 57°C (Blankenship and Craven 1982). Hence, pasteurization of milk and cooking of beef and poultry should destroy any *C. jejuni* in these products. Svedhem, Kaijser, and Sjögren (1981) suggested heating food to at least 60°C for 15 min to prevent infection. Gill and Harris (1984) recommended minimal cooking of poultry at 190°C for 20 min to eliminate *C. jejuni*.

The organism is more sensitive than salmonellae to gamma radiation (Lambert and Maxcy 1984). Competitive exclusion in young chicks worked similarly to that for salmonellae (Soerjadi-Liem, Snoeyenbos, and Weinack 1984).

Brucellosis

The number of cases of brucellosis has generally decreased in the United States. It has become a disease of people who handle cattle and swine. This includes veterinarians, farmers, and packinghouse workers. The workers in the kill department are most subject to contamination, but workers handling the carcasses and meat also have become infected.

In 1975, there were twenty-four cases of brucellosis caused by the ingestion of unpasteurized dairy products, such as raw domestic milk and imported cheese. The consumption of raw dairy products continues to cause about 10 percent of the cases of brucellosis (Bryan 1983).

Q Fever

Coxiella burnetii causes Q fever, a rickettsial zoonosis, in humans. Cattle, sheep, and goats are the usual reservoirs of the organism. Usually, transmission to humans is by inhaling aerosols derived from contaminated animal products (including excreta). The organism is a hazard to people working in meat-packing plants and handling raw meat (McKelvie 1980). Infected cows shed the organism in their milk. The pasteurization of milk was adjusted so that the treatment inactivates *C. burnetii*.

Yersinia

Yersinia enterocolitica is a member of the family Enterobacteriaceae. For most strains, the temperature growth range in nutrient broth is 1 to 44°C (Sutherland and Varum 1977), and, at 25°C, the pH range for growth is about 4.6 to 9.6 (Stern, Pierson, and Kotula 1980).

The organism produces several disease syndromes in people (enteritis, terminal ileitis, mesenteric lymphadenitis, erythema nodosum, polyarthritis, septicemia, and metastatic abscesses in various organs). The gastroenteritis is characterized by diarrhea (which may be bloody), abdominal pain, fever, and vomiting. The illness may last from two days to several months. The illness may mimic appendicitis and, in some outbreaks, appendectomies have been performed before yersiniosis was suspected (Black et al. 1978; Shayegani et al. 1983). The organism has been isolated from the appendices of patients (Fukushima et al. 1981).

Y. enterocolitica can be found almost anywhere in nature, but only certain serotypes are involved in human infection. These serotypes are prevalent in swine (Fukushima et al. 1983; Walker and Grimes 1985), and have been isolated from dogs (Fukushima et al. 1984). Hence, although the organism has been isolated from various animals and food products derived from animals, these isolations are important only if they are serotypes that can cause human yersiniosis.

The ability of the organism to grow at refrigeration temperatures, makes refrigerated animal foods, such as milk, a potential hazard. Milk that was either improperly pasteurized or contaminated after pasteurization has been involved in various outbreaks (Black et al. 1978; CDC 1982c; Tacket et al. 1984).

A fermented soybean product, tofu, was a vehicle for outbreaks (Aulisio et al. 1983; Tacket et al. 1985b). Besides foods, the organism is found in water and can be acquired by direct transfer to people from animals or from other people.

Since fever and bloody diarrhea are symptoms of gastroenteritis, an invasive type of infection similar to shigellosis would be suspected. A plasmid is associated with invasiveness and other virulence factors (Kay, Wachsmuth, and Gemski 1982).

Some strains of *Y. enterocolitica* produce an enterotoxin similar to *E. coli* ST, which stimulates the activity of guanylate cyclase (Inoue et al. 1983; Rao et al. 1979). The results of Schiemann (1981) indicated that the enterotoxin has no role in pathogenesis. Hence, the exact role of invasiveness and enterotoxin needs further evaluation. Also, the extent of involvement of persiniae in human illness is not known.

The methodology for *Yersinia* has been discussed by various researchers (Bercovier et al. 1984; Hill, Payne, and Aulisio 1983; and Sack 1984).

Listeria

The first human illness due to *Listeria monocytogenes* was recognized in 1929. The organism is present in soil and has been detected in many species of animals. The infection of humans usually occurs orally, such as

with contaminated food, or possibly by direct contact with infected animals. About 1 percent of humans excrete the organism.

The organism is a small Gram-positive rod with a tendency toward a diplobacillary form. Although it is aerobic, it grows better at reduced O_2 and increased CO_2 levels. It is more motile at 25°C than at 37°C. The cells can multiply at pH 9.0 and at 10 percent NaCl. It can grow at 4°C, which makes it a problem in refrigerated foods. It ferments a variety of sugars, producing acid but no gas.

There are several manifestations of listeriosis, including septicemia, (which in pregnant women can lead to abortion or stillbirth), endocarditis, pneumonia, conjunctivitis, pharyngitis, cutaneous papules, and pustules, urethritis, and meningitis. Septicemia may result in the invasion of organs such as the spleen, liver, or adrenal gland. The generalized infection may mimic other diseases such as typhoid fever or infectious mononucleosis.

A rather large outbreak of listeriosis was described by CDC (1985a). According to this report, the outbreak occurred in California due to consumption of contaminated cheese, with eighty-six reported cases and twenty-nine deaths. Of these twenty-nine deaths, there were eight neonatal deaths and thirteen stillbirths. The other eight were non-neonatal. There are indications that forty-eight or more deaths may have occurred in this outbreak.

There have been other recent outbreaks of listeriosis. In 1983, fourteen persons died in Massachusetts after consuming pasteurized milk. Although considered to be heat sensitive, the organism is believed to be able to survive pasteurization by living parasitically within white blood cells present in milk.

In August, 1985, soft cheese produced in Ohio was found to contain *L. monocytogenes* and fortunately was recalled before any illness occurred.

Although most of the problems with *Listeria* have been associated with the dairy industry, fowl are considered to be a natural reservoir of the organism. Fortunately, most poultry products are cooked sufficiently to kill the organisms before consumption.

Others

Although not listed as major causes of gastroenteritis, several organisms produce enterotoxins. These include species of the genera *Aeromonas* (Buchanan and Palumbo 1985; Goodwin et al. 1983; Jiwa 1983; Ljungh, Eneroth, and Wadström 1982), *Citrobacter* (Wadström 1976), *Enterobacter* (Klipstein, Engert, and Short 1977), *Klebsiella* (Klipstein, Engert, and Short 1977), and *Kluyvera* (Farmer et al. 1981). *Clostridium difficile* and its enterotoxin (toxin A) have been associated with a severe gastroin-

testinal illness in patients who have ingested certain antimicrobial agents (Mulligan 1984; Thelestam and Florin 1984). Some strains of *Clostridium sordellii* also produce an enterotoxin (Yamakawa et al. 1983). The extent of the role of these organisms as agents of foodborne illness has not been fully evaluated.

FUNGAL ILLNESSES

Fungi are a diverse group of organisms that include molds, yeasts, rusts, smuts, and mushrooms. These organisms can cause many diseases in plants, animals, and humans.

In humans, the fungi can cause mycoses, allergies, or toxicoses. Mycoses are diseases resulting from the invasion of living cells by the fungi. Allergies are diseases resulting from the development of hypersensitivity to fungal antigens. Toxicoses consist of illnesses due to ingesting toxic fungal metabolites formed in the food (mycotoxicoses) and the mycetisms caused by ingesting toxic fungal fruiting bodies.

Besides producing toxins, the fungi can degrade the food so that it is deficient in certain nutrients, or the mycosis can upset the metabolism of the animal so that a nutrient deficiency occurs. Our concern is primarily with mycotoxicoses.

Mycotoxins

Mycotoxins are secondary metabolites produced by fungi and can cause unnatural or deleterious biological changes in plants, animals, humans or microorganisms. The mycotoxins may be contained within the spore or fungal thallus, or they may be secreted into the growth substrate.

Mushroom poisoning was experienced by people in ancient times. Cases of mushroom poisoning still occur in the United States and elsewhere (Kendrick and Shimizu 1984).

Ergotism, or St. Anthony's Fire, was reported in the Middle Ages. This illness is caused by eating cereals infected with *Claviceps purpurea*. This fungus can grow and produce toxic alkaloids on wheat, barley, rye, oats, and wild grasses. About forty alkaloids have been isolated from the fungus. These alkaloids can cause constriction of peripheral blood vessels with violent pain, intense burning, and gangrene of the extremities, or they may cause convulsions and hallucinations, as well as abortion in pregnant women. Some of the ergot alkaloids are important therapeutic agents (Esser and Düvell 1984; Řeháček 1984). The last major outbreak of ergotism occurred in France in 1951. The maximum level of ergot

allowed is 0.1 percent dry weight in barley, oats, and triticale, and 0.3 percent in wheat and rye (USDA 1978).

Alimentary toxic aleukia (ATA), or septic angina, has occurred in the Soviet Union at infrequent intervals. Symptoms of this illness include a burning sensation in the mouth, stiffness of the tongue, and diarrhea, nausea, vomiting, and perspiration. After these symptoms, there is a quiescent period, followed sometime later by leukopenia, weakness, hemorrhages of the skin and mucous membranes, necrotic areas in the mouth, throat and skin, gangrenous pharyngitis, fever, and recovery or death. The fatality rate varies from 2 to 80 percent.

This illness (ATA) was especially severe during World War II. The Soviets did not have farm labor to harvest the grain in the fall. When the grain that was harvested during the following spring was consumed, it caused outbreaks of ATA. Investigations revealed several toxin-producing fungi on ovewintered grain, with *Fusarium* predominating.

Also, during the period of World War II, rice that Japan imported from other Asian countries was responsible for outbreaks of an illness called "yellow rice disease," which caused several deaths. The illness was associated with the invasion of rice by *Penicillium islandicum, P. citrinum,* and *P. citreoviride.*

In 1929, Fleming described a substance from a *Penicillium* mold that he called penicillin. During the search for other antibiotics, several substances were found that are more toxic to animals and humans than to microorganisms. Even with the finding of these toxic substances, there was little activity in mycotoxin research until 1960, when over 100,000 turkeys died in England. An investigation revealed that the deaths were due to peanut meal, in which *Aspergillus flavus* had produced mycotoxins called aflatoxins.

With an increased interest in fungal toxins, more than 150 species of fungi have been reported to be capable of producing substances that are toxic. The number of mycotoxins has been reported as over 100 (Pollock 1983), about 200 (Wyatt 1980), and several hundreds (Stark and Demain 1980). New mycotoxins are found each year. Many genera of molds can produce mycotoxins, but fortunately most molds produce no toxins. Not all species of a mycotoxin-producing genus, and not all strains of a mycotoxin-producing-species, are toxigenic.

TYPES OF TOXINS. Apparently most mycotoxins do not affect humans. However, those that might are listed in Table 6.24. A few, such as aflatoxins, citrinin, luteoskyrin, patulin, penicillic acid, rugulosin, and sterigmatocystin are carcinogens (Stark and Demain 1980). Although a discussion of every individual mycotoxin is not possible in this text, it seems desirable to investigate a few of those that might affect humans.

TABLE 6.24. SOME MYCOTOXINS THAT MIGHT CAUSE HUMAN ILLNESS

Mycotoxin	Some Producing Organisms
Aflatoxins	*Aspergillus flavus, Aspergillus parasiticus*
Citrinin	*Penicillium citrinum, Penicillium vividicatum*
Cyclopiazonic acid	*A. flavus, Aspergillus tenuis, Penicillium cyclopium*
Ergotoxins (alkaloids)	*Claviceps purpurea*
Luteoskyrin	*Penicillium islandicum*
Ochratoxins	*Aspergillus ochraceus, P. viridicatum, Penicillium verrucosum*
Patulin	*Penicillium expansum, Penicillium patulum, Aspergillus clavatus, Byssochlamys nivea*
Penicillic acid	*Penicillium martensii, P. viridicatum, P. cyclopium, Penicillium puberulum, Penicillium palitans*
Roquefortine	*Penicillium roqueforti*
Rubratoxin	*Penicillium rubrum*
Sterigmatocystin	*Aspergillus versicolor, Aspergillus nidulans*
Tenuazonic acid	*Alternaria tenuissima, Alternaria alterata*
Trichothecenes	
Deoxynivalenol (vomitoxin)	*Fusarium graminearum (Gibberella zeae)*
Fusarenon-x	*Fusarium nivale*
Neosolaniol	*Fusarium solani*
Nivalenol	*F. nivale, Fusarium roseum*
Fusartoxin T-2	*Fusarium tricinctum, F. solani, F. roseum*
Zearalenone	*F. graminearum (G. zeae), F. tricinctum, Fusarium culmorum, Fusarium roseum*

Aflatoxins. Investigations of the deaths of turkey poults in England, the deaths of trout in a fish hatchery in the United States, and the deaths of ducklings in Africa revealed that feeds containing a toxic substance were involved. A feed constituent, peanut meal, and extracted fractions from this meal showed a toxicity similar to that involved in killing the turkey poults. Molds isolated from peanut meal included *Aspergillus flavus*. Fractions isolated from growth of *A. flavus* were found to be toxic and were called aflatoxins.

A chromatograph of the extract of moldy feed was prepared and observed with ultraviolet light. Four fractions appeared. Two were blue and two were green. The toxic substances were called aflatoxin B_1, B_2, G_1, and G_2, the letters corresponding to the fluorescent color, and the numbers denoting the relative mobility. Thirteen of these compounds are known to occur in nature (Ciegler 1978). They all possess a coumarin nucleus fused to a bifuran moiety. Eight aflatoxins (B_1, B_2, B_{2a}, M_1, M_2, P_1, R_0, and Q_1) have a pentene ring, while four (G_1, G_2, G_{2a}, and G_{m1}) have a six-member lactone ring. The other toxin was referred to as parasiticol (B_3) (Ciegler 1975) and is similar to the B series, except that the pentene ring is opened. Eight of these aflatoxins are shown in Figure 6.13.

Aflatoxin B_1 is the most abundant aflatoxin and is considered to be the most toxic. When given orally to ducklings, the LD_{50} in mg/kg for

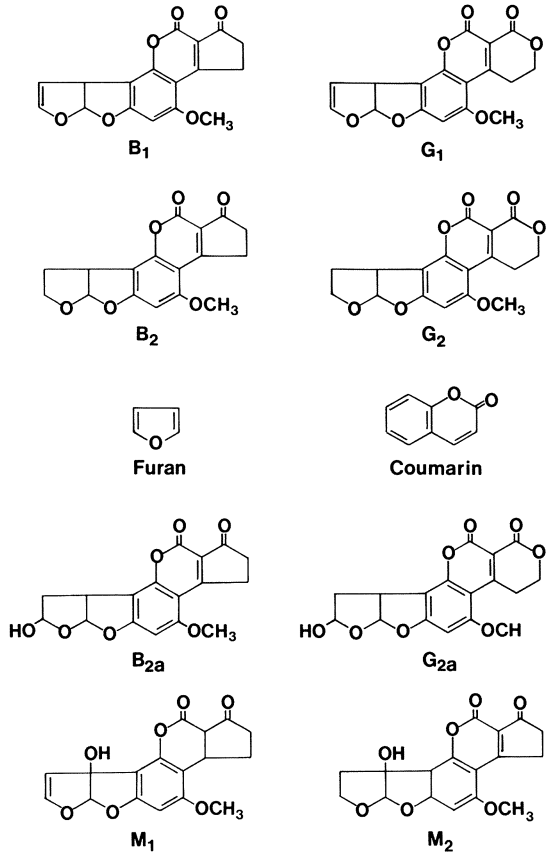

Figure 6.13. Formulae of aflatoxins. They contain a coumarin molecule and bifuran ring.

some of the aflatoxins is 0.36 (B_1), 0.33 (M_1), 0.78 (G_1), 1.7 (B_2), and 3.5 (G_2) (deWaart 1973). B_{2a} and G_{2a} are not toxic to ducklings, even when given at very high levels. Thus, there may be justification for not calling these substances toxins.

Aflatoxins can cause a response in microorganisms, cell cultures, plants, and animals. There may be acute or chronic effects, depending upon the dosage and the frequency of exposure to the toxins. The effects of aflatoxins can be toxigenic, mutagenic, teratogenic, or carcinogenic. The current concept is that B_1 is activated by liver chromosomes to a 2,3-epoxide before it can exert its carcinogenic and mutagenic effects in animals.

The effect of aflatoxins on animals is influenced by the type of afla-

toxin, species of animal, age, weight, health, and diet. Environmental stresses may alter the susceptibility of the animal.

Very young animals are more susceptible to aflatoxicosis than are mature animals. Species of animals vary in their resistance. In farm poultry, ducklings are the most susceptible, followed by turkey poults, goslings, and domestic chicks. For farm animals, the order from high susceptibility to resistant is young pigs, pregnant sows, calves, fattening pigs, mature cattle, and sheep.

Acute aflatoxicosis has been characterized by hemorrhage in tissues, anorexia, hepatitis, and death of animals. The liver is the primary tissue that is affected. However, the spleen, pancreas, kidneys, and other tissues may be involved. Aflatoxin also causes fragility of the capillaries, so there is a greater tendency for bruising (Huff et al. 1983). The clinical signs of B_1 toxicosis in goats included decreased feed consumption, some loss of body weight, mucopurulent nasal discharge, dyspenea, coughing, lethargy, icterus, diarrhea, and subnormal body temperature 24 to 48 hr before death (Clark et al. 1984). They found that the clinicopathological changes included increases in red blood cell count, packed cell volume, hemoglobin concentration, serum bilirubin concentration and serum activities of aspartate aminotransferase, isocitric dehydrogenase, and ornithine carbamyl transferase. At necropsy, goats fed aflatoxin B_1 showed evidence of ascites, pale livers, petechial hemorrhages, nasal discharge, and icterus, as well as bile duct proliferation, hepatocytic karyomegaly, hepatocellular degeneration, pneumonia, rhinitis, and proximal renal tubular nephrosis (Miller et al. 1984). Similar effects have been noted after B_1 was fed to other animals (Applebaum and Marth 1983; Ketterer et al. 1982; Wyatt et al. 1985). Reduced egg production was noted when B_1 was fed to pullets at 0.7 mg/kg body weight, or at higher levels (Exarchos and Gentry 1982). The effect of aflatoxins on rats was reduced when cabbage or cauliflower was included in the diet (Boyd and Stoewsand 1981; Boyd, Babish, and Stoewsand 1982).

Aflatoxin is carcinogenic in ducklings, rainbow trout, rats, and ferrets (Detroy, Lillehoj, and Ciegler 1971). They stated that some reports indicate that aflatoxin induces carcinomas in pigs, hamsters, guinea pigs, mice, and sheep. A monkey developed a cancer due to aflatoxin ingestion (Gopalan, Tulpule, and Krishnamurthi 1972). Although the carcinogenic activity is primarily in the liver, aflatoxin might induce tumor formation in other organs.

There is ample evidence that aflatoxins produce acute and chronic illness in animals. What is their effect when ingested by humans? The deliberate feeding of aflatoxins to human subjects has not been reported. The only information concerning humans is from instances in which the food, naturally contaminated with aflatoxins, has been consumed. Thus,

evidence for human aflatoxicosis is not based on scientific experiments. Information accumulated after an incident is often incomplete.

Aflatoxin B_1 was detected in specimens of stools, urine, liver, brain, and kidney obtained from Thai children who died of encephalopathy and fatty degeneration of the liver (Shank et al. 1971). In another outbreak, caused by consuming moldy maize, 106 people died and 291 showed signs of hepatic dysfunction (Krishnamachari et al. 1975).

Attempts have been made to associate the high incidence of liver cancer in humans in certain areas to the presence of toxigenic molds. Newberne (1974) listed evidence associating aflatoxins with cirrhosis of the liver, as well as hepatoma, hepatitis, and Reye's syndrome. Stora, Dvorackova, and Ayraud (1983) further associated aflatoxin B_1 with Reye's syndrome and the deaths of five children. At autopsy, aflatoxin B_1 was detected in the liver at levels of 120 to 810 μg/kg. They reported fatty degeneration of the liver, kidneys, myocardium, and fibers of the striated muscles. Wray and Hayes (1980) reported liver cancer in a patient whose serum contained aflatoxin B_1. In a study of the diets of ninety people with liver cancer, and ninety controls in the Philippine Islands, researchers found that the mean aflatoxin load per day was 440 percent greater for people with liver cancer than for the controls (Bulatao-Jayme et al. 1982). Dichter (1984) considered the current levels of aflatoxin exposure from consuming peanuts and peanut products, and, on the basis of epidemiological data of populations in Africa and Thailand, estimated some fifty-eight cases of liver cancer annually in the United States. We are exposed to aflatoxins from other products and activities, so the rate may be higher. There is surely sufficient evidence, if not from scientific experiments, that aflatoxins can cause human illness, including liver carcinoma.

Any food that becomes contaminated with spores of toxigenic *A. flavus* or *A. parasiticus* is capable of supporting growth of the mold and, if held in an environment favorable for growth and toxin production, can become a source of aflatoxin. Foods found to contain aflatoxins are listed in Table 6.25. In the United States the main concern has been with aflatoxin on peanuts, corn, and cottonseed.

Peanuts can become contaminated with mold and aflatoxin during growth, harvesting, or storage. Peanuts grown in soil used consistently for peanuts tend to have a higher accumulation than when crops are rotated. The type of soil and rainfall following dry weather influences pod splitting of peanuts (Graham 1982). The infection of peanuts also is associated with mechanical damage during or after harvesting. It is essential that the harvested peanuts be dried to 8 percent or less moisture, and stored under conditions so that the moisture content will not increase.

When contaminated peanuts are pressed to remove the oil, most of

TABLE 6.25. FOODS AND FEEDS FOUND TO CONTAIN AFLATOXIN

Almonds	Meat
Almond paste	Milk
Bakery products	Millet
Barley	Noodles
Beans	Nutmeg
Beer	Oats
Black pepper	Palm kernels
Brazil nuts	Peas
Bread	Peach seed paste
Capsicum pepper	Peanuts
Cayenne pepper	Peanut butter
Cheese	Peanut meal
Chili powder	Pecans
Cocoa beans	Pepper corns
Cocoa	Pistachio nuts
Coconut	Raisins
Coconut oil cake	Rapeseed
Coffee beans	Rice
Copra	Rue
Corn	Sesame
Corn grits	Shea-nuts
Cottonseed	Sorghum
Cottonseed meal	Soybeans
Cowpeas	Soybean meal
Dried chili pepper	Soy sauce
Dried fish	Spaghetti
Dried milk	Sunflower flour
Egg white	Sunflower meal
Figs	Sunflower seeds
Filberts	Sweet potato
Flour	Various leafy foods
Garlic	Walnuts
Hazel nuts	Wheat
Locust beans	Wheat flour
	Wine

the toxin remains with the meal cake, and only small amounts remain with the oil. Clarification of the oil removes much of the residual toxin. During refining, a hot alkali wash and bleaching treatment given the oil removes most, if not all, of the remaining aflatoxin. Thus, peanut oil is considered to be safe. However, the peanut cake or meal can be highly contaminated.

Aflatoxin has been detected in corn prior to harvesting (Lillehoj et al. 1984; Shotwell and Hesseltine 1983). There is a direct correlation of insect damage to mold contamination and the presence of aflatoxin in corn kernels (Lillehoj et al. 1980; Widstrom 1979).

Ninety percent of all corn harvested is used as animal feed, and the remainder is used for breakfast food, grits, corn meal, starch, sugar, syrup, and alcohol. There is little, if any, carryover of aflatoxin to the

main edible products of the wet-milling process (starch, sugar, syrup). However, the residuals (steepwater, gluten, germ) retain the toxin (Bennett and Anderson 1978). The aflatoxin is not carried through the distillation process for alcohol production (Lillehoj et al. 1978). It is found in the spent grains that are used for animal feed.

The consumption of aflatoxin-contaminated feed may result in the deposition of the toxin in meat (Stubblefield et al. 1983; Trucksess et al. 1983), milk (Stubblefield et al. 1983), and eggs (Trucksess et al. 1983). When dairy cows ingest aflatoxin B_1, about 2.2 percent (Patterson, Glancy, and Roberts 1980), 1.6 percent (Price et al. 1985), or 0.6 percent (van Dijk, O'Dell, and Bodine 1984) is converted to aflatoxin M_1, which is found in the milk. Aflatoxin M_1 is about as potent a toxin as B_1 (Brown 1982). Since milk is the primary diet of infants and young animals, the presence of aflatoxin in this food can be especially hazardous. Aflatoxin in liquid milk is concentrated about eight times when the milk is dried. It was reported that aflatoxin M_1 from milk was found in cottage cheese (Applebaum and Marth 1983), in yogurt and buttermilk (Wiseman and Marth 1983) and in brick and Limburger-like cheese (Brackett et al. 1982).

From the list of foods found to contain aflatoxin, it is evident that there can be a carryover of aflatoxin from the field crop through processing to human food. An example of carryover is wheat to wheat flour and then to bread, spaghetti, and noodles.

Citrinin. In 1931, a yellow compound was isolated from *P. citrinum* and named citrinin (Fig. 6.14). Several species of *Penicillium* and a few species of *Aspergillus* can produce this mycotoxin. Primarily, citrinin is a nephrotoxin, causing kidney degeneration and necrosis in various laboratory animals, hogs, and poultry (Mehdi, Carlton, and Tuite 1984).

Citrinin has been found in various cereals and is thought to be involved in the yellow rice disease in the Far East. Although it does not appear to be a carcinogen, it apparently increases tumor production in laboratory animals by certain carcinogenic agents.

Cyclopiazonic Acid. This mycotoxin (Fig. 6.14) is produced by several *Penicillium* and *Aspergillus* species including *P. camemberti*, which is used in the production of Camembert cheese, and *A. flavus*, a producer of aflatoxin. LeBars (1979) detected low levels of this toxin in the crusts of eleven of twenty samples, but not in the interior of four samples of Camembert cheese. Also, the toxin was reported in twenty-one of twenty-seven samples of loose shell kernel fractions of peanuts at levels of 32 to 16,525 μg/kg and at lower levels in sound fractions (Lansden and Davidson 1983).

The effect of this toxin on humans has not been determined, but it

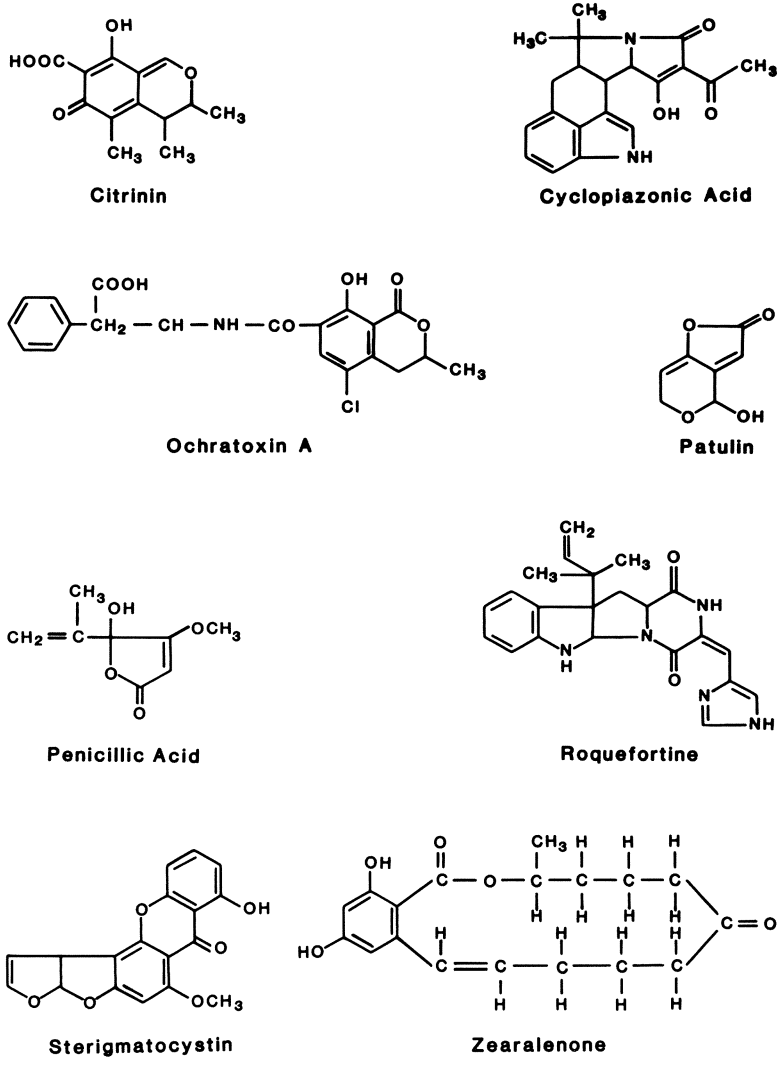

Figure 6.14. Formulae of various mycotoxins.

was found to be mutagenic to two strains of *Salmonella typhimurium* using the Ames test (Sorenson, Tucker, and Simpson 1984).

Luteoskyrin. This mycotoxin is produced by *Penicillium islandicum.* In reviews by Mislivec (1981) and Stark (1980), it is listed as a carcinogen to rats and mice. When given orally, it causes liver cirrhosis and hepatomas. It binds to DNA and is mutagenic to yeast. In areas in which rice is a

major constituent of the diet, the presence of this toxin is associated with a high incidence of liver cancer in humans.

Ochratoxin A. This is the most common and toxic member of a group of related toxins (Fig. 6.14). It was first discovered as a product of *Aspergillus ochraceus*. Also, it is produced by other aspergilli and penicillia (see Table 6.24). This toxin has been found in various grains (wheat, barley, oats, corn), beans, peanuts, and as residues in animal carcasses. In hogs, the residues are greatest in the kidney and declining amounts in lean meat, liver, and fat (Madsen, Mortensen, and Hald 1982). In Denmark, hog carcasses are condemned if the ochratoxin residual in kidneys exceeds 25 μg/kg.

The ingested toxin can cause various syndromes in many types of animals. However, it affects pimarily the kidneys (nephrotoxin). The effect of ochratoxin A on humans has not been determined, but there may be a relationship of renal disease of humans to the consumption of food containing this toxin.

Patulin. This mycotoxin (Fig. 6.14) is produced by some 19 species of fungi, including those listed in Table 6.24. Being produced by *P. patulum*, it was given the trivial name *patulin*. Also, it has been named *claviformin, expansin*, and *clavicin*. It inhibits a wide variety of bacteria and for a time was used as an antibiotic. However, side effects of intestinal distress, vomiting, and diarrhea ended its use as a therapeutic agent. The toxin is mutagenic (Mayer and Legator 1979), terotogenic (Ciegler, Beckwith, and Jackson 1976) and carcinogenic (Stark and Demain 1980).

This toxin has been found in fruit products, especially apple juice (Brackett and Marth 1979a; Scott and Bullerman 1975). During the alcoholic fermentation of apple juice, patulin is converted to other chemicals (Stinson, Osman, and Bills 1979).

PR Toxin. This mycotoxin is produced by *Penicillium roqueforti*, the mold used in the production of Roquefort, blue, and Gorgonzola cheese. Scott (1981) stated that it was one of the most acutely toxic metabolites of *P. roqueforti*. It is mutagenic to bacteria and cytotoxic to various cell lines. The toxin causes degenerative changes in the liver and kidney of rats. When it was injected into rats, researchers observed breathing difficulties, motor incoordination, and flaccid paralysis (Polonelli et al. 1978). Chen, Chen, and Wei (1982) concluded that the effects of PR toxin in animals are increased capillary permeability and direct damage to the lungs, heart, and kidneys. The effect on humans has not been determined.

Although it is produced by *P. roqueforti*, the toxin was not produced in blue-veined cheese (Polonelli et al. 1978). The toxin is not stable in cheese.

Penicillic Acid. This toxin is produced by species of *Penicillium* and *Aspergillus* including those listed in Table 6.24. According to the reviews of Mislivec (1981) and Stark and Demain (1980), it causes malignant tumors in rats at the site of subcutaneous injection. It is cytotoxic to various cell cultures. The ingestion of a sufficient amount of penicillic acid can induce liver necrosis in mice and fatty liver degeneration in quails. The effect on humans has not been determined. According to Northolt, Van Egmond, and Paulsch (1979a), penicillic acid has only a low toxic effect.

This toxin has been isolated from corn and beans and is thought to be a potential health hazard to humans.

Roquefortine. This is a group of indole alkaloids (Fig. 6.14) produced by *Penicillium roqueforti* and a few other penicillia. Roquefortine has been detected in commercial samples of blue cheese at an average level of 0.8 μg/g and as high as 6.8 μg/g (Scott and Kennedy 1976).

From the amount of roquefortine found in blue cheese, Scott (1981) stated that there is no potential acute human health hazard. The effect of this toxin on humans is not known.

Sterigmatocystin. This mycotoxin (Fig. 6.14) is produced by *Aspergillus versicolor* and some fourteen other molds (Davis 1981; Terao 1983). This mycotoxin is structurally related to aflatoxin B_1 and is mutagenic and carcinogenic. It attacks mainly the liver and kidneys, causing liver cancer in rats. In mice, it affects the lungs and blood capillaries. It causes severe necrosis of the kidney and the heart of monkeys. Sterigmatocystin naturally present in feed caused bloody diarrhea and death in dairy cattle (Vesonder and Horn 1985).

In nine of thirty-nine samples of naturally moldy cheese, researchers detected sterigmatocystin in the cheese surface at 500 to 600 μg/kg (Northolt et al. 1980). In brown rice, sterigmatocystin was found at levels up to 21.8 μg/g (Takahashi et al. 1984). It has also been isolated from other cereals and pecans (Davis 1981). So far, outbreaks in humans have not been reported (Terao 1983). Perhaps this is due to its poor absorption from the digestive tract.

Trichothecenes. The trichothecenes are chemically related toxins (sesquiterpenoids) that are produced by various species of *Fusarium, Cephalosporium, Myrothecium, Stachybotrys, Trichoderma, Trichothecium*, and *Verticimonosporium*. There are more than 40 known trichothecenes. The ones found occurring naturally include those in Table 6.24, deoxynivalenol (vomitoxin), fusarotoxin (T-2), neosolaniol, and nivalenol.

The trichothecenes were implicated in the human illness ATA in Russia. They also apparently are involved in cancer of the esophagus (Marasas, van Rensburg, and Mirocha 1979). They have been involved in various symptoms and syndromes in animals such as feed refusal, vomiting,

growth depression, hyperestrogenism, intestinal tract inflammation, bloody stools, lesions of various organs, skin lesions, tumors, and death (Hoerr et al. 1982; Trenholm et al. 1984). These toxins inhibit the synthesis of protein and DNA. Animals vary in their resistance to vomitoxin. Without serious adverse effects, swine can ingest 2 mg/kg of feed, poultry at least 5 mg/kg of feed, and cattle 6 mg/kg of feed (Trenholm et al. 1984).

The tricothecenes have been found in various grains and also in feeds (Côté et al. 1984; Osborne and Willis 1984; Teich and Hamilton 1985).

Residues of ingested trichothecenes have been found in muscle tissue (Robison et al. 1979), eggs, or milk of animals. This indicates that these agents may be ingested by humans when consuming animal products. Although deoxynivalenol in grain is not regulated in the United States, the FDA has issued a "level of concern" of 2 μg/g of whole grain, and 1 μg/g of finished feed. It is regulated in Canada at these levels.

Zearalenone. Also called F-2 toxin, this mycotoxin (Fig. 6.14) is produced by various *Fusarium* (Table 6.24). It is estrogenic, causing infertility in animals, especially swine (Chang, Kurtz, and Mirocha 1979). No effect on humans has been reported due to ingestion of the naturally produced toxin. The toxin is found in various grains, but most commonly in corn (Mirocha et al. 1979).

EFFECTS OF MYCOTOXINS. The ingestion of mycotoxins may cause either short-term or long-term effects. Short-term or acute effects are rapid and sometimes fatal. Long-term or chronic effects include genetic and birth defects as well as cancer, which may occur after a number of years of ingestion of mycotoxins. Some effects are described in the previous section of individual types of mycotoxins. The biological effects of mycotoxins on living systems were reviewed by Bullerman (1979) and Hayes (1978). One study described a case of acute mycotoxicosis in which the symptoms were a throbbing frontal headache, a feverish feeling, nausea, vomiting, diplopia, weakness, bloody diarrhea, and tremors (Cole et al. 1983). After some 30 hr the symptoms disappeared. Although not involved in this case, other acute symptoms can include chills and skin irritation.

The main chronic symptom is cancer. Of the mycotoxins that seem to be carcinogenic, the toxins of primary concern are the aflatoxins.

Because foods may contain more than one type of toxigenic mold and some of the molds can produce more than one toxin, the symptoms of a mycotoxicosis may be a combination of multiple toxins, or there may be synergism or antagonism among the toxins. Antagonistic reactions between patulin and rubratoxin were described by Kangsadalampai, Salunkhe, and Sharma (1981).

PRODUCTION OF MYCOTOXINS. The ability of a fungus to produce and accumulate toxins is dependent upon such factors as genetic potential, environmental conditions (substrate, moisture, temperature), light, aeration (oxygen and carbon dioxide present), inhibitors, competitive growth, and the time of contact between the mold and the substrate. Being secondary metabolites, to be produced, the mold must grow. However, growth is not synonymous with the presence of mycotoxin, since some substrates suitable for growth are not suitable for toxin production. After growing and producing the toxin, the mold might die and not be recoverable from the food.

Molds can grow over a wide range of temperatures. Even cold storage does not prevent growth or toxin production. The optimum temperature for growth of *A. flavus* and *A. parasiticus* is 35° to 38°C. However, maximum aflatoxin production occurs at lower temperatures (24° to 30°C). Aflatoxins are not produced below 8° to 10°C (Bullerman, Schroeder, and Park 1984). The lowest temperature for growth of *A. flavus* was 15°C (Niles, Norman, and Pimbley 1985).

The production of ochratoxin by *Penicillium cyclopium* or *P. viridicatum* can occur at 4° to 31°C (Northolt, Van Egmond, and Paulsch 1979b). However, the temperature range for ochratoxin production by *Aspergillus ochraceus* was 12° to 37°C. Reportedly, *P. viridicatum* grew at 5°C but did not produce ochratoxin (Damoglou, Downey, and Shannon 1984). It did produce the toxin at 10°C. More than fifty times as much ochratoxin was produced at 25°C than at 12°C (Häggblom and Ghosh 1985).

According to Ciegler (1978), *Fusarium tricinctum* produces T-2 toxin best near freezing temperatures, and *Penicillium martensii* produces penicillic acid faster at 20° to 30°C but accumulates more of the toxin at 4° to 10°C.

In some studies, temperature cycling resulted in higher levels of toxin production than if a constant temperature were used (Bullerman, Schroeder, and Park 1984).

Even with other environmental conditions being optimum, the minimum and optimum temperature for the production of mycotoxin is quite variable and depends on the genus, species, and strain of mold, as well as the mycotoxin that is elaborated.

Moisture satisfactory for growth and toxin production may be present in foods prior to or during harvesting, before they are dried adequately. Improper storage of dried foods can result in a moisture level adequate for mold growth. During harvesting, damage to the protective covering of foods increases the susceptibility of the food to invasion by fungi. High-moisture corn is more readily damaged during harvesting than is low-moisture corn. Most molds have a minimum a_w for growth of between 0.70 and 0.80. The minimum a_w for growth of *A. flavus* is 0.78 (Bullerman,

Schroeder, and Park 1984). The minimum a_w for growth on peanuts varied with the condition of this food (Diener and Davis 1970). The limiting a_w was 0.83 (broken immature kernels), 0.84 (sound mature kernels), or 0.86 (kernels from unshelled peanuts). In most cases, molds can grow in a wider range of a_w and temperature than needed for the production of mycotoxins (Magan, Cayley, and Lacey 1984; Roland and Beuchat 1984a).

The effect of pH of the medium on aflatoxin production is related to the type of substrate, the acids or bases used to alter the pH, and other environmental factors. Jarvis (1971) observed that aflatoxin producers did not grow well below pH 4.0. Maximum production of aflatoxin occurs at pH 5.5 to 7.0 (Buchanan and Ayres 1975).

Substrates with a high concentration of carbohydrates favor aflatoxin production (Diener and Davis 1969). Since plant products have a higher level of carbohydrates than animal products, the majority of foods containing aflatoxins are plant products (Table 6.24). Aflatoxins in animal products usually are due to residues from the mycotoxins in feeds ingested by the animal.

Payne and Hazler (1983) reported that when used as the sole source of nitrogen, proline or asparagine supported more toxin production than did either tryptophan or methionine. The production of ochratoxin increased as the protein concentration of the substrate increased (Häggblom and Ghosh 1985).

Trace metals influence mycotoxin production. Both KH_2PO_4 and $ZnSO_4$ were found to be essential for growth and formation of aflatoxin (Reddy, Viswanathan, and Venkitasubramanian 1979). Increasing the zinc level from 0 to 10 μg/ml increased the amount of aflatoxin over 1,000 fold (Marsh, Simpson, and Trucksess 1975). The aflatoxin content of feed samples correlated significantly with the zinc content (Jones, Hagler, and Hamilton 1984). However, researchers reported that the addition of $ZnSO_4$ to autoclaved soybean meal inhibited aflatoxin production (Hensarling et al. 1983). The addition of sodium phytate relieved this inhibition. The production of patulin by *Penicillium urticae* required manganese (Scott, Jones, and Gaucher 1984).

Various substances can inhibit mold growth and mycotoxin production. The inhibitors include SO_2 (Roland and Beuchat 1984b), caffeine (Buchanan and Lewis 1984), butylated hydroxyanisole (BHA) (Lin and Fung 1983), potassium sorbate (Bullerman 1985), and ferulic acid (Bilgrami, Sinha, and Singh 1981); but the most inhibitory substances seem to be cinnamon and cloves, as well as eugenol extracted from cloves, thymol extracted from thyme, and cinnamic acid from cinnamon (Bullerman, Lieu, and Seier 1977; Hitokoto et al. 1980; Llewellyn, Burkett, and Eadie 1981). Besides inhibitors, some chemicals tend to stimulate myco-

toxin production (Fanelli et al. 1983, 1984; Gareis et al. 1984; Tice and Buchanan 1981).

Generally, molds are aerobic, so it might be expected that oxygen is needed for growth and aflatoxin production. Reducing the oxygen concentration or increasing the carbon dioxide or nitrogen concentration reduces aflatoxin production (Clevström et al. 1983; Paster, Lister, and Chet 1983). The inhibitory effect of CO_2 is enhanced as the temperature and relative humidity are lowered.

The time needed for aflatoxins to appear varies with the environmental factors. The toxins may appear in 24 hr, peak production may require one or two weeks or longer and, with adverse conditions, it may not be produced.

Competitive growth of fungi can result in inhibition of aflatoxin production. *Scopulariopsis brevicaulis, Nocardia*, and *Aspergillus niger* detoxify aflatoxins. *Rhizopus oryzae* metabolizes aflatoxin (Jarvis 1971). *A. chevalieri* and *A. candidus* reduced or prevented aflatoxin formation by *A. parasiticus* (Boller and Schroeder 1974). *A. niger* inhibited aflatoxin production by *A. parasiticus* (Misra, Sinha, and Singh 1981). The presence of T-2 toxin enhanced the production of aflatoxin by *A. parasiticus* (Fabbri et al. 1984).

Even in pure cultures of *A. flavus* or *A. parasiticus*, the aflatoxin level falls after reaching a maximum. This seems to be true, even though some strains appear to be nondegraders of the aflatoxin. Perhaps there is some nonspecific chemical mechanism for degradation of the toxin.

METHODOLOGY. It would be an endless task to discuss all of the existing methods for determining mycotoxins in food and feed. There are certain steps involved in almost any analytical procedure. They are sampling and sample preparation, extraction of the toxin, removal of lipids, cleanup, separation, and quantitation. Depending upon the product, some steps may be excluded and additional steps such as removing specific interfering substances, may need to be added.

Since mycotoxins are not distributed homogeneously, the first problem is to obtain a representative sample. One report stated that a bin of corn can be sampled adequately only while it is being filled or emptied (Davis et al. 1980). Their reasons for errors included inadequate sample size, biased sampling procedures, inadequate sample comminution, and improper subsampling for analysis. The proportion of error due to sampling becomes greater as the aflatoxin concentration becomes lower (Schuller, Horwitz, and Stoloff 1976). A random selection of samples is more representative of the lot than selecting a sample from one location (Waltking 1980).

Two types of tests have been developed for the assay of mycotoxin:

physicochemical and biological. Although some simple tests have been suggested for screening foods and feeds for the presence of aflatoxins, at least one research team believes that there is no reliable screening technique (Calvert et al. 1983).

The simplest test that can be used for some commodities involves scanning grain or seed with long-wave ultraviolet light (365 nm) and watching for a bright greenish yellow (BGY) fluorescence. For corn, there were significant associations between kernel moisture, BGY fluorescence, and aflatoxin (Lillehoj et al. 1983). Of the corn samples with no BGY fluorescence, 98 percent had less than 20 ng/g (20 ppb) of aflatoxin, the action level established by the FDA (Shotwell and Hesseltine 1981). However, the BGY test is not intended to be a quantitative method. The fluorescence is not due to the presence of aflatoxin, but rather to fungal-produced kojic acid. In cotton, Marsh and Simpson (1984) suggested that the fluorescence was caused by interaction of the kojic acid and peroxidase from the fiber. Bothast and Hesseltine (1975) reported the BGY test could be used as a presumptive test for aflatoxin in wheat, oats, barley, corn, and sorghum, but it was not satisfactory for peanuts, rice, or soybeans. Lee and Cucullu (1978) questioned the value of the BGY test for cotton.

A fluorometric-iodide method for screening corn for aflatoxin was described by Davis and Diener (1979). The system was altered to be faster and more convenient by Davis, Guy, and Diener (1981).

Before extraction of the toxin, the sample may need some preparation, such as milling or grinding, to reduce large particles. The toxins are extracted with an organic solvent. They are soluble in methanol, chloroform and acetone, but are only sparing soluble in water. The solvents used for extraction depend upon the food or feed being analyzed (Whitaker, Dickens, and Giesbrecht 1984). Further cleanup, such as removing lipids and other interfering substances may be accomplished with column chromatography systems. Then the aflatoxins can be separated by thin-layer chromatography (TLC) and detected with long-wave UV light (AOAC 1985; Shannon, Shotwell, and Kwolek 1983; Trucksess, Nesheim, and Eppley 1984).

Confirmation of the presence of aflatoxin can be accomplished chemically (Cauderay 1979; Van Egmond and Stubblefield 1981), or by mass spectral methods (Haddon et al. 1977; Rosen, Rosen, and DiProssimo 1984). The extraction and TLC method is sensitive to 2 to 4 μg/kg of aflatoxin (Romer 1973).

Many variations of the physiochemical assay of aflatoxin have been reported. In one system, a minicolumn or florisil tube is used for separation. A direct readout of aflatoxin concentration in the tube can be obtained with a Velasco fluorotoxin meter, or the column can be observed

with long-wave UV light (Holaday 1981; Shotwell and Hesseltine 1981; Velasco 1972).

Systems using TLC have been used to detect ochratoxin A (Howell 1982; Letutour, Tantaoui-Elaraki, and Ihlal 1983), patulin (Ough and Corison 1980), sterigmatocystin (Van Egmond et al. 1980), trichothecenes (Trucksess, Nesheim, and Eppley 1984), and zearalenone (Howell 1982; Swanson et al. 1984).

Liquid chromatographic systems have been used to determine aflatoxins in various foods (Hisada et al. 1984; Tarter, Hanchay, and Scott 1984; Yousef and Marth 1985). Systems using high-pressure liquid chromatography (HPLC) have been suggested to separate and determine the various aflatoxins (Cohen and Lapointe 1981; Francis et al. 1982; Takeda 1984).

Chromatographic systems (liquid, column gas, HPLC) have been developed to detect citrinin (Gimeno and Martins 1983), deoxynivalenol (Bennett, Megalla, and Shotwell 1984; Chang et al. 1984), luteoskyrin (Takeda et al. 1979a), ochratoxin A (Ehrlich and Lee 1984; Howell and Taylor 1981), patulin (Gimeno and Martins 1983; Möller and Joffesson 1980), T-2 toxin (Cohen and Lapointe 1984), and zearalenone (Bennett, Megalla, and Shotwell 1984; Chang and DeVries 1984).

Systems using tandem mass spectrometry to detect and identify aflatoxins have been developed (Grove, Plattner, and Peterson 1984; Plattner, Bennett, and Stubblefield 1984). The inhibition of protein synthesis of culture cells by T-2 toxin was suggested as a method to detect this toxin (Thompson and Wannemacher 1984).

Immunoassays for mycotoxins were reviewed by Chu (1984). A radioimmunoassay (RIA) for T-2 toxin in corn and wheat was sensitive, accurate, reproducible, and relatively simple (Lee and Chu 1981). The sensitivity of the RIA was 10 ng/ml for T-2 toxin (Fontelo et al. 1983) or 5 ng/ml for zearalenone and zearalenol (Thouvenot and Morfin 1983). An enzyme-linked immunoassay (ELISA) gave more consistent data, relatively lower standard deviations, and lower coefficients of variation than did RIA for determining aflatoxin B_1 (El-Nakib, Pestka, and Chu 1981). An ELISA was reported to be simple, sensitive, and specific for aflatoxin M_1 in milk (Hu, Woychik, and Chu 1984), and for M_1 at 10 to 50 ng/kg (ppt) in various dairy products (Fremy and Chu 1984).

ELISA systems were effective for the analysis of ochratoxin (Lee and Chu 1984) and T-2 toxin (Gendloff et al. 1984). Monoclonal antibodies to certain toxins have been produced (Hunter et al. 1985; Woychik, Hinsdill, and Chu 1984), which will aid in the immunoassay for specific toxins.

There are several systems for the biological assay of aflatoxin. In the chicken embryo test, specific amounts of aflatoxin deposited in a fertile

egg will destroy the life of the embryo. Typical lesions develop in the embryo with subacute levels (less than 0.1 μg/kg) of aflatoxin B_1.

The effect of aflatoxin on ducklings can be used for assay purposes. Aflatoxins inhibit cell cleavage in fertilized mollusk (*Bankia setacea*) eggs without preventing nuclear division. The resultant cells are multinuclear. Over 60 percent mortality of brine shrimp (*Artemia salina*) is obtained in 24 hr by 0.5 μg/ml (aflatoxin B_1). With 1 μg/ml, greater than 90 percent mortality occurs. At a concentration of 1 μg/ml, aflatoxin B_1 is lethal to zebra fish larvae. With a concentration of 0.15 μg/ml, the water flea (*Daphnia*) is killed in 40 hr. Schuller, Horwitz, and Stoloff (1976) stated that the chick embryo bioassay is the most useful of these biological systems.

The reduction of bioluminescence of *Photobacterium phosphoreum* was suggested as a possible assay system for some mycotoxins (Yates and Porter 1982). The cytotoxicity of trichothecenes on HEp-2 and Chang cells was more sensitive than the TLC system (Robb and Norval 1983). Swiss mouse fibroblasts were adapted to a biological assay of triothecenes (Abbas, Shier, and Mirocha 1984). The sensitivity was 0.1 ng/ml for T-2 toxin and 5 μg/ml for zearalenone.

The FDA requires that the presence of aflatoxin B_1 in certain foods be confirmed by negative ion chemical ionization mass spectrometry instead of the chicken embryo bioassay.

CONTROL OF MYCOTOXINS. The systems for controlling aflatoxins in foods or feeds are essentially the same as for controlling any microbial toxin. We might prevent contamination of the food by toxigenic *A. flavus* or *A. parasiticus* and inhibit their growth and toxin production. We can analyze food for aflatoxin and, if present, remove, destroy, or detoxify the toxin.

Prevent Contamination. The ubiquitous nature of toxin-producing fungi makes it difficult to grow crops without subjecting them to potential contamination. One solution might be the selection or development of plant varieties that resist such contamination. Irrigation of crops may reduce mycotoxin contamination by lowering the stress on plants caused by insufficient rainfall. No aflatoxin was detected in peanuts harvested from plots treated with gypsum (Mixon, Bell, and Wilson 1984).

Prevent Fungal Growth. The fungi attack damaged seeds more readily than they attack sound seeds. Control of insects and using care in harvesting can reduce the number of damaged seeds. The removal of damaged seeds and foreign material before storage will help control mold growth and mycotoxin production.

The development of a low moisture content in the seeds, and storage at low RH is perhaps the simplest and best method for controlling growth

and mycotoxin production. If there is improper ventilation during storage, localized areas can develop with sufficient moisture for fungal growth. These areas can be caused by sweating, movement of moisture through the food, and biological activity of seeds.

Cold storage will prevent growth of the aspergilli and production of aflatoxin, but other toxigenic molds can produce mycotoxins at low temperatures (below 10°C). Hence, cold storage alone is not beneficial from an overall viewpoint.

The addition of certain chemicals will inhibit or reduce the production of aflatoxin. Davis and Diener (1967) found that soaking peanuts in a solution of *p*-aminobenzoic acid reduced the production of aflatoxin by 50 percent. Potassium sulfite and potassium fluoride also inhibited aflatoxin production by the molds. Other effective chemicals are discussed in the section on production of mycotoxins.

Destroy Organism. Spores of *A. flavus* do not survive a 45-sec treatment with ultraviolet light (Bean and Rambo 1975). Gamma radiation at levels of 0.25 to 1.0 Mrad inactivate fungi in stored products.

According to Doyle and Marth (1975), the conidia of *A. flavus* and *A. parasiticus* strains have a D_{55} of 3 to 29 min and D_{60} of 8 to 59 sec at pH 7.0. Hence, it might be possible to heat certain foods prior to storage to destroy the organism. However, the foods would then need to be stored in a manner that would prevent recontamination.

Remove Toxins. Aflatoxins can be removed from food by segregating obviously contaminated grains or kernels, or by extraction of the food with solvents. These procedures can be used only for certain foods or feeds.

The aflatoxin content of a batch of peanuts may be confined to a few highly contaminated kernels. When peanuts are removed from the shell, they are sorted to separate any moldy, discolored, shriveled, or damaged raw peanuts. The aflatoxin content of peanut lots is reduced significantly by this segregation procedure (Telford 1982). The difference in density was used to separate contaminated from sound corn (Huff 1980), as well as wheat (Huff and Hagler 1985).

The removal of mold growth from products such as cheese does not remove all of the aflatoxin, since aflatoxins tend to diffuse away from the mold mycelia. Hence, not only must the mold be removed, but also any food material that may contain diffused aflatoxin.

Examination of seeds with UV light for greenish-yellow fluorescence, and segregation of the fluorescent seeds, should aid in reducing the aflatoxin content of remaining seeds. Schade and King (1984) reported that almond kernels that fluoresce violet-purple under long-wave UV light contain high levels of aflatoxin. By separation of these kernels, the aflatoxin content can be lowered.

For some types of foods, solvents have been used to extract the aflatoxins. Due to the distribution of toxin throughout the peanut, solvent extraction of whole kernels is ineffective. However, with crushed peanuts or peanut meal, extraction can be used. Besides peanut meal, the extraction can be useful for other oilseed meals. Detroy, Lillehoj, and Ciegler (1971) reviewed the systems, solvents, and results of the extraction procedures for removing aflatoxins. Activated charcoal avidly adsorbed aflatoxin from a liquid medium (Decker and Corby 1980).

When animals eat feed containing aflatoxins, residues may be detected in various tissues. Researchers reported that feeding contaminated pigs an aflatoxin-free feed resulted in a significant reduction of aflatoxin in organs and tissues in one day, and no aflatoxin was found in any tissues after four days (Furtado et al. 1982).

Inactivate Mycotoxins. Either chemical or physical treatments can be used to inactivate or detoxify aflatoxin. Various methods of detoxification have been reviewed by (Dollear 1969; Doyle et al. 1982).

Chemical treatments that have been reported for inactivation include hydrogen peroxide, chlorine gas, sodium hypochlorite, benzoyl peroxide, sodium perborate, sodium hydroxide, ammonia, methylamine, chlorine dioxide, formaldehyde, nitrogen dioxide, ozone, and bisulfites.

Ammonia is effective in inactivating aflatoxins in cottonseed meal, peanut meal, and corn (Bothast et al. 1982; Lee, Koltun, and Stanley 1984; Norred 1982). Ammoniation inactivates the aflatoxin, reduces the carcinogenicity to rainbow trout, and does not reduce the nutritive value of treated corn (Brekke et al. 1977). Although ammoniation reduced the amount of ochratoxin A in barley, it did not influence the daily weight gain or feed efficiency when fed to hogs (Madsen, Hald, and Mortensen 1983). The process is not acceptable for foods for human consumption (Wood 1982).

Treatment of peanut meal with NaOH reduces the aflatoxin content. A combination of calcium hydroxide and formaldehyde reacts with aflatoxin in highly contaminated meal, so that it is acceptable for feed use (Codifer, Mann, and Dollear 1976).

Sodium hypochlorite (NaOCl), or bleach, has been effective in inactivation of aflatoxin (Draughon and Childs 1982). However, researchers have found that the reaction of chlorine with aflatoxin B_1 forms another carcinogen (Castegnaro et al. 1981). Sodium bisulfite was more effective than either NaOH or ammonia in inactivation of aflatoxins B_1 and B_2 (Moerck et al. 1980). According to Hagler, Hutchins, and Hamilton (1983) bisulfite reacted with B_1 to form B_1S, but aflatoxin B_2 or G_2 were not susceptible to the action of bisulfite. When vitamin C was present in apple juice, patulin disappeared (Brackett and Marth 1979b).

Oil, as extracted from oilseeds (peanuts and cottonseed) may contain small amounts of aflatoxins. The oil is treated with alkali and bleach to purify it. This processing reduces the aflatoxin content from 812 μg/kg to less than 1 μg/kg (Kensler and Natoli 1969).

The primary physical treatment used to reduce toxicity of aflatoxins is heat. Roasting peanuts at 150°C for ½ hr reduces aflatoxin B_1 about 80 percent and B_2 about 60 percent (Lee, Cuculla, and Goldblatt 1968). Waltking (1971) roasted peanuts at 204°C and found an average loss of 40 to 50 percent aflatoxin B_1 and G_1 and 20 to 40 percent of B_2 and G_2. Oil roasting of peanuts at 163°C for 3 min reduces B_1 and G_1 by 65 percent (Lee et al. 1969).

Roasting soybeans at 180°C reduced levels of aflatoxin from 40 to 73 percent (Hamada and Megalla 1982). Normal cooking of rice destroyed about 49 percent of the aflatoxin, while pressure cooking with excess water reduced aflatoxins by 81 percent (Rehana, Basappa, and Murthy 1979). El-Banna and Scott (1984) found that cooking fava beans or wheat reduced ochratoxin A by only 16 to 20 percent. About 20 percent of ochratoxin was lost by frying pork at 150° to 160°C (Josefsson and Möller 1980). Aflatoxins and at least some other mycotoxins are considered to be heat stable.

Biological systems might be used to degrade aflatoxins (Mann and Rehm 1976). *Corynebacterium rubrum, Aspergillus niger, Trichoderma viride,* and *Mucor ambiguus* degrade aflatoxin B_1 to aflatoxin R_0. Before such systems can be used in industry, much more research is needed, as well as approval by the FDA. Aflatoxin R_0 reportedly is less toxic than B_1 to ducklings, but is carcinogenic to trout.

Control by Regulation. The production of aflatoxin by toxigenic molds cannot be controlled by a regulation of a government agency. However, it is possible to regulate against the use of food or feed if it contains aflatoxin.

Aflatoxins are considered to be carcinogens. The Delaney clause to Section 409 of the Federal Food, Drug and Cosmetic Act prohibits establishment of a tolerance for additives that "induce cancer when ingested by man or animals."

Instead of tolerances, the FDA has used administrative guidelines, the acceptable level depending upon the ability to detect aflatoxin. In 1964, a control program was established requiring the analysis of shelled peanuts by a certified laboratory. Peanuts with more than 40 μg/kg (ppb) could not be used in feed. In 1965, this was lowered to 30 μg/kg, and in 1969, to 20 μg/kg. Present methods enable the detection of aflatoxin at levels of 2.5 μg/kg or less. Ideally, there should be no aflatoxin, but realistically this would eliminate a large amount of food from an already overtaxed food supply. In 1975, the FDA proposed a formal tolerance of 15

μg/kg for aflatoxin in peanuts and peanut products, but after more than ten years, this lower level has not been adopted. Dichter and Weinstein (1984) estimated that lowering the tolerance level to 15 μg/kg would be cost effective, and that lower levels of 10 or 5 ppb would not be.

Besides peanuts and peanut products, FDA surveillance activity is directed at any food that might contain aflatoxin and action is taken against any food or animal feed with 20 μg or more aflatoxin per kilogram. There is an administrative guideline of 0.5 ppb for aflatoxin M_1 in fluid milk in interstate shipment. This lower level was set because milk is consumed in large quantities by children. For laboratory animals in toxicology studies, the FDA and Environmental Protection Agency specify that diets contain less than 5 ppb aflatoxin. Animal feed is included in the guidelines because aflatoxins consumed by animals may be found in the resultant food (milk, meat, eggs). However, during years when corn has contained levels of aflatoxin above 20 ppb, the FDA has made exceptions in certain cases (Hamilton 1984; Labuza 1983).

In Australia, the permitted levels of aflatoxin are 15 ppb for peanuts and 5 ppb for other foods (Pitt 1982). The only other restrictions are various tolerance levels of deoxynivalenol (vomitoxin) in wheat in Canada (Morrison 1982a, 1982b), and ochratoxin A in the kidney of hogs in Denmark.

The continual surveillance and the elimination of aflatoxin-contaminated products from the food or feed supply constitutes a system of controlling these toxins.

VIRAL HEALTH HAZARDS

For most foodborne illnesses caused by bacteria, the organisms must proliferate in the food for illness to occur. Viruses that cause human illness cannot multiply in food. Thus, a virus that contaminates a food either survives or becomes inactivated. This means that viruses tend to be present only in very low concentration. This has made it difficult to detect viruses in foods. The isolation and identification of viruses require special techniques.

Bryan (1973) listed four viral diseases in which there is evidence of foodborne transmission. These diseases are infectious hepatitis, poliomyelitis, Bolivian hemorrhagic fever, and Russian spring-summer encephalitis. He also listed eight viral diseases that might be foodborne, but proof is lacking. These include reovirus infections, serum hepatitis, and nonbacterial gastroenteritis. Cukor and Blacklow (1984) stated that only two viruses, rotavirus and Norwalk virus, are important etiological agents of human gastroenteritis. Besides gastroenteritis, enteroviruses may in-

fect other systems and cause paralysis, encephalitis, menigitis, pleurodynia, myocarditis, or herpangia (Moore 1982; Phillips et al. 1980).

There were thirty-three confirmed outbreaks of viral foodborne illness from 1977 to 1981 (CDC 1983b). This represents only a small part of the total outbreaks of foodborne illness. However, many viral illnesses are neither reported nor confirmed. According to Cukor and Blacklow (1984), acute viral gastroenteritis is second in frequency only to the common cold. Estes and Graham (1979) stated that epidemic viral gastroenteritis is the leading cause of mortality among infants and children in underdeveloped countries.

To cause an illness, the ingested viruses must be able to survive the acidity of the stomach, digestive enzymes, and also bile in the duodenum. The number of viruses needed for infection is not known. However, swine became infected with ingestion of only 250 plaque-forming units (PFU) (Cliver 1981). In many cases, little is known as to which host cells are involved, how the viruses pass through the mucous membranes, or other factors that influence the initiation or spread of the infection.

Viruses causing human diseases have been isolated from various domestic animals. In some cases, the same virus was isolated from animals and from people who had close contact with the animals. Human and animal viruses have been recovered from raw and heated milk, dairy products, meat, eggs, oysters, mussels, clams, and crabs (Goyal, Gerba, and Melnick 1979; Larkin 1981; Wait et al. 1983). Raw or only partially cooked foods are those primarily implicated as vehicles for virus transmission. The usually long incubation periods and problems of isolating viruses has made it difficult to implicate food in viral diseases.

The first viral illness reported to be transmitted by foods was poliomyelitis. Raw milk was the predominant food vehicle for this illness.

Viruses also are found in sewage and polluted water. They can contaminate various inanimate objects, which can act as sources to contaminate other items, such as food.

Survival of Viruses

Mahl and Sadler (1975) studied the survival of various viruses on hard-surfaced inanimate objects (glass, stainless steel, tile). Although inactivation was observed, some viruses persisted for eight weeks at room temperature (25°C) and low RH (3 to 7 percent). At 37°C and 55 percent or 93 to 96 percent RH, survival time varied from one day to eight weeks, depending upon the virus type.

The stability of viruses in foods is determined by the type of virus, the moisture and pH of the food and the temperature of storage. Enteric

viruses survive longer than influenza viruses (Cliver, Kostenbader, and Vallenas 1970). At low temperatures, poliovirus is extremely stable in food with pH 7.0 or greater. As the temperature of storage is increased from 4°C, the inactivation also is increased. They found that enteroviruses persisted in low-moisture food for over two weeks at room temperature, and more than two months in the refrigerator.

Herrmann and Cliver (1973a) inoculated ground beef and Thuringer sausage with a coxsackie virus. When ground beef was held at 4°C, about 85 percent of the viruses survived for eight days, but less than 1 percent survived for fourteen days. Kantor and Potter (1975) reported that a poliovirus and echovirus persisted in high numbers and were virtually unaffected during the commercial production of salami and cervelat sausages.

Larkin, Tierney, and Sullivan (1976) irrigated lettuce and radish crops with sewage sludge and sewage effluent inoculated with poliovirus. This virus persisted for up to thirty-six days on these crops. They suggested that a cycle of infection could be established if contaminated effluent or sludge were used on crops in the food chain. Enteric viruses survived at least five months in food stored at −20°C.

Inactivation

Viruses can be removed from materials or be inactivated, by essentially the same systems used to control bacteria. The action of sunlight, water-treatment processes, physical and chemical inactivation by suspended and dissolved materials, bacterial antagonism, as well as simple dilution, serve to decrease the viral concentration in water. In a rather extensive survey, viruses were not detected in chlorinated drinking water (Clarke et al. 1975).

Just as viruses (bacteriophage) can affect bacteria, certain bacteria may inhibit some viruses. Hydrogen-peroxide-generating bacteria are antagonistic to poliovirus (Klebanoff and Belding 1974), culture filtrates of *Proteus mirabilis* contain a substance inhibitory to Sindbis and vesicular stomatitis viruses (Mahdy and Bansen 1974), and proteolytic bacteria inactivate coxsackie virus type 9 (Herrmann and Cliver 1973b).

Viruses can be inhibited or inactivated by various chemical agents, such as β-propiolactone, guanidine hydrochloride, quaternary ammonium, phenolic and iodophor compounds (Gaustad, McDuff, and Hatcher 1974), ozone (Burleson, Murray, and Pollard 1975), glutaraldehyde (Saitanu and Lund 1975).

Viruses are more resistant to gamma radiation than are other types of microorganisms. Considering the high doses needed, the use of radiation to inactivate viruses is not practical.

Thermal inactivation seems to be an acceptable and practical method of controlling viruses in food. Many viruses are inactivated in 5 min when heated to 65°C. Therefore, it is sometimes assumed that properly cooked or pasteurized foods are not a concern of public health unless the product is recontaminated. However, researchers have reported that the time used to cook crabs may not be sufficient to inactivate viruses (DiGirolamo et al. 1972). Filppi and Banwart (1974) inoculated ground beef with poliovirus type 1 and heated the mixture at various temperatures. From their data, it is evident that meat, when highly contaminated with a potentially pathogenic virus, might be a source of infection, even if heated to 70°C. Other researchers found no surviving poliovirus 1 or coxsackie virus B-2 in hamburgers broiled to either 71°C or 76.7°C (Sullivan et al. 1975).

Cliver (1971) reported that inoculated polio 1 virus was not completely inactivated in oysters by stewing, frying, baking, or steaming. Pasteurization treatments, such as in egg products or milk, cannot be relied upon to destroy large numbers of heat-resistant viruses.

Methodology

The analysis of food for viruses requires techniques different from those used for bacterial enumeration. Viruses are determined by inoculation into host cells for enumeration, by observation with an electron microscope, or by immunological methods, either by determining viruses directly or by determining antibody produced by a host in response to a viral infection.

Since viruses, when present, are usually at a low level in food, a large quantity must be analyzed. After making a suspension of the food sample, the larger food particles are removed by filtering through glass wool or cheesecloth, or by slow centrifuging (Tierney et al. 1973). After removal of the food, the viruses are concentrated. This is accomplished by membrane adsorption, adsorption to a precipitable salt, filtration, centrifugation, electrophoresis, or dialysis (Guttman-Bass and Nasser 1984; Sullivan et al. 1984).

To determine viruses in the concentrate, the plaque-forming units that develop on a susceptible tissue culture are counted. Bacteria present in food can interfere with the growth of cells in a tissue culture. Therefore, if they are not removed, they must be inhibited. This is accomplished by adding antibiotics that inactivate the bacteria but not the tissue culture or viruses.

Due to the cell specificity of viruses, the more types of tissue culture cells that are used, the more likely that most types of viruses can be detected and isolated.

In some cases, procedures using the electron microscope have been used to detect viruses, viruslike particles, or virus-antibody complexes.

Serological systems for the detection and identification of viruses include fluorescent techniques, RIA, hemagglutination inhibition, immunoperoxidase, counterimmunoelectrophoresis, latex agglutination tests, enzyme immunoassay, immunocytochemical and ELISA (Angarano, Laddago, and Materi 1984; Herrmann, Hendry, and Collins 1979; Sambourg et al. 1985; Smith 1985).

Diseases

Although several human illnesses are caused by viruses, only those that may be considered a foodborne illness are discussed. These illnesses were reviewed by Gerba, Rose, and Singh (1984).

HEPATITIS. There are three types of viral hepatitis: hepatitis A, hepatitis B, and hepatitis non-A, non-B. Hepatitis A, sometimes called infectious hepatitis, is caused by a picornavirus which is transmitted by the fecal-oral route. The virus can be grown in cell culture (Deinhart and Gust 1982). The illness is not chronic, and no chronic carriers are known. Hepatitis B (serum hepatitis) is transmitted through blood or blood products or close contact with a hepatitis B-positive person. Hepatitis B may result in a persistent or chronic infection. This virus has not been grown in cell culture (Deinhart and Gust 1982). In 1983, the total number of cases of hepatitis reported in the United States was 56,469. Of these cases, 38 percent were hepatitis A, 43 percent hepatitis B, 6 percent hepatitis non-A, non-B, and 13 percent unspecified (CDC 1985b). This was the first year that the incidence of hepatitis B was greater than that of hepatitis A.

Since hepatitis A virus is spread by the fecal-oral route, waterborne and foodborne outbreaks can occur. People with the infection who handle food are a primary source for spreading the virus. Even then, only about 6 percent of the cases are foodborne or waterborne (CDC 1985b).

The eating of raw shellfish (oysters, mussels, or clams) is an important source of the hepatitis virus. This is due to the potential contamination of water with raw sewage and the collection and concentration of the virus by the shellfish. However, since infected food handlers are a primary source, various foods have been implicated as vehicles of the virus. These foods include dairy products, meats, salads, bakery goods, and fruits, which were contaminated during preparation and received no further cooking prior to serving.

A series of outbreaks of hepatitis A due to eating raw clams was described by CDC (1982d). Some 6 to 72 hr after consumption, gastrointes-

tinal distress (diarrhea, abdominal cramps, nausea, vomiting), which lasted from one to three days, was experienced. From twenty-one to thirty-seven days after ingestion of the clams, symptoms of hepatitis A were evident. These symptoms include bile in urine (dark urine) and jaundice. In an outbreak resulting from a family reunion of forty-five people, fourteen cases of clinical hepatitis were identified from nineteen to forty-four days. The illness may be rather mild, allowing cases to be ambulatory. In other cases, liver injury occurs, resulting in cirrhosis. A severe infection can cause death.

During hepatitis, the level of serum glutamic-oxalacetic transaminase (SGOT) or serum glutamic-pyruvic transaminase (SGPT) is at least twice as high as the normal laboratory standard, and there is a total level of serum bilirubin over 2 mg/100 ml. Also, immunoglobin M hepatitis A antibody (IgM anti HAV) is produced (CDC 1981b). This antibody usually appears in the serum before or shortly after the onset of symptoms and persists for two to three months. Detection of the antibody can identify early and subclinical cases. A team of researchers developed an ELISA system to detect IgM anti HAV in serum (Locarnini et al. 1979).

The hepatitis A virus is present in the feces some two weeks prior to symptoms.

The disease can be attacked from four areas. First, people who have been exposed or might be exposed to the virus can be given immune serum globulin. This has been an effective deterrent for the secondary spread of the illness. Second, the beds from which seafoods, such as shellfish, are gathered should be clean and not polluted with raw sewage. Harvested shellfish should not be eaten raw. However, there are some people who insist on consuming these animals in the uncooked state.

The third means of control is to inactivate the virus in a food product by heating, radiation, drying, or chemicals. Unfortunately, little is known about the exact treatment to inactivate hepatitis virus A. Clams added to a pot of boiling water, heated until they opened and then eaten, were incriminated in an outbreak of hepatitis. This virus evidently is not inactivated by limited heating. However, adequate cooking should inactivate the virus.

Inasmuch as most of the foodborne outbreaks seem to be caused by a food handler contaminating a food with no further cooking, the fourth area of control is the most important. The infected food handler must be controlled. This is not easily accomplished, since the virus may be excreted in stools during the incubation period, from seven to ten days prior to the onset of symptoms and by asymptomatic persons. However, those clinically ill persons should not be allowed to work with food until they are clearly convalescent. Employees can be screened by the analysis of the SGOT or SGPT levels of their serum. If the levels are more than

twice the normal values, then the employee should not work with food. Better personal hygiene of food handlers, including more frequent hand washing and less handling of food and ingredients, would help prevent the spread of not only infectious hepatitis, but also other illnesses.

ROTAVIRUS. Human rotavirus is a major cause of gastroenteritis in infants and young children throughout the world (Albert, Bishop, and Shann 1983; Al-Nakib et al. 1980; Coiro et al. 1983; Mata et al. 1983; Oishi et al. 1979; Paniker, Mathew, and Mathan 1982; Rodriguez et al. 1980). The illness is most prevalent and severe in children from six months to two years of age. After the second year of life, rotavirus infections requiring hospitalization decline significantly. By age five, most children have acquired serum antibodies to rotavirus.

The incubation period is two to four days, and the symptoms include diarrhea, vomiting, fever, and abdominal pain. The duration of the illness is two to ten days. Due to loss of fluids and severe dehydration, death can occur, especially in children less than one year old. In temperate climates, the incidence of the illness is highest during the winter and is virtually absent in the summer. This is the opposite of salmonellosis, which is more prevalent during the summer. Some possible reasons for wintertime prevalence have been suggested (Brandt et al. 1982). In tropical areas, the incidence of rotavirus tends to be influenced by rainy or dry seasons.

The transmission of rotavirus appears to be by the fecal-oral route; however, the respiratory route also has been suggested (Foster et al. 1980; Oishi et al. 1979).

Adults that contact infected infants often become infected with rotavirus as evidenced by serological tests. Most adults have only mild symptoms or are asymptomatic; however, a few may have a severe illness.

NORWALK VIRUS. This name was given to the causative agent involved in an outbreak of nonbacterial gastroenteritis in Norwalk, Ohio, in 1968. Particles 27 nm in size were observed in stool filtrates with the aid of electron microscopy. Similar particles have been observed in other outbreaks. Other characteristics of the viruses were described by Cukor and Blacklow (1984). According to Dolin (1978), six agents (Norwalk, Hawaii, Montgomery County, W, Ditchling, Cockle) are all similar, but there are multiple antigenic types. At the present time, Norwalk virus is the most prominent. None of these agents has been grown on cell cultures.

Foodborne, waterborne, and person-to-person (both primary and secondary) modes of transmission occur (Goodman et al. 1982; Kaplan et al. 1982). The illness appears during all seasons and affects all age groups.

The incubation period is about 30 to 36 hr, and the duration of the illness is 24 to 48 hr. The illness is rather mild, with symptoms including diarrhea, nausea, vomiting, cramps, headache, and fever. Foodborne outbreaks have involved green salad (Griffin et al. 1982; Lieb et al. 1985), raw oysters (Eyles, Davey, and Huntley 1981; Gunn et al 1982), cake frosting (Kuritsky et al. 1984) and chicken sandwiches (Pether and Caul 1983).

Control

The control of viral infections involves essentially the same effort as other foodborne illnesses. There is a need for harvesting shellfish only from satisfactory waters and practicing good personal hygiene.

The viruses do not multiply in food and tend to become inactivated during storage. They can be inactivated with chemical and physical agents in a manner similar to bacteria.

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7

Indicator Organisms

Indicator organisms have been used since 1892, the year Schardinger tested water for what is now called *Escherichia coli,* instead of testing for *Salmonella typhi.* It is difficult to detect *S. typhi* in a water supply. The presence of *E coli* indicates that there might have been contamination from sewage and that *Salmonella* or other intestinal pathogens might be present.

Since the original studies, indicator organisms have been used to determine an objectionable microbial condition of food, such as fecal contamination, the presence of potential pathogens or potential spoilage of foods, as well as the sanitary conditions of food processing, production, or storage. Tompkin (1983) stated that the reason for using indicators is to monitor raw materials, processing conditions, and products at various stages of processing and distribution to forewarn of potential microbiological problems.

According to Stadhouders, Hup, and Hassing (1982), the organisms used to assess foods for the possible presence of certain organisms capable of causing foodborne illness should be called index organisms. However, these researchers stated that for many foods, a distinction between indicator and index organisms is not possible. Mossel (1983) called these indicator organisms "marker organisms," but in this text, they will be called indicator organisms.

Certain criteria are used to determine the value of an indicator organism. If such an organism is to indicate possible fecal contamination or potential pathogens, it must be associated exclusively with feces or closely allied with the pathogens.

The indicator should not be present as a natural contaminant of the material being analyzed. If it is usually present on a food product, its detection would not necessarily indicate the presence of either fecal material or pathogens. Hence, for these two criteria, knowing the sources of the indicator organisms is important.

The organism should be easy to grow and differentiate. There should be a relatively simple, accurate, rapid, and standard test to enumerate the indicator. If the procedure to determine the indicator is more difficult or less accurate than that for pathogens, then we might as well analyze for the pathogens.

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If the indicator multiplies very rapidly at storage temperatures of the food, the actual degree of contamination may be obscured. If it dies very rapidly, its value as an index of contamination is lowered. The indicator organisms should withstand processing treatments of the food in a manner similar to pathogens. If the indicator is less stable, it may disappear before the pathogen. If it is more stable, it may persist long after the pathogen is destroyed.

Many organisms or groups of organisms have been suggested as indicator organisms. Fecal microorganisms include the enteric bacteria, viruses, and protozoa. In general, viruses and protozoa are more difficult to enumerate than are bacteria. Bacteria such as coliforms, *E. coli,* Enterobacteriaceae, enterococci, pseudomonads, clostridia, staphylococci, and the aerobic plate count have been suggested as indicator organisms. How well do these bacteria or groups of bacteria meet the general criteria needed for a good indicator? We need to know the usual source, the association with enteric pathogens, methods for enumeration, growth conditions, survival during storage and processing, and the significance of these organisms in foods or a particular food.

COLIFORMS

As a result of the successful use of coliforms as indicators in water, they have been employed as indicators of possible fecal contamination of foods.

Whenever organisms are given a group terminology, there is a question as to which organisms are included, or at least a definition is needed. Coliforms include all aerobic and facultatively anaerobic Gram-negative nonsporeforming bacilli which ferment lactose with gas formation within 48 hr at 35°C. For techniques that are used in the membrane filter method for water analysis, an alternative definition is that coliforms include all organisms that produce a colony on eosin methylene blue agar with a golden-green metallic sheen within 24 hr of incubation at 35° to 37°C.

With the procedures of the U.S. Food and Drug Administration (FDA 1978), the definition of a presumptive coliform is an organism that produces gas in lauryl sulfate tryptose (LST) broth in 48 hr at 35°C. A confirmed coliform is a presumptive coliform that produces gas in brilliant green lactose bile (BGLB) 2 percent broth in 48 hr at 35°C.

The coliform group includes *Escherichia coli, Citrobacter freundii, Enterobacter aerogenes, Enterobacter cloacae,* and *Klebsiella pneumoniae.* There are a few strains of other species that ferment lactose and might be included in a coliform determination.

Sources

Coliforms are common inhabitants of the intestinal tract of people and animals. However, bacteria in this group are of both fecal and nonfecal origin.

The organisms are found in undisturbed soil and can be at high levels in fecal-contaminated soil. Under certain conditions they tend to die in the soil, but with proper nutrients and moisture they can increase in number. Soil dust can disseminate coliforms into the atmosphere. Rain carries the surface contamination from soil to streams, rivers, or lakes.

Coliforms are found on all types of plant material (foliage, roots, flowers). *Klebsiella* predominated in samples obtained from forest environments and from fresh farm produce by Duncan and Razzell (1972). Most of the fresh vegetables these reseachers examined had coliform counts of 10^6 to 10^7 per gram. All thirty samples of lettuce examined by Käferstein (1977) contained coliforms (from less than 10 to over 10^5/g).

Coliforms are found on the shells of freshly laid eggs and can penetrate through the pores if the egg surface is damp. The organisms can be found in raw milk through contamination of the teat canals by feed or manure. Coliforms cause infection of the mast cells (mastitis) of cattle, from which they can be shed into the milk. Milk can be contaminated after milking by air, people, or equipment.

Coliforms are present on the feathers of live poultry and on the hide, hooves, and hair of other animals. Upon slaughter, the organisms can contaminate the meat from these sources or from intestinal leakage during evisceration.

Shellfish grown in polluted areas concentrate the organisms so that they are contaminated at levels higher than that of the water.

Coliforms are widespread and can be detected on many types of food products, especially those of animal origin.

Methods for Detection

Many methods have been used to detect coliforms. The fermentation of lactose on or in a medium is the primary requirement for an organism to be considered a coliform.

The procedure suggested by FDA (1978) consists of using the MPN method by inoculating tubes of LST broth. Growth from those tubes showing gas production (due to fermentation of lactose) after incubation at 35°C for 48 hr is inoculated into tubes of BGLB (2 percent) broth. These inoculated tubes are incubated at 35°C for 48 hr and observed for gas production (due to fermentation of lactose). The production of gas is considered to be confirmed positive for coliforms.

Many of the reported coliform investigations are performed by plating samples with violet-red bile (VRB) agar. Higher counts are often obtained using an agar, since anaerogenic strains will be counted. These strains are not positive in the MPN tube test. Not all of the organisms that ferment lactose on VRB agar prove to be coliforms. This may be because of lactose-fermenting organisms other than coliforms that are not inhibited by the selective agents (bile salts and dyes), because the carryover of food material may reduce the inhibition of the selective agents, or because the sampled food can furnish other carbohydrates besides lactose for the organisms to ferment. These factors also can affect the results of the MPN test. Processed food material often contains stressed coliforms that require a noninhibition medium for repair. Hartman, Hartman, and Lanz (1975) suggested using a VRB2 agar to increase counts of stressed coliforms in foods. A similar procedure enumerating injured coliforms was suggested by Ray and Speck (1978). They plated the sample with trypticase soy agar and overlayed the surface with VRB after a 1-hr incubation at room temperature. For acid-injured coliforms, Reber and Marshall (1982) reported higher counts with the VRB2 system than with VRB. A petrifilm VRB system compared favorably with the VRB agar plates and the MPN method for enumerating coliforms (Nelson, Fox, and Busta 1984).

An enriched sulfate-aniline blue medium was reported by Wright (1984) to yield higher coliform and *E. coli* counts than did media containing bile salts. To overcome the natural variations in bile salts media, a chemically defined minerals modified glutamate medium (MMGM) was developed. Other researchers reported that MMGM was superior in sensitivity and favorable in specificity to other media tested, including LST and BGLB broths (Abbiss et al. 1981).

The membrane filter (MF) technique is used to enumerate coliforms in water. After filtering a standard volume of water, the filter pad is placed directly on a differential Endo-type medium for incubation, or a preliminary enrichment in LST broth can be used. One advantage of the MF system is that organisms can be concentrated from low-density material. Inhibition of coliform growth and interference in the production of the typical sheen due to growth of noncoliform bacteria has been observed (Burlingame et al. 1984; Clark 1980). Anaerobic incubation reportedly overcame part of this problem (Franzblau et al. 1984).

Besides water analysis, the MF technique can be used for certain food products. This would be especially advantageous if a component of the food is a carbohydrate that could be fermented by organisms other than coliform organisms, since filtration will remove these carbohydrates.

The hydrophobic grid membrane filter system to enumerate coliforms from food has been adopted as official first action by AOAC (Entis

1983). Resuscitation of injured organisms is accomplished by placing the filter with food sample onto tryptic soy agar supplemented with 0.15 percent magnesium sulfate (Brodsky, Boleszczuk, and Entis 1982).

The impedimetric system has been adapted for use in determining coliforms in water and food (Firstenberg-Eden et al. 1984; Strauss, Malaney, and Tanner 1984). Although this system is more rapid than the MPN system, Tenpenney, Tanner, and Malaney (1984) suggested that it is more expensive and the results deviated considerably from those obtained with the MPN system.

Growth and Survival

Coliforms include psychrotrophic types capable of multiplying at 3° to 10°C, or refrigerator temperature. They grow well on most carbohydrates and proteins.

Members of the coliform group may persist in water, soil, or foods for long time periods. Usually, the organisms are not able to withstand the pasteurization of milk. Therefore, the presence of coliforms in pasteurized milk indicates either improper pasteurization or recontamination after the heat treatment.

The organisms are rather hardy under natural conditions and resist drying. Coliforms do not withstand the rigors of freezing or frozen storage very well. The ultraviolet light in sunlight is bactericidal to coliforms.

Significance of Coliforms

There are good and bad aspects of using coliforms as indicator organisms. Since members of this group are found in the intestines and discharges of people and animals, their presence could be due to fecal contamination, and enteric pathogens could also be present. Unfortunately, since some members have nonfecal origins, the presence of these members in a food would not indicate fecal contamination or potential pathogens. One can rationalize this predicament by saying that using nonfecal coliforms as indicators gives an extra margin of safety. Finstein (1973) called the coliforms the most conservative indicator.

Since coliforms can multiply outside the animal body, their presence in high numbers in a food product may not be indicative of original contamination, but of improper handling, which allowed multiplication of the organisms.

A good indicator should be present if an enteric pathogen is present. In one study, data showed that 11.4 percent of protein feed supplement samples with a coliform count of less than one per gram contained salmonellae (Loken et al. 1968). Of 607 samples with less than ten coliforms per gram, ninety-two (15.2 percent) contained salmonellae. Only 42 per-

cent of the samples with over 100 coliforms per gram revealed salmonellae. Hence, there is some question as to the correlation of the presence or absence of coliforms to salmonellae. After examining their data from raw and ready-to-eat foods, researchers stated that the coliforms cannot serve as an indicator of potential pathogens in these foods (Solberg et al. 1976).

Smith (1971) stated that persons with severe diarrhea often do not excrete coliforms, but primarily excrete the enteric pathogen. He questioned the use of coliforms as an index of contamination.

Coliforms in milk products generally are considered to be of animal origin and tend to indicate the hygienic standard of the product and its keeping quality, rather than the presence of human pathogens. The more carefully the milk is produced and handled, the lower the coliform count. Hartley, Reinbold, and Vedamuthu (1968) stated that the coliform count is of only limited value with fresh, high-quality milk, and of no value for older or low-quality milk.

No statistically significant correlation has been found between enteric viruses in clams or clam meat and the water from which the clams were harvested (Wait et al. 1983). The number of coliforms in shellfish fluctuated with rainfall (Hackney, Ray, and Speck 1980).

The value of the coliform group as an indicator is both questionable and acceptable. In some products, the coliforms may indicate fecal contamination from either human or animal sources or from soil, poor sanitation of equipment, faulty pasteurization techniques, or recontamination after pasteurization or cooking. Because of their ubiquity in nature, the ability to multiply outside the animal body, and the nonfecal habitat of some strains, the coliform group might not be a reliable indicator of the presence of pathogens in food. However, strains of non-*E. coli* coliforms, in addition to strains of *E. coli,* are enterotoxigenic (see Chapter 6).

FECAL COLIFORMS

To overcome the criticism of nonfecal coliforms as indicator organisms, there have been suggestions to determine only the fecal coliforms. In 1904, Eijkman discovered that intestinal coliforms produce gas in glucose broth incubated at 46°C, and nonfecal coliforms fail to grow.

The fecal coliforms can be defined as Gram-negative facultative rods that ferment lactose at 44.5°C. The fecal coliforms consist primarily of *E. coli,* but a few *Enterobacter* and *Klebsiella* strains can produce gas in lactose broth at 44.5°C (Duncan and Razzell 1972).

Sources

The fecal coliforms are relatively specific for fecal material of warm-blooded animals. Fecal coliforms were reported on fresh produce (Duncan and Razzell 1972). Also the effluent from a potato-processing plant as well as four fruit and vegetable canneries contained fecal coliforms (Menon 1985). Fecal coliforms can be found in animals and plants and foods derived from them, as well as in contaminated soil and water.

Methods

The usual procedure is to inoculate growth from positive presumptive coliform tubes into tubes of EC medium (Hastback 1981) and incubate these at 44.5°C for 24 hr. The incubated broth is examined for gas production, and the number of fecal coliforms estimated with an MPN table.

The direct inoculation of a sample into EC broth does not reveal all of the fecal coliforms. An enrichment in a presumptive broth is needed to attain a level of cells that will produce gas when inoculated into EC broth and incubated at 44.5°C.

With this test, a small portion of fecal coliforms is excluded and a few nonfecal coliforms are included in positive tubes. The use of both LST (48 hr) and EC (24 hr) broths requires a total of 72 hr for incubation. However, Hastback (1981) found that incubation of LST for 48 hr, as compared to 24 hr, in most cases increased the count less than 2 percent.

An A-1 broth was described by Andrews and Presnell (1972) for the determination of *E. coli* in raw seawater. The inoculated tubes of A-1 broth were incubated at 44.5°C for 24 hr. This system has been modified so that the inoculated tubes are incubated for 3 hr at 35°C (to allow repair of stressed cells), followed by 21 hr at 44.5°C. Although this modified A-1 procedure was found to be satisfactory for analyzing various substances (Hunt et al. 1981; Standridge and Delfino 1981; Varga and Doucet 1984) and was given official status by the AOAC in 1980 for detecting fecal coliforms in seawater, one study stated that the reproducibility of recovery of fecal coliforms depends upon the food being analyzed and recommended that the technique be used only as a screening procedure (Andrews et al. 1981).

A membrane filter (MF) method can be used. After filtration, the MF is saturated with M-FC broth and incubated at 44.5°C for 24 hr. Blue colonies, observed with low-power magnification, are counted as fecal coliforms. Researchers developed media and systems for resuscitation of stressed cells for detection of fecal coliforms with the MF method (Du-

four, Strickland, and Cabelli 1981; LeChevallier et al. 1984.) A hydrophobic grid membrane filter system has been tested for fecal coliforms and *E. coli* detection (Entis 1984). The method was recommended for adoption by AOAC as official first action.

Besides the elevated temperature test, the IMViC tests can be used to differentiate coliforms, according to the reactions as listed in Table 7.1. There are some problems associated with the IMViC system. Both the *Klebsiella* and *Enterobacter* are IMViC − − + +. *Klebsiella* can be found in feces, industrial wastes, or nature, while *Enterobacter* usually is found in nonfecal sources. Further, this biochemical testing requries the isolation of a sufficient number of colonies from an agar surface so that the tests are meaningful. These time-consuming and laborious procedures are not adaptable on a routine basis in many laboratories.

Other systems for detecting fecal coliforms have been suggested, tested, and reported. These include fluorescent antibody (Abshire and Guthrie 1973), coliphage (Kenard and Valentine 1974), gas chromatographic system (Newman and O'Brien 1975), Millipore coli count sampler (Richards 1979), and electrical impedance (Silverman and Munoz 1979).

Growth and Survival

Although the fecal coliforms are associated primarily with the intestinal contents and discharges of humans and warm-blooded animals, they can multiply in an acceptable environment outside the intestines. Bagley and Seidler (1977) reported that 49 percent of the environmental *K. pneumoniae* grew in EC broth at 44.5°C, and 16 percent were fecal coliform positive.

The fecal coliforms are destroyed by pasteurization or normal cooking temperatures. The survival of frozen fecal coliforms depends upon the strain of organism and the substrate (Splittstoesser, Stewart, and Wilkison 1982). When inoculated vegetable homogenates (four types) were

TABLE 7.1. USUAL IMVIC REACTIONS OF COLIFORMS

	Reaction			
Type	I	M	V	C
Escherichia (Type I)	+	+	−	−
Escherichia (Type II)	−	+	−	−
Enterobacter	−	−	+	+
Klebsiella	−	−	+	+
Citrobacter	−	+	−	+

I = indole, M = methyl red, V = Voges-Proskauer, C = citrate utilization.

held at −10°C for 200 days, the survival of fecal coliforms (six strains) ranged from 0.014 to 75 percent. The *E. coli* strain (0.014 percent survival in broccoli) when rerun, survived storage at a level of 7.7 percent.

Significance of Fecal Coliforms

The fecal coliforms are more closely related to fecal contamination than are the total coliforms. This should be considered an advantage; however, critics state that testing for only fecal coliforms does not give a margin of safety that is important when using indicator organisms.

The organisms can originate from improperly sanitized working surfaces in a processing plant. In these cases, their presence would reflect the quality of sanitation and not the direct pollution of the product.

No salmonellae were found in fresh or stored shellfish with a level of fecal coliforms below 230/100 g (Hood, Ness, and Blake 1983). Higher levels of fecal coliforms were limited in predicting the presence of salmonellae.

Researchers found that the presence of *V. cholerae* in shellfish was not adequately indicated by fecal coliform standards for shellfish-growing waters and market shellfish (DePaola et al. 1983).

The absence of fecal coliforms in disinfected sewage effluents does not assure the absence of viruses (Berg et al. 1978). The level of fecal coliforms in oyster-growing waters does not reflect the level of virus contamination (Ellender et al. 1980).

ESCHERICHIA COLI

E. coli is the most prominent fecal coliform. Hence, we might expect that there is little difference between *E. coli* and fecal coliforms. Since the main habitat of *E. coli* is the intestinal tract of humans and warm-blooded animals, some microbiologists believe that only the presence of *E. coli* is indicative of fecal contamination of food.

Sources

E. coli is a common inhabitant of the intestinal tract of humans and animals. The number of this organism varies in different animals, but an animal will excrete from 130 million to over 18 billion *E. coli* each day. Hence, there is a close association between *E. coli,* fecal material, and, possibly, enteric pathogens. It is difficult to completely prevent contamination of animal carcasses with *E. coli* during slaughter.

The organism can be found on plants and plant foods, although usu-

ally not in great abundance. The inoculation of cells onto dry soil resulted in a 99 percent reduction in 24 hr (Chandler and Craven 1978). The survivors persisted for eight weeks. The survival of *S. typhimurium* cells was similar to that of *E. coli.*

Marcus and Amling (1973) detected *E. coli* on 23 percent of the samples of pecans harvested from orchards in which animals had grazed, but from only 4 percent of the samples obtained from ungrazed orchards.

Hall, Brown, and Lewis (1967) analyzed various foods for coliforms and *E. coli. E. coli* was found in cheese and cheese products (75 percent of the samples), fish, and seafood (30 percent), raw and frozen vegetables (15.8 percent), prepared and convenience foods (50 percent), raw meats (76.5 percent), and sandwiches (16.3 percent). Only thirty of seventy-four raw beef patties contained ten or more *E. coli* per gram (Surkiewicz et al. 1975).

Bettelheim et al. (1974) suggested there is a difference in the serological groups of *E. coli* isolated from humans and from animals. The strains of *E. coli* isolated from meat generally resembled animal strains. Typically, human serotypes found on meat are believed due to contamination during handling. According to the review of Mason and Richardson (1981), the types or strains of *E. coli* in human feces are changing constantly. Some strains persist for several weeks before being replaced, while other strains are merely transients. Hinton, Linton, and Hedges (1985) reported a similar finding for *E. coli* in the feces of calves.

Methods

The *Bacteriological Analytic Manual* (FDA 1978) test for *E. coli* consists of inoculating growth from presumptive positive coliform LST broth into EC broth and incubating at 44.5°C. Growth from those EC broth tubes showing gas production is streaked onto EMB agar and incubated (35°C for 18–24 hr). Typical *E. coli* colonies (nucleated with or without a sheen) are tested using the IMViC reactions and Gram stain. The designation in Table 7.1 is used to classify the *E. coli.*

In a review, Silliker (1982) stated that all the imprecision of the coliform test is inherent in this MPN method. It is also long and laborious, requiring up to eight days for completion.

A rapid method requires inoculating and incubating LST broth at 44.5°C, and then streaking on EMB agar. Typical *E. coli* colonies on EMB are confirmed with the IMViC tests.

Although incubation of the EC broth for 48 hr will yield slightly more *E. coli,* the number of false-positive cultures increases. Therefore, to obtain optimum specificity, the incubation period should be limited to 24 hr.

The MPN method (FDA 1978) and an MF system (Anderson and Baird-Parker 1975) were compared for enumerating *E. coli.* One team of researchers reported higher recovery of *E. coli* from beans, cheese, and ground beef with the MF system (Sharpe et al. 1983), while another team reported higher recovery from estuarine water and shellfish with the MPN method (Motes, McPhearson, and DePaolo 1984). Apparently the type of food influences the method of choice for the enumeration of not only *E. coli* but also other types of microorganisms.

A rapid *E. coli* assay using 4-methylumbelliferone glucuronide (MUG) was developed by Feng and Hartman (1982). Glucuronidase, an enzyme produced by *E. coli* and some strains of salmonellae and shigellae, hydrolyzes MUG to a product that is fluorescent when viewed with long-wave ultraviolet light. They found this system to be sensitive and rapid for the detection and enumeration of *E. coli.* Further work has revealed the MUG assay to be accurate, rapid, simple, and inexpensive to perform (Alvarez 1984; Koburger and Miller 1985; Petzel and Hartman 1985; Robison 1984). However, the assay may not be useful for all foods, or certain modifications are needed for it to be valid (Koburger and Miller 1985; Petzel and Hartman 1985).

Survival

The type of food influences the rate of death of *E. coli* in frozen storage. In orange concentrate, the organism may survive only one day or as long as one week. In a product such as frozen gravy, the death rate is lower and *E. coli* may survive frozen storage for several months. Some cells of most strains tested survived at least 200 days in frozen vegetable homogenates (Splittstoesser, Stewart, and Wilkison 1982).

E. coli and salmonellae decline in numbers at a similar rate in many refrigerated foods. The reduction in *E. coli* and salmonellae is similar during spray-drying. The thermal resistances of *E. coli* and salmonellae are similar. In filter-sterilized water, the survival of *E. coli* and *S. typhimurium* are essentially the same over normal seasonal temperatures (Mitchell and Starzyk 1975).

Significance

The source of *E. coli* is generally the intestinal tract and feces of warm-blooded animals and people. Hence, it should be an excellent indicator organism. On the other hand, if fecal coliforms lack a desired margin of safety, it is an even greater problem with the use of *E. coli* alone as an indicator.

The determination of *E. coli* includes any problems associated with

the enumeration of coliforms plus the added necessity to do additional procedures to confirm and complete the tests for these specific organisms. Thus, the tests for *E. coli* are not as acceptable as for those for coliforms, and the direct enumeration of pathogens might be considered.

According to Evison and James (1973), *E. coli* is a reliable index of fecal pollution of water sources in temperate climates. Other researchers reported that with increasing contamination of coastal water with *E. coli,* the proportion of positive salmonellae samples also increases (Gärtner et al. 1975).

In many food products, the survival of *E. coli* is similar to that of salmonellae. However, many pathogens may persist after *E. coli* is destroyed. On the other hand, the presence of *E. coli* does not mean that enteric pathogens also are present. Lovell and Barkate (1969) found that more than 92 percent of the samples of crayfish contained *E. coli,* but only 3 percent contained either coagulase-positive staphylococci or salmonellae. Conversely, salmonellae have been found in food samples with no *E. coli.* The presence or absence of *E. coli* in oysters was not indicative of the presence of enteric viruses (Fugate, Cliver, and Hatch 1975).

The finding of high coliform or *E. coli* counts in a food has been the basis for seizure by regulatory agencies on the basis that the food "consists in whole or in part of a filthy, decomposed or putrid animal or vegetable substance."

ENTEROBACTERIACEAE

Those enteric bacteria in food that do not ferment lactose may be more important in food than those that do ferment lactose (coliforms). Therefore, it has been suggested that the entire family Enterobacteriaceae be determined and used as an indicator group.

If the use of coliforms as indicators is criticized because of nonfecal origins, some members of this family deserve to be criticized. Some Enterobacteriaceae occur as saprophytes in nature, some are parasites for plants, causing blights and soft rots, many are found in the intestinal tract of animals, and others are potential pathogens of humans and animals. Thus, the direct relationship of the organisms of this entire family to fecal contamination or potential pathogens is questionable.

Researchers found Enterobacteriaceae on poultry carcasses at various stages of chilling (Cox et al. 1975). *Salmonella* and *Shigella* were not found. As pointed out, this does not mean that these organisms were not present. The most prevalent isolate was *Escherichia,* accounting for 87 to 96 percent of the detected Enterobacteriaceae. Thus, the use of *E. coli* or fecal coliforms as indicators would reveal essentially the same informa-

tion as the Enterobacteriaceae. There are many foods in which the coliform count and Enterobacteriaceae count are almost the same.

The Enterobacteriaceae most frequently detected on various foods were *E. coli, Enterobacter agglomerans, Serratia liquefaciens, Klebsiella pneumoniae, Enterobacter cloacae, Serratia marcescens,* and *Erwinia herbicola* (Cox et al. 1979; Mercuri et al. 1978; Oblinger, Kennedy, and Langston 1982; Stiles and Ng 1981). The Enterobacteriaceae are of little value as indicators of contamination or growth of enteric pathogens on refrigerated raw meats (Shaw and Roberts 1982). There was no relationship between Enterobacteriaceae counts and *Salmonella* in broilers (Mercuri et al. 1978).

The methods suggested to enumerate the organisms were evaluated by Drion and Mossel (1977). They stated that testing two samples of 1 g of food for Enterobacteriaceae and allowing no positives gives the same degree of safety as testing 25- to 60-g samples for salmonellae and allowing no positives.

FECAL STREPTOCOCCI ENTEROCOCCI

Because coliforms—fecal coliforms and *E. coli*—are not very resistant to heating or freezing, other organisms have been suggested as indicators. The fecal streptococci and enterococci are potential indicator organisms.

The enterococci are in Lancefield's group D of the genus *Streptococcus.* Since references are made to the group D streptococci, the enterococcus group, and the fecal streptococci, some distinction should be made between these categories.

S. faecalis and *S. faecium,* along with two subspecies of *S. faecalis* (*liquefaciens* and *zymogenes*) comprise the enterococcus group. These organisms, plus *S. bovis* and *S. equinus* are the group D streptococci. Buchanan and Gibbons (1974) included *S. suis* and *S. avium* as members of group D. The fecal streptococci include all of these group D streptococci, and *S. mitis* and *S. salivarius.*

S. avium has been listed in group Q, but it also possesses the group D antigen, is prevalent in chicken feces, and is also present in animal and human feces. In most cases, the growth of *S. faecalis, S. faecium,* and the group Q streptococci are not distinguishable. Thus, the group Q streptococci (*S. avium*) may be enumerated and reported as enterococci. *S. avium* fits most of the characteristics of the enterococcus group, except it neither tolerates 0.1 percent methylene blue, nor hydrolyzes arginine.

The definitions and the organisms that are included are not necessarily those listed in some of the literature. The streptococci have been studied for many years, but the status and interrelationships of the species